SNOWMOON

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Chapter 1

Meldan, Veridia · 3724 SNOWMOON 3

Gladias was walking along a foot path, with tall trees towering on both sides. A series of houses made out of large stone bricks lay behind the trees on both sides, medieval in style but without any of the junk, dust on the ground, or other imperfections that a real medieval structure would have had a thousand years earlier. A few of the houses featured small signs advertising clothing, flowers or food available for sale. Each house seemed to meld seamlessly into the scattered leaves, plants and trees that surrounded it.

He looked around, breathed in and briefly stopped to enjoy the nature, dense enough and quiet enough that he almost forgot that he was in the middle of the Kalimar district of Meldan, the capital city of Veridia.

Suddenly, a small drone flew over his head. Gladias instinctively checked his watch. Almost immediately, a green checkmark lit up. Even before he could fully process what had happened, his watch had engaged in a zero-knowledge authentication protocol, and received a cryptographic attestation verifying that the drone was a toy, and its navigation algorithm had a safety override that would prevent it from coming within five decimeters of a human without permission. His watch had learned that the drone was not a threat - and nothing else.

The drone flew off, and continued above a sidewalk branching off to the left. For a moment, Gladias could briefly see its shape: a small ball, with feather-like rotors on three sides. And the very next

moment, a young boy jumped up, and grabbed it with his hand while it was still in the sky.

"Five points for me!", the boy shouted excitedly.

Gladias let his attention wander back to peacefully observing his surroundings. The boy who had just caught the ball was part of a family with four children, the younger two each riding a tricycle.

On another sidewalk branching off to the right, Gladias could see a grandmother with a small child. As they approached a bench and seemed to stop, a couple that was already sitting on the bench pre-emptively stood up to offer their seats. The grandmother thanked them, and sat down with her child.

And playing into Gladias's ear this whole time, there was an audio program from Emerald, the local AI running from Gladias's hand device, explaining the nuances of airborne virus transmission and how to calculate how much a given change to a building's ventilation or density might affect total city-wide cases in an epidemic.

* * *

As Gladias walked forward, the houses soon gave way to thick green shrubbery. As soon as the shrubbery cleared, the foot path started gently curving upward. Smoothly, and suddenly, the path transitioned into a sky bridge above a busy road, with many cars and autobuses and a few autonomous delivery vehicles passing beneath him. A chilly breeze blew into his face.

"Pause, I want to enjoy the view", Gladias said to Emerald, out of habit whispering the sentence so softly that only his throat muscles moved and no sound at all left his lips. The silken cloth band tightly covering Gladias's neck picked it up perfectly.

He stopped in the middle of the sky bridge, turned to his left side, and looked forward. The busy road beneath him continued for about two kilometers, and then turned rightward. Beyond that point, he could see a mountain covered with trees. A few apartment buildings about fifteen storeys tall, decorated and shaped like over-sized medieval castle towers though with far more windows on the sides,

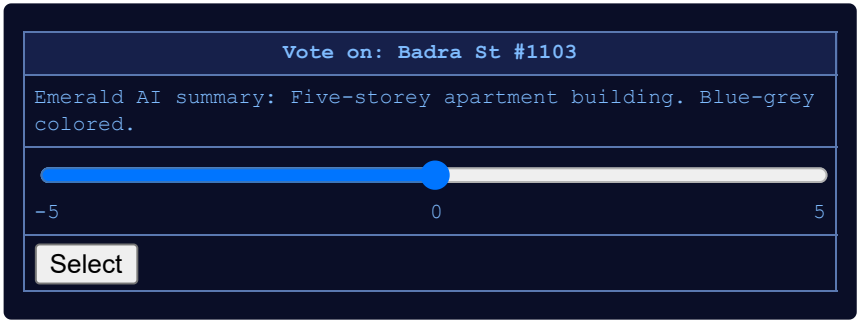
poked up high above the trees in several places on the lower half of the mountain slope.

Gladias turned his face left once again, and took one last look at the natural world. The nature in Veridia is beautiful, he thought. He then turned his face to the right, where the road continued past the sky bridge with a much higher density of buildings. Today, his job was to offer opinions on the human world.

Gladias began to walk forward once more, and stepped off the sky bridge. On the left side, there was now a row of five-storey apartment buildings, interspersed with walking paths and small shops selling various kinds of goods. Both the walking paths and the sidewalk were protected from the sun by tall green trees.

On the right side, there was now a tall soundproof barrier, painted with green and blue colors, barely visible behind another even thicker row of tall trees.

Shortly after he began walking beside the soundproof barrier, his hand device buzzed.



Gladias immediately knew what he was being asked to do. As a citizen of Veridia, he had been randomly selected by cryptographic sortition to vote on how much he liked the building.

The goal of the voting, Gladias remembered from the Veridian civics lessons of his youth, was to determine the public aesthetics tax - a major component of the composite land tax - that each building would have to pay. Developers paid a lot of attention to this. In

Veridia, making buildings that the public found appealing was not just a chance to show off their craft - it was profitable.

Gladias looked at the building, the blue-grey colored Badra St #1103. The structure seemed in-character, not too different from the other buildings on the same street. The style was modern, but tasteful, nothing like the soulless rectangular concrete buildings that he was accustomed to seeing in pictures of places like Freetown. It had footpaths on both sides, and both footpaths were covered by trees - just like the other buildings nearby.

This was hard, Gladias realized. To get some inspiration, he waved his hand device, and took a look at the land tax rates that some of the *other* nearby buildings were paying. First, the building that he was supposed to vote on.

Badra St #1103	
Base land tax:	1.00%
Aesthetics tax:	0.18%
Other rubric taxes:	0.33%
Total land tax:	1.51%

He then turned to his left, and zoomed his hand device in to point at the first few genuinely different buildings he could find: one building made with an architectural style mixing wood and stone, another with a garden on the roof, and a third, smaller but more luxurious looking, that was shaped like an octahedron, and had a beautifully arranged square garden along the perimeter of the territory with a maple tree brimming with red leaves at each corner. All three paid aesthetics taxes significantly lower than Badra St #1103, and one was even slightly negative.

He then zoomed in to another building that he could barely make out in the distance. At this distance, the trees thickly lining the streets

cleverly covered it up, but his hand device easily locked on to it and downloaded both a close-up image and the tax percentages.

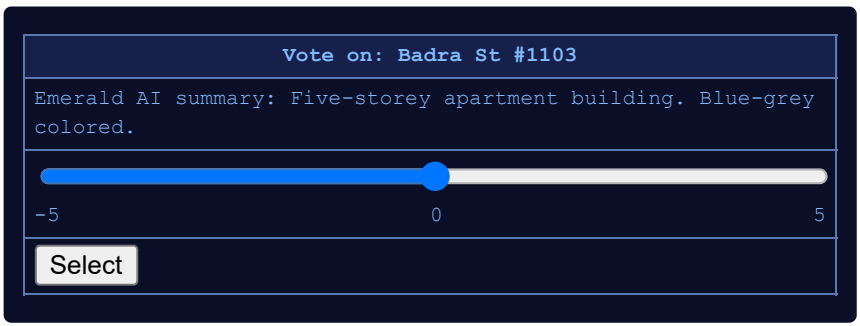
The building itself was a corporate headquarters for an aviation company. The company had had the bright idea to make their office look like an aeroplane. It was impressive architecture, he thought, but he would certainly demand some compensation if he had to walk by it every morning.

He checked its taxes.

Clearhill Ave #980	
Base land tax:	1.00%
Aesthetics tax:	0.59%
Other rubric taxes:	0.30%
Total land tax:	1.89%

Well deserved, Gladias thought. Though he was glad that Veridia's institutions let people with tastes that were so different from his express themselves, even if they had to pay for the privilege.

Gladias's mind snapped back to his task. He was voting on the blue and grey building right in front of him. He took a look at the interface again.



This was quadratic voting, he remembered. All his votes would be automatically shifted, stretched or squeezed so that the total average of the votes becomes zero, and the average of the squares of the votes becomes one. So there was no point in voting up on everything, down on everything, or overly extreme on everything. It was best, *mathematically provably best*, to vote not just the *direction* you felt right, but also with the *strength* you felt right - not more, not less.

Gladias's mind struggled. The aesthetics taxes given to each building seemed ... already fair? Maybe the aeroplane building was perhaps being treated a little *too* harshly. Allowing such a building to stand is not just a concession to people's freedom to do their own thing on their own land, it also brought genuine variety to Meldan. Maybe it's okay to have in what was already a corporate district.

Then he reminded himself that he was not voting on the aeroplane building, he was voting on Badra St #1103.

Right, he thought. What decision to make? Support or oppose?

After thinking for a moment, he finally decided by raw instinct, and pushed the slider only a slight amount toward the right.

He hit "Select".

That was just a warmup, he thought - a random fluke that he had been assigned an aesthetics vote as a Veridian citizen right before one of his more consequential votes as an Acolyte in the Order of Steering. That vote had been hard. The vote that he was about to do would be far more challenging - and there, the system would be keeping score.

He continued walking, and after a minute he arrived at an opening in the soundproof barrier: the entrance to the concert that he was attending.

* * *

Gladias held his hand device near a circle with a glowing green outline on the entrance gate. He waited briefly as his hand device produced a cryptographic proof, and then a machine embedded in the entrance gate verified it.

"Verified, welcome", the machine said.

The gate learned that someone holding a valid and unused credential had arrived - and nothing else.

Gladias passed through the gate, and headed toward the main concert area.

He could see a few of the attendees wearing the same signature Veridian Privacy Robe that he was wearing. This was a loose-fitting dark purple dress that covered the full body from the hood covering the wearer's head down to the ankles. Below the ankles, privacy robe wearers all wore shoes of the same dark purple color.

The other attendees were mostly wearing simple shirts, either plain colored or featuring slogans of their favorite bands.

Some were moving forward with clear intent and efficiency, just like Gladias. Others were dancing. Others were chatting with each other. And still others were walking forward calmly, simply trying to passively enjoy the show.

Gladias emerged into a large open field surrounded by a high soundproof barrier - the same barrier that Gladias saw before, but from the inside.

Gladias pushed through the crowd, as the noise became louder. He pulled on a strap on his privacy robe's front face cover, tightening a belt that separated the area below his nose from the area above his nose. The program on airborne virus transmission that he had been listening to only minutes earlier had primed him to take care to not get sick.

He soon got to the concert stage area, and was able to clearly hear the gnarly voice of Dreadknot's lead singer.

WE MEET THEM WITH OUR FURY IN OUR HEARTS
AND FLAY THEIR FLESH WITH SHINING STEEL AND DARTS

Gladias took out his hand device again.

Most people attending the concert were there to enjoy the music and hang out with their friends. For Gladias, this was certainly not his *ideal* entertainment - his preferred music was of a much softer and

gentler variety - but he did not *dislike* it either. More than anything else, he found the concert interesting in an anthropological way, exploring the interests and emotions of people he would otherwise have little interaction with.

But ultimately, Gladias was at the concert for quite a different purpose. Gladias checked the rubric on his hand device:

Basic summary

Tier	Criteria	Tax rate
1	• Content strongly promotes two or more of the following: personal development, scientific and technical education, kindness toward others, healthy family relationships, good conflict resolution practices, civic duty • Does not meet any Tier 4-5 criteria	0%
2	• Content weakly promotes one or more of the following: personal development, scientific and technical education, kindness toward others, healthy family relationships, good conflict resolution practices, civic duty • Does not meet any Tier 4-5 criteria	10%
3	• Content does not meet any Tier 4-5 criteria	20%
4	• Content weakly promotes one or more of the following: violence, lawless behavior, use of anti-recommended substances, gambling or risky investment, excessive display of wealth, poor conflict resolution practices, religious or ethnic hatred	30%
5	• Content strongly promotes two or more of the following: violence, lawless behavior, use of anti-recommended substances, gambling or risky investment, excessive display of wealth, poor conflict resolution practices, religious or ethnic hatred	40%

Tax rates increased by 2.5x on proceeds from Veridian national Quadratic Funding and Graph Funding.

View relevant precedents Back

In Veridia, very few things are strictly *banned*. If Dreadknot wanted to, they could have all their songs repeatedly shouting "Hail the Arctic Emperor" and "we hope all government officials get painfully murdered" over and over again, and they would never be shut down or go to jail. But what Veridia *does* have is intricate systems of taxes and subsidies, maintained by a government-linked but decentralized collective called the Order of Steering.

One part of this collective, the Keepers, voted on tax rubrics that determined what the taxes are. Another part, the Sentinels, audited individual businesses, each audit assigning a particular business a position on a particular rubric. Gladias was an Acolyte: an Order member in training.

And today, Gladias's task was to evaluate Dreadknot's position on the tax rubric for social impact of public performances.

THE SMOKE OF CITIES BURNING IN THE NIGHT
WE LEAVE NO PLACE TO HIDE FOR ALL IN SIGHT

Gladias clicked "View relevant precedents", and scrolled through them. He pressed a side button. A small popup appeared on the top right: "Emerald listening mode active". He did not speak; instead, he simply let Emerald listen to Dreadknot sing. But even after five minutes of listening, only a few relevant precedents showed up.

Gladias switched the mode to "analysis" and kept it listening for another five minutes. Emerald listed some conclusions that sounded reasonable, but Gladias could tell it was unreliable. There was just not enough source data on this kind of music for the Emerald AI to give good answers, and ultimately, it was Gladias's job to figure out when to defer to Emerald and when to let his own instinct have more of a say. Here, instinct it was.

Well, the song was definitely violent.

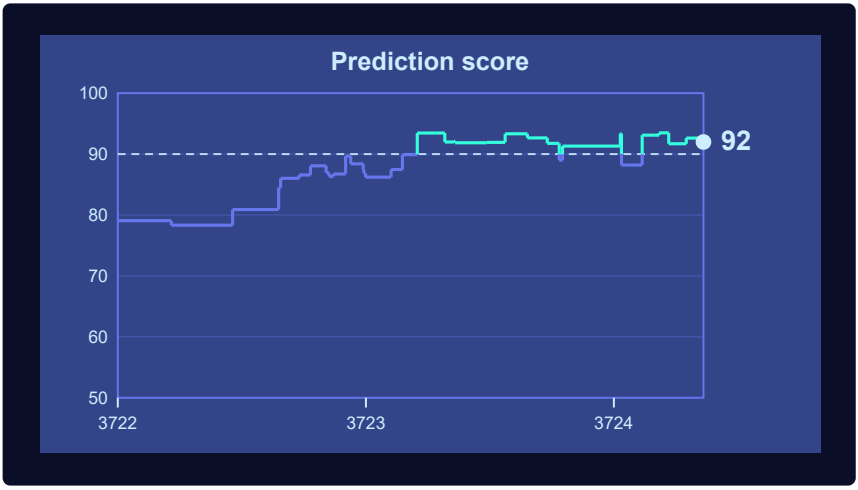
But on the other hand, over a century of Veridian civics lore made a distinction between *advocating* violence, and merely *showcasing* it for entertainment value. Gladias could see and hear the audience dancing,

cheering and enjoying the music. They seemed to be having fun, not rising up in hatred against anyone.

But on the *other* other hand, Gladias knew that Sentinels in recent times paid less attention to that distinction than before, and seemed to value a softer and gentler society almost as an end in itself.

Gladias was not quite sure if this was what the Keepers who created that rubric intended, or if it was the right call for society. In a few months, he would be a Sentinel in full standing, and would have more leeway to decide. But for now, his freedom to decide was limited: the system was keeping score.

Gladias looked down at his hand device.



Ten percent of the time, any audit made by Acolytes would be repeated by Sentinels. An Acolyte's score reflected how well that portion of their votes predicted the Sentinels' judgements. Today, Gladias's score was ninety two. If it stayed above ninety, he would get in and become a Sentinel - or, if he changed his mind, a Keeper. But if his score dropped by even three points, his career would be on hold for at least another year.

The rest of the screen was filled with chat messages. There were nine Acolytes assigned to this audit, split into three groups of three.

The other two members of his group could not agree. One was insisting on Tier 3 - a sort of neutral judgement - and the other was harsher, demanding Tier 4.

Gladias sighed. The other two groups were probably thinking the same way. If we simplify and say every other Acolyte was a coin-flip between Tier 3 and Tier 4, then by the binomial formula, the probability the other six would also be evenly split was six choose three - 20 - divided by 64, or about thirty-one percent. Now if the other Acolytes' votes were *not* coin-flips ... eh, calculating that exactly is too hard, but it's definitely lower, let's just say twenty percent. A twenty or thirty percent chance his vote would significantly affect Dreadknot's tax rate for the next year. Oh great. Actual responsibility.

Gladias looked around once more at the crowd. The first song finished, and he heard them cheering. He felt a burst of courage. He would take the ten percent chance of a hit to his prediction score. Without thinking further, he pressed Tier 3.

For a brief moment, he felt proud of his decision.

"HEY EVERYONE, IT'S AN ORDER MEMBER!", someone behind him yelled.

* * *

Gladias's body froze in shock for a moment. Before his mind could catch up, his eyes had already scanned the venue, and discovered the nearest exit.

He immediately understood what had happened. The security of the Order, particularly preventing influence and bribery, rests on the privacy of its members. Order members are strictly forbidden to reveal their assigned task. To make sure that Order members take their privacy seriously, they were kept on their toes by a sort of deliberately engineered game. Anyone can connect to the decentralized cryptographic network that manages the Order's operations, and send in a guess about what a particular Order member's assigned task is, along with a small deposit. If their guess is correct, the Order member's salary is docked, and the discoverer gets half as a reward.

Before he could react any further, another man nearby, also wearing a privacy robe, broke and ran toward the fire exit on the right. Everyone within ten meters took out their hand devices and pointed them at him, recording video. Gladias immediately understood what had happened here too: a decoy. Some member of the public altruistically diverting attention to himself, to give Gladias a chance to escape.

Gladias turned around, intending to run. But before he could take a step, he saw right in front of him a large burly man, pointing a hand device straight at his face.

Gladias froze. At first, he thought that he was done for. His mind raced through the possibilities, looking for any possible way out. Finally, Gladias the thoughtful philosopher gave way to a different Gladias, more imaginative and mischievous - a Gladias that had not been awake since recess in middle school.

Gladias looked toward the right.

"Wait, is that a new screed from the Arctic Emperor?", he asked.

The man, confused, looked to where Gladias was looking. At that moment, Gladias snatched the man's hand device from his arm, and suddenly turned around.

"Delete all audio, images and video from the last longhour. Reboot."

He began to run. A few ticks later, the device responded, "4 files deleted", and briefly showed screenshots of some photos and videos. The one on the bottom right was of Gladias's face. The screen turned black.

Gladias threw the hand device up into the air, hoping that his pursuer would choose not losing his device over chasing him. He continued running.

Despite being slowed down by his shoes, which had high heels to raise his height to the same level as other robe wearers and deliberately uneven soles to randomize his gait, Gladias managed to quickly reach the exit. Distracted by the rest of the commotion, the crowd seemed not to notice.

He exited the concert venue, and hopped onto a local autobus.

* * *

The autobus was nearly empty. To his left, Gladias saw two other people, both wearing privacy robes. Were they male, female, young or old, he could not tell. To his right, he saw a teenager wearing ordinary clothes and a backpack, who seemed to be coming home from school.

His eyes wandered to the advertisements on the inside walls of the autobus.

The first advertisement was a poster reminding Veridians of the importance of lifelong learning. On the left side of the poster was a picture of a classroom with a child, a middle-aged woman, and a much older man, all reading textbooks together. On the right side of the poster was the text:

> *School lasts until age 20. Education is every Veridian's responsibility and duty for a lifetime.*
>
> *Learn about Veridian Graph Funding-subsidized education programs for all ages today.*

The second advertisement was for Hydrafill, a brand selling water and various kinds of drinks. Nudged gently by Veridian Steering taxes, they had over the last twenty years become almost completely sugar free.

On the left side of the advertisement, was the classic yellow and green colored Hydrafill bottle. On the right side of the advertisement was the text:

> *Hydrafill*
>
> *Our drinks are proudly free of over 2,000 known poisons and anti-recommended substances.*

Gladias smiled.

He wished that it had been the Hydrafill advertisement that he had been selected to vote on, and not the blue and grey building. He would have slid the slider almost all the way to the right for this one, he knew.

We need more fun in the world.

* * *

Fifteen minutes later, Gladias stepped off the autobus. After a short walk along a street, he arrived at his house. He opened the door.

"Are you home, Seila?", he shouted.

"Was just writing up my blog post", she replied, and came out of her room to greet Gladias.

"So, what do we want for dinner?"

"Well, the Dzego food truck is coming in seven minutes, how about we have that again?"

"Sure! I assume the number ten for you again?"

"Yep!"

"Okay, I'll have the number thirteen this time."

Seila took out her hand device, and began to place the order.

They heard loud footsteps coming down the stairs.

"Dad is home!", the younger boy, Zven, shouted. Zven jumped down the stairs three steps at a time, and ran up to hug Gladias, who obliged.

The older teenage boy, Febric, walked down more slowly, carrying his hand device in one hand, attached by a cable to a larger compute box in the other.

"We're ordering from the Dzego food truck. What do you guys want?", Gladias asked.

"Number four!", Febric answered.

"Number five, I win!", Zven shouted.

"What about Lily and Hreda?"

"Lily's in the bathroom. Hreda will be back from after-school number theory in a longhour. Come on, you know she always gets the number nine."

The bathroom door opened, and Lily slowly walked out, her eyes looking to the ground.

Seila sighed. "What happened?"

"Sixty percent on the history exam. Whatever..."

Seila turned, and put her arms around Lily. "It's okay to not be good at everything", she said softly.

Lily turned her face further away from Seila and the family. She ran forward, and sat down on the couch.

Seila sighed, and ordered Lily the number eight.

Seila clicked order. Gladias, Seila, Zven and Febric sat at the table and waited.

On the couch, Lily took out her hand device, and began intently watching and quickly tapping on the screen, devoting all her attention to some game that Gladias and Seila could never make out.

This was Gladias and Seila's fourth last dinner with the kids - of whom Zven and Lily were theirs, and Febric and Hreda were from their co-family, Uncle Vil and Aunt Daia. In four days, Uncle Vil and Aunt Daia were going to come and pick all four kids up, and take care of them for the next half of the year.

Gladias sighed. He would miss the kids. But at the same time, he was looking forward to the six months of freedom.

"Are we still going to stay with Uncle Vil for six months?", Zven asked.

"Of course!", Seila replied.

"Yay, I love Uncle Vil!"

"For me Vil is *Dad*!", Febric made a point of reminding. He thought for a moment longer. "Uncle Glad and Aunt Seila, I love you both too!"

"And I love you."

A doorbell rang. Gladias walked to the door, opened the door, and got a bag full of food boxes. He set the bag on the table, and took out and gave each person the food box with their number on it. Gladias, Seila, Febric and Zven all started eating.

"Any plans once Vil is here?", Gladias asked.

"I'm going to visit Freetown for a month or two, see Jahn, maybe figure out some blog post to write on the United Cities. I've spent so much time reading about their co-governance, actually looking forward to seeing more how it all connects to what's on the ground."

"Nice! I'm going to go on some long walks in the forest and take a deep dive with the bot on the environment and biology rubrics. Still have four more audits to do before I can become a Sentinel!"

Gladias and Seila both resumed eating. Quietly, Lily walked up to the table, and began to open her food box as well.

At that moment, Gladias's hand device buzzed. He took it out. A single notification popped up.

From		Message	Time
Anonymous Rep score ≥ 200 Verified ✓	→	I saw you at the concert, I know that you are in the Order.	60259

And a few ticks later, another:

From		Message	Time
Anonymous Rep score ≥ 200 Verified ✓	→	Don't worry, I'm not intending to report you for the bounty. But I do have a request that I would like to make.	60266

Chapter 2

Pafogai Du, Dzego · 3724 SNOWMOON 5

Fin and Zei got off the autobus and walked down Hun Min street.

Hun Min street was one of the more lively commercial and industrial streets in Pafogai Du. On both sides of the street, they could see signs with colored blinking lights advertising all kinds of electronics: hand devices, watches, compute boxes, lighting, batteries, sensors and more. Among the stores were miniature factories, building various sophisticated devices and in a few places even computer chips. The buildings were all one or two storeys tall, and many signs were on top of stairways leading underground.

On the left, there were two adjacent stores. One was selling sensors of all kinds - cameras, microphones, biological and chemical sensors, air quality monitors, and of course, sensors designed to detect sensors. The other was selling anti-transmission foil - a common material used by default in many kinds of rooms, to block wireless signals. Privacy and transparency, monitoring and anti-monitoring - the duality of keeping the Dzegojan safe.

Between the two stores, a sign read in large letters:

zui fia kun zun

"Underground safe place".

"gie fe kiu kai ci bin hu", Zei thought to himself. "Grow roots, no head". The strategy that Dzego had been employing for half a century to grow and stay protected.

Fin and Zei both kept walking, enjoying the shade under the trees between the road and the sidewalk.

A drone buzzed as it flew overhead. A moment later, their watches buzzed softly once their software had run the authentication protocol and confirmed that it was a friendly security drone that came from a batch that had passed all inspections and was running friendly algorithms. The whole chain of events - the flyby, the authentication, the confirmation - finished within hundreds of milli-ticks, and Fin and Zei both barely noticed that any of it had even happened.

"How was the Minpentai game last night?", Fin asked.

"I got third place! Finally figured out a wall structure that reflected their gliders right back at them, completely wreaked havoc on their territory."

"How about you?"

"Studying for the geography exam in boring school."

"Boring, I take it?"

"Yeah, we're covering the United Cities, and they're just making us memorize all the co-governance percentages. Freetown gets five percent votes in Devanvil, Devanvil gets fifteen percent votes in Redshire, Redshire gets..."

"Do you have any new opinions on how effective that whole system is?"

"Oh no, critical governance studies only becomes a topic in university."

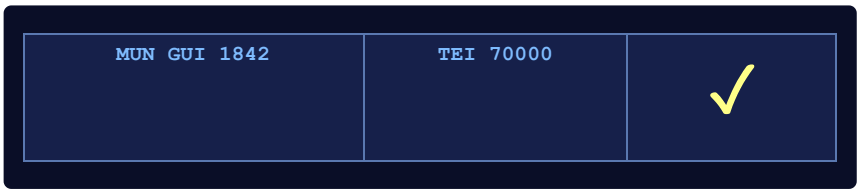
"The same way that chemistry is a topic in high school?"

Fin laughed. "Are they gonna cover an entire year of critical governance studies in ten days in DU now?"

DU - "Decentralized University", or *dzu hu sun du* in Dzegoban - the place they were going. Zei remembered that they needed to make sure they got there on time, and checked his watch.

"Time just hit sixty-nine thousand. The exact location of the classroom should be getting decrypted and broadcasted any moment now."

The watch buzzed. Zei got a notification.



Zei stopped, and showed his watch to Fin.

"Huh, yellow checkmark. Looks like they've upgraded to a new cryptographic proof system, gotta make sure we sync our codes when we get to the lesson today."

"Again?"

"Guess so. I hear the Arctic Empire is getting more aggressive, so we're stepping up security."

"Okay, anyway, it says Mun Gui 1842. We... just passed 1990 Hun Min. So it's behind us."

"Let's just right left here on 20th, and then right again on Mun Gui?"

Zei checked over the math in his mind. Named streets were spaced two hundred meters apart, H to M was four streets. House numbers were just meters, so to get from 2000 to 1842 they had to walk one hundred fifty eight meters. Eight hundred and one hundred fifty eight together is ... nine hundred fifty eight meters to walk. The lesson starts at seventy thousand, so they had a little under a thousand ticks, or more colloquially, ten minutes. A slow walking pace is one meter per tick, and Fin and Zei were fast. So they were fine.

"Sounds good!"

They began to walk briskly, to make sure they could get there with time to spare.

A few minutes later, they arrived. "Oh Fin, there you are!", a young woman shouted. Fin and Zei were led down two flights of somewhat dusty stairs.

When they got to the bottom, another man was waiting, and stopped them. "Change of plans, it's in the opposite room on floor -1

now."

Zei sighed.

The four of them walked back up a flight of stairs, entered a large room, and sat down beside each other on one of the rows of chairs.

The walls were covered with a thick, hastily organized layer of anti-transmission foil. Coated on top of the foil was a classic aesthetic of Dzego: pictures of cute cartoon animals playing with electronic gadgets and chemistry lab equipment, with a backdrop of green trees and blue skies. At the front of the room, there was an inflated cartoon-style pig on the left and a hamster on the right. The pig, in its left hand, was holding a calculator. The hamster, in its right hand, was holding a vial of some green liquid. With their other two hands, the two together held up a poster with DU's motto:

pa jan li sun zo
zai gau fe pa jan

go li sun zo
bai gau fe go

"One man learning will set one man free. A nation learning will make the nation great."

Behind them, the instructor walked in, and closed the door. Zei checked his watch. 70,001.

* * *

"Welcome to physics class", the instructor said.

"Today, I'm going to try to introduce an important topic that I think almost everyone gets confused about: thermodynamics."

The more attentive students put away their hand devices, stopped talking, and got ready to listen. The others kept talking at first, but then quickly quieted down.

A girl beside Zei reached under the table edge and ran a finger along it without looking - a habit, not a serious search. Another student habitually waved her hand device around the room, trying to pick up on any unknown electronic devices. Nobody commented. Nobody ever did.

"Who here has heard of the word 'entropy', in the context of information theory or cryptography?"

About three quarters of the students raised their hands.

"And who here has heard of the word 'entropy', in the context of the second law of thermodynamics - that it always goes up, and one way that happens is that heat always flows from hot to cold, never the reverse?"

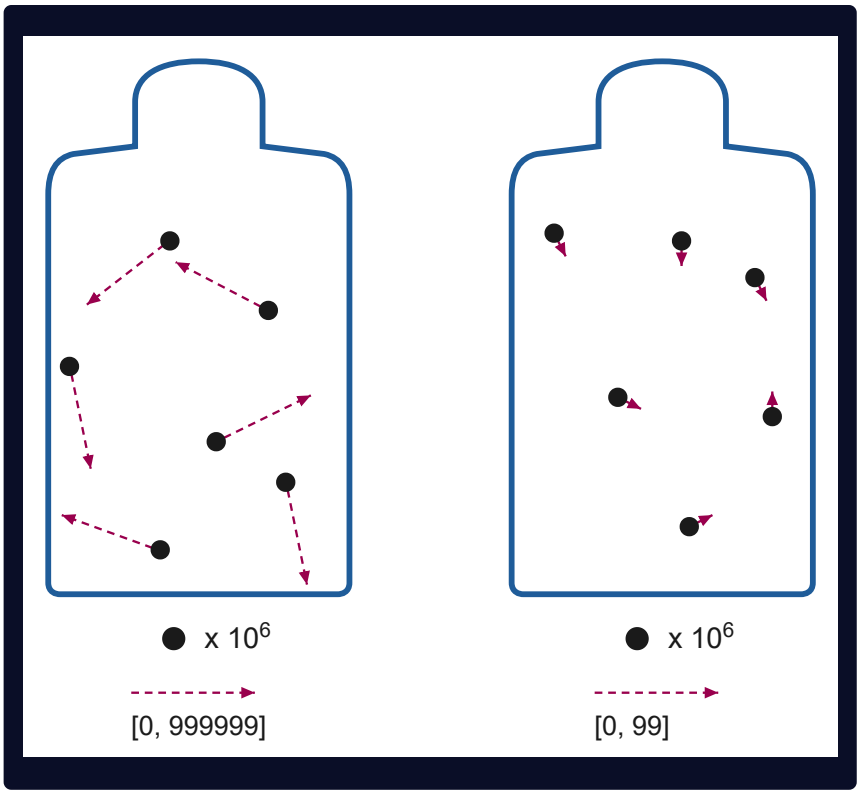
Almost every student raised their hands. Entropy was a favorite topic of science fiction.

"Now who here thinks they can explain the connection between the two?"

No one raised their hand.

"Alright. Let's begin."

The instructor pressed a button on his watch, and a slide showed up on the front wall:



"Imagine that we have two jars of gas. Each jar has a million molecules. But in the jar on the left, the molecules are moving much faster. Suppose, in the jar on the left, you can represent the velocities with six-digit numbers, from zero to 999,999. And in the jar on the right, the molecules are moving slowly, so you can represent the velocities with two-digit numbers, from zero to 99."

"Can someone tell me, how many digits do you need to fully represent everything going on in the jar on the left? Assume that they already know it's gas and there's a million molecules, the only free variable is the velocities."

"Six million!", about ten students shouted in unison.

"Good. Now, what about the jar on the right?"

"Two million!"

"So let's recap. From your point of view, as the observer, you know that these are both jars of gas, you know what kind of gas, and that there are a million molecules. But you don't know the velocities of each individual molecule. So for the jar on the left, *the amount of information that you do not know* is...?"

"Six million digits!"

"And for the right?"

"Two million digits!"

"So eight million digits in total, information that you do not know. Now, what happens if we put the two jars together and let the gas mix? What will we know about the velocities there?"

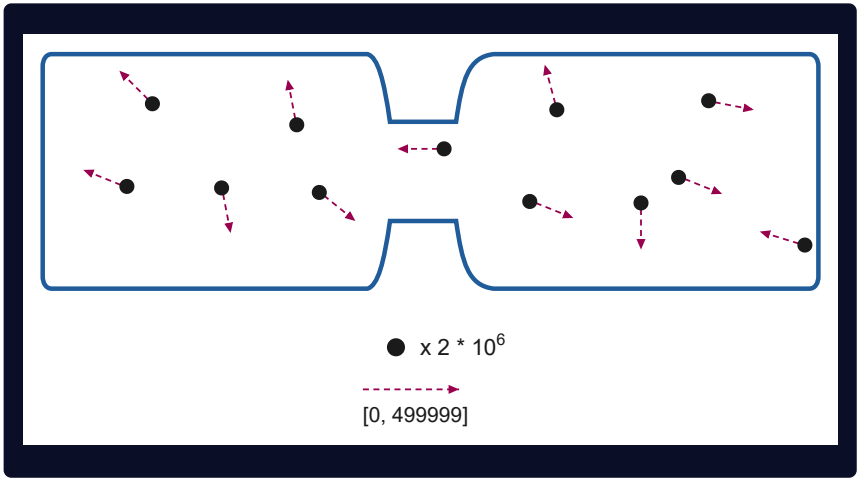
Zei raised his hand.

"Yes, you there?"

"The molecules will bounce off each other randomly. When they settle down, because of conservation of energy, the velocities end up somewhere in the middle, between zero and five hundred thousand."

"Yes, excellent! Now technically, energy is proportional to velocity squared, and we can argue about how these velocities are represented, and by the way we're also completely ignoring positions, but let's just keep it simple and say zero to five hundred thousand."

The instructor tapped his watch again. Next slide.



"Now, how many digits do we need to represent the information we don't know about this new system?"

Another student raised his hand.

"Go ahead."

"From zero to five hundred thousand is more than five digits but not quite six digits, if we go by logarithm, it should be about five point seven for each molecule."

"So what's the total?"

"Eleven point four million digits."

"Very good! So the conclusion is: by mixing the two jars, we have lost knowledge. We know less about the gas than we did before. And we can quantify *how much* knowledge we lost: it's three point four million digits."

"This is what we mean, when we say that entropy goes up, when heat flows from hot to cold. For each tiny increment of heat, heating up a cold thing adds more unknown digits than cooling down a hot thing by that same increment takes away, and so the total unknown goes up. Every possible action that you can do to these bottles will, on average, make the total unknown either stay the same or go up - and if you're doing anything useful or making any mistakes, it will go up. This is the second law of thermodynamics."

The instructor paused.

"Now, how do we prove this?"

None of the students answered, but many of them seemed to lean forward with curiosity.

"Let's prove by contradiction. Suppose we did have a magic procedure that could go in reverse: start with the merged jars with eleven point four million digits of unknown, and bring us to the separate jars with eight million digits of unknown."

Zei raised his hand once again.

"Can I try this?"

"Go ahead."

"I remember the laws of physics of our universe are fully time-reversible, right? So if you can do something in one direction, you can always run the same steps in reverse."

"Exactly. Now why does that time reversibility, at that level of atom-by-atom physics, imply that temperature equalization is something you cannot reverse? Isn't that a paradox?"

"Well, if we *could* separate medium temperature into hot and cold, then we would be able to do unlimited data compression."

"Like, imagine you had a file with eleven point four million digits of data. You would make a really perfect double-jar, and have a gun that shoots out the gas molecules with exactly the right velocities. Then, you would use your process to un-mix the gas, then split the jar, and measure the velocities at the end. You would get eight million digits out."

"And because our universe's physics is time-reversible, you would be able to do the same thing in reverse: take those eight million digits, put them into the gun to shoot out gas molecules at the right velocities, apply the exact same un-mixing procedure backwards, measure it, and you would get out whatever your original eleven point four million digits were. The output of the time-reversed process has to equal the input of the original process. And so we get a fully general-purpose way to represent any eleven point four million digits inside of eight million digits. And if we could do that, we could do it again and again, and represent the whole planet in eight million digits, or even make it smaller and do it in eight digits. And that's impossible!"

"Exactly. Or in other words, you can't go from not knowing more things to not knowing fewer things in a universe with time-reversible physics. In a universe with time-irreversible physics, you *could*, because you can just destroy all the things you don't know."

"Just like the Arctic Emperor!", Zei wanted to shout. But he kept quiet. It's bad form to openly explain a joke that is so obvious, that half the students probably already have on their minds. The instructor and the students briefly paused to appreciate the joke and smirk and enjoy a few moments of mental communion. The instructor then moved on.

"But in a universe with time-reversible physics, there's always things you don't know, and that lack of knowledge can only be shuffled around, and inevitably increased. The best we can do, is go from not knowing things we care about, to not knowing things we don't care about. Like going from not understanding thermodynamics, to not knowing the velocities of the molecules that are coming out of the projector showing these slides as waste heat."

"That was an amazing job, kid. What was your name?"

"Zei."

"Well, Zei, you got it per-"

A loud noise erupted from a nearby room. The whole room shook.

A fraction of a tick later, part of the wall and the floor came crashing down.

The next thing Zei knew, a fragment of stone hit him in the arm.

* * *

Zei found himself on the ground, his knocked-over chair beside him.

His mind flashed back to the man who had redirected them last minute to a different room, one floor above where they had been supposed to go.

He understood immediately. All of the extra precautions that DU had been taking - the cryptography, the sudden changes in location, the anti-transmission foil - they were not just a game. And the Arctic Empire was not just a unit in geography class. This stuff was *real*.

"Is everyone okay? Anyone injured?", the instructor yelled.

Zei thought for a moment, trying to figure out if his arm was injured enough to be worth complaining. He tried raising and lowering his arm. It did not respond.

The next thing he felt was two arms grabbing on to him, and pulling him up off the ground and onto a chair. Zei turned his head. It was Fin - to Zei's relief, completely uninjured.

Smoke filled the room. Every student's watch, among its many features including an air quality monitor, started beeping loudly. The students tapped their watches, and soon the beeping died down.

Slowly, the students helped each other get back up, as the instructor walked around to check if there was anyone who seriously needed help.

The instructor got to Zei, and Zei silently pointed to his left arm. The instructor took out his hand device, and scanned it. AI-generated output filled the bottom of the screen.

"Are you in pain?", the instructor asked.

"Some, but I can handle it."

"It looks like a broken bone. Get it checked out at the hospital today. The sooner you go, the sooner you will get better."

"I'll take you", Fin added. "Gotta be sure you'll be good for your game at the end of the month."

The instructor kept walking. The room was filled with smoke. The air quality alarms were still blaring, but Fin and Zei could make out the instructor asking different students "are you okay?" and a student responding "I'm fine".

The air filters activated to full power, and soon the smoke cleared. Fin and Zei looked at each other. For a few moments, the students and the instructor sat in an awkward silence.

Zei pushed off the table with his uninjured arm, and stood up. "I think we know much less about this wall than we knew two minutes ago", he joked.

A few students smiled or gently laughed. He had judged the audience's mood well.

"No one else hurt, are you all sure?"

"I'm fine", about ten people replied.

"Alright, I think we've had enough entropy for one day. Let's go home."

* * *

Fin and Zei stayed in the classroom longer than the others. Zei sat back down on his chair. Fin grabbed a broom, and pushed all the debris to one side. Soon, only they and the instructor were left.

"Wow, the Arctics are getting aggressive. They didn't use to go after physics education."

"Whoever moved our room last-minute was a genius. Just saved our lives there."

"Unfortunately I expect it will only get worse from here."

Fin and Zei looked at the instructor in shock.

"Remember, their game is to permanently keep down our level of heavy industry and technology. Let us live, but only at the level of civilization where we can never be a threat."

"And our counter-game is 'gie fe kiu kai ci bin hu': grow roots, no head. Use distributed manufacturing to build all our tools. Let the official government keep going, slow and corrupt just like it was when we lost the war a hundred years ago, but use cryptography to coordinate the decisions that matter anonymously, without needing any exposed individual leaders at the helm. All of DU already runs this way. Soon big parts of our power generation, agriculture and medicine will."

"It's been effective. After the last few years, we're finally almost at the level where we can stand up for ourselves without needing any centralized heavy industry at all. The problem is, it looks like they're catching on."

"Dze go ba fau gie", Fin and Zei said in unison, calmly but defiantly.

"Dzego will rise again."

Chapter 3

Meldan, Veridia · 3724 SNOWMOON 8

Gladias was walking forward along a sidewalk. On each side of the road, there were small shops one after the other, selling various goods from flowers to clothing to electronic equipment. He crossed a street.

On the other side of the street, the road sloped down, and transitioned seamlessly into a tunnel, lit just enough that it was possible to comfortably see and read, but not more.

Gladias got to a Y-shaped intersection. Pasted on the wall, there was a poster, attached at the corners with four pieces of tape. He stopped right in front of the poster to take a look.

From a comfortable Veridia to a strong Veridia

* * *

Veridia is a beautiful country, you are told in school. You see it every day: the trees, the mountains, the houses and buildings all trying to eke out another few basis points reduction in land tax.

But you know, in your heart, that there is a problem. When did Veridia last go to space? Twenty years ago. When was Veridia's tallest building built? Forty years ago. Why? Because all your institutions - the Steering system, the Graph Funding mechanism, and the very *culture* - are built around making people comfortable and happy in the present, with no concern for tomorrow.

Meanwhile, look at the Arctic Empire. Where is the housing, the transportation, the industry being built the fastest? In the Arctic Empire. All the scientists working to make people live longer? In the Arctic Empire.

What does Veridia say to those scientists? Living longer is not that important. Death and birth are okay, they are the natural cycle of life. Yes, that's what they believe. The natural cycle of life is the best they have for morality - by that same twisted logic, any injustice could be natural, like a tiger eating a rabbit.

Veridia's vision of nature is like a plant. Beautiful, but never rising up above its stature. Our vision of nature is like a lion. If this vision appeals to you, if you want Veridia to be stronger, to reach for the stars, to aim beyond incremental public health and seek eternal life, then Veridia needs you, the next generation of leaders, to *take charge*. Don't mind the mob. Build the future. Join us, and we will build it together.

With regards,

Lord Ephelion

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Arctic Empire consul to Veridia.
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"Yay, more Arctic propaganda", Gladias thought. Of course, the Arctic Empire is free to speak just like everyone else, but he wondered what kind of Veridian would feel inclined to vandalize the public environment and put up a printout of their message in the Veridian tunnels. He started mulling over ways to rip the poster down.

Suddenly, Gladias felt a tap on his shoulder.

Gladias turned his head.

* * *

"Oh, is it you, Mov?"

"No, it's the Arctic Emperor. I'm here to literally suck out your soul."

"Okay fine... Wait, I'm like one of a hundred people walking toward the Greater Plum Harbor wearing a privacy robe, how did you figure out where I am?"

"You put me on your trust list a year ago and never took me off, so our local AIs had permission to enable location sharing. Your AI knew that we were supposed to meet before I shadow you when you go see the mysterious gentleman, so it unlocked location sharing with me for today."

"Aaaah, I see ... something something contextual intimacy or whatever the new buzzword is."

"Yep, probably didn't even need the two-party garbled circuit feature to do that."

"So what's been on your mind?"

"Well, I'm assigned to one of the openness rubrics on internet media this year. We've been talking a lot in our Keeper group about bumping up the taxes on API restrictions. I talked to a citizens' assembly about the topic a month ago, and they're really unhappy about how things are going."

"Why is it such a big issue now?"

"Bluewhale announced a month ago that they're cutting their API and are only letting verified hardware interface with the platform. They say it's to prevent automated spam, but basically everyone in the assembly thought they just want to block alternative interfaces so they can control people's attention more and extract more money."

"Do we know exactly what kinds of things they are doing? Are any of their algorithms public?"

"Nah, their algorithms are totally opaque, and they're already paying the max tax bracket for that."

"I see..."

"So yeah, many of the Keepers think we should bump up more taxes on that kind of behavior. So either they stop, or, well, they keep going and the internet sucks more, but the gov gets enough money that we can pay for everyone to get better healthcare. The problem is, Bluewhale has been driving a hard campaign against it. One Keeper was not careful about their privacy, she and her views got found out recently, it went public, and her husband lost his job only a few days later. No idea if it's connected, but, well, this is why we gotta run this whole tax system with all these cryptographic networks randomly selecting Keepers, anonymous discussion and voting, and privacy robes."

The mention of Bluewhale brought Gladias back to his younger days, when he had learned programming by writing an alternative client for the other major social platform, Silverchat. That had only been possible, he knew, because the API was open, and the platform allowed developers to make alternative clients and users to export and migrate their data - all decisions that had undoubtedly been motivated, or at least justified to the bean counters, by trying to lower their tax rate on the hardware and software openness rubric. That experience had been what motivated Gladias to contribute back, and get involved in Steering himself.

"Anyway, speaking of privacy, what's the plan with your 200 rep anon friend?"

"Well, he told me that he thinks the whole Steering mechanism is fundamentally broken. I figured I would hear him out. Often even people who are crazy are at least ten percent right in some way that's interesting. And hopefully figure out some clue about how he found me?"

"And... what if this is a trap, to get even more compromising info on you?"

"Well, if he really wanted to hurt me, he could have just submitted my identity and task into the network and pulled me out of the running for Sentinel status for this whole year. But anyway, figuring out this kind of thing is what you're here for."

Mov then opened up his privacy robe, and took out a large package from an inside pocket.

"Here, before we get into the rest of the business, put on a fresh privacy robe. Just in case there's some piece of dirt that makes yours trackable. The Arctics are getting better at smuggling their surveillance devices in here, faster than we're getting better at detecting them. We gotta step up our paranoia."

"Surely if it's at this level we should have just not met in person at all."

"Come on Glad, you're not just someone about to join me as a Sentinel, you're a friend. I enjoy it when we see each other - while we still can."

* * *

The tunnel sloped up, and the outdoor light once again shone into Gladias's eyes. Both sides of the path were covered by trees.

He passed by two children standing beside a water fountain and talking. He could make out one of them saying something about a recent visit to Northglade with his family, and the other mentioning that he liked their Helisport team.

Soon after he had walked far enough that he could no longer hear their conversation, his watch buzzed.

From		Message	Time
Seila	→	[4 images]	44029

He took a look.

Seila was sending some photos from Freetown. The first was of the downtown of the city. The second was of a major park, taken from in front of an entrance gate. On the left, he could see a poster with some writing. Bringing his watch closer to his eyes, he could make out a price list.

Gladias pressed a button, responding back with a heart symbol.

Turning his attention away from his watch screen back to the street, he noticed that he had now entered a more urban and more commercial part of Meldan.

To his right, he saw a series of office buildings. The ones further away shone with a clear glass and steel construction, though at that distance they were only partially visible behind the tree cover. The ones close by were under construction, all covered with large tall barriers shielding the construction site from public view.

On the first barrier, he saw a poster reminding Veridians to be kind toward their families and their community.

On the second barrier, he saw another Hydrafill advertisement. On the left side, the advertisement showed the classic Hydrafill water bottle, covered with yellow and green colors.

And on the right side, it showed the text:

Hydrogen hydroxide is:

- *A commonly found substance in toxic sewage water*
- *Deadly in large doses*
- *Frequently administered to prisoners undergoing torture*
- *A source of skin damage if held for a prolonged time in its solid form*

But did you know that you can buy YOUR OWN BOTTLE of fresh drinkable hydrogen hydroxide today?

Hydrafill, buy it now.

Gladias smiled again. The construction had been going on for a long time, and he was grateful for both posters that were covering up the construction sites. He vastly preferred looking at Hydrafill advertisements *or* Veridian virtue posters to construction cranes and half-built concrete buildings.

He then turned his attention to the long castle-like wall, mixing grey and purple colors, that was on his left.

In front of the wall, there were two rows of trees, with a sidewalk in between them.

Behind them, a large poster mounted on the wall:

"Learn about what Greater Plum Harbor is doing for your community!"

The text was in front of a background that featured robots picking up garbage, drawn in a child-friendly cartoon aesthetic. On the bottom right corner there was a two-dimensional barcode. Gladias scanned it out of curiosity.

Greater Plum Harbor Public Activities

Environment	• Cleanup • Pollution monitoring (water, air) • Animal welfare research	Learn more
Health and medicine	• Large-scale studies in effects of food, environment and clean air improvements • Healthy eating, exercise, other behavior and lifestyle research	Learn more
Education	• Writing advanced education curricula targeting ages 4-18 • Youth math and programming competitions	Learn more
Art	• Beautiful Cities research • Publicly available exhibitions and museums • Movies and shows, three in Veridian Top 20 last year.	Learn more
Load more		

```
All research outputs and tooling done by GPH are open source!
Partially funded by Veridian Quadratic Funding
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Gladias saw two people sitting on a bench, each drinking a tea and looking at the billboard, one with a hand device on his lap.

"Have you ever been in there?", the first man asked.

"No I haven't, never managed to get a visa", the second replied.

"It's a fascinating place, it looks totally different inside than the rest of Veridia, it's almost like a country inside a country."

Gladias stopped, and faced the billboard to avoid arousing suspicion. Overhearing a conversation was slightly impolite, but understanding public sentiment was part of his duty as a future Sentinel. Citizens' advisory assemblies helped, but they were never quite as good as an authentic conversation by people who did not realize that anyone was listening.

The second man looked down at the hand device that the two were sharing. "Oof, check what taxes they pay."

"What taxes do they pay?", Gladias asked Emerald, whispering silently so only his neck band could make out what he was saying.

Emerald immediately replied.

"Sales tax, averaging 5.2%, half of the city average. Total land tax is 1.48%, slightly greater than the city average. Graph Funding and especially Quadratic Funding revenue recovers roughly half of this."

Gladias turned his attention back to the conversation.

"I don't get it. These guys are just peacefully off in a corner, by themselves. I've never been there, and they've never bothered me. And they even do all this work on the environment, education, healthcare, their museums. They can't even just work on the science they're actually good at, they have to do all these populistic things to push their taxes down and get some of the money back from Quadratic Funding, things that they're not even good at. And even after all this? So many people hate them, and pretend their wall is ugly just to push their land taxes up. What's the point of worrying so much about the Arctics when we're so good at being our own worst enemy?"

"At least here that's the worst we can do to them. If this was Devanvil or Inglewore, GPH would be fighting to not get shut down entirely."

Gladias had not been inside Greater Plum Harbor for nearly a decade, so this made him all the more curious to see it again. He kept walking.

* * *

Gladias held his watch near the green circle on the entrance gate, and waited for about one tick.

"Proof verified, welcome", the machine answered in a robotic but approving tone.

He walked through the gate.

He could immediately see why the two people on the bench had described Greater Plum Harbor as being so different from the rest of Veridia.

The buildings inside Greater Plum Harbor were more uniform than those outside, all with similar circular and rectangular shapes, a mixture of white, grey and purple colors, and roughly five to ten meters in height. The trees were all of a few types, and carefully planted an equal distance apart on the sidewalk. Beyond having distinct names and distinct quarter-meter-wide logos, the stores and restaurants were not trying at all to distinguish themselves, instead accepting their visual role as small elements of the Greater Plum Harbor's whole. The buildings were noticeably more futuristic, more glass-and-steel, and with more sharp edges than the usual structures in Veridia.

Gladias kept walking forward, following the directions on his watch. After five minutes, he arrived at a tunnel entrance, and went down, briefly passing through a space where there were a large number of people walking.

Just after he passed through the crowd, his watch buzzed.

Air	
CO_2	965
PM2.5	22.1

Devices	
Attested ✓	55
Unknown ?	6

Two problems, Gladias thought.

First, too many people, not enough ventilation or filtration. Gladias pulled on the strap on his privacy robe's front face cover, separating the areas above and below his nose.

Second, no idea what those unattested devices are. These days, almost every device was attested with a zero-knowledge proof verifying that it was produced in a particular batch by a particular manufacturer. Six out of fifty five was an unusually high number unattested. Unfortunately, there was no time to figure things out. It could be a kid's science project, Gladias thought, it could be that it was simply normal in Greater Plum Harbor to have devices without attestation - after all, from Veridia's perspective, the land was all just one giant business's private property - or it could be a hidden spy camera, or worse. He increased his speed slightly, and kept walking.

After taking several more turns through the maze, he was once again outside, this time in the middle of a hidden courtyard. There were several tables and chairs, with people eating and drinking water. Gladias found a table in a booth that had a cloth curtain protecting it. He went in and sat down.

There was a decorated lamp in the middle of the far end of the table, with one large flower on each side. On the floor, pressed against the corner of the booth, Gladias noticed a white box. He leaned in to

look more closely, and his watch buzzed. PM2.5 readings were below 1. Must be an air filter, he thought. He took the hood off his privacy robe.

He tapped his watch against a green circle in the middle of a table, and a moment later heard a confirmation sound.

Two minutes later, a robot with a smiley face slid up to the table, carrying water and a salad. Gladias took both and put them on the table, and the robot slid away.

Five minutes later, another man wearing a privacy robe stepped in.

* * *

"So, we meet at last."

"You track me down, you tell me you know things I need to hear but only want to tell me in person, and you bring me to a far corner of GPH that I did not even know existed. And so far I don't even know your name."

"Ah, my name is Delwart. Pleased to meet you."

"And I'm Gladias."

"I knew that. And your wife is Seila, your children's names are-"

"Are you trying to intimidate me?"

"Actually no, I guess now that I think about it, that was more showing off than anything else. I do apologize."

The two went quiet for a couple minutes. Both took off the face covers and hoods from their privacy robes. Gladias started to drink his water and eat his salad. Delwart tapped his own watch against the green circle on the table, and a minute later his own tea, and some kind of rectangular nutrition bar along with a fork, arrived.

Gladias leaned forward again.

"So what do you want?"

"Do you feel happy with the Steering system lately?"

"It has its flaws like any system, but in general it feels fine?"

"I see, I haven't broken your wall yet. Okay, let's take the Dreadknot concert for example. They're a nice band, they're great fun, right?"

"Yeah, I like them."

"But a bunch of the Sentinels are trying to push them up into Tier 4, because their lyrics are not gentle enough."

Gladias nodded in agreement.

"You were brave, voting Tier 3. Most Acolytes would cave to what they think the Sentinels want. You didn't. That's rare."

"But let's take another example. What do you think about GPH?"

"It's clearly a nice place. Not like Veridia, but beautiful and impressive in its own way. It's not for everyone, but I can see why a lot of people like coming here."

"Well, do you know what their Steering taxes are?"

"The Steering taxes are okay, but they really get dinged on the land tax."

"Exactly. A well-considered argument about how much this place benefits Veridia is worth less than a barely thought-through 'I don't like you'."

"So what do you propose?"

"Freedom. Actual freedom. Stop trying to nudge everyone, and let people's true selves come out."

"And how can I help?"

"Are you planning to be a Sentinel or a Keeper if you get accepted in two months?"

"Was thinking Sentinel so far."

"Be a Keeper. And just vote to delete every rubric you see."

"That's it? That's your plan?"

"See, there's a lot more Sentinels these days than Keepers. And Sentinels are the people everyone is trying to influence. Why? Because Sentinels get direct connections to specific things in the world. Keepers adjust the global rules. But there's a limit to how much any single rule can affect any single business, and there's like twenty Keepers deciding each on each rule. Being a Sentinel, you have a much more tangible impact. And it's more fun."

"My program is different. I'm trying to round up as many Keepers as I can, because what I want is systemic change. It's not about one concert or one enclave that has too much purple in it. It's about

releasing Veridians from this tyranny of people's prejudices and letting them be themselves."

"Do you want to remove Graph Funding too?"

"I think Graph Funding definitely overreaches in a bunch of ways too, but my focus is Steering."

"A lot of the open tech Graph Funding supports has been pretty key to helping Dzego modernize to defend itself, do you not think that's important?"

"Look, I like their people, but really we're playing a dangerous game cutting so close to funding a potential war. We should focus on our own problems first."

Delwart stopped speaking, and began to drink his own tea and eat his nutrition bar. Gladias took the moment to ponder Delwart's words, while continuing to drink his water and finishing his salad.

"It's an interesting thought, that the Steering is just manipulating people too much. I don't know if I fully agree with just dropping it all to zero, and especially killing the Graph Funding - which we would have to do, if the Steering taxes went to zero."

Gladias paused for a few ticks, and looked at Delwart. Delwart looked back. They each took another sip of their respective drinks.

"But okay, even if I might agree on some parts, why should I listen to you?"

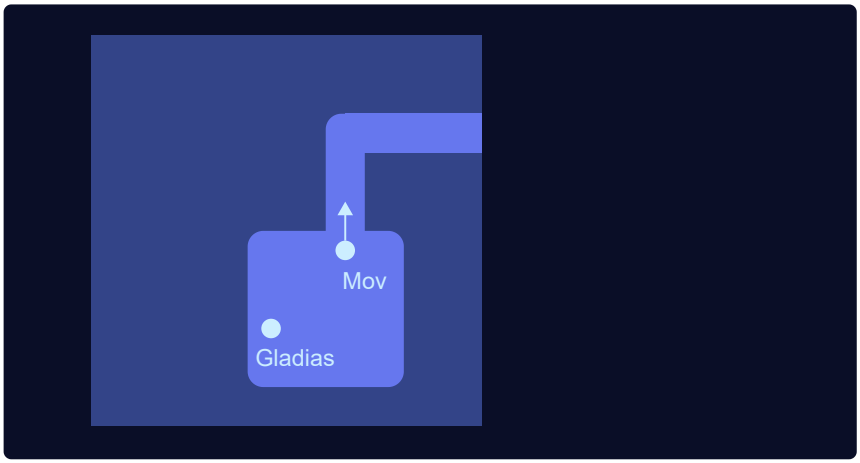
"Well, I know you, I know where you live, I know how you think. And if I can find out all that, imagine what kinds of things Bluewhale might be able to do. Or, for that matter, the Arctics. I can help protect you. But I will need your help too."

"Anyway, the deadline for you to submit your final track decision is in about a month. Don't go for being a Sentinel, be a Keeper. Let's just start with that."

Delwart stood up, and exited the booth.

Gladias kept the crack in the middle of the curtain slightly open, and watched Delwart go. Just before Delwart disappeared from view, another person wearing a privacy robe exited the adjacent booth, and followed Delwart.

Gladias checked his watch.



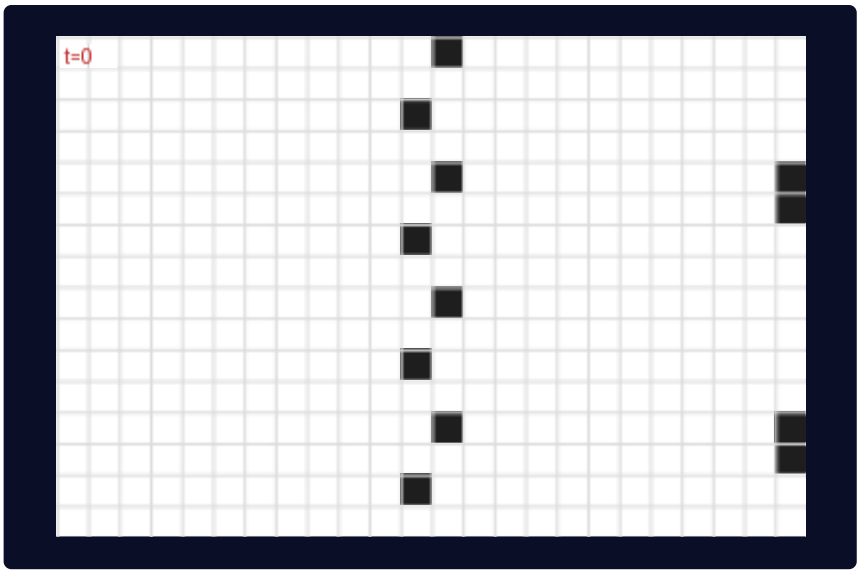
Chapter 4

Pafogai Du, Dzego · 3724 SNOWMOON 24

Fin and Zei were sitting at a table in an outdoor courtyard. Some of the nearby tables were empty, and at other tables they could see a few groups of teenagers and a few older couples sitting and eating.

Through the opening at the entrance, Fin could see across the street a teenager climbing up a wooden ladder that had been put up hastily against a public lamp post. Very soon, he was already at the top, and he immediately got to work unscrewing a security camera. Once he finished, he put the camera into a black pouch, and then climbed back down.

Zei tapped his hand on the table. Fin's eyes snapped back, and he once again started looking intently at an animation on Zei's hand device.



"Okay, so what am I seeing here?", Fin asked.

"This was from my last Minpentai game. We were playing with the 'rotate one eighty if three' rule. Someone discovered this new way to breach the usual wall construction, and flooded their opponent's territory with garbage. I'm trying to figure out some countermeasures."

"What's the endgame? Are you trying to get to a perfect wall, or is it rock paper scissors all the way down?"

"Well, remember, the rules are time-reversible. So any wall that can be created in some way can be destroyed in some way."

"Huh?"

"Remember when we were still in middle school, and we thought that in those movies the good guys or the bad guys or whoever were stupid for putting the shield generator outside of the shield so the other side could find it and destroy it?"

"Yeah."

"Turns out they weren't dumb, we are. Because the laws of physics are time-reversible, nothing can be invincible. Anything that can be created can be destroyed. You just have to take the path that creates it

and do the same thing in reverse. So a shield that can't be destroyed is not something that can be created. There has to be a weakness somewhere - it's just a matter of how well you hide it."

"Are you trying to get into defense or something?"

"Hey, I'm just trying to win the championship. This year is the last year I can participate, and if I win, go to the nationals."

"You mean this year is your last chance to impress Bai?"

"Hey-"

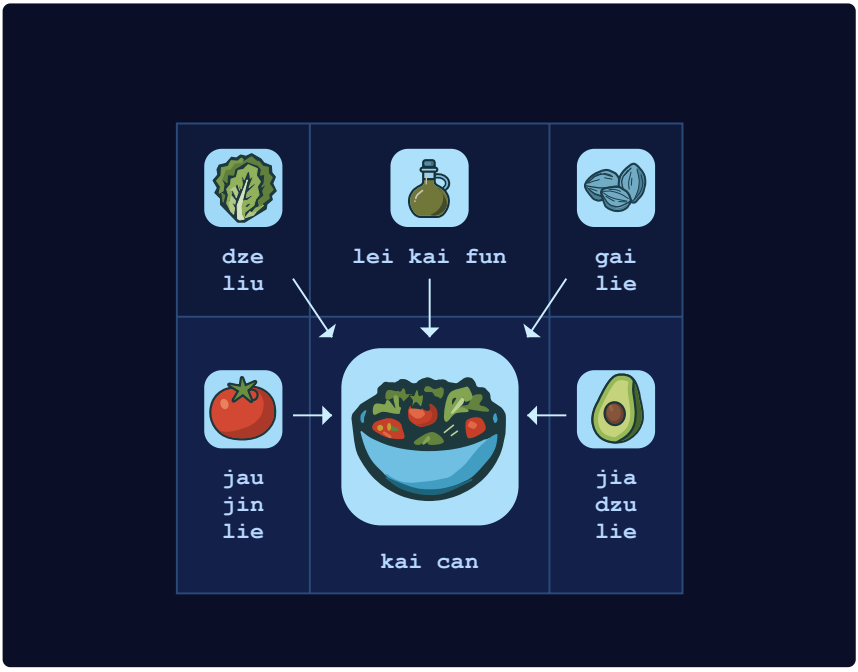
A robot slid up to them.

"jie mo kai tu jie hei ja ma?", it asked in the classic deliberately cute Dzegoban typical of restaurant robots. "Do you want to eat or drink?"

Fin looked at the screen on the robot.



He clicked on "kai", and then on a picture of a salad.



Fin clicked again, and then clicked a few more buttons to order tea. The robot beeped in acknowledgement, and then slid over to get closer to Zei.

"Wow, everything is getting so picture-based now!"

"Well my last DU class was in linguistics so finally I get to be the genius here."

Fin took a small paper booklet out of his backpack. On the front cover, there was the Bansunpei logo: two horizontal arms, one from the left and one from the right, meeting in the middle to hold their hands. Above the logo was the title: "po so dze go ban de sia" - "the conception of Dzegoban version nine".

"Tell me more?"

"You know how after we got the number of roots down to 256, the word was that we're going to keep going, and soon enough express

everything through the four cardinal elements - earth, air, fire and water?"

"Yeah?"

"Turns out the Bansunpei wants to give up on that now. Stick to 256, and push the next version of Dzegoban in a different direction."

"And what's that?"

"Go visual. So first of all, they're actually seriously thinking of bringing back hieroglyphics."

"WHAT?"

"Switching to the Veridian alphabet made sense back then because all the old symbols were hard to draw. But now we have computers. The problem isn't output, it's input. And the trick is, they're figuring out new symbols where some parts of the symbol encode the whole pronunciation, so you can look at the symbol and know exactly how it's pronounced if you forgot. And the rest of the symbol still tries to cover the meaning."

"Okay...?"

"There's a bigger change in how they are thinking. Language so far has been very one-dimensional, because speech is one-dimensional, you're saying one word after the other, and you write what you speak. But if you think about parts of 'language' that are not tied to speech, say, you like math - so think of the quadratic formula."

"Minus b plus minus square root of b squared minus four a c all over two a?"

"Think how ugly that looks if you had to write it down and show it to someone on a screen. Like, imagine someone reading those words on a screen right now and how that must make them feel. Now think of the equation, the way we normally write it down. I still don't remember the whole thing, but I remember it looks nice! It's a pretty picture. The fraction line splits things into top and bottom, there's a root, and it's easy to see what's inside the root and what's outside."

"So they want to make all of written Dzegoban like that?"

"Yep."

"You know, there's an old result: if you take the same text, and translate it into every language, the lengths of the translations are different. But if you take the translations, and compress them - literally, any off-the-shelf compression used in any of your compute devices - the lengths end up almost the same. So it's all the same amount of information. The question is always: how do you organize it so that it's the most accessible and useful?"

Zei pondered for a moment, thinking back to the thermodynamics lecture from a month ago. Information is information, the only question is how to arrange it. Zei nodded.

"So that's what the Bansunpei is A/B testing right now all across Dzego. Altogether, the priests came up with six different styles of visualization, and they're giving out reputation points if you adjust any of your writing or signs to their new ideas or report on your experiences reading things put up by others."

"Wait ... they're doing *that* level of nationwide organizing, all just to optimize our language yet again?"

"I guess so..."

* * *

Zei walked toward the half-kilometer-tall pyramid - the most visible landmark at the southern edge of Pafogai Du. From the outside, it looked like a smooth clad stone pyramid with a square base, like something that had been built by a great civilization a thousand years ago. Interrupting the stone were two rows of trees along a path, starting from the position where Zei was standing ane leading to a point halfway to the top, and another row of trees going horizontally around the pyramid at exactly the halfway height.

But the looks were deceptive. Building a pyramid that tall would be far too expensive, even for Veridia, or even the Arctics. This was a pre-existing mountain with pre-existing caves, which was refurbished to look like a pyramid, with the caves converted into living and working spaces. A portion of the Dzego government was located there, along with many major businesses - though space was expensive.

Zei stopped at a streetlight, and crossed Dzai Gun street: the last street before Pafogai Du officially ended, and the point at which a straight well-maintained path leading directly to the entrance of the mountain pyramid began.

Zei opened and closed each finger on his injured arm, then repeatedly spread out his fingers and put them back together. He simulated some button pressing motions. His arm *felt* fully healed, though Zei knew that his reaction time would still be a little bit slower.

The street curved up with about a ten percent slope, roughly doubling Zei's calorie consumption as he walked along it. And Zei could feel it. His mind subconsciously stopped running through strategies for the upcoming match, and switched to simply marveling at the beauty of the pyramid.

When Zei was somewhat further up, he briefly looked back, and marveled at the view of Pafogai Du: a sprawling series of low-rise buildings interspersed with trees, the flickering lights of various small shops and factories producing and selling all kinds of goods, all inside a perfectly regular grid of streets - sixteen streets two hundred meters apart from north to south, and thirty one streets one hundred meters apart from east to west. Some of the roads snaked between being fully above ground when they could afford to, and the portion for cars going below ground when the sidewalks needed the whole space to accommodate pedestrians.

Zei turned forward again, and kept climbing. About five minutes later, he arrived at the main door.

Zei's watch buzzed.

zan		tie	tei
Fin	→	jie fe hen dzi	74160

"Wish you luck", Fin had messaged him.

As instructed by some signs on the side of the main door, Zei turned left, walked about thirty meters along the horizontal path covered with trees, and entered the mountain pyramid through a smaller side entrance.

On the inside, the tunnels looked like the corridors that one might see inside a modern building. Each corridor was about two meters wide, and three meters tall; the lower two meters were a rectangle, and the upper meter a semicircle. The stone was polished clean, and looked like the interior of any other well-maintained building, though without the windows. Occasionally, Zei would pass by rooms with open doors. He could not tell which tunnels, and which rooms, were refurbished caves and which were dug out from scratch.

He followed the signs, turning right and down, then left, then straight and down for about a hundred meters. Finally, another right turn.

After about ten meters, he saw the door. On the sides, a series of stone bricks, curving near the top into an arch. At the center of the top, there was a symbol: two horizontal arms, meeting in the middle with clashing fists.

He walked in. An older woman noticed him and turned to him.

"Oh hi, Zei!"

"Ah, Mu, good to see you again!"

"Excited for the quarter finals?"

"Yep!"

"Well be careful. Remember, you're going to be part of a team now. You're a math guy, not a diplomat. You didn't even remember my birthday!"

"Hey, the teams are randomly assigned! No relationships to manage, it's more like math than diplomacy."

"You'd be surprised. I think it makes the real-time diplomacy all the more important."

Another woman walked in, about the same age as Zei.

"Ah, Zei, we meet again."

Zei's facial expression immediately turned into a frown. Bai had beaten him three out of three times over the past years, and was hardly magnanimous in victory. He turned away.

Bai walked up closer, and put her hand on Zei's shoulder. "So, you afraid of me, Zei? You know I enjoy making you feel afraid. Taking your excitement about getting some prize money to buy a compute box for your mom, and snatching it all away, getting myself a vacation in the beautiful south islands of Redshire. I should send you pictures."

Zei clenched his fists.

"See you in battle! Or maybe you'll lose to someone else this round before you even get to a game against me." She laughed, and slowly walked away.

A bell buzzed. A robotic voice, more stern than the coffee shop robots, instructed all players to get to their stations.

Zei walked into one of the elevator pods on the wall. The door closed, and he could feel the elevator start to descend. A tray opened up, and a robotic voice, this time cute like the robots bringing food and drinks, asked Zei for all electronic devices. Zei took out his hand device, took off his watch, and put both on the tray. The tray retreated back into the side of the elevator.

A quarter minute later, the elevator door opened, exposing a giant room with over a thousand people watching. A path lit up, showing the way to a giant game console with a chair.

Zei gazed in awe at the room. This was Zei's first time there - the games before the quarter-finals had been played in much humbler circumstances. He stared, amazed that a high-school programming game was able to secure this kind of space. Then his mind snapped back - he would ask Mu about it later, he thought. Right now, he wasn't there to enjoy, he was there to compete and win.

Zei walked forward. He could make out the silhouettes of three other players walking forward in the distance, but it was too dark to see who they were. He got to the chair, and sat down.

Drones about the size of a hand device buzzed as they flew around Zei, scanning for any kind of equipment that could be used for

cheating. If Zei's hand device or watch had been with him, they would have automatically done the same in reverse - demand and verify an attestation from the manufacturer and two inspectors that the drone's firmware was running an anti-cheat scan and was not there to kill him. But here, Zei was without his electronic support, and so he had to rely on trust.

Half a minute later, the drones were satisfied, and flew away. A tray slid out from under the game console, offering glasses attached to a long cable. Zei picked up the glasses. He slid them under and behind his arm, so that the cable would rest behind him, and then put the glasses on.

A loud automated voice, once again stern, announced: "MU GU GEI FA". Starting in fifty ticks.

Zei calmed down and tried to put his stress behind him, and rehearsed once again in his head the tactics. Walls, gliders, rocks, techniques for extracting usable negentropy from rocks, sideways attacks. Though every game there's always some kind of new rule, and the most important tactic of all was knowing how to adapt.

"LE MU GEI FA". Starting in twenty-five ticks.

Zei closed his eyes briefly, and breathed.

"PA GU GEI FA"

"Welcome to the first quarter-final round of the eighteenth year of the Minpentai tournaments. Tonight, four games will be taking place, one after the other. By midnight, we will know who is advancing to the semi-finals, and who will have to find another career. Or try harder for next year!"

"Today, fated by chance to be on team Ja, we have Pan Munbau and Dza Dzingo."

"And on team Zau, we have ... Zei Leimin and Bai Jahen."

Zei froze.

* * *

Zei could see the game field, which at the beginning appeared completely black. He saw in the corner of his display his *symbol*: a

pattern of dots that, were a copy of it anywhere on the board, would allow him to see everything within thirty squares.

He drew a few copies of his symbol in the bottom right corner of the board, where he was allowed to put down squares during the initiation phase. As soon as he did, nearby parts of the game field became visible. Almost simultaneously, he also became able to see a few parts of the field at the bottom left. Bai had put down her symbols too.

Zei saw Bai setting up walls and glider factories - gliders that would fly right to a nearby rock formation, and then reflect one glider back into Bai's territory where a mirror wall would send the glider right back, but also shoot another glider upward.

He began to do the same. He experimented with a few glider and wall formations in his sandbox, and after confirming that they work, deployed them to the field.

"MU GU GEI TAU FA"

The battle begins in fifty ticks.

Normally, at this stage, the players would be speaking intensely by voice chat, coordinating their strategies. Between Zei and Bai, neither spoke a word. The closest there was to communication was Zei intentionally not targeting gliders toward a few rock formations, Bai taking the hint and taking those rock formations herself, and similarly leaving some rock formations to Zei.

"LE MU GEI TAU FA"

Twenty-five ticks.

Zei suddenly realized: there had not yet been a new rule announced. The rules were the same that he had played many times before. Would the rule be announced at the start? Or was something else going on?

No time for speculating, Zei thought. Have to focus on the game.

"PA GU ... SO ... BI ... ZE ... HA ... MU ... FO ... SHI ... LE ... PA ..."

"TAU FA"

Gliders started flying.

Larger spaceships flew out from Zei and Bai's bases.

For the first few hundred turns, there was no combat - it was simply a matter of expanding into the field and gathering resources. Gliders flew out, interacted with rock formations, and created more gliders. Spaceships flew out, acting as large slow-moving gliders that eventually crashed into a rock formation in a controlled way, leaving behind copies of each player's symbols. The portion of the game field that Zei could see increased.

Zei and Bai's territories reached each other. A few stray gliders from Bai accidentally flew into Zei's territory, and wrecked one of his factories. Zei cursed silently, and spent his next intervention turn putting up more walls facing Bai. When putting up one of his walls, his finger slipped, and he accidentally created a glider that began to fly back toward his own base. Zei cursed silently, but then felt a sigh of relief when the glider harmlessly crashed into a rock formation.

A thousand turns later, the two sides' gliders and ships met. At the same time, the first intervention turn came. Here, all players were able to put down more squares near any copy of their symbols. Zei frantically adjusted his forces to more effectively attack Dza.

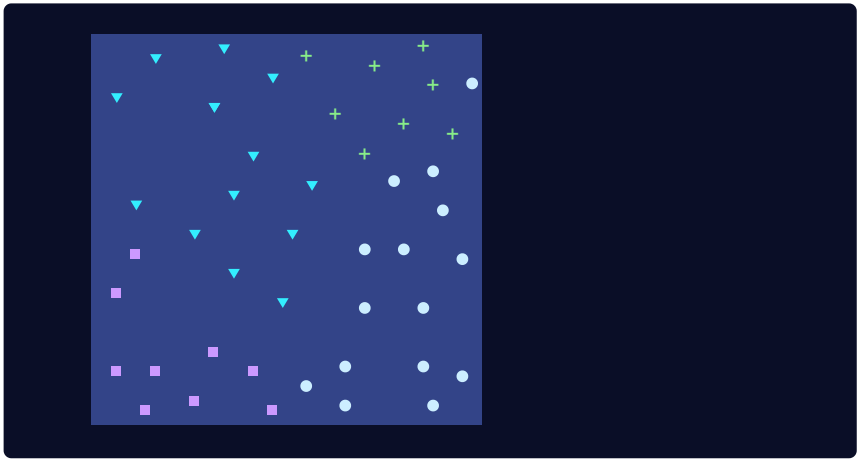
At the front, Zei put down a "moving wall" design that he had perfected - a sort of large glider that flew half as fast, but was optimized for reflecting more basic types of gliders right back at the sender. Right behind his three moving walls, all aimed at Dza's corner of the game board, he put his symbol-carrying spaceships.

On Bai's side, her gliders and Pan's flew into each other's territories with roughly equal frequency. Bai had some success in sending gliders deep inside Pan's territory, turning some rock formations into a chaotic mess that blasted gliders everywhere, and destroyed some of Pan's symbols. Pan, meanwhile, was more effective at pushing into the center, and then rotating back, attacking some of Bai's formations from behind.

Zei then rolled out his next move: sending three unusual-looking gliders deep into Dza's territory along the edge. The gliders hit the wall that Dza had set up around his home base in the northeast corner, and came together into Zei's symbol for exactly one turn - Zei's next

intervention turn. Zei quickly removed junk and put up walls around the symbol, and three gliders aimed in strategic directions. He now had a base right at the core of Dza's territory.

The audience was able to see a large detailed map of everything that was happening onscreen. Fin, checking from far away on his hand device, saw the summary view which showed the locations of each player's symbols:



Zei continued attacking Dza. Remove one of the two players completely, he thought, and that takes out half the activity in the intervention turns - always the biggest source of surprise.

Meanwhile, more of Bai's symbols disappeared, as Pan launched an attack going into the middle of the bottom of the field, seeking to cut Zei off from Bai.

Watching from the sidelines, Mu sighed. She knew Zei and Bai did not get along, and their side was suffering from it. This was the "price of anarchy", she thought - the amount that a group lost from its members independently pursuing their own separate goals without coordinating. She watched intently. The next minute, she knew, would determine if they could figure it out.

Watching from his hand device kilometers away, Fin felt his own frustration. Zei wanted to win, Fin knew, but almost as much, Zei wanted to hear Bai beg.

Zei stared at the board, concentrating his mind and trying to figure out his next move. After what seemed like an eternity, he decided to change his tactics.

Zei continued the attack on Dza, but intentionally did not target any of his symbols. Instead, he attacked a northern part of the eastern edge of Pan's territory, and spent a whole intervention turn to put up walls and build glider factories there. He put up *more* walls, just to let the gliders build up and bounce back and forth.

Pan and Dza did not see any of this. Instead, they saw Zei having greatly wounded Dza, but still struggling to finish him off, while Pan cornered Bai more and more toward the southeast. Emboldened, Pan launched an attack toward the east, seeking to sow chaos in the weakly defended territory in between Zei's home base and the front in the northeast against Dza.

Zei saw that Bai was struggling. But the communication line between the two remained silent. She was still too proud to beg.

Then, Zei saw that Pan was sending against Zei the same type of moving wall that Zei had used himself. He realized that he could not beat Pan alone. And so, he used his intervention turn to try a new strategy.

First, Zei sent some gliders to destroy Pan's symbols in the center of the field. Then came the real attack. Zei, Bai and the audience saw a spaceship flying diagonally northwest, from Zei's homebase, straight toward Pan's. With all of Pan's symbols cleared from the path, Pan and Dza were oblivious to this. What Pan saw was that his attack had successfully split Bai and Zei, and pushed Bai back all the way to her home base, surrounded on all sides. Dza still had five symbols, miraculously holding on against the force of Zei's primary attack.

Zei checked the clock. Sixty turns until the next intervention turn. His spaceship was almost at Pan's base.

Forty turns.

Finally, Zei said the first words that went across his team's audio channel in this entire game.

"Bai, next intervention turn is your chance to shine."

Twenty turns.

Zei's spaceship crashed against the remaining intact southern wall of Pan's home base. It briefly fluttered into chaotic shapes.

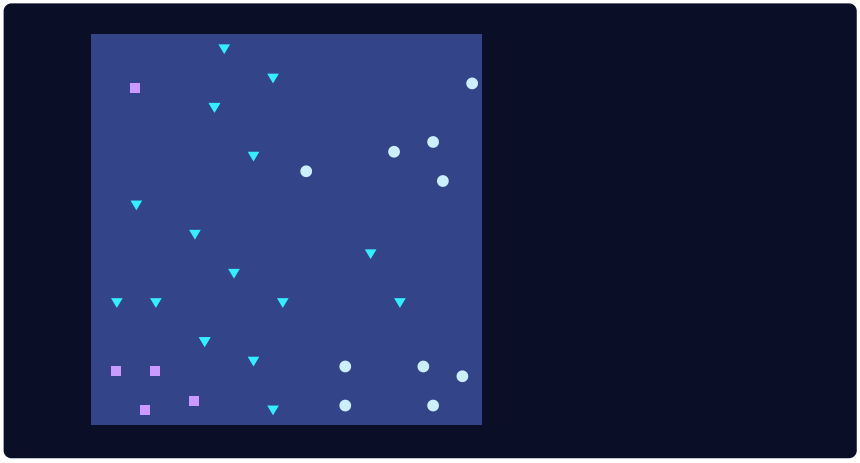
Then, seemingly miraculously, for one single turn, the chaos came together into a symbol.

Bai's symbol.

Zei could hear a gasp - the second message that went across his team's audio channel in this game.

Zei used his intervention turn to remove his own wall. Dozens of gliders flew in all directions, simultaneously targeting all of Dza's remaining symbols from multiple angles. Bai struggled in her sandbox view, made a few attempts at copying Zei's trick from over a thousand turns earlier. On her third try, she succeeded, and she repeated her moves live. The symbol was walled off, and gliders were flying in all directions.

Two hundred turns later, all of Dza's remaining symbols had been eliminated.



By total symbol count, it was fifteen to fifteen - exactly even. But now, Dza was out, and Zei and Bai could do twice as much together in their intervention turns as Pan could do alone in his.

Pan made a last-ditch effort to send new kinds of gliders to destroy everything he could in Zei's home base. He copied Zei's signature tactic, sending a spaceship to try to set up a symbol there. The plan partially succeeded. But the keystone, setting up the symbol, did not. Zei had already put up the countermeasures, and blocked the spaceship in its tracks.

Meanwhile, Bai and Zei continued to wreak havoc on Pan's territory, Bai attacking the west and Zei attacking the north and east. A thousand turns later, most of the game field had become a high-entropy wasteland. The main untarnished territories were Bai's home base, a fraction of Zei's home base, and Zei's side of what had been the battle front against Dza. Total symbol count: Zei five, Bai five, and Pan four.

At that moment, Pan resigned.

The audience cheered.

* * *

Two elevators opened up in the waiting room, and Zei and Bai simultaneously stepped out. Mu was waiting for them.

"Congratulations, you two. Next stop is the semi-finals in fifteen days."

"I'm confused. Everything in this game was the same as before. What was the new rule?"

"Ah. I thought you would ask that."

"A few days ago, right before the priests went into the dungeon to decide on the rule sets, I suggested to one of them that they should make things interesting, and be more creative. The doors stayed sealed for five days - twice as long as usual. And now we know why. The rule change was: the team selection was not done through cryptographic randomness."

Zei's eyes opened wide.

"This time, the game was a psychological trial. Despite having the teammate you wanted to cooperate with the least, despite making a new all-time record for the fewest messages exchanged in the audio channel during an entire game, when the moment came, almost without talking at all, you cooperated. And so you passed."

Mu paused.

"The priests will be pleased."

For a moment, Mu and Zei silently stared at each other.

Breaking the silence, Bai walked up.

"I'm sorry."

"Come on, you gotta give me something more substantive than that. Say, next time you take a vacation to Redshire, bring me along."

Then, realizing that the suggestion he just made might give Bai an unintended impression, he quickly added:

"And my mom."

Chapter 5

Freetown, United Cities · 3724 RAINMOON 11

Zei walked out through the door of the air harbor, and held the door open behind him.

"Come on, Mom!", he shouted.

Zei's mother, Min, hurried forward and came out through the door. Bai followed along behind, carrying a shopping bag.

The three of them followed the signs to the taxi stand, and immediately put their luggage into the taxi's luggage trunk, and opened the doors to get in.

"Welcome to Freetown, you both!", Bai exclaimed.

"Glad to be here", Min replied. "Actually my first time abroad."

"So how does this compare to Redshire, since I see you've been everywhere by now?", Min followed up.

"Oh I think Redshire is nicer. Was a shame that the Arctic drones made them cancel all the flights for the past five days."

"That really was too bad. But so far I think Freetown is great! So that means whenever I make it to Redshire I won't be spoiled, and I'll feel just as excited about Redshire then as I am about Freetown now!"

The conversation quieted down, and the three of them stared out the windows of their taxi. They were driving along a highway, and there were buildings of all shapes and sizes - some short, some tall, some stone, some concrete, some glass and steel, some beautiful, some ugly. Some halfway under construction, seemingly abandoned. And among them a few genuinely charming places, small parks with

benches and carefully arranged trees to make the rest of the city seem invisible to someone inside.

A few small copters buzzed in the sky overhead, some carrying individual passengers and others carrying goods - most likely food for the wealthy and extremely impatient.

In the distance, there was one area that seemed noticeably nicer than the others, that was surrounded by a high grey and purple colored wall. Bai and Zei, sitting on the right, both turned their attention toward it.

"Probably the local Greater Plum Harbor", Bai mused.

The taxi swerved off the highway to take an exit. It then continued driving along a street, winding through several roundabouts.

For a few minutes, the people and the insides of the buildings looked considerably richer than anything in Dzego. Suddenly, they passed under a highway, and on the other side the structures looked significantly more run-down, the restaurants and stores more basic, and more people outside were driving motorcycles.

As they passed through Freetown street by street, Zei, Bai and Min continued looking, observing the surroundings from inside their safe cocoon.

After five more minutes of driving, the taxi turned, and drove into a garage. It stopped, and the door opened.

"Thank you!", Min shouted, and opened the trunk. The three of them took their luggage, and entered the hotel.

Zei slapped his watch against a green circle on top of a one meter tall pole, which was labeled "Check in" in the center.

After one tick, the circle beeped.



Zei, Bai and Min's watches all buzzed in unison, confirming their room keys.

* * *

Having put his luggage into his room and taken a shower, Zei stepped out of the elevator and once again entered the hotel lobby. Bai was already there, looking outside a window.

"Mom said she needs to sleep, we should go explore", Zei suggested.

Bai nodded. The two exited the hotel, and entered Freetown's great outdoors.

"So why did you agree to come with me anyway?", Bai asked.

"Well it was obviously the lowest-cost opportunity I would have to go abroad for a long time", Zei replied.

"We both won our games in the semi-finals easily - my opponent was just not prepared for the sudden rule change four thousand turns in - and I saw you won even faster. So the finals are between me and you."

"And so it's a unique situation. Normally, there's a prisoner's dilemma: neither of us can take a break and enjoy anything, because if either of us does, the other will cram harder right before the big game and gain an advantage. But this time, that's not true - because we can both take a break together, so we're sure that neither of us is doing anything else."

"And the surplus is high - both Mom and I get to go abroad for the first time in our lives."

"Now why did you agree to take me?"

"Pretty much the same, now that I think about it. Just without the game theory math."

"Though for me what you call the surplus is spending time with you."

Zei groaned.

* * *

Zei and Bai arrived at a building that looked noticeably larger and nicer than its surroundings.

"Freetown Economics Institute", a sign read.

Right below the name of the institute, the sign listed a schedule of admission fees, featuring a formula that was a function of two variables: increasing with the square root of how much time you spend inside, and decreasing the higher your academic or intellectual reputation score rises.

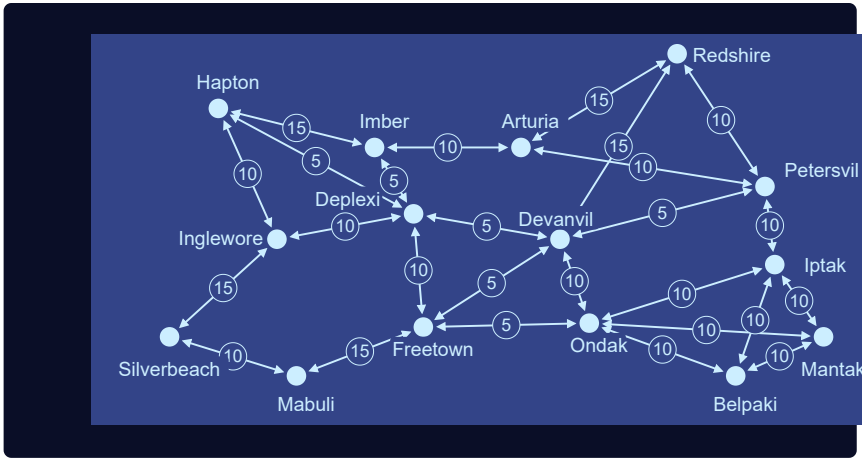
Academic and intellectual reputation score was difficult to get. The kind of reputation you could zero-knowledge-collateralize to get loans or check into hotels, that was easy to get; it was only hard if you had failed to pay back loans or trashed hotel rooms multiple times. Academic and intellectual reputation, on the other hand, required doing serious work.

There was an entrance gate with turnstiles and glowing green circles. Zei and Bai tapped their hand devices, and walked through.

The inside of the building was a single spiral-shaped sloped floor that looped around several times. If one walked along the outside wall, the slope was barely noticeable - a circumference of perhaps three hundred meters that one could walk around to climb four meters.

In the middle, there were elevators and staircases, for those who wanted to go up or down faster.

Zei and Bai leisurely walked along the floor, and soon saw a restaurant. Two people, a woman and a man, were sitting, drinking tea and eating a dish with vegetables and rice. There was a large screen, at that moment playing a lecture explaining the United Cities' co-governance system.



"When two cities are in a co-governance relationship, they each elect a percentage of the other's parliament."

"The percentage is based on a pre-agreed number. Whichever city is larger, as computed by a geometric average of population and GDP, elects that pre-agreed share of the smaller city's parliament. Whichever city is smaller elects a share of the larger city's parliament equal to the pre-agreed share, divided by the square root of the ratio in the two cities' sizes."

"You might ask, why square root? Well, it turns out that the utility of a large share of governance, held by a larger participant, is superlinearly larger than the utility of a smaller share-"

Zei stopped listening, and let his mind ponder, trying to work out the math from first principles.

Bai started walking toward the two people sitting at the table.

Zei, still thinking about the math, followed Bai. They kept walking until they were almost standing over the eaters.

"Hi, are you guys busy?", he blurted out awkwardly.

"Ah no, not at all", the woman answered.

"Please come sit down with us, I'm a tourist, always happy to meet locals and understand this place better."

"Ah, we're also tourists. From Dzego", Zei replied. "I'm Zei."

"And I'm Bai."

"I'm Seila", the woman replied. "And this is my ... friend, Jahn."

Zei and Bai sat down.

"So where are you from?", Bai turned to ask.

"Veridia", Seila responded.

"And I'm actually a local", Jahn added.

"So what brings you here?"

"Well, I'm a blogger, I focus on economics and culture. My husband's and my turn to take care of our children just finished, they're with our co-parents now, so we're both taking a break -"

"Well, not really a break, still working. But traveling and exploring."

"So how has Freetown been so far?"

"Well, free in some ways. Not always in others."

"You don't see as many privacy robes here as in Veridia or Dzego, but that's because people just go around in cars, all the interesting stuff tends to happen in these enclosed compounds."

"How is Dzego?"

"Same old. Studying, playing Minpentai. I'm trying to figure out if I'm going into physics or cryptography."

"Tell him about your arm", Bai whispered, loudly enough that all three of the others could hear.

"Oh yeah, my arm broke when our classroom got bombed a while back."

Seila and Jahn froze in surprise - first at the revelation, and then once again at Zei's total nonchalance in relating the news.

"Are the attacks getting worse?"

"They're starting to go after classrooms more. Have also been hearing about more and more surveillance drones getting snuck in."

"So is Dzego going to, like, add its own counter-surveillance network to keep up?"

"Well, the problem is, the Arctics got spies inside all the agencies, so if the counter-surveillance network can see too much, then it becomes their own weapon."

"That's why we've been trying to figure out the cryptography. Each surveillance camera running an algorithm, revealing data only in those cases where it's suspicious, posting a hash to a cryptographic network every time it reveals so we can track how often it's happening, and then a whole system of hardware attestations and randomized infrared scanning to make sure that the cameras are actually running the code that they say they are."

"There's even a rule that anyone can go and unscrew any camera in public and inspect it themselves. There's entire clubs that do it and publish their reports online. It's pretty advanced stuff. I wish I could understand how all the math behind these encryptions and proofs and attestations works."

"Come on, our whole DU chapter knows you're a hotshot at understanding thermodynamics, even better than the teacher. You can totally re-spec to cryptography and figure this all out."

"Are you *trying* to distract me from focusing on the Minpentai finals?"

A robot slid up to them, with a menu on its screen.

Bai and Zei started looking, and clicked buttons in turn to order some tea, as well as a similar rice and vegetables dish to what Seila and Jahn were eating.

"How much of your time is this whole arms race taking up?", Jahn asked.

"Of our time? Well, a few percent of us are full-time on it, for the rest, it's not supposed to be our job, beyond keeping up with how to stay safe. Of our happiness, our spirits, our souls? That's a different question."

Jahn and Seila sighed.

A few minutes later, the robot came back with their drinks, and a few minutes after that the robot came back again with their food. They ate, at times engaging in light conversation learning more nuances of Dzego culture - Minpentai, the Dzegoban language, the people's strong will to see the nation rise - and at times silently. The screen,

continuing to explain co-governance and show various diagrams and mathematical formulas, now went entirely unnoticed.

* * *

From		Message	Time
Gladias	→	[now calling]	58130

Seila tapped on the watch.

She could make out Gladias's face.

"How is your time in Freetown?", Gladias asked.

"Just had lunch with Jahn, and then these lovely kids from Dzego came by."

"How were they?"

"They were wonderful. Though it's so sad what the Arctics are doing to them - one of the two got his arm injured recently."

Seila decided to change the subject.

"How have you been?"

"Good."

He paused.

"I'm pondering something."

"What is it?"

"Someone suggested to me recently - that instead of going for being a Sentinel, I should be a Keeper instead."

"Huh!"

"What do you think about that?"

Gladias visibly paused for a few moments, to ponder his answer.

"It's the more important thing for Veridia, to really make sure that the rubrics themselves are set right. And I feel like the work will be more interesting."

"Well my first instinct is that I actually support the decision."

"You know, I've been thinking about Dzego for these past fifty minutes or so after those kids left, and then Jahn had to go for a meeting soon after."

"They benefit a lot from our Graph Funding, right?"

"Indeed they do."

"What about the rubrics though? Do the rubrics encourage companies to make any of their products more helpful to Dzego in any way?"

Gladias thought for a few moments. "Not really, at least I can't think of anything."

"Well maybe you should become a Keeper. And this could be one of your projects! Figure out how to make or adjust the rubrics to support Dzego better."

Seila paused for a moment.

Gladias stared in response, his eyes wide open.

"Yeah become a Keeper definitely. It's exactly the right choice."

"It'll be fun. As they say, Dze go ba fau gie!"

Chapter 6

Meldan, Veridia · 3724 RAINMOON 12

Gladias walked slowly toward the door of his house, keeping his head down so that his privacy robe with its hood could double as shielding from the cold rain. A small roof covered the area right in front of the doorstep. Once he got to the doorstep, he slowly took off his hood. He picked up a food box - the usual number ten from the Dzego food truck - from a small table. He then tapped his watch against a glowing green circle beside the door.

The door opened. Gladias sighed, entered, and let the door close behind him. He went down a set of stairs to the basement, and put the food box down on a table. The lights turned on, a gentle background noise simulating sounds from nature slowly ramped up, and another door automatically closed behind him.

Gladias took off his privacy robe, though keeping his neck band on, and threw the privacy robe onto the back of a chair to dry off. He sat down on another large soft chair that inclined back about forty-five degrees.

His watch buzzed as his local network authenticated it and let it connect.

He opened his mouth slightly and whispered an instruction through his neck band to open a call with Mov.

Almost immediately, Mov picked up.

"Are you in a secure location?", Mov asked.

"I'm whispering into a subvocalization neck band, and your audio is coming in to me by earpiece. We're inside a sealed basement room, the

doors and the walls covered with anti-transmission foil, inside of a locked and well-insulated house. There's real rain outside, and the sound of fake rain inside. Secure enough for you?"

"These days you never know."

"Well, if you want I can change the fake rain to a crowd babble so it's even harder to strip, *and* put on headphones."

"Fine, no need. Your setup is better than most as it is."

"I'm submitting my track decision today. I'm going to become a Keeper."

For a few ticks, there was silence.

"When did you decide?"

"A few days ago."

"You've been set on becoming a Sentinel for at least the past year."

"The Keeper role is where the real ability to make change lies. The rubrics themselves are more important than-"

"It was him, wasn't it? The mysterious gentleman, Delwart?"

For a few ticks, silence again.

"Does Seila know?"

"Of course she does, I told her two days ago."

"Before me."

"Well of course, she's my-"

"But you still haven't told her ... why you changed your mind, have you?"

"Can you just tell me what you learned from your snooping already?"

"Okay fine."

"So remember what I told you last time - when Delwart left the meeting, I followed him, and it turned out that he just walked over to some totally different tea shop in GPH, and he had another meeting there fifty minutes later. He left through some different door, so I lost track of him there."

"Yeah I remember."

"Well guess what, the story goes further. So I threw my surveillance bat into an innocuous spot in both places - both of those got caught, no

luck there. But I also made a point to re-visit both spots back and forth the next three days. Both spots kept quiet for two days. But on the third day, I caught sight of Delwart again. He always wears a new privacy robe every day, he re-randomizes his insoles to change his gait and his height, but he has one weakness that let me catch him."

"What is it?"

"His behavior. He looks around much less than the average person, he feels really focused on where he's going. I sacrificed a surveillance bat a couple of times to make distracting noises. Most people turn their heads at least a little bit; Delwart does not."

"And so that's how I managed to find him again, walking from one tea shop to the next."

"So he's having these meetings this whole time?"

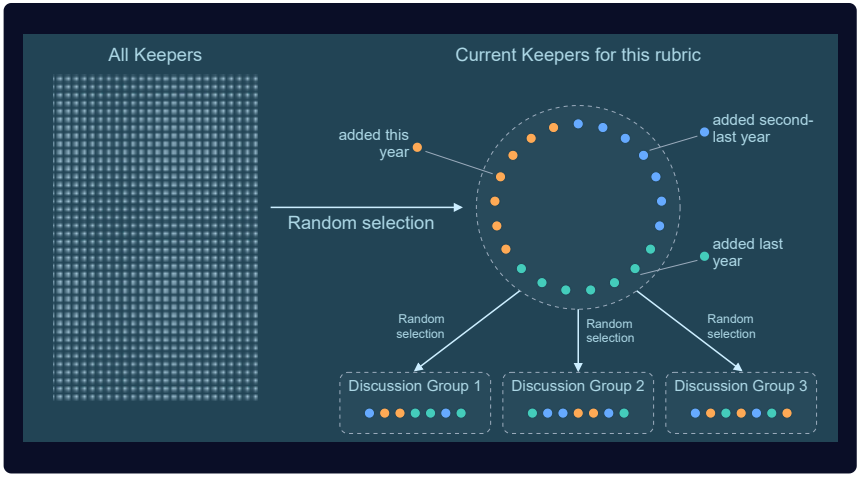
"Yep. Maybe you're not the only person he's trying to convince to become a Keeper."

"So what do we do?"

"It seems like there's only one thing we can do right now. Go along with what Delwart says, and keep our eyes open."

"Anyway, you should get some rest. Remember, we have that citizens' assembly to go to tomorrow."

Gladias ended the call. He took out his hand device, and returned to studying the Keeper governance structure.



Similar to the Sentinels, Gladias noted, but bigger: Sentinels were three groups of three, Keepers were three groups of seven. Courts were not part of the Order, but they were also similar, he remembered: two pairs of two separately deliberating before they cast their votes, plus one as a tiebreaker, the median of their five views determining the judgement. The Keeper groups having more people than Courts and Sentinel panels made sense: a rubric mattered more than any one single decision. The Courts being smaller made sense too: five judges were just for one round and you can always appeal.

Gladias slid back and forth between various screens, and scrolled through the current rubric set planned for evaluation:

	Rubric title	Summary
5	Emergency preparedness in physical spaces	Reduced tax bracket for presence of secure shelters, durable emergency supplies, full or partial grid-independence. Increased tax bracket for unsafe or fragile building design.
6	Openness in hardware	Reduced tax bracket for respecting user right to repair, publishing usable schematics, designs and other sources, and following open standards. Increased tax bracket for deliberate incompatibility, restricting modifiability and related anti-competitive practices.
7	Clean indoor air	Tax bracket proportional to observed average air quality indicators - CO2, PM2.5 and other mainstream measures. Average gives higher weight to higher-occupancy time periods. Applies to spaces that host more than 20 people.

Gladias kept scrolling through the Keeper documentation. After about thirty minutes, he laid back on the chair, tired. He asked Emerald to come up with a quiz related to the information he viewed so far to help him study. Emerald acknowledged, and began working in the background.

He then put down his hand device, mustered the energy to get off the chair and onto his bed, and went to sleep.

⁂

3724 RAINMOON 13

Gladias exited his house, wearing a privacy robe without putting on the face cover or hood. He turned right, and began walking down the street.

On both sides of the street, there were rows of houses just like his. Trees lined both sidewalks to provide shade. Most of the houses had one of a few similar default styles, with beige and light blue and grey

colors, and appeared to have been constructed within the last twenty years.

About one in ten houses, however, had some form of quirk to it. The first that Gladias passed by had a balcony that looked like a futuristic spaceship. The second did not have a regular house at all, and instead featured three smaller houses that were clearly refurbished shipping containers. They *looked like* something that he would live in, Gladias thought, and he even enjoyed looking at them, but they made it impossible to perceive the neighborhood as being uniform.

The third was a house with a coffee shop on the side. Several people were standing outside and drinking tea. Gladias remembered one of Seila's blogs from years ago. In Inglewore they would never allow this, he thought - housing districts were for housing. The cost: in Inglewore, if you lived in the wrong place, you had to walk one and a half kilometers to get to the nearest coffee shop. Here, he had just walked to one within three hundred meters without even intending to.

Finally, Gladias arrived at his destination: a small community and event center that was located at the first street corner. He saw others coming in, some walking, and a few taxis slowing down and stopping on the street beside the building. He turned right, and went through the same entrance that everyone else was going through.

"Mov, there you are!", he exclaimed, speaking in a voice that was simultaneously silent so that only his neck band could pick it up, and aggressive so that his neck band interpreted it as shouting.

"Ah good to see you Gladias!"

"So what is the topic of the assembly today?"

"They're discussing what kinds of public spaces we want to have more of, trying to create more opportunities for in-person social interaction. They're worried we're not getting enough of that now that everyone is talking now just on Silverchat and Bluewhale but also to Emerald and the other AIs."

"Makes sense."

They heard a whistle. The other participants started walking into one of the meeting rooms. Gladias and Mov both followed them.

Inside the room, there were five tables, each with ten chairs, and a stage area at the front with a podium for one person to speak. As soon as Gladias and Mov entered, they could feel a gentle breeze as ceiling-mounted fans pushed air downwards and replaced it with new fresh air sucked in from the outside. When Gladias looked upwards, he also saw what appeared to be faint violet lights shining down from the ceiling. Must be far-UVC disinfection lights, he thought.

They picked one of the tables, and sat down.

An older teenage girl sat down beside Gladias, and introduced herself.

"Hi, I'm Pelae."

"I'm Gladias."

"Nice to meet you."

About half a minute later, a man wearing a privacy robe without the hood or face cover walked up to the podium. After pausing for a moment, he began to speak.

"Welcome everyone!"

The room quickly went silent.

"For those who are new to this assembly, welcome and thank you for coming. I'm Alagael, I've been a Herald for seven years. This is the fourth session of this semester's assembly, the topic is how we can improve Veridia's public spaces to support better in-person social interaction."

"And for those who are new *and* those who have been with us the whole time, thank you very much for attending."

Out of the corner of his eye, Gladias saw Pelae take out and put on a neck band, turn towards him, and start whispering. A few moments later, his watch buzzed. It had registered that Pelae was trying to ask him something, and opened a conversation channel.

"Remind me, what do Heralds do?", Gladias heard her ask through his earpiece.

Before Gladias could formulate a response, Mov had already begun to cover it.

"Heralds are Order members - Keepers or Sentinels - that get randomly selected each year to retire, to be a sort of ambassador helping to create better understanding between the Order and the public. I think Graph Funding members or judges can become Heralds too."

"The goal is to provide some transparency. Keepers and Sentinels are very private to protect themselves from manipulation, but they don't want people to get the wrong ideas and start distrusting the whole thing. So Heralds act as a bridge. They're very transparent about how they think and how they did their work - but only after they no longer have any power, so there's no risk from anyone influencing them."

"So do all Heralds run assemblies?"

"It's up to each person's choice. Some give speeches, others write books, others help organize assemblies. The most antisocial ones just write books or moderate sub-forums on Silverchat."

"Heralds are also good people to talk to if you want to figure out if you're interested in joining the Order", Mov added.

"Do you want to join?", he added, and then bit his tongue, realizing that he had come uncomfortably close to committing the faux pas of revealing to a stranger his status as an Order member.

"Nah, I've already decided. I'm going into chemistry."

"Nice!"

"My friends think I won't enjoy it and I'll have to work really hard. But I like the challenge."

They turned their attention back to the podium.

"Alright, so I want everyone at each table to take turns, and share a positive experience you've had of a public space that you've spent time in that you appreciate. We'll do this for half a longhour. I will go around the tables, I'm happy to answer any questions."

"Let's begin!"

Alagael stopped speaking, and slowly walked off the podium toward the side of the room.

At the table that Gladias was sitting at, a middle-aged woman softly raised her hand. The others nodded, offering her the chance to speak first.

"I really appreciate Galanar Square, the one that's on the border of Kalimar about a kilometer south of here. I always go there to drink tea and have food and look at the birds and trees. It's such a nice space. Good trees, good tea, good shade, there's always interesting people to talk to coming by."

Gladias jumped in to the conversation.

"I like how there's also a playground right beside it. My family used to go there a lot when our children were younger, we would drink tea, and leave the kids to go and play."

The woman and several others nodded.

"There's also a mini library and science museum there!", Pelae added.

As soon as she finished, they heard footsteps. Alagael had walked by and was standing over them.

"Alagael", Pelae asked, "how do the results of these assemblies actually turn into outcomes?"

"Good question. Lots of people watch what the assemblies are doing. Keepers check the results when they're debating their tax rubrics, figuring out what features of these properties they want to incentivize."

"Sometimes Keepers quietly join the assemblies directly. Who knows, maybe there's even a Keeper or an Acolyte sitting at the table right now", he added in a joking tone.

For a few moments Alagael paused, and then opened his mouth again to say one last thing.

"Also Graph Funding members, and anyone participating in quadratic funding. It's a good signal to watch."

At that moment, Gladias's watch buzzed, and the watches of everyone in the room buzzed at the same time.

Gladias checked his watch. Some flu virus had been detected by the sensors inside the air filters in the room.

Almost immediately, the ceiling fans became faster and the violet lights brighter. About half of the participants in the room took out masks, some cloth and some elastomeric respirators, from their pockets and put them on. The other half simply did nothing and watched. Many participants also put on neck bands, intending to whisper through them for the rest of the conversation, just in case they were the ones that were sick - it reduces transmission if a sick person does not speak loudly. The process seemed natural and automatic, and everyone seemed to respect each other's choices.

Another woman spoke up. She was whispering through her neck band, but to Gladias her voice felt just as clear and natural through his earpiece as if she had been speaking directly.

"There's a similar square about a kilometer further west, Tapaia Square. It has similar things - you can sit down, drink tea, there are shops nearby. But the playground is much more basic, my son says it's not even comfortable. And I always get the feeling that that one is less lively than Galanar. So maybe the playground really matters."

"Tapaia's library is also like that. All the books are either some boring business books, or things in foreign languages nobody understands, either Old Belpakian or something from Haragmir."

Mov jumped back into the conversation.

"Ah yeah, I know what's going on there. There's a rubric that incentivizes ground-floor active use, but a lot of people are gaming the system - putting things in that theoretically qualify but that nobody actually wants to use. I suspect whoever runs Tapaia thinks that their main customer base doesn't like being around kids."

"That rubric really needs to be adjusted..."

Pelae and the other woman both nodded.

Mov sat back, and the conversation continued. As the others continued to answer each other's questions, Gladias and Mov both mentally took notes on what the common threads were in the

preferences that people were expressing. In the background, Emerald was taking notes as well.

* * *

Soon after leaving the assembly and saying goodbye to Mov, Gladias arrived at a restaurant. The door opened in front of him. He walked in, and then sat down. A robot slid up to him, and showed him a menu. He pressed a few buttons, ordering water and a salad.

He looked around the restaurant. Four others were sitting down, all beside the window to enjoy the view - two people alone, and one group of two.

One of the two people sitting alone was wearing some kind of earbuds and a neck collar, and about twice a minute he moved his mouth for a few ticks. It was either a conversation, Gladias thought, or some kind of AI-assisted tutoring. The other person sitting alone did not seem to be doing anything, and was simply enjoying the view. The pair sitting together were both women, seemingly in their thirties, and were having a conversation. Gladias could make out some of their words from a distance - something about their sons' upcoming Helisport game.

After a couple of minutes, a teenage boy, not quite an adult, walked in. Gladias recognized him instantly: Febric.

"Uncle Glad, good to see you here!"

"Febric!"

Gladias reached out his arms for a hug. Febric obliged.

"You on your way to school?"

"Yep, off to school. Exams are in two days!"

"You feeling confident?"

"I think so! We're doing the trees exam today, I coded two different kinds of bounded-depth trees from scratch from memory yesterday, as well as the hash-based trie. Once you actually understand it, it's so fun!"

"Are Vil and Daia taking good care of you guys?"

"Oh they're amazing. Really caring, as always. They cook really good food for us every day. And Dad bought me a new compute box!"

A waiter handed Febric a tea in a cup with a sealed lid.

"I'll pay for it", Gladias called out. "You should get to school, good luck!"

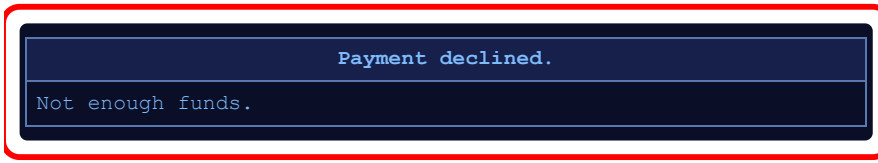
"I miss you, Uncle Glad!"

Febric ran out of the restaurant, turned left, and disappeared from view.

A few minutes later, Gladias's water and salad arrived. Gladias slowly ate, and then walked up to the counter to pay.

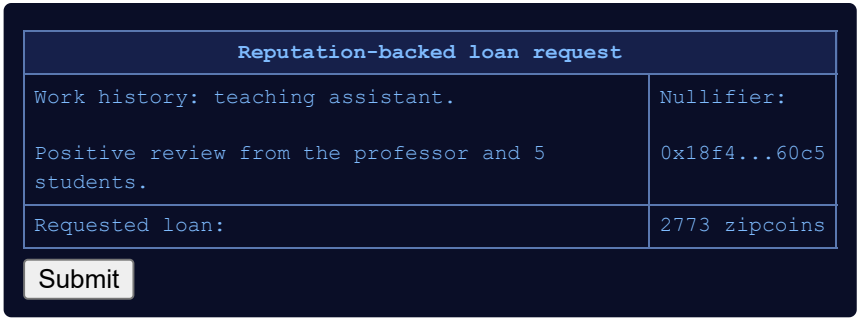
The outline on a green circle on the tabletop lit up. Gladias tapped his watch against it.

The outline turned red.



Gladias quickly scanned through the wallets in his hand device, and realized that he had forgotten to send himself funds. The bulk of his money was only accessible either from home, or, if absolutely necessary, through social recovery via combined signatures from four of six keys held by pre-selected family members and friends, none of whom knew who the other recoverers were.

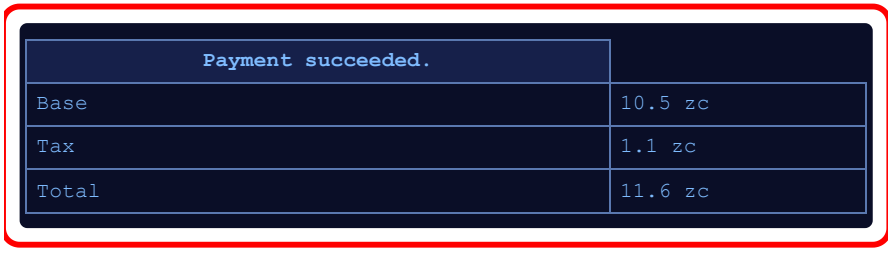
He thought quickly. He asked Emerald to take out a loan collateralized by his reputation. His hand device spent a tick collecting nullifiers and generating a zero-knowledge proof, and displayed it on Gladias's watch for review.



The only information that would be publicly visible was the nullifier and the amount borrowed. The nullifier was a pseudorandom hash generated in a deterministic but unlinkable way from the public reviews of his work. It would get published to the cryptographic network, preventing the same reputation from being used to collateralize a loan twice. The proof would confirm that the nullifier was valid and the reputation it was representing was sufficient to take out those funds, and reveal nothing else. Gladias's name would not be visible to anyone.

Gladias tapped "Submit", and the transaction flew off into the cryptographic network. Two ticks later, it confirmed and the 2773 zipcoins appeared in his wallet.

Gladias tapped his watch again.



In Veridia, and across much of the world, even the sales tax payments were zero-knowledge. Only Gladias and the restaurant knew that a 10.5 zipcoin order had been made. The sales tax was sent to the government in real time, automatically. The whole scheme was

enforced not by auditing spreadsheets, but by random inspection - people coming to order, and verifying that a tax-paying transaction type was used any time that the restaurant requested payment.

Gladias thanked the waiter, and slowly walked out of the restaurant.

* * *

After leaving the restaurant, Gladias continued walking along the street, and after another few hundred meters turned right. After passing by a few houses, the path melted seamlessly into a forest. After a few hundred meters more, a series of houses made out of large stone bricks began to appear on both sides.

Gladias had taken this road through the Kalimar district many times before.

He kept walking, went across the sky bridge, and into the tunnel. He got to the Y-shaped intersection. This time, the poster with Lord Ephelion's message was gone. After stopping for a moment, for what might have been only the second time in his life, Gladias turned left at the intersection. He kept walking forward.

The path soon went back above ground. It snaked under a few highways, though covered by trees the whole time - *especially* when the highways were overhead.

The amount of graffiti on the bottom of the highways, as well as the amount of dirt under them, felt like more than usual. Gladias wasn't sure if the cleaning was less effective recently, or if it was simply this route that was less well-maintained for whatever reason.

Soon, the road began to curve up, and transitioned into a straight, two-hundred-meter-tall outdoor staircase on the side of the mountain. Gladias had only been here once before: when he was inducted as an Acolyte.

For about ten minutes, Gladias climbed the stairs. The intensity of the exertion, together with the privacy robe, made Gladias feel warm and almost sweaty, despite the outdoor cold. Finally, he made it to the top. The path once again turned into a tunnel. The tunnel soon turned

into stairs, and finally after climbing two flights of stairs, open daylight appeared once again. Gladias was now standing in the center of an open area on the top floor of a tower, surrounded by stone walls. He could see the sky above him. Beside him, there was a green circle, larger and glowing more brightly than usual. Sitting on seats around the edge of the tower, there were ten Order members, all wearing privacy robes.

Gladias tapped his watch against the circle. Various lights along the walls, lights that he did not notice before, lit up in a unique green shade.

Almost immediately, an Order member spoke.

"Welcome, Acolyte."

"You can call me-"

"We're not supposed to know, it's the cryptography's job to make sure this procedure gets applied to the correct profile."

"Then I guess you can call me by whatever the digits of the nullifier in my proof are."

A few Order members' bodies tensed up somewhat uncomfortably. They were not expecting this style of engagement for what was supposed to be a solemn and mostly silent procedure. But a few others leaned forward, seeming to welcome the variety.

"Very well, three-seven-two-eff. Have you made your choice?"

"I have."

"And that choice is?"

"I am applying to become a Keeper."

For a few ticks, silence.

"Three-seven-two-eff, you may tap your hand device against the circle once again to confirm your decision."

Gladias tapped his hand device against the circle.

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Request confirmed.
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Chapter 7

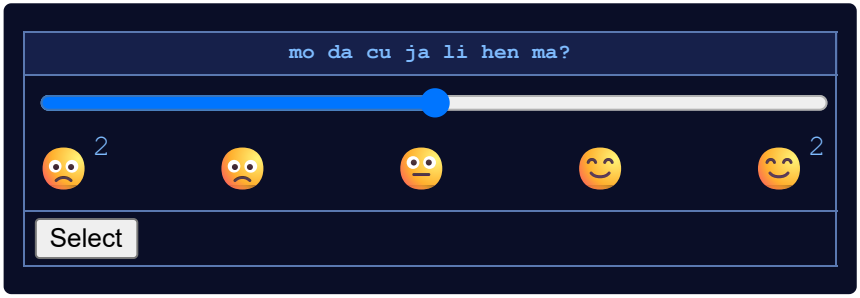
Pafogai Du, Dzego · 3724 RAINMOON 23

Zei was sitting alone at a table in a courtyard, drinking a tea and eating a rice and vegetable dish. In his left hand, he held his hand device, reviewing Minpentai strategies.

A robot slid up to him.

"mo da cu ja li hen ma?", it asked. "Is the food and drink good?"

The screen on the robot turned into a voting view.



Zei thought for a few moments, and moved the bar about three quarters of the way to the right.

The robot beeped, and slowly slid away.

When Zei looked up again, he could see Fin running up to him. Fin arrived, and sat down.

"Where have you been?", Zei asked.

"More linguistics training", Fin replied. "Looks like the Bansunpei think I'm actually a good fit for them."

"How is training for the big game?"

"Did about five games against the bots in the past longhour", Zei replied. "I feel like this is already hitting diminishing returns. It's all going to hinge on what the rule change this time is."

"So in my last two games I asked the bots to come up with random rule changes, and played against them. Problem is, I won easily both times: the bots play better than me on stock Minpentai, but they are much worse at adapting to any new situation they don't have training data for."

"Can't you make a pipeline where once they come up with the new rule, they self-play finetune to get good at it, and then you play against that?"

"Well if you can go help spin that up for me in the next two longhours, I'd much appreciate it. Game is tomorrow, remember!"

Fin thought briefly. "If I do that I'll have to tell the Bansunpei I am skipping tonight then."

"'Po so dze go ban' can wait, the game against Bai is tomorrow."

"Please?"

Fin thought for a few ticks more. "Okay fine, I'll tell them I'm skipping. But I'll have to make it up with them next month."

⁂

3724 RAINMOON 24

Fin and Zei walked up the slope toward the mountain pyramid. Zei went through Minpentai strategies in his mind, as the tiring climb helped him concentrate, suppressing his mind's desire to wander toward anything else.

Suddenly he felt a tap on his shoulder.

Zei turned around. It was Bai.

"So, excited about today's game?", she asked. "I think it'll be a good finish for you, I don't think your mom quite realizes how impressive it is that you are going to come in second across the whole city."

Zei's mind raced for a retort.

"You know if you were nicer to Zei", Fin snapped back, "he might pay for *your* next trip to Redshire or Freetown from the prize money when he wins the nationals."

"Maybe bring along your mom too."

Bai's grin faltered for a moment, then returned.

"Thank you, Fin."

"But seriously, let's just concentrate on giving everyone a good game."

Suddenly, the ground shook slightly. A few moments later, from their high vantage point already a third of the way up the mountain, they could see a cloud of smoke emerging in the distance, somewhere in the outskirts of the city.

"Whoa, what is that?", Fin asked.

Zei took out his hand device, and pointed it at the cloud of smoke.

"Let's check."

Zei held his hand device pointed at the cloud of smoke and the area below for a moment, and then put it down. "Geolocated it. It's the largest computer chip manufacturing facility in the city. Remember, the Arctics have been trying hard to stop us from progressing any more in our technology."

"Looks like this facility was government-run. I imagine they'll just say it was an accident or something."

"Heh", Fin smirked.

"What part of the 'bin hu' in 'gie fe kiu kai ci bin hu' do these people not get? No head. You're not supposed to centralize stuff this much."

"Again, this is government. Half of them don't understand what they're doing, and many of the rest are probably on the payroll of the Arctics themselves in some form. All of the actually smart and effective stuff in Dzego has been outside of government *or* the big corporations for decades."

"But anyway, we have lots of smaller, distributed underground fabs these days. We'll be fine."

"Okay, guys", Bai interjected. "We have to actually get to the game."

The three of them resumed climbing. After a few more minutes, they arrived at the main door, and entered the tunnels. On the third turn, Fin turned left to go to the audience section, and Bai and Zei turned right. They walked through the door, and saw Mu waiting for them.

"Welcome back to you both", Mu said. "Excited to see how you both perform."

"I hope the rule change this time is going to be good."

"Oh, it will be. You'll see."

A robot slid toward them, and gave them three glasses of water. Zei, Bai and Mu each took a glass.

Mu raised the glass. "To our future," she said. "Dze go ba fau gie."

Zei felt slightly confused - "Dzego will rise again" was a beautiful thing to say, but it seemed more appropriate for either an actual battle or some kind of industrial or educational activity than for a Minpentai championship game. But he nevertheless instinctively replied back, at the same time as Bai.

"Dze go ba fau gie."

The three of them drank their water.

They sat down and rested for a few minutes. Then, two elevator doors opened.

* * *

"MU GU GEI FA", the loud automated voice announced. Starting in fifty ticks.

Zei breathed in and out, and recited his strategies in his head for the last time.

"LE MU GEI FA".

Zei scanned the audience with his eyes. He could barely see most of them, but in the aggregate he could make out more excitement than he saw in both of his previous two games.

"PA GU GEI FA".

Zei breathed in again, and held his breath for the last remaining ten ticks until the announcement.

"Welcome to the final round of the eighteenth year of the Minpentai tournaments in Pafogai Du. Tonight, we will witness the most decisive game in our entire city, between contestants whose playing skills have both been rated by AI as better than the national champions of only five years before."

"Today, we have our two finalists, two brilliant minds whom we saw fight side by side as teammates a month earlier: Zei Leimin and Bai Jahen."

"And today, I am also pleased to announce the new rule for this round. Both players will have access to a powerful personal AI, which will be able to assist them in their gameplay."

A new tab suddenly opened up on Zei's display.

"However, the players and their AIs will have full read access to the thought transcripts of each other's AI."

Zei took in the information, and quickly began to formulate his thoughts about how he would use the AI. He asked it to set up standard glider and wall formations, in his own style. To his surprise, the AI knew what his style was.

He saw Bai's AI transcript in a separate tab to the side; Bai was doing the same, and Bai's AI also knew her style.

"MU GU GEI TAU FA", the loud automated voice blared. The battle begins in fifty ticks.

The AIs kept optimizing their glider and wall formations. At this point, both AIs started designing countermeasures to the strategies in the other's transcripts.

Zei started planting a few structures by hand, speeding up his work by copying and pasting pieces of what the AI had already made, looking attentively at Bai's transcript to see what strategies Bai's AI had not yet built countermeasures to.

"LE MU GEI TAU FA"

Zei started planting one last major structure by hand: a symbol-carrying spaceship. There were already four created by his AI, but he figured that he would make one all on his own, so that Bai would not see it, and with a different style of moving wall in front of it. He aimed

the symbol-carrying spaceship so that it would reflect off a wall that his AI had already constructed, and then move upwards along the middle of the game field.

"PA GU..."

Come on, Zei thought, finish the symbol-carrying spaceship.

"SO ... BI ... ZE ... HA ..."

Zei put down a few last squares.

"MU ... FO ..."

And... done.

Zei let out a long breath.

"SHI ... LE ... PA ... TAU FA"

Gliders started flying. Zei's symbol-carrying spaceship started moving west from the southeast corner. The portion of the game field that Zei could see increased.

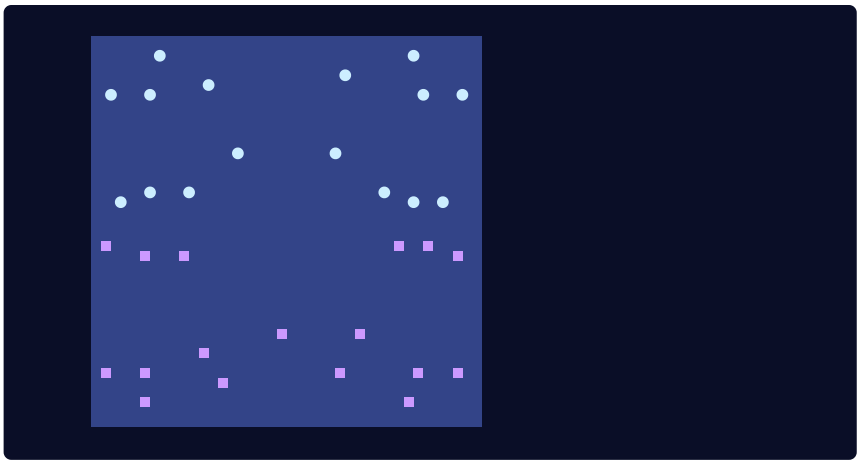
Both AI transcripts were focused intensely on exploiting their territories, building more gliders, spaceships and walls, and expanding.

After a few hundred turns, Zei and Bai's gliders began to crash into each other, and then into each other's spaceships and walls.

And a few hundred turns after that, their symbol-carrying spaceships got within sight of each other.

Shortly after that, Zei's symbol-carrying spaceship reflected off the wall in the south, and began moving north.

On the big screen, the audience watched.



The first intervention turn came. Zei let his AI do the work adjusting structures near the frontline, as the consequences would be immediately visible to Bai anyway.

He scanned Bai's AI's transcript for weak points. He found one: it was not thinking at all about glider attacks coming down the diagonals. He toned down the amount of resources allocated to the AI, and used those resources to manually send a large number of gliders down both diagonals, targeted so that one would pass right after the other, to strike both of Bai's home bases. He also sent a symbol-carrying spaceship, targeted to pass the center hundreds of turns after both sets of gliders.

The game continued.

The battle along the front lines in the middle was surprisingly even-handed. Both AIs were playing near-optimally, and there was little opportunity for trickery because both sides could see each other's moves and thoughts. However, Bai did put more resources toward the frontline especially on the west side, and as the hundreds of turns went on, it began to show on the board. Soon, one of Zei's symbol-carrying spaceships on the west was breached, and it exploded into a storm of gliders flying in all directions.

Zei kept playing, optimizing the minutiae of the game board. He knew that he would not get to see the results of his attack for a long

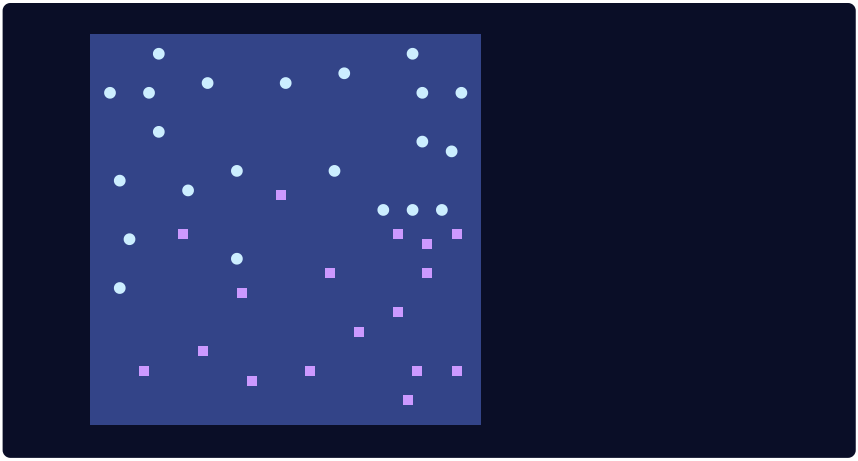
time.

The second intervention turn came.

The best that Zei could come up with was constructing more factories beside his symbol-carrying spaceship that was slowly flying north along the middle of the field. Aside from the attack down the diagonal, this was the one structure that Bai had no way of knowing about - unless his AI decided to ignore Zei's instruction to focus exclusively on the domain it was assigned to, and blurted out the spaceship's position mid-transcript. Zei decided to ignore that possibility; not much he could do about it anyway. He put up more walls along the west side.

More turns passed. On the west side, Bai continued to advance. Some of Bai's gliders got into Zei's base in the southwest, and destroyed two of his symbols. Zei watched as his symbol-carrying spaceship made its way toward the northwest.

Then, suddenly, gliders flew at the spaceship from the west, north and east side. It quickly crumbled, and exploded into rubble and a dozen gliders. At that point, Zei realized what had happened. Bai's AI's lack of thought about the diagonals had not been a mistake at all. Rather, Bai had deliberately directed the AI to focus on the other directions, and she had put up a strong multilayered defense along the diagonals herself. It had been a trap, and Zei had fallen squarely into it.



Mu felt a pang of disappointment in Zei. Seventeen symbols to twenty-two, she thought. How could he have fallen into that trap, and not come up with anything interesting himself? He had one spaceship going north in the middle, but so what? Bai's bases were well protected against attacks along the diagonals, and Zei's spaceship did not have excellent chances of reaching the north side. Unless Zei came up with something totally unexpected, she figured, this would turn into a game of attrition, which would only shift further in Bai's favor.

An unexpected set of gliders knocked out another two of Zei's symbols, one slightly southwest of the center of the field, and another slightly west of the center near the south end. Another set of gliders unexpectedly hit Zei's southeast base, but bounced off harmlessly against now well-prepared walls.

Sitting on the other side of the room, in less comfortable public seating, Fin gazed with fear at the gameboard as well.

Finally, something inside of Zei's mind clicked.

"Take my fourth westernmost symbol", he told the AI. "Take the best sixteen strategies that you can come up with. Construct near that symbol a factory capable of executing each of those strategies. It should be activated by a glider coming in from the southwest, and the exact strategy chosen should depend on the type of glider and the time

it hits. Then, near my second westernmost symbol, make that glider. Allocate eighty percent of resources this intervention turn to this."

The AI started thinking in detail, implementing the factory, and the glider.

Bai's AI started thinking about countermeasures to all sixteen strategies, but soon hit a limit: it could not optimize defenses against all of them at the same time.

Bai's AI also focused on increasing defenses across her own western flank. Zei's *first* gamble paid off, he realized: Bai herself did not realize just how much damage she had done on the west side, and just how little Zei had done to compensate. And so neither she nor her AI realized that by this point, Zei's fourth westernmost symbol could be in the middle of the field.

Then, Zei went to work on his *second* gamble. He used the resources he had not allocated to the AI to add a brick south of his factory to block the glider that the AI was sending, and add two of his own gliders coming in from the south.

The intervention turn ended. Part of the factory had been built during the intervention turn, the rest continued to be constructed in real time out of nearby rock formations by gliders.

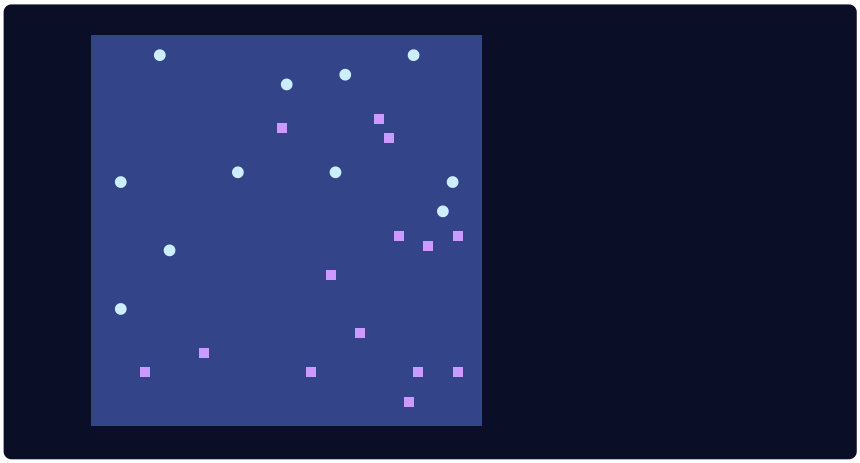
A few hundred turns later, the glider sent by Zei's AI from the west was blocked by the brick that Zei had put in place. A few hundred turns after *that*, the gliders that Zei sent manually crashed into the factory, and activated it.

The factory picked a strategy that was none of the four that Bai's AI had focused on planning for. And it was coming from a place that Bai had not expected: the halfway point between the center of the board and the northern edge. The factory quickly started generating symbol-carrying spaceships that began to move toward the east, and a storm of gliders that moved north, and then bounced off of rock formations and Bai's own walls to move toward the west. Bai, and her AI, were caught completely by surprise.

By the time the next intervention turn came, Zei's spaceships had gotten well behind Bai's lines on the east. Zei's AI took the opportunity

to design a storm of gliders to attack Bai's flanks from behind, and Zei added a few of his own gliders to attack Bai's northeastern base - building them himself so that they would not appear in the transcript.

Five hundred turns after that, the game board looked very different.



By this point, Mu's solemn head shaking and Fin's expression of fear had both turned into excited smiles.

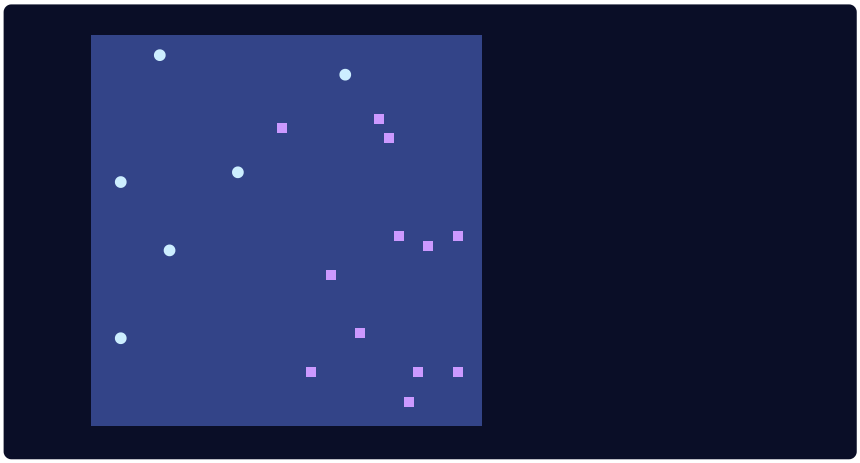
Fourteen to eleven. Zei was winning again. And he had a better position.

Soon, the next intervention turn came. At this point, Zei figured, he would just repeat the same strategy, though a more near-term version of it: immediately create many glider factories in the northeast, and have their exact style and direction be set off by a trigger that would only appear in ten turns. Bai could not see the exact internal logic of the trigger, and Zei designed that piece himself, so that it would not appear in the transcript.

Bai attempted to copy Zei's strategy. Zei spent half his resources generically shoring up his defenses, without attempting to be particularly clever in outsmarting Bai. He was reasonably safe, he

knew: Bai simply did not have any positions behind his lines the way he had behind Bai's.

Five hundred turns later, both attacks came to fruition. Zei's succeeded, Bai's failed. Bai's only significant accomplishment was destroying the remaining symbols in Zei's southwestern base.



Twelve to six.

Bai resigned.

The audience cheered.

* * *

"Congratulations, you two-", Mu began to speak, standing in the waiting room along with Zei and Bai.

"Congratulations, Zei. You only had to wait until our last season to beat me."

"You played the best game of your life today, Bai."

Mu paused for a few moments.

"The priests did not even know what you two would come up with. They told me that they wanted to make the game a little more ... *realistic* - but they were as curious as anyone else to find out what that means."

"Realistic?", Bai exclaimed incredulously. "How is it at all realistic that AIs can read each other's minds?"

"It sounded crazy to me too. But the more I think about it, the more I've come to see their point."

"Look at the Veridian Steering system, for example. It's one of the most opaque parts of the government - even the Keepers don't know who each other are - and yet one of the most trustworthy and incorruptible parts precisely for that reason. Similar for the Courts - anonymous, randomly selected committees out of a large pool."

"But the Parliament is transparent, and so when it comes to the Parliament's direct decisions - large-scale infrastructure, foreign policy, healthcare, all the things that it's funding - everyone keeps exploiting and outsmarting it. Same with the quadratic funding - actually, it's even worse. Keepers, Sentinels and Court judges are not fully insulated - it is still Parliament committees that are approving them - but at least there's a layer of indirection there, and those positions are also gated by prediction scores."

"That's the lesson. In the real world, it's not just thinking well that matters, it's also hiding your thoughts - sometimes, even hiding your thoughts from your allies or the people you're meant to serve."

Zei and Bai both stared at Mu.

"Anyway, you've both performed extremely well today."

A robot slid in carrying two bags, one with the number 1 on the side and the other with the number 2.

Mu handed the first bag to Zei and the second bag to Bai.

Zei and Bai both sat down and started looking at the contents inside their bags. A plaque, a scratch-off one-time-scannable code containing some cryptographic tokens and credentials, and smaller gifts: a cup, a handwriting instrument, and an ozone-based electric water disinfection pen.

Bai began studying all of the gift items in detail. Zei immediately started to scratch off the cover from the scannable code, and took out his hand device to prepare to scan it. He had been hoping to be able to

buy his mom a new compute box for many months now, and he still could not believe that now he could finally actually get it.

Mu stood up.

"I hope you both have a good rest", she said, as she turned and began to leave the room.

As soon as Mu was gone, Zei noticed that he got one gift that Bai did not: a dark purple letter, with an inscription on the front.

dun fe tin di zei de kin kie

"For Zei's eyes only"

Zei stopped scratching the cover on the scannable code, and instead put the letter into his pocket.

Chapter 8

Meldan, Veridia · 3724 RAINMOON 26

Gladias's watch buzzed as he was walking along the street.

From		Message	Time
Seila	→	[5 images]	34125

He opened up his hand device to see the images more clearly.

Seila was sending photos from Iptak. Gladias was immediately curious. Neither he nor Seila had ever before been to "the aks", as the Veridians liked to call the cluster of Ondak, Iptak, Mantak and Belpaki. The architecture was visibly different, more brick and wood, the colors more red and brown than what he was used to seeing not just in Veridia, but even in pictures of Dzego, or for that matter any of the United Cities that Gladias had been to or seen pictures of.

The signs were in a language that was half alphabet, half pictorial - the sort of thing that Dzegoban had been not too far from before they stripped that out in order to streamline the language.

He pressed a button, responding to Seila with a heart symbol.

"I'm looking forward to your blog", Gladias whispered through his neck band, letting Emerald automatically route that message to Seila.

"I'll be back in a few days!", Seila replied.

At that point, Gladias arrived at the end of the path, at an intersection with the sidewalk along a street. Right in front, there was

a sign listing the autobuses that stopped at that point. Gladias saw the number of the autobus that he was meant to take. He stood and began to wait.

Gladias's watch buzzed, and a notification appeared onscreen.

From		Message	Time
Mov	→	Remember that your trial in today is not just about knowledge. It's also about making sure the senators like you.	34576

Gladias sighed, and let his arm drop back down to beside his waist. He continued to wait.

* * *

A few minutes later, the next autobus arrived. Gladias checked the number. It was exactly the autobus that he needed. He quickly got on, and the door closed behind him.

Out of the corner of his eye, he saw more people than usual looking intently at their hand devices at that moment. Curious, he stepped forward and quietly looked at what one of the other passengers was looking at.

"Veridia National University", the header at the top of the view on the passenger's hand device said. "Live debate between Professor Sylka, president of VNU, and Lord Ephelion, Arctic ambassador to Veridia".

Sylka was speaking, and Gladias could make out the subtitles.

"Gross economic output alone measures one part of human progress", she said - though Gladias could only see and not hear. "But it becomes a very limiting and impoverished perspective if you start viewing it as the entire goal."

Gladias was curious enough that he whispered to Emerald through his neck band to put up an audio livestream of the debate.

"There is so much more that matters. There is the citizens' health, their emotional well-being, and how healthy the bonds in the society are."

As Sylka finished speaking, the audio livestream connected. Gladias stepped back from the passenger, somewhat embarrassed that he had momentarily violated the passenger's privacy by looking at his screen, and relieved that he was now receiving the feed in a completely ethical way.

"What good is a large car if people feel unhappy because they have no purpose in life? What good is a large house if people feel alienated in their society, and don't have friends to invite to it?"

"What good are the latest electronics, if the people at the bottom use them to hack and steal, the people at the top use them to invade people's privacy and surveil, and everyone all the time uses them to get outraged on social media platforms?"

"Such questions are the entire reason why Veridia has mechanisms like Steering."

Sylka paused. Lord Ephelion started speaking.

"There has now been a century-long tradition of people trying to make better metrics than gross economic output. But what we have seen is that especially over the long term, the countries that perform best on any of those metrics are precisely the ones whose economic output is highest."

"And there is a deep reason for this. Historians have long understood that many phenomena in society move in cycles. The ancients talked about a cycle: flourishing, pride, arrogance, downfall, suffering, rebirth, and then flourishing once again. They said the same thing about governance, and how it inevitably cycles between technocratic, democratic, chaotic and strongman."

"Whenever we talk about the inevitability of rises and falls, it is precisely these 'softer' traits like 'societal cohesion', 'freedom' and 'well-being' that move in this fashion. On those dimensions, the dominant transformation is a rotation: good creates the generator of

the bad, which creates the bad, which creates the generator of the good once more."

"And let us be real: these 'softer' traits that people are targeting, they're chosen by committees of priests - selected not for competence, but for echoing whatever ideology strikes people's fancy at the time."

"There is only one thing that stands out from this pattern and offers a truly lasting improvement: the relentless march of economic growth, enabled by the unceasing progress of science, technology, and capital accumulation. While the emotional vibrations in a society cycle between the pride of success, and the desperation of collapse and defeat, with every cycle, people are smarter and richer than before."

Gladias could see the eyes of a few of the people on the autobus watching perking up.

"The Order members are chosen by a sophisticated-"

"Order, please. Ephelion has the floor.", the moderator interjected.

"And so if you want to truly improve a society, in a way that provides lasting value for the long term, you do not focus on the thing that moves in cycles. You do not focus on accelerating the rotation. Instead, you focus on increasing the largest eigenvector. You focus on affecting the thing that has a thousand-year record of compounding on itself, the vector on which good reliably leads to more good. And you let the rest follow the natural laws as it may."

Ephelion stopped, and Sylka cleared her throat, preparing to speak. At that point, Gladias realized that the autobus door was open, and it was already at his destination. He quickly ran out, and Emerald automatically paused the livestream to help him concentrate.

His mind lingered on the debate for a moment longer, and he noticed with faint irritation how easily Ephelion's sentences arranged themselves into an argument that seemed to make sense on its own terms. A gifted demagogue, he concluded. He then refocused on what lay ahead.

Gladias started walking forward along a path covered with trees on both sides. After a few minutes, he saw a large circular building ahead. This was the parliament. The structure was notably different from the

intentionally distant and mysterious headquarters of the Order of Steering: it was on ground level, the construction mixing medieval and modern aesthetics, with multiple entrances on all sides.

He kept walking forward, and soon arrived at the entrance.

* * *

"Welcome, Gladias, to your selection hearing."

"You have been an Acolyte for four years, and you have reached the ninety-fourth percentile in your prediction score. You have applied to join the Order of Steering in full standing - though whether you're applying to be a Sentinel or a Keeper, that's something that only you and perhaps your most trusted advisors will ever know."

"Today, we will administer the second part of the admissions test: the approval of the Parliament committee. Are you ready to begin the hearing?"

A robot slid up to the chair in the middle of the room where Gladias was sitting, carrying a cup of water. Gladias took the cup, and moved it to the cup holder on the side of the chair.

"I am ready."

"Then let the questions begin."

Twenty senators were sitting on large chairs arranged along the whole edge of the circular-shaped room, spread about two meters apart from each other. In between some of the chairs, there were air filters. Above them, there were bright lights aimed downward, some bright yellow lights to light up the room, and a few faint violet-looking lights to keep the air clean.

In a particularly distinguished seat was the Chairman of the committee. About five senators pressed down on the buttons on their table, requesting to speak.

The system automatically selected one, and lighting around the edge of his seat turned green.

"Senator Verdow, you may ask your question."

"Gladias, tell us about your background."

"Well", Gladias stammered. "I was born thirty-nine years ago. Fairly middle class family, nothing too fancy. I went to a normal school, got good grades, but never got top of the class or anything."

"Why not?"

"I think I was just not motivated to. I tried very hard for one year, got an average of ninety-seven percent. But then I realized that the main emotion I was feeling, was the fear that if I screwed up even once, my beautiful record would be gone. I even avoided taking the harder classes. At the end of that year, I just felt like that was a game not worth playing. So I put in less effort, and started programming, and then studying law, in my spare time, because those things interested me, and I ended up doing decently well in the programming competitions."

"Why did you choose to become an Acolyte?"

"The first hobby project I gave myself when I seriously got into programming as an adult was writing an alternative client for Silverchat. After I finished, I tried to do the same for Bluewhale. And it was so much harder, because they were more closed and proprietary in many ways. That was when I realized just how much value Silverchat's openness was bringing to me that Silverchat itself would never capture. And I had been studying rubrics only a few days before, and it clicked - this, the aspects of a good society that the market alone misses, is exactly why Steering is so important. And so I felt motiv-"

A timer buzzed.

"Next question! Papli."

"What's your favorite band?"

"Err... I don't really have one."

"Favorite game?"

"Don't have one either."

"Favorite sport?"

"I remember enjoying playing Colorball when I was a kid? The school game that Helisport ended up being based on."

"How can we possibly trust you to judge on our society's behalf, when you're so out of touch with so much of regular Veridian people's

lives?"

"Isn't this why citizens' assemblies and broad listen reports exist, and why a Keeper group has months to listen to both before it makes its decision on a rubric? And isn't this why Keepers and Sentinels are allowed to specialize? I probably won't be deciding on anything that has to do with sports."

"What do you think of Dzego?"

"I like their food. I only started learning more about their history recently. My wife met a couple of Dzegojan kids and they seemed nice. In general, I feel for their plight, I hope we could find a way to support them more."

"Aha, so you want us to support Dzego, and you want us to make our social networks more open so that some super smart kids can play with them more easily. But I haven't seen you talk once about something that a regular Veridian would care about, that affects their lives."

"You don't think increasing hardware openness allows more mechanics to join the market, which makes owning things cheaper-"

"Well what about the salaries of the mechanics themselves!"

Gladias started formulating a response in his head. Economics was a basic part of the curriculum for anyone training to be an Acolyte, and so he could immediately recognize the fallacy.

On the one hand, every dollar of price increase in repair was a dollar taken from a customer but given to a mechanic, so from that view the good and the bad balanced out. But on the other hand there were second order effects: reducing the cost of ownership of goods means that people can use the money they saved to buy other things, which means more jobs get created somewhere else. Ignoring this was the fallacy of all naive economic interventionism done in the name of politics: favoring the seen at the expense of the unseen.

It took him about three ticks to formulate the structure of this argument in his head. But then he realized that it would take him about twenty ticks to speak it - when the timer on the clock had only four ticks left. Gladias gave up, and simply did not respond.

The timer buzzed.

Gladias began to sweat nervously.

"Senator Ancus?"

"What was the last thing you voted on for the aesthetics tax?"

"A blue and grey building on Badra street. It was hard to decide, it was just average, nothing out of character in a bad way, but also nothing exceptional. Whenever I pass by it, there were always families coming in and out, people even sitting on the benches outside. Children occasionally selling food they make. But that's normal for Meldan these days. So I just gave it a very tiny vote of approval."

"What was the last thing you *wished* you could have voted on?"

"A Hydrafill billboard. The one that talked about their drinks being free of over 2,000 poisons and anti-recommended substances."

"Why?"

"The ad was fun. I still remember when I was young, and all the ads were boring or just painful to look at. They were just trying to get inside your head, that's all they cared about. It's especially bad when they do that on public transit; if the public transit experience is bad then even more people switch over to cars and we get more pollution and congestion and that's bad for everyone. I was glad that Hydrafill tries so hard to be better than that."

"Papli", Ancus shouted, whipping his head toward Papli. "Don't you think families enjoying the outdoor environment, good public transit, and fun are all things that regular Veridians value a lot?"

Papli did not answer.

"Keep the conversation between the asking Senator and the applicant", the Chairman interjected.

"I think so. Maybe Papli would say Hydrafill's style of humor is pretentious smart people humor or something like that."

"But I think he underestimates Veridians, and just how smart many of them are. I was on the autobus coming here, when Sylka was debating Ephelion on how to measure social progress, and people seemed really engaged when she was speaking."

"That's just the smug smart people's version of enjoying watching people fight. Helisport is actually *more* peaceful", Papli interjected. "At least the Quadratic Funding funds Helisport games. We should get rid of you guys and just make a negative version of Quadratic Funding and use that to calculate the taxes on all the big businesses."

"Order!", the Chairman shouted. "Next question."

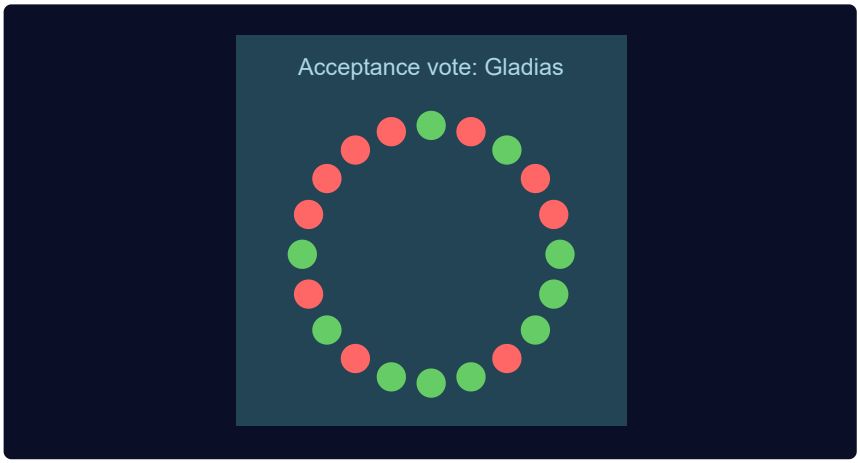
The questioning continued for another half longhour, as each Senator got a chance to ask.

After the last Senator asked Gladias his question, and the timer buzzed, the Chairman stood up.

"Time for the vote!"

The YES and NO voting buttons were clearly visible on the table, so Gladias could make out any Senator's vote just by looking at them.

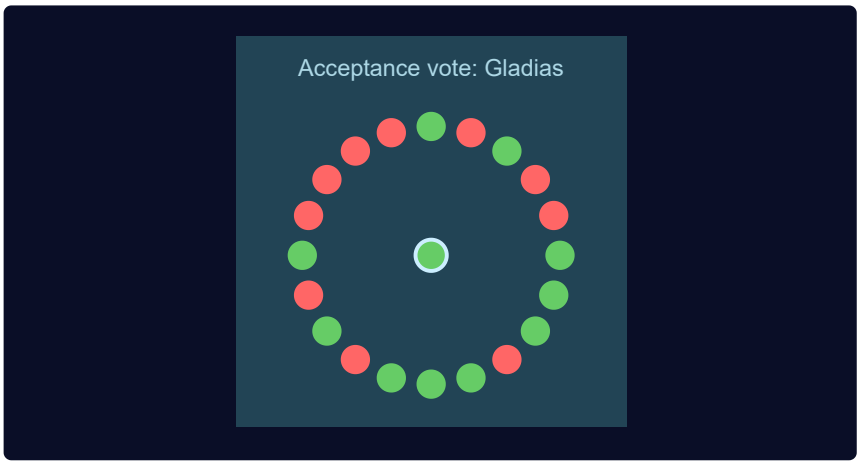
The Senators started pressing down on their buttons. A different noise buzzed every time a vote was made. Within ticks, every vote had been cast. Gladias looked toward the screen near the top of the room.



"Wow", the Chairman exclaimed. "Ten to ten, exactly. By the rules of Acceptance voting, I am the deciding vote. And so I vote..."

He pressed down on one of the two buttons on his desk.

"Yes."



"Welcome to the Order, Gladias."

* * *

Once Gladias left the building, he checked his watch. There had been loud buzzing while the acceptance session was in progress, and he had been ignoring it.

From		Message	Time
Vil	→	Sorry to bother you, but it's about Lily. She just failed a test in civics class. I think she would benefit from seeing you and Seila.	45183

Seila's still in Iptak, Gladias thought. He would have to go himself.

Gladias did not bother asking Emerald for a route to Vil's house. He knew it himself, it was not too far. He started walking, his mind filled mostly with concern for Lily's wellbeing, and slightly - perhaps more than he was willing to admit - frustration that his triumph was so

quickly replaced by dread, a feeling that he was faced with a problem he had no idea how to solve.

"I'm coming", Gladias whispered into his neck band - once again, not out of any strong desire to conceal his privacy at that moment, but because by that point he was simply accustomed to whispering. "Get me the usual food I get. I just got out of my acceptance hearing so I'll be hungry."

From		Message	Time
Vil	→	Did you at least get accepted?	45501

"I did. Just barely though."

After twenty minutes of walking, Gladias arrived at Vil's house. Vil opened the door for him, and then sat back down in his chair. Lily was sitting on the couch, crying.

Gladias sat down on the couch beside Lily, and hugged her.

Gladias waited for a few moments, and began to speak.

"It's all right", he whispered. "Scores aren't everything. Vil, Daia, Seila and I all know the beautiful person that you are inside, and we love you."

Lily turned her face away, unhappy.

"We can help you. We just have to find some way to engage with the material that you can relate to."

"You've already tried", Lily replied softly, sobbing. "I got fifty-nine percent on this test. FIFTY-NINE PERCENT. Third-worst in the class."

"Well, do you think the smartest kid in your class can bake a birthday cake for Seila the way you did six months ago? Would he bring food to Hreda every day for a month the way you did when she was recovering from that accident years ago?"

"You need to realize that you're a beautiful person already. That's where self-confidence comes from, and that's where you can get your

motivation to improve."

"These schools really need to get better at connecting these topics like civics to topics that matter in your life. I only went down my path to become an Acolyte because I saw how the Steering incentives were helping to open up the online world so I could participate and write code for it. You will have your moment too."

"If you get above seventy-five on your next civics test", Vil said, "I'll take you on my next trip up to Northglade."

Gladias winced slightly at the crudeness of the incentive - though he figured in the back of his mind that it was worth a try.

At that moment, the doorbell buzzed. Vil stood up and walked toward the door, and opened it. It was the food.

Vil picked up the food, and put it down on the table. Vil and Gladias together began to open it.

Lily slowly stood up, and also joined the table. The three of them began to eat.

* * *

Gladias's watch buzzed.

From		Message	Time
Mov	→	[call requested]	59063

He picked up.

"Mov, how are you?"

"I have some interesting news to share. But first of all, you got in, right?"

"Just barely."

"In is in."

"So what's your update?"

"Looks like we're not going to be Keepers after all. The cryptographic random gods selected me to become a Herald starting next term."

"Wow."

"Yep. So as of today, I have no more formal power."

"What do you expect you will do as a Herald? I can't imagine you actually running those assemblies or giving speeches or anything like that. Are you just going to go moderate a sub-forum on Silverchat?"

"You know me too well. But also, I'll have more time to help you."

* * *

That evening, Gladias called Seila.

"I got in!", he said excitedly.

"I knew you would!"

"It came quite close. Ten to ten, the Chairman broke the tie."

"A win is a win. Ever since you told me you were going to go Keeper, I've been so excited about all the good you can do from that role."

"I'll try my best!"

Chapter 9

Pafogai Du, Dzego · 3724 RAINMOON 28

Bai and Fin were sitting together in a courtyard, eating and drinking. Tears ran down Bai's face.

"You know, Zei is really quite a nice kid", Fin stated softly. "If you could just be nicer to him, he could help you so much."

"Nicer, like you?", Bai replied, distressed.

Fin hesitated. He did not know how to respond.

"You know how I felt when Zei brought his mom along on the trip to Freetown with me?", she asked.

"Sad because you wanted to have the trip alone with him, and you could tell that's not what he wanted?", Fin asked.

"NO!", Bai shouted.

"I'll give you a hint. I felt the same after the game a few days ago when you suggested I bring my mom along for the next trip."

"You don't like your mom?"

"You never asked me where or who my mom even is! Not once, in the more than ten times we've met."

"Well, sorry. Who is your mom?"

"My mom is DEAD! My dad too."

"Killed in an explosion in a lab when I was still a small kid. You know who did it."

Fin's face dropped.

"I stayed in my aunt's place. But she basically ignored me, just gave me a room, and two meals a day, and that's basically it."

"So I had to survive with my own bare hands. Learn how to make my own money. Do side jobs even while I was in high school."

"So I hope Zei forgives me for those times when I didn't let him get his prize money to give his mom a compute box."

"I don't even have a mom! All I have is my achievements and my success. And now Zei has taken that away from me too."

She went quiet, and sat back in her chair. She had not planned to say any of that. After a brief pause, she spoke again.

"But those tactics of Zei's, they were beautiful. I was so proud when I put up such perfect defenses along my diagonals and tricked Zei into attacking there. I felt like that was the smartest thing I had done in my whole life. And then he came out and did something even smarter."

"I do want to understand you guys better. What drives you, how you think."

Bai and Fin both took a brief break to drink some of their respective teas.

"Let's talk about something else. Here, do you want to see what they have in mind for the next version of Dzegoban?"

Bai's eyes perked up.

"Sure, why not", she replied.

Fin took out his booklet, with "po so dze go ban de sia" - the conception of Dzegoban version nine - on the cover.

"Dzegoban is a masterpiece", Fin explained. "Always navigating the tough balance between being easy to learn, logical in syntax, but also nice to speak and read, and useful for talking about the things that people care about in the real world."

"The first versions, started only a decade after we lost the war against the Arctics last century, were simple and uncontroversial. Move from the pictorial script to an alphabet, get rid of the complicated declensions and conjugations and irregular verbs."

By this point, Bai was listening intently, her eyes full of curiosity.

"Then came the harder tradeoffs, on the grammar. Dzegoban grammar is very logical, if you want it to be. Take a simple sentence, like in elementary school: 'giu jan li mo fe jia dzu lie'. The man eats the

avocado. 'li' is the separator between subject and verb, 'fe' is the separator between verb and object. A computer program can parse that, very unambiguous."

"You can get more complex. 'lo mo fe jia dzu lie zo hen jun'. It's healthy to eat an avocado. 'lo' and 'zo' are the opening and closing brackets. And colloquially, you can leave out the 'lo'."

"But then you get ambiguity. Let's unpack 'jia dzu lie' - black center fruit. Is it a fruit with a black center, or is it a fruit that is both black and 'central' in some sense? Theoretically, we have the separator for that: 'ji'. We could say 'jia dzu ji lie', and it would be less ambiguous. But you can never remove the ambiguity completely. Papaya seeds have a black center too."

"So you have to balance regularity with practicality. Being easy and pleasant to speak when you talk about things that are common in people's lives. So the Bansunpei priests tried to optimize the regularity of the meaning space, even while knowing they would never - could never - fully succeed."

"If all you know are the 256 roots, you can understand more than half of what people are saying. Not all, but much better than the olden days, when you could not tell what some complicated word means at all unless you had seen that exact word before."

"Why are they putting so much work into this? Why not focus more on improving the economy, or technology, or the military or something?"

"Well, if we work on those things, the Arctics bomb us. Language is harmless. I think even the Arctics find Dzegoban impressive. Also a couple of times the Bansunpei took time to reform the timing system - that's how we got a hundred ticks a minute, a hundred minutes a longhour and ten longhours a day - and we got that to stick across the whole world. That, even the Arctics adopted."

"Do you guys have a secret textbook with a planned version nine of *that*?"

"Yeah, at some point we want to replace the meter with the lightfoot. Light travels at one billion lightfeet per tick. One lightfoot is

two hundred fifty-nine point oh two oh six eight three seven one two millimeters."

"Apparently, people in Inglewore like it because for so long people made fun of their older measurement units for being complicated and illogical, and now we're coming out with a unit that's very close to their older distance unit in length, that's directly based on the very base physical constants of the universe. So in some sense, we're saying that they were right all along."

Bai grinned.

"So anyway, the next version of Dzegoban-"

Bai cut him off.

"Maybe next time. This is all making me curious, I feel like I should go study the history of Dzegoban on my own at some point."

"Good instinct! I promise, it gets even more fun once we get into version nine. Zei doesn't see the value of all this himself, he's so focused lately on his games."

"I'm sure he is. And I don't think it will stop, it's clearly paid off very well for him."

"Maybe I should play some games with him for fun - or I guess for him, for practice - at some point."

"Oh, and thank you for distracting me and cheering me up!"

* * *

Zei walked through the tunnel, and down the stairs, following the directions that his watch gave him. Every time he got to a fork, he made sure to look in all directions, taking in his surroundings. He had never been this deep into the depths of the mountain pyramid before.

Eventually, he got to a sealed door with a green circle beside it. He tapped his watch against the circle. The door slowly opened.

Behind the door, there was a narrow and long room, and at the end of the room there was a large table. At the table, a man was sitting wearing a shadowy cloak - not a Veridian privacy robe, but the dress of the Minpentai priests.

Zei walked across the room slowly, and sat down across from the priest.

"Welcome", the priest said softly.

"I am honored to be here", Zei replied. "But ... who are you?"

"I'm Den. High Council member of the Kungaupei. Thank you for coming."

"Kungaupei? We have a secret society for defense too? Growing Dzego a whole secondary shadow army?"

"Of course, are you surprised?"

Zei hesitated.

"I was impressed by your performance in the Minpentai game a few days ago."

"But I wasn't just impressed because you won. I studied the transcripts from the game. I was impressed because of the kind of strategy that you played."

"Tell me, have you studied any cryptography?"

"I... know the basics? Factoring is hard, you cube modulo n to encrypt, and only someone who knows the p times q that make up n can invert the cubing to decrypt."

"No no, that's popular science cryptography. Nobody has been using the factoring-based stuff for decades now, even elliptic curves are falling out of fashion because everyone's worried that the Arctics are going to come out with a quantum computer any year now."

Zei did not respond. He stared back, in a mix of appreciation for the update to his knowledge, and curiosity about where this whole conversation was going to lead.

"Okay, well I think in your last game you've come really close to inventing big parts of cryptography from scratch."

"Think about obfuscation protocols. You want to give someone a program, that they can execute, without being able to see the internal details of the program. It's what our modern cryptographic networks - like the ones that handle payments and reputation and manage governance for Dzego's secret societies - all rely on."

"But how do you make a single protocol that can do that? Hide the program? It's easy to see how to hide the data, and then do addition and multiplication on hidden data, especially when it's all done by hiding it in big vectors under linear transformations and adding errors on top. But how do you hide a program? What kind of a space of possible programs do you even hide a program in?"

"The solution - well, okay, a small part of one type of solution - is that you create a meta-program. A program that, depending on its input, can run any possible program. And then you've reduced the program to just being a set of numbers that you're doing addition and multiplication to."

"It's one of the most basic tricks in computer science. Our computers, the physical hardware, are meta-programs themselves. But you figured out a really basic version of that - just sixteen possible algorithms, melded into one factory - and used that, plus a hidden glider to provide the input, to hide your intentions."

Zei nodded.

"Well thank you for the compliment. But ... I'm curious now, where are you going with this?"

"Well, Zei, I want to recruit you into defense."

Zei's eyes and mouth opened.

Defense. Without thinking, Zei opened and closed each finger of his left arm. That arm had been broken and taken a month to heal, all because of a thermodynamics lecture. What Den was suggesting was not a lecture.

But his life had been saved by exactly that kind of work - a secretive, well-coordinated room change at the last minute. Maybe this was his chance to do the same for someone else.

"I do want you to continue Minpentai into the nationals. I think it will be good for you. But for your education, do cryptography. I think it will also help you in your games."

"And of course I'll stay in touch the whole way, and when you're ready, I will start suggesting some tasks for you."

Zei responded with silence once again.

Tasks. He noticed how quickly Den had shifted into the register of assuming that Zei would go along with his plans. Part of him was already coming up with reasons to say yes. Another part was watching his enthusiasm slowly build up, and did not entirely trust it.

"I have an idea", Den began. "Do you know how chip verification works? Like, how I'm automatically verifying that all your electronic devices are friendly right now?"

Zei began to reply as soon as Den finished. By sheer luck, Den had decided to grill him on one of the areas of applied cryptography that he *had* already researched.

"When a device is manufactured, a physical unclonable function is produced as part of the process - a secret key that gets generated though chemicals settling down as the chip cools down after it's produced. Not even the manufacturer knows what it is. The manufacturer prods the device to get its public key - the p times q, or I guess now I know it's not a p times q at all, it's something totally different. It then signs that key with its key, and publishes that to the cryptographic network."

"Okay I guess you can call a function a key, close enough. Go on."

"As a user, you can ask the device to sign a value. The device will do that. You can then verify its signature against the public key that you download from the cryptographic network, to check that it actually came from that manufacturer."

"And then you have public inspectors. They buy devices randomly from the manufacturer, and use some kind of scanning to destructively inspect them. Verify that the chips actually have the logic that they are supposed to have."

"Why the cryptographic network? Why not just sign it?"

"So that you know how many there are, avoiding attacks where either the manufacturer is cheating or someone hacks their key, and they start signing over even more public keys than they should. Also it means you know when the signature was made, so you know it was made when the device was manufactured. And if the manufacturer's

key somehow gets stolen, you can tell if the signature was made before or after that happened."

"And how do I verify your device?"

"You ask it for a signature wirelessly, same way I verify it."

"How do I know it's not just forwarding responses from a different device somewhere else?"

"Here, because we're deep inside a mountain. Generally, I don't know."

"How does the chip inspection process work?"

"That I don't know."

"How do the inspectors know that they're getting devices from the same pool the public is getting, and not devices specifically chosen by the manufacturer to fool them?"

"I don't know."

"Let's think about a different case first. Take the public security cameras, the ones on the lamp posts or the outsides of the buildings or the autobuses. Every Dzegojan has a right to occasionally randomly pick one, unscrew it and analyze it themselves. That's how we know that they're only recording and transmitting when they're supposed to, and no one in the supply chain from the people manufacturing them right up to the people installing them can bug them and leak any data to a stalker or an insider trader or an Arctic."

"That's the random inspection process - assuming at least some of the people who make inspecting cameras their hobby actually choose randomly. So that's for security cameras, that are already in public so anyone can just choose a random one of them to go and unscrew. Now, what do we do here? How do we do random inspection for computer chips that you buy from a manufacturer?"

"Okay, you tell me."

"Well, I'm glad there's still a lot that I can teach you. How about we go and visit one of these factories right now?"

Zei thought for a moment. It was not yet a commitment, he reasoned, and it would be something he had never seen before.

"Sure, would be happy to!"

Zei and Den walked out of the room, through the tunnels, and to a smaller exit from the pyramid. The exit led directly into the interior of a large rectangular-shaped black car. The pyramid door and the car door were exactly the same shape, and were pressed tightly against each other.

Zei and Den got in.

As the door sealed shut, Zei noticed anti-transmission foil lining the sides of the car. The car had windows, but they seemed to show a dimmed version of the world outside; from the outside, he knew, they would show almost nothing at all.

The car started moving, and a minute later it entered a tunnel.

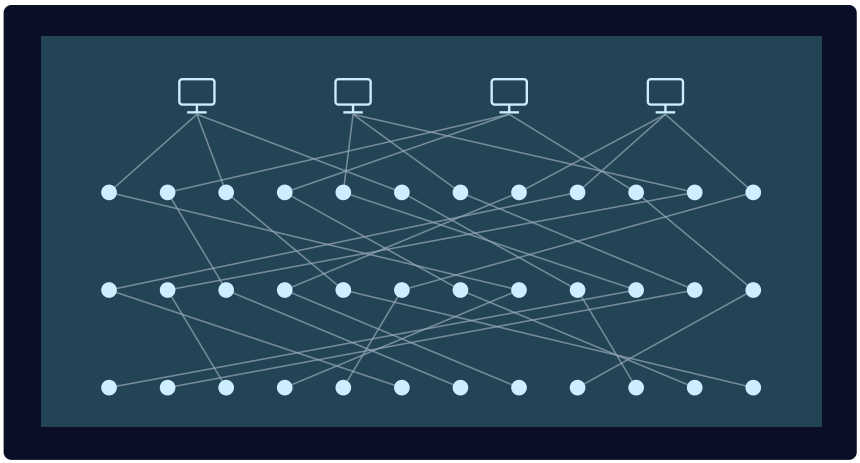
At that point Zei realized that no one in the world now knew where he was.

* * *

Zei and Den got out of the black car. The car's door now opened directly into the open door of a staircase leading down to a basement, shielding them from even a tick of being visible to the outside. The two walked down the stairs, and eventually entered a large room.

They saw thousands of empty white boxes, with labels on the side containing a two-dimensional barcode, and a machine automatically putting chips into boxes as they came out of a conveyor belt from another room.

Den opened up his hand device, and showed a diagram to Zei.



"You ever seen this before?", Den asked.

"Some kind of diagram of the internet?"

"It's a mixnet. It's how transactions get onto the cryptographic network without people being able to see who sent them."

"Each message from each sender gets wrapped with three layers of encryption. That top-level message gets sent to a node on the first layer."

"The node on the first layer unwraps the top layer of encryption, and sees another encrypted message, along with the address of which node in the second layer they are supposed to send it to."

"The node on the second layer unwraps the next layer of encryption, and sees another encrypted message, along with the address of which node in the third layer they are supposed to send it to."

"Finally each node on the third layer unwraps that layer of encryption, and gets the address of the destination, plus the actual payload. And of course, usually, the payload itself is encrypted."

"Wow, I didn't realize that's how it works. So you need three nodes to collude to unmask the sender of a message."

"Not just three nodes - three *randomly selected* nodes. And if you're paranoid, you can make the inner payload be yet another

envelope, requesting to send the message right back through the system in the next round."

"So this is what the packaging step is doing. Those boxes are highly opaque and tamper-proof sealed, and the two-dimensional barcode is the encrypted payload. It's a mixnet, but for delivery of physical goods."

"If you buy a product through this system, only the last of the three nodes knows your address, and they don't know who the sender was or what they're delivering. And if you're *double paranoid*, you can set your address to a public drop-off point and pick it up yourself."

"Inspectors buy through the mixnet. And so as a consumer, if you buy through the mixnet yourself, you're getting your chip from the same pool that the inspectors are buying from - the manufacturer has no way to give an honest chip that passes inspection to the inspectors, without almost certainly giving one to you too. They can't tell which chips - or whatever other physical goods are passing through this thing - are going to whom."

Den then walked forward toward a door. Zei followed him.

Den put his watch against a circle with a glowing green outline. The door opened.

They walked through a hallway, took a few turns, and then went through another door. They entered a room with a large microscope.

"This is the inspection facility of this fab. They inspect both their own chips, and those produced by others."

"They do multiple types of three-dimensional scanning, at different wavelengths, to make sure that the physical chip exactly matches the specification."

"I'm impressed. Say thank you to whoever gave you permission to give us a tour."

"I will. So Zei, I remember you had another question."

"Yeah I remember. Though it wasn't about manufacturing."

"Indeed - it was about how someone verifying could know that responses are not being forwarded from a different device. I can tell you the answer for that too. My hand device has a protocol by which it

repeatedly contacts each device a million times, and expects the whole protocol to finish within a tenth of a tick. And so wherever the signature comes from, it has to be within fifty lightfeet away - almost thirteen meters - actually, even closer, because of the latency of the chips themselves."

"So Zei, what do you think?"

"You said I can still do the nationals, right?"

"I *want* you to do the nationals."

"Okay yes, I'll join - or at least, I'll commit as far as switching my studies to cryptography and see how that goes."

Den nodded.

When Den kept walking forward, Zei's mind wandered - first, to the large microscope that was directly in front of him, and then to other matters.

He started thinking about what he would tell his mom that afternoon. He could talk about Fin being busy, about himself going for a walk. But he could not tell the full truth. The Kungaupei forbade it - and his mom had already watched him heal from one bomb.

Chapter 10

Freetown, United Cities · 3724 WINDMOON 24

Jahn's taxi pulled up at the Freetown Economics Institute, where he worked as a researcher, and he got out of the car. He entered the building, and tapped his hand device to pay to get through the turnstile.

Out of the corner of his eye, he noticed that the entrance price was ten percent higher than before.

He walked to the restaurant section, and saw most tables full. Only one table had empty space, and there was already a woman sitting there.

"Can I sit down here?", Jahn asked.

"Of course."

Jahn sat down.

"I'm Jahn", he offered.

"I'm Utaku. I'm from Belpaki, I'm doing my doctorate program here at the Institute."

A robot slid up to Jahn, offering a menu.

Jahn quickly scanned through, and ordered a dish with vegetables and fish, and a glass of water.

He noticed that there too, the prices were higher than before.

"Wow, the prices are all up now."

"Haven't you been following the news?"

"Not since I got back from a meeting in Hapton."

"Huh, well it should be news in Hapton too. Everyone's been talking about raising their defense budgets."

"Arctic scare?"

"Yeah, their drone activity has been going up. Though there's surprisingly many politicians who seem totally nonchalant about it. Only Redshire and its neighbors seem to be really taking it seriously."

"Well, somehow, also Freetown."

"Why us? I'm glad we are, but like half the United Cities are closer to Redshire than we are, they should be worrying more!"

"Well, Devanvil is taking it seriously - after all, Redshire chooses twelve percent of the seats in Devanvil's parliament. But yeah, Deplexi, Ondak, Imber, not at all."

"It is weird. Remember, the government here is constitutionally banned from collecting more than the basic land tax and some rent from its natural resource holdings. So it's not even the government that's taking the problem seriously, it's the megacorps."

The robot came back, carrying Jahn's food and water. Jahn picked up the plate of food and put it onto the table. He then picked up the glass of water, and immediately took a sip.

"They're now donating three percent of their revenues to defense, and even colluding to punish companies that don't. Mogna was unwilling to contribute to defense - they said something about corruption, though everyone thought they were just being cheap - and then the charging station makers came together and threatened to lock out Mogna's cars from their grid, ninety percent of charging stations across Freetown, unless they joined in. Yesterday Mogna folded."

"That's ... harsh. If any industry tried that in normal times, they'd get half of Freetown counter-colluding, maybe funding five new competitors against them."

"Normally yes. But hey, times are different now."

⁂

MELDAN, VERIDIA

"You off to your first Keeper meeting?", Seila asked Gladias.

"My first two Keeper meetings, back to back. First one is on cultural values in public events and performances - actually the exact one that I judged Dreadknot on way back then."

"Oh, what's the topic for cultural values in events going to be?"

"In the chat I mentioned how I thought the rubric was encouraging people to be too bland to not offend anyone. Surprisingly, a couple others agreed. I think I'll try to make a big push to fix that."

"Veridia ba 'fun' gie!", Seila joked.

"Isn't it the 'gie' that you're supposed to replace? 'fau' just means 'again'."

"Aaah right."

"But yes. Veridia will be fun again."

"What about the second one?"

"The hardware openness rubric. The topic is whether or not to modify it to exempt military and surveillance products."

"Huh, what are the tradeoffs of that?"

Gladias thought carefully about how to phrase this. He thought for a moment about whether or not he should mention Delwart, but then decided against it.

"Well", he began, "some of the Keepers think we should be staying out of funding war - it's a negative-sum game, if both sides have better bombs and battle drones, the result is the same, just more things get blown up and more people get hurt."

He was pleased with himself. He had a feeling that Seila might be sensitive about anything that could hurt Dzego, and would plead for him to take whatever position benefited Dzego the most. He braced himself, preparing for Seila's next question.

"But, funding ... isn't that more of an issue on the Graph Funding side?"

Gladias internally sighed with relief.

"Well remember, we make some weapons too. Also surveillance products. And the rubrics influence how much of the secret sauce we keep to ourselves."

"I see..."

"Well, tell me how the meetings go!"

* * *

Gladias walked down a hallway, and entered a courtyard. He found a series of booths with thick sliding doors. He went to the rightmost one, as he had been instructed.

He slid the barrier open. It took a significant effort for him to do so. He could tell immediately that this was a much more proper booth than the cloth-covered booths from the Greater Plum Harbor. As the barrier was opening, he could see its internal structure: a layer of anti-transmission foil on the inside, followed by layers of soundproofing, and other layers he could not recognize.

At that moment, he also noticed a constant humming sound that was being emitted from the outside of the booth on all sides. White noise to further mask anything that makes it through the soundproof walls, he realized. If the noise is at forty decibels and the walls are good enough to attenuate sound by fifty, then it can mask up to ninety decibels coming from the inside - perhaps seventy if you want a safety margin. That's enough to completely mask any conversation.

After fully opening the barrier, Gladias looked inside the booth. There was a table stretching from near the entrance to the opposite wall, and a long bench on each side.

There were five others already sitting down, all wearing privacy robes.

"Welcome! You must be number Four", one of them exclaimed.

"Indeed I am."

"I'm number Twenty in this circle, a third-year. We have two second-years - though Eleven has not arrived yet - two third-years, two other first-years, and now also you."

"Glad to meet you all."

Gladias sat down on the bench beside Twenty. Twenty had his hand device open, and Gladias could see him scrolling through recent audits measuring various businesses' statuses on the cultural values rubric, and the resulting decisions.

A few moments later, another Keeper walked in. He slid the barrier shut behind him, and sat down.

"So, cultural values rubric", the other third-year began.

"Cultural values *in events and performances* rubric", the first-year, One, corrected. "We're not doing social media or other online things, that's a different rubric. And we're not doing architecture and public billboards, that goes into the aesthetics taxes."

"We all know that, you don't have to show off here", the third-year replied, with a joking tone calibrated to try to make the point without offending One.

"How do we want to do this? Maybe we should first come to a common understanding of what the problems we're trying to solve are?"

"Let's go through some examples", a second year who had not yet spoken up replied.

Gladias saw his chance to intervene.

"How do you guys feel about Dreadknot?"

"What do you mean?"

"Their taxes got bumped up to tier 4 in the last audit."

Gladias began to read from the rubric.

"Content weakly promotes one or more of the following: violence, lawless behavior, use of anti-recommended substances, gambling or risky investment, excessive display of wealth, poor conflict resolution practices, religious or ethnic hatred."

"Do we think Dreadknot is guilty of promoting violence, lawless behavior or hatred?"

"Their last song was about destroying cities and flaying people's flesh. Why not?"

"Do we really think that those words are going to make anyone want to flay anyone's flesh in real life? It's a song. I remember we went through this with games a few years ago, and the science ended up saying that they can slightly increase a few aggressive behaviors but there's no evidence that they actually lead to violence."

"I remember that too. And my read was that the conclusion of the evidence was harsher than that."

"Do we want to dig up and go through those studies together?"

A few other Keepers nodded. They started whispering instructions at their hand devices.

Gladias took the opportunity to continue.

"But even aside from all that, do we really want Veridia to be a society where the artists have to constantly worry about this and all the songs, and the games, and the movies, become boring and pale?"

"They don't have to constantly worry. That's the whole point of Veridia's system of regulation. We're not banning them. We're just setting their taxes slightly higher instead of slightly lower. So along with the ten things that they have to worry about a medium amount already, like what their fans will like, what might or might not offend their corporate sponsors, and so on - things that won't break them, but still make a difference - we ask them to think about one extra concern: the public interest."

Gladias decided to pursue another line of argument.

"I actually like this rubric in general. I think it's great that the Northglade Science Center is getting rewarded for being such a wonderful place for kids to learn about chemistry and biology and rockets."

"But violence is part of life. The human race started out as monkeys fighting over bananas. When we were still living in forests and caves, a quarter of our men died because someone else killed them. The epic poems they make us study and memorize in schools have violence. Every hero of every Great Book did something violent. I don't think we should be discouraging Veridians completely from engaging with such a central part of our human nature. If we steer the public media away from violence everywhere and always, people will get *all* their ideas about violence from the dark corners of the internet, where nobody gets taxed, and that will be even worse."

"Are there any studies that try to measure this?", Twenty interjected.

"I don't know if you *can* measure this with studies", Gladias replied.

The other third-year again began to speak.

"This is good, I feel like we've already surfaced some important questions about not just the science, but also our underlying goals and values."

"I think we should re-read the studies from back then about games. But also, we should see what the Veridian people think about this. Can we get Emerald to do a broad listen report on what people have been saying about Dreadknot on Bluewhale or Silverchat in the past year?"

"I don't think what people say on social media is necessarily a good guide to what's good for people", Eleven interjected.

"Okay, but let's take some time right now to read."

Gladias nodded. A few other Keepers nodded after him.

"Just give me a minute. Bluewhale always resists letting you read messages on their platform in any way other than their interface, I have a few hacks to get around that."

Almost exactly a minute later, a new window appeared on each Keeper's hand device, showing Emerald think in real time as it fetched and took notes on what people were saying about violence and Dreadknot on Bluewhale and Silverchat.

The Keepers all began staring intently at their hand devices - some reading the video game studies, others watching Emerald think.

* * *

Gladias was standing outside, and sipping a cup of warm water. His first Keeper meeting, seventy-five minutes long, had finished, and he was taking the twenty-five minute break.

His watch buzzed.

From		Message	Time
Personal reminders	→	Next Keeper meeting starts in five minutes.	39500

It was time to go.

Gladias turned the corner and walked the last few steps toward the booth.

He soon arrived, and slid the barrier open.

There were two other Keepers already sitting inside, one on each side. Both were wearing privacy robes, with their hoods and face coverings still on.

"Hello to you both!"

"Hello! I'm Eighteen. Which number are you?"

"I'm Six."

"And I'm Nine."

Eighteen thought for a few moments, and then stood up.

"Wait, before anyone else comes here, would you all mind standing up again?"

Gladias and Nine both stood up, and briefly exited the booth and looked in from the outside, with the door open.

"Alright, let's take a look."

Eighteen switched his hand device to a different mode, and began waving it around the room. He also took out a drone, and pressed a few buttons on his hand device. The drone began flying around the room near the edges, both above and below the table.

Eighteen and Nine watched silently as Eighteen did his scan, and two more Keepers arrived and quietly joined them.

"Hmm, nothing transmitting, not picking up any powered electronics with thermal either."

A minute later, the drone beeped unexpectedly just above the middle of the table at the far end.

Eighteen jumped to that position with his hand device, and started looking and poking at the table intensely. A few moments later, he

stood up triumphantly holding up in his hand a needle-like device, with a bulb a few millimeters wide at its tip.

"What is that?", Gladias asked.

"Spying device. There was a small bump on the table, and this was under it. Definitely takes some skill to insert one of these and then fix up the table this well afterward.", Eighteen replied.

"Wow."

Nine stepped forward and looked at the device briefly.

"I doubt it was meant against us", he said. "Normally, the goal of people who do this is to get information about upcoming business decisions, and trade based on that in the financial markets. Usually, smaller businesses who are less careful. The Order is quite safe against this. Our Keeper group is only one of three discussion groups, so even if you do overhear us all you're only getting one third of the perspective. And we're all speaking through our neck bands so the sound is going between us digitally."

"Actually this one had a camera. It can read your lips and your neck, so there might still be enough information to pick up some words, especially if you're a beginner."

"How do you know?"

"That's how the drone detected it, infrared light reflected off the camera lens. My hand device caught it too with the nonlinear junction detector, but I may well have missed it if the drone did not tell me where to look."

The last two Keepers arrived.

Eighteen stepped out, explained the situation, and the group looked at the listening device together. A few Keepers scanned it with their hand devices. When they were done, Eighteen snapped the listening device in half.

Eighteen led the group to another open booth a few meters away, and slid the booth door open.

He spent two minutes checking the new booth for bugs, again using both his own hand device and his drone. Once he was done, he

motioned with his hand, and all seven Keepers entered and sat down on the benches.

One of the newly-arrived Keepers cleared his throat, as though preparing to speak.

"Maybe let's get to business?", the Keeper asked.

"Sure!", Gladias replied.

"So we're supposed to be talking about the openness rubric, and how to adjust it for military hardware."

"Yep, I read the docs."

"And I've had some conversations with Zero over there. His argument is that we should just exempt military hardware completely. I'll let him explain why."

Zero heard his number, and quickly raised his head.

"So normally, open hardware is good, because other people can build on it, and everyone benefits, right?"

"Right."

"But think about what happens with anything military. There isn't actually a net benefit to the world from everyone having better weapons. If everyone has twice as good swords and twice as good shields, the two cancel out."

"If everyone has twice as good swords but shields stay the same, then the benefits to the two sides in any war still cancel out, but more people get hurt."

"But if everyone has twice as good shields but swords stay the same, isn't that good?"

"Well, maybe", Nine interjected. "But actually it's even more complicated than that. What if it's a war of attrition, so if everyone has more defense, the same number of people get hurt, it just takes longer? Or what if increasing defense reduces deterrence, so more wars happen?"

Gladias was surprised: the argument was eerily similar to an argument that Delwart had expressed to him recently.

"Basically, the game tree is just so chaotic that we can't reliably be sure of anything at all. So I think we should just stay out. And instead

just increase Steering taxes on all military hardware, open or not, push people to make more useful and pro-human and fun things instead."

A third-year interjected:

"I don't think it's true that we can't be sure of anything. Look at the United Cities. I think it's pretty clear that a big reason they all managed to stay independent from each other for centuries, and figured out things like co-governance to nudge each other to cooperate better without just turning into one big blob, is that they have so many mountains and peninsulas on their continent. So if everyone has more defense, more people can keep their freedom and sovereignty. I think we can be quite confident of that."

Eighteen raised his hand again. The others turned to him, and he began to speak.

"I don't think we can look at the situation like it's a black box like that. Let's look at where we are now. The Arctic Empire's technology is the most advanced. Ours is second. On military tech, Dzego and the United Cities are furthest behind."

"And so when we release open-source military-relevant anything, it doesn't benefit the Arctics at all, and it's marginally good for us because it makes it easier for us to make more."

"But for Dzego, it's a lifeline. They're not set up to invent their own proprietary stuff at the top end, but they're really good at building and adapting from open-source plans. Twenty years ago, when we got our programming languages and formal verification to the point where our devices basically stopped having critical errors in their code that could let someone take it over from the outside, the fact that it was open allowed the Dzegojan to adopt those techniques, and protect themselves much better from Arctic sabotage."

"But do we really want to play this game", Zero continued, "being so confident in our view of who's good and who's bad? Computer security was defensive, fine. But the next technology we invent might be offensive. Won't we just accelerate the arms race even more for everyone?"

"I'm not even that convinced the Arctics are that big a threat. Sure, they're waving more drones around the northern United Cities now, but I don't think the intention is to attack, I think they're just trying to impress us with their power so we become attracted to their ideology."

"Do *you* think we should be more receptive to their ideology?"

"I think we should keep an open mind. Just remember, the Dzegojan are definitely trying to manipulate us too."

For a few moments, the discussion went quiet. Gladias let it rest. He was still trying to understand how everyone else was thinking before forming his own views.

"Let's all think about this more on our own. We'll keep talking about this, we still have a few more discussions, the last one is in four months. And then the actual vote is only in six months."

⁂

PAFOGAI DU, DZEGO

Bai and Zei sat together at a courtyard table, both staring at their hand devices.

"Fin not joining us this time?", Bai asked.

"No", Zei replied with a strong hint of disappointment. "He's busy with the Bansunpei."

The hand devices buzzed. It was their intervention turn. Zei arranged a few gliders and symbol-carrying spaceships. A few hundred turns passed.

Suddenly, as the spaceship hit a rock formation and was in the process of reflecting, gliders from Bai hit it. The spaceship exploded into a storm of gliders.

"What?", Zei shouted.

"Caught you, I had a symbol along the path, right in the position where you could not see it. So I could see the edge of your spaceship coming."

Zei groaned. That had been his last-ditch attempt to rescue the game. More gliders from Bai came in, and shredded through both of

his remaining symbols in the southwest corner.

Zei clicked a button to resign.

"Good game", Zei said, not yet with total sincerity, but with more sincerity than he had in any game where he had lost to Bai before.

"You know, I am going to need your help in the nationals."

"Happy to. As long as..."

"Yes, I know, you want to visit Veridia now."

"Anyway, thanks for explaining your thought process in that championship game. Really made me think."

"Of course."

"Of course you enjoy reliving the moment you defeated me, just like I enjoyed beating you right now."

Zei sighed.

"Anyway, I have to go. The Kungaupei councilor, Den, is asking to see me again."

* * *

Zei entered the long room, and immediately saw Den and Mu sitting on the other side.

"Thank you for coming."

"Of course. I went through the hash-based signatures you recommended. It's a really fun set of constructions, especially the stateless tree-of-tree-based ones."

"Do you see why, when I looked at your style of Minpentai, I knew that you would do well in this?"

"I think so!"

"You are truly amazing. You've rushed through a third of the cryptography lectures in less than a month, and got a perfect score on both of the first two tests. I can't think of anyone who did better."

Zei's eyes gleamed with satisfaction at receiving the compliment.

"Zei, do you know why I called you in?"

At this point, Zei realized one possible implication of Den's words, and his cheerful facial expression suddenly dropped into a fearful one.

"Are you going to convince me to go into this full-time? Like, abandon the Minpentai nationals?"

Mu laughed. "So I guess my hunch was wrong - this part, you still don't get."

"Huh?"

"Have you ever thought about why the Minpentai game is so serious, so much ceremony, the priests going into the dungeon for days, and so well-funded?"

"It's a sport. Sports are taken seriously and well-funded in many countries."

"Not sports of the mind. Their audience isn't large enough to fund it just based on advertising and national prestige and all those other reasons."

"So why then?"

Mu leaned forward.

"It's because Minpentai isn't just a sport."

Zei stared.

"It's military training."

"The whole point of the sport, and why we support it so much, is to raise the next generation of defenders."

"We need people who can think quickly, and think strategically under pressure."

"People who can compete with their foes, but also cooperate with their allies, even allies that they do not personally get along with."

"People who can understand a battlefield, including the battle itself, all the way up to grand strategy and resources and logistics, and all the way down to the level of surveillance and secrecy and trickery, and ultimately entropy itself."

"People who can deal with suddenly changing rules."

"People who are able to micromanage, but are also able to set a plan in motion and watch it unfold."

"And people who can work with the latest technology - including AI - but also understand and work around its limits."

The realizations hit Zei all at once. At first, he had wanted to ask a basic question: why pretend it's a game, why not just have a regular military with regular recruitment? But as soon as he put together the question, he already knew the answer: the Arctics would put all their energy into targeting anything like that. Minpentai was only safe because it pretended to be just a game. But even then, Zei knew, the ruse could not be safe forever.

"Why are you telling me this now?"

"Because we have to give you *some* reason why we're now telling you to work hard on cryptography *and* continue to the nationals. If we didn't tell you, you might get suspicious, eventually figure it out for yourself, and then do something stupid, like excitedly blurting out your discovery to a dozen other people."

Zei nodded.

"I won't tell anyone. Not even Fin."

Den and Mu nodded.

"And when the nationals come, you'll want a sparring partner who knows your style."

No one said a name. No one had to.

"We appreciate your support", Den replied.

"Dze go ba fau gie", Mu added.

"Dze go ba fau gie."

Chapter 11

Meldan, Veridia · 3724 BLOOMTIME 15

Seila, Daia and Vil were standing together in a large grass field, with many other families nearby.

From a distance, Daia caught sight of Gladias approaching. She waved at him. Gladias noticed, and with a smile on his face walked toward them more quickly.

"Happy to see you here!", Daia exclaimed.

"Well I can't miss my son's first real Helisport game", Gladias answered.

"So how have you been?"

"Well, lots of Order business. Can't talk about it publicly, you know. Going well though!"

"How about you?"

Vil started to speak.

"Well, lots of documents for the lawsuit against Divergent. It's been taking up a big part of my time for the last two months."

"Remind me, what happened there?"

"Ah sorry, I've been trying not to burden you guys with the details, I know you have so much on your plate already."

"Come on, Vil, your life is important to us."

"Well, remember about half a year ago I lost my job at Jump because they went bankrupt. A month later, another one of the employees reached out to me, and I learned from him what really happened. See, the official story had been-"

"I remember this in the news. Jump was struggling, and as a last-ditch effort to save themselves, they made a risky bet on the financial markets, buying a derivative on property prices in a region that they were about to announce they would start serving. But it didn't work - not enough people cared, then they got unlucky and prices went down, so the whole thing collapsed."

"Exactly. But here's what I learned. The only reason why Jump was able to make such a large bet on a niche market without insane slippage is that a hedge fund - Divergent - had taken the other side of the bet. And the primary owners of Divergent had quietly bought a majority share in Jump for very cheap."

"If housing prices in that region had gone up, Jump would have gained, Divergent would have lost, for them it would have been a net zero. And if housing prices went down - which they did - Jump loses, and Divergent gains, also a net zero. But here's the asymmetry: if Jump loses - which they did - then Jump declares bankruptcy, and so they get out of paying end-of-year bonuses plus a year of severance for their employees. So they gave themselves a free option on our backs."

"When we found out about this, we sued - all the employees together sued, as a class action. It's been tough; Divergent even managed to intimidate some media sites into not covering it. But I feel like so far the judges are seeing through their tricks. So I'm optimistic."

They heard a loud whistle, and turned to face the direction that the sound came from.

It was the instructor. A large truck had been moved into the center of the grass field.

"Helisport game starting in five minutes!", the instructor yelled.

The instructor pressed a button, and the back door of the truck began to open gently, tilting backward on its bottom hinge. After a few ticks, it had turned into a ramp.

Some personal copters, simple chairs with rotor blades on top surrounded by shielding, automatically rolled out of the truck.

Some children, around ten to twelve years old, ran toward the truck, half wearing yellow shirts and the other half wearing purple shirts. Among those wearing yellow shirts was Zven. They grabbed their copters and began strapping into them.

The instructor walked into the truck, and soon came out with a much smaller box. He opened the box, and four bright red drones, shaped like balls a few centimeters wide with soft feather-like rotors on top, sprang out and flew into the sky.

"Welcome to today's Helisport game", the instructor said. "Today we're going to do a limited game, playing only with redballs. If you catch a redball, bring it back to me, and your team gets one point. First team to get to six points wins."

By this point all ten children had strapped into their personal copters, and the rotor blades were beginning to spin - just enough to overcome about three quarters of the earth's gravity. The kids ran around excitedly, and some jumped, reaching about two meters off the ground before gently gliding back down.

"Everyone know what to do?", the instructor asked.

"Yes!", many kids shouted in unison.

"Then we will begin the game in five ... four ... three ... two ... one ... blast off!"

The personal copters flew into the air.

"You know, I'm surprised they still do this in schools", Gladias said to Seila. "If this were invented today, they would definitely say it's too dangerous."

"It was only invented less than ten years ago."

"In Dzego."

Zven caught sight of a redball, and began flying toward it. A few moments after, two kids wearing purple shirts also saw the redball, and began to chase it right behind him.

The redball suddenly shifted direction to the left.

Zven and the other kids also turned left. Zven's copter's rotor shield bumped into the rotor shield of one of the other kids' copters, and the two bounced harmlessly off each other.

Zven aimed his copter to fly higher. He fell slightly behind the kid he had just bumped into, while the third kid flew forward, and was soon directly under him. The air sucked Zven down, and soon his feet bounced into the grid on top of the other copter's rotors.

Zven jumped off quickly, and he and the other kid both tumbled down a few meters before they stabilized and flew off in separate directions.

"The AI on these things is getting really good now", Gladias murmured.

The redball suddenly changed direction again, this time going upwards. Zven immediately set his rotor to full power and climbed upward. The kid he had just bumped into was still a few meters below him, and the other kid did not notice the redball's change of direction in time, and so Zven was ahead.

Zven saw the redball get closer, and then closer again. Now it was only one meter in front of him.

The redball suddenly turned off its rotors, and began to drop. Thinking quickly, Zven kicked the ball back up just as it dropped to the level of his feet, and moved his copter forward. He stretched out his hand, and caught the redball in his palm.

He then aimed the copter to fly toward the instructor. The rotor slowed down - though it did not stop completely. It sped back up when Zven's altitude was decreasing at the safe maximum of five meters per tick - the speed that you would hit the ground after a fall from a height of slightly under two meters. The instructor watched Zven land, and stretched out his hand to receive the redball that Zven handed over.

"First catch of the game by Zven, first point for team yellow!"

He let the redball fly once again. It quickly shot forward, in the opposite direction from where Zven was standing.

Zven flew back up, and continued his search for more redballs.

"How are Hreda, Lily and Febric doing?", Gladias asked Daia. Seila heard the question, and turned her head toward them slightly to hear the answer.

"Hreda is off in Northglade doing a math competition. We've been trying to get Lily more interested in art, though that hasn't been really successful so far."

"I also talked to Zven for a bit and he showed me his homework. I was surprised. I saw the questions they were giving him for arithmetic, and they seemed just slightly ... less advanced than what Febric got at his age?"

"Nice catch by Dommus!", the instructor shouted. "And nice catch for Gale! Two points for team purple"

Seila joined into the conversation.

"Well, Glad, you're a -" she stopped herself, briefly remembering that it would be improper even to specify Keeper - "an Order member now. Anything the Order can do to make Veridia's education better?"

"You know the schools aren't Order business, the public ones are run by the education committee, staffed by parliament. Though there have been people thinking about trying to support more private ones, support them out of Graph Funding, figure out some rubrics to give them the right incentives."

"Maybe you should try pushing for that in your Keeper meetings?", Seila replied, whispering into her watch as a substitute for a neck band. Better to just have this conversation privately than have to choose each word carefully to avoid giving out Gladias's position, she thought.

Delwart would be supportive, Gladias thought. Though he would be even more supportive of private schools with *no* rubric micro-nudging attached.

"Another nice catch by Zven! Second point for team yellow."

Gladias and Seila's attention went back to the game. There were three groups of three kids each in the sky, each group chasing what they assumed must be a different redball. Zven flew up alone, and did not try to join any of the groups. Instead, he seemed to simply wander the sky, his eyes constantly watching the horizon to try to find where the fourth redball might be.

One of the kids tumbled out of the sky, though the rotor's AI automatically broke his fall. He hit the ground, and then a few ticks later stood back up and went back into the sky.

A minute later, a kid landed beside the instructor, and handed him a redball.

"Another nice catch by Gale! Third point for team purple."

The redball flew back into the sky, and both Gale and the other two kids who were chasing the redball with him once again flew after it.

Two more minutes passed. Another kid landed beside the instructor.

"Nice catch by Dela! Third point for team yellow."

One more minute later, two more kids landed at around the same time:

"Nice catch by Plat! Fourth point for team yellow."

"And a nice catch by Meny! Fourth point for team purple."

Zven continued his solitary quest to search for the fourth redball, which all the other kids seemed to have completely forgotten the existence of.

"Nice catch by Egardo! Fifth point for team purple."

"Another nice catch by Dela! Fifth point for team yellow."

He began to fly closer to the ground.

Gladias and Seila saw Gale shooting ahead of his two competitors, getting ready to catch the redball. It was only one meter in front of him.

Gale's hand stretched out. He prepared to grab hold of it.

Then, suddenly, another kid bumped into him, and he veered several meters to the right. That kid attempted to grab the redball instead, but it quickly flipped direction and flew backwards, right in between the two of them. Gale and the other kid both attempted to grab it with their hands as it passed by, but both missed.

Gale regained sight of the redball first, and quickly flew towards it. His two competitors soon followed, though they were at least ten meters behind.

Gale and the redball both gained altitude, and Gale began to get closer to the redball. Five meters. Four meters. Three meters.

Meanwhile, out of nowhere, Zven seemed to fall to the ground, the fall broken as usual by the copter's AI.

High in the sky, far above the other players, Gale raised his hand triumphantly, holding the redball inside a clenched fist.

Many in the audience cheered. Gale maneuvered around the other two students who were following him, and began to drop back down at the maximum speed of five meters per tick.

Suddenly, Gladias and Seila's eyes shifted toward Zven. He had gotten out of his copter, and was running toward the instructor.

When Gale was still about thirty meters in the air, Zven opened up his hand and gave the instructor the fourth redball, with broken rotors, that he had found on the ground.

"And, nice catch by Zven. Sixth point for team yellow. Team yellow wins the game!"

Gladias sighed with relief.

Seila beamed with pride for Zven.

"He took the path less traveled", she quietly recited, quoting an old Veridian author, "and that blazed a trail for all."

* * *

Gladias slid the barrier open, entered the booth, and slid the barrier shut again.

Three other Keepers were already there. The other three, Gladias thought, must still be on their way.

This was the third circle of Keepers that Gladias was assigned to, and the topic was air quality. Here, his assigned number was two.

Gladias immediately took out his watch, and whispered into his neck band to check the air quality of the current room.

Air	
CO_2	906
PM2.5	4.4

Devices	
Attested ✓	16
Unknown ?	0

"How are we already at 906 with just four people here?", he asked. "Filters are working great, but I guess they haven't yet figured out how to give these secure booths any ventilation?"

"Yeah, pretty much everyone is really behind standards. We've been saying that we really have to increase the taxes on this bracket across the board."

The CO_2 reading on Gladias's watch refreshed. It now showed 918.

"I think people will figure this out on their own", another Keeper replied. "It's just a matter of the technology being open and deployable enough. We've been making these filters twice as cheap every two years for the past decade."

"Well I feel like we have to push harder", Gladias replied.

"What do the indoor air quality tax rubrics say now?"

"It's a land tax component. Zero if your air quality is perfect, an extra third of a percent per year if the average occupied room reads worse than a thousand. The limits get adjusted up if you have filters or UVC."

"Only a third of a percent? For the difference between theoretically perfect and abysmally bad?"

"A third of a percent of the land value, per year. For many businesses, that's like a sixth of their revenue."

"There's so many things that go into land tax. Public aesthetics. Accessibility - both the kind that's about being friendly to disabled people, *and* the kind that's about letting people walk across your land freely and not fencing it off. What percent of the lot you're building on. Ground-floor active use, though that one is controversial and some people think it should just be folded into aesthetics. Emergency preparedness. Fire safety. There's just not enough space for everything."

"And come on, Veridians are not that sick. If you look at the broad listen report we ran a month ago, indoor airborne disease resistance is just not one of Veridians' top ten concerns."

"The ground-floor active use isn't just about aesthetics, it also contributes to walkability, reduces congestion-"

"Fine, though I think we should just target those things directly."

"Anyway, that's not our rubric to decide. Air quality is. You're missing the tail risk."

"What tail risk?", Gladias interjected.

"Super-pandemics. Like, deliberately released by the Arctics."

"Is there any evidence of them working on anything like that?"

"They're a super opaque society, the only evidence we have that the Arctics even know Veridia exists is that we have to listen to Lord Ephelion giving his sermons all the time."

"So what are the possible plans on the table?"

"Increase the maximum to two thirds of a percent. And then make the case to the funding organs - Graph Funding, Quadratic Funding, parliament allocations - that they should support more open-"

Another Keeper interjected:

"Quadratic funding? That round is a mess right now. Someone released a program you can download onto your device and prove that you donated to a particular project researching social media polarization - actually not a very good one - and get a reward. You donate ten zipcoins and get back twenty, they get a hundred including the bounded matching. Lots of people participated. And then

yesterday some insider at Bluewhale blew the whistle, apparently that was all their doing."

"Bluewhale? But why would they do that? They don't even control the project that was getting that funding."

"It doesn't matter where it goes, if it was hiring people from Iptak to come dig holes and then fill them back up again they would still have done it. Their goal is to drain funding from anything that interfaces with open platforms. Starve the open ecosystem, the closed one wins by default."

"Fine, we'll stick to Graph Funding. Go through our channels to suggest to the Graph Funding people that they support more open source tech for air filtering, cleaning, wastewater scanning, all of it."

"Wastewater scanning too? You want to break everyone's privacy even more? What if the Arctics hack in and take all that data, and then they'll find out who lives where even if they move homes, and when each person is healthy or sick?"

"I thought you were the one who keeps saying there's no evidence the Arctics are actually doing those kinds of crazy things?"

"Isn't there some cryptographic way to get the data we want out of the scans, without ever collecting the data itself in cleartext in any one place?"

"Well, if you can find a genius who can figure out something like that, let us know."

* * *

Gladias got off the autobus, and pulled on the strap on his privacy robe's front face covering. His face was still covered, but the belt between the areas above and below his nose was now no longer tight. Less virus protection - outside the unusually crowded indoor space that was the autobus, he no longer needed it - but more comfortable and easier to breathe. He began to walk back home.

He passed by the same restaurant where he had bumped into Febric months earlier. On the restaurant's outer wall, he saw an advertisement. On the left side of the advertisement, there was a

Hydrafill mascot, clothed in Hydrafill's classic yellow and green colors, hugging a small child, and using his back to shield the child from a jet of sludge-like green-colored water. On the right side of the advertisement, there was the text:

> *Heavy water, with its atoms of hydrogen replaced by the neutron-bearing isotope deuterium, is known to be toxic in large doses.*
>
> *We studied our competitors' water, and found that on average, it contains 145 parts per million heavy water.*
>
> *Our water comes from fresh Veridian NORTHERN ICE, bravely extracted only thirty kilometers away from the Arctic border, and it contains 132 parts per million heavy water.*
>
> *Hence, our water is TEN PERCENT LESS TOXIC than our competitors. Only WE care about your health. Nobody else does. Hydrafill wants to heal and protect you. Our competitors think you deserve to drown in sludge and die.*
>
> *Trust Hydrafill. Buy Hydrafill today.*

Gladias smiled. He kept walking.

Suddenly, someone grabbed and hugged him from behind.

Gladias turned around.

"Seila!"

"Finally caught you on your way home."

"How was the rest of your day?"

"Finished up my post on urban economics in Freetown. It's online, you should go see it."

"Ooh, I'm excited."

"Jahn gave me some gossip recently, about how a lot of things are changing now that people there are getting more spooked about the Arctics."

"Do you have any theories about why people in Veridia seem so... nonchalant about the whole Arctic situation?"

"I guess the Arctics don't really affect people's lives here much? They just publish their stupid sermons from time to time and people watch and listen out of curiosity."

Gladias groaned.

"Even inside the Keepers, the situation is so strange. The older ones seem to understand there's a problem. But the newer ones ... not all, but many of them think we should just not care?"

"What's weird is that the corporates in Freetown seem more resistant to ... whatever is causing that stupidity. They're basically the only city taking the problem seriously - except for Redshire, and the cities partially run by Redshire's parliament through co-governance."

"I guess more than anything, the newer Keepers seem suspicious of Steering and Quadratic Funding and Graph Funding and government being able to do much of anything in general."

Seila sighed.

They continued walking. After a few minutes, they arrived at the door to their house. Gladias put his watch against the circle, and the door opened. They walked in.

Seila opened up her hand device.

"Dzego food truck in one minute. The usual I assume?"

"Yep, how did you know?"

Seila laughed.

Seila clicked a few buttons, and ordered.

The door to the house was not yet closed, and Seila could already see the food truck coming towards them from a few hundred meters away. They both waited for the food truck to arrive.

Seila went out, and picked up the food directly from the food truck. She walked back into the house, and closed the door behind her.

She put the food on the table, and immediately unpacked it.

They both sat down, opened their food boxes, and began to eat.

Chapter 12

Sadzu Du, Dzego · 3724 GRASSTIME 22

Zei looked out of the window of the train, as the train began to slow down.

At first, he saw trees. Then, he began to see buildings with greater and greater density. Finally, after a few minutes, he saw the station.

A sign slowly passed across the window.

sa dzu du

"Main city", Dzego's helpfully-named capital city.

The train came to a stop, and Zei could hear the doors open. The other passengers stood up, and started to gather their belongings.

Zei decided he was not in a rush to leave first, and would maximize the quality of his time instead. For about two minutes, he finished his game of Minpentai against a bot.

Fin had never gotten around to writing the harness to fine-tune the AI to play well on custom rulesets, and so Zei was still playing against the stock AI.

At that moment, the other AIs resigned. Zei had beaten them one-on-three. To compensate for their lack of skill, he had adjusted the game to only give him one corner and three other AIs the other three corners, and put them all on the same team against him. Even despite this, Zei had won. He realized that the three-on-one did not even help the AIs that much - playing in that style was not in their training data, so they made basic mistakes and cannibalized each other's negentropy.

Zei put away his hand device and sighed.

By this point, the other passengers were almost all off the train. Zei grabbed his bag, and stood up and began to walk out of the train as well.

When he stepped out, there were six people wearing Minpentai priest robes waiting for him.

"Welcome to Sadzu Du, Zei", the lead priest, an older woman about the same age as Mu, said to him.

"Excited to be here", Zei quickly replied. "Only my second time to the capital."

"I'm Su by the way. I'm one of the organizers of the nationals."

"Well nice to meet you, Su."

"So what's on my itinerary these next few days?"

"Check in to the hotel, then orientation tomorrow morning. Thirty-two contestants, the brightest young minds in Minpentai across all of Dzego. You'll get to meet them all."

Su led Zei to a room that contained many doors. One of the doors slid open, and on the other side Zei could see the inside of a black car, identical to the one that Den had used to take him to the chip fab a few months earlier.

"Unfortunately, I'll have to leave you here, Zei. There's another contestant arriving here on a train in half a longhour. But have a good ride and have a good rest."

"Thank you so much."

Zei got into the car, and the door closed. The car started moving.

Zei stared outside the window of the car, marveling at the buildings and trees in the city. Parts of it looked not too different from Pafogai Du, though here there were fewer electronics stores, and instead a wider variety of venues. Some medicine stores.

On the window of one of the stores, Zei saw a sign:

ti ka dia shu fe cen su jui hui

"Get your longevity package here"

Soon the health stores gave way to some stores selling artistic furniture and other household items.

And after that a few traditional-looking restaurants and tea houses, in a style mimicking Dzego's own medieval era.

For a moment, the buildings on the right side stopped completely, and Zei could see a park. Behind the park, Zei could see several mountain pyramids, each about half as tall as the mountain pyramid south of Pafogai Du.

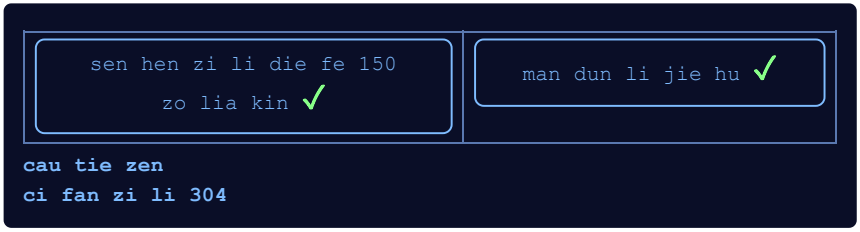
A minute later, the park ended, and the car crossed a bridge across a small river. Zei could see the buildings get denser further ahead.

The road slowly descended underground, and turned into a tunnel.

A few minutes and a few turns later, the car stopped. When the car's door opened, it was pressed tightly against another door, completely shielding Zei from being visible from outside the car and the building he was entering.

Zei stepped out. He was in a hotel.

He walked over to the nearest green circle, and tapped his watch against it.



Reputation above 150 verified. Payment not needed. Approved, room number is 304.

Zei's watch buzzed, confirming his room key.

Zei entered his room, and took a shower.

He promptly went to sleep.

* * *

An alarm buzzed.

Zei tapped on his watch to check what time it was. 30,003, the watch said. Only half a longhour before the commencement ceremony. He tapped a few buttons on the room service menu, and ordered a fruit and vegetable juice.

He put on his clothes.

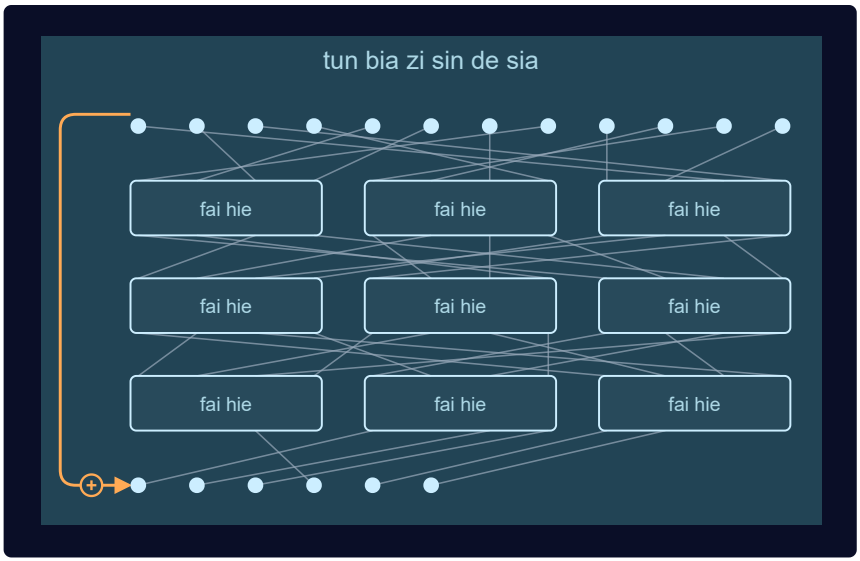
Three minutes later, the juice arrived, and Zei quickly drank it. He immediately felt energized - or at least, energized enough to take the trip to the venue.

Without wasting any more time, he opened the door, and went downstairs.

A black car was already waiting for him. Zei walked through the lobby, and immediately stepped through the hotel door and car door that were already open and pressed tightly against each other. The door closed.

As the car began to move, Zei took out the cryptography textbook that Den had recommended to him in one of their meetings, and continued reading it where he had left off.

The section was on hash functions. In the middle of the page, there was a large diagram.



"Conception of a hash algorithm".

Zei read the accompanying text.

Hash algorithms, it said, were there to create a shrunken fingerprint of a large piece of information, that still uniquely identifies the original. The goal is for it to be infeasibly hard to find two different pieces of data that have the same hash.

Hashes inherently destroy information - representing a larger thing with a smaller thing is their entire goal. But there is a paradox: in the internal workings of all the best hash functions that we build, the largest component is an *entropy-preserving* transformation: a permutation.

The core building block maps inputs to outputs one-to-one. And not only that, it can be *efficiently* run both forwards and backwards. The shrinking, and the irreversibility, comes from a simple layer on top: a deletion, and an xor of part of the original input, added to the end.

Why do this? Why build irreversibility on top of reversibility? The reason is subtle. If you build irreversibility directly, then you cannot control when entropy is destroyed. If you apply an irreversible round

function enough times, then about half the entropy collapses, bringing the protocol's security down along with it. And so even when destroying information is your final goal, it's best to cleanly separate out the piece that's going that - destroy information in one precise spot in the algorithm, on your own terms.

Zei put away his hand device, and leaned back.

It's all entropy, isn't it, he thought to himself.

A minute later, the car suddenly slowed down, and turned left. To his right, Zei could see an area blocked off by safety tape with various large vehicles further inside.

"What happened?", Zei asked in a voice loud enough that the driver would unambiguously recognize that the question was meant for him.

"Explosion in an advanced biology research lab. Probably the Arctics, as usual. They're going to spend days cleaning up to make sure everything is safe."

A few minutes after that, the car slowed down and came to a stop. The door opened, leading straight into a hallway.

* * *

Zei walked through the tunnel, and down the stairs.

He saw a door in front of him. The door looked similar to the door to the Minpentai room inside the mountain pyramid in Pafogai Du: a series of stone bricks, curving near the top into an arch, and the symbol at the top: two horizontal arms, meeting in the middle with clashing fists.

However, this door was noticeably larger, more well-maintained, and the decorations more grand.

Zei tapped his watch against the green circle, and the door opened. He walked through.

He entered a room with about fifty chairs, with about twenty contestants already sitting down. On the stage, he recognized Su. Beside her, there were three other priests.

Zei sat down. Over the next fifteen minutes, other contestants sat down in their own chairs.

An older teenage boy sat down beside Zei.

"Hi, I'm Deluin", he said.

That's not a very Dzegojan name, Zei thought.

"I'm Zei."

"Where are you from?"

"Ah, I'm from Redshire. My name gave me away, didn't it?"

Zei laughed. "Do you get that reaction from everyone?"

Deluin laughed along with Zei.

"De lu hin, they call me here. 'A fork with the appearance of'. Lovely name, don't you think?"

"Are you studying here? Did your parents immigrate?"

"Ah, I'm a foreign contestant. They brought in four foreign contestants this time, to make things more fun."

"Are you excited?"

"Absolutely. Actually my first time in Dzego, super cool place."

A bell rang, and the contestants quickly quieted down.

"Welcome to the national Minpentai championship", Su began. "I am honored to see so many brilliant contestants this year, and I hope that everyone arriving had a good evening and a good morning."

"I am pleased to introduce our lead organizer, Taskmaster Jin."

One of the other priests stepped forward, and began speaking.

"This is a special moment for all of us in Dzego, not just for us and for you, but for millions of Dzegojan who watch the games from across the country, as they will get to watch the brightest minds from Dzego's new generation face off against each other."

"I'm also honored to announce that this year, we have four distinguished foreign guests participating along with us."

"Today, we will have a walk through the Minpentai museum, right here inside this mountain pyramid. Tomorrow, we will have some tourist excursions around Sadzu Du, and in the evening you will get a chance to rest. Two days after, the first championship games will begin."

"There will be six rounds of qualifying games, with an eight-day-long break in between each game."

"Out of the thirty-two of you, the four players who win the most games will advance to the semifinals. If there is no win threshold that exactly four players pass, we will break ties by the geometric average of the number of turns that you survived before losing your losing games and the inverse of the number of turns that you needed to win your winning games, counted up to a maximum of eight thousand turns. Thus, there is no incentive to resign before eight thousand turns. If there are still ties, we will insert an additional qualifying round."

"I hope you can enjoy the games, enjoy everything that Sadzu Du has to offer, but also, just as importantly, enjoy each other. Here you are not enemies, you are temporary competitors. Off the arena, you should consider each other friends, and remember that we are all part of a shared project together. Dze go ba fau gie."

"Dze go ba fau gie", all the contestants replied in unison.

"Now, please follow Su. She will lead the excursion through the museum."

⁂

3724 GRASSTIME 25

zan		tie	tei
Bai	→	[dia lau shau]	23098

Zei woke up. Bai was requesting to call.

Zei tapped the watch.

"How are things going over there?", Bai asked.

"This is a bit early for a call, isn't it? It's still 19 minutes before my alarm was supposed to wake me up."

Zei paused for a moment, and decided to answer Bai's question.

"Things have been well. Very fascinating exhibition at the museum three days ago, and got to see a bunch of Sadzu Du the next day. Met

someone from Redshire too."

"A contestant?"

"Yep, they've decided to internationalize it a bit this year."

"So how's Pafogai?"

"Ask me something I can answer. I've been staring at the mountain pyramid you'll be playing in from across the river for the last longhour."

Zei sat up.

"You're HERE?"

"Den's people got hold of me and told me to come and take notes on all the contestants' strategies."

"Your first game is today, right?"

"It is indeed."

"Want me to come over and help you out, or at least watch?"

"I... think I'm fine. And I doubt you can get a ticket to the live stadium."

"Not if you let me in."

Zei groaned.

"I'll watch from afar then. We're all rooting for you!"

"How is Fin? I haven't heard from him in days now."

"Busy with Bansunpei stuff I guess. He hasn't said anything to me in days either."

"Alright then. Well, I gotta start preparing. I hope you enjoy your day!"

"Best of luck to you in yours, you'll need it!"

"Oh, and, let's play a practice game tomorrow."

Zei once again ordered a fruit and vegetable juice, and put on his clothes. A few minutes later, the juice arrived, and Zei quickly drank it. He opened the door, went downstairs, and entered the car that was waiting for him.

Twenty minutes later, the car arrived at an entrance to the pyramid.

Zei stepped out, and began walking through the tunnels.

After a few hundred meters of walking, and a few turns, he once again arrived at the large stone brick door. He entered, and this time immediately turned right, into a hallway that he had not even noticed when he was first there three days ago. He followed the signs, and soon got to a waiting room.

He saw Su, and the other contestant.

"Welcome Zei! Meet Bo."

"Nice to meet you, Zei."

"Nice to meet you too, Bo."

"It will be a fun game today, I'm sure of it."

A robot arrived, and gave all three of them a cup of tea. They each drank their cup.

"So what now? We sit here for a longhour without internet while the others play, so we don't learn the new rule ahead of time?"

"Exactly. And no hand devices or watches allowed either. The robot will be coming to pick them up in a minute. You can watch last year's matches on that screen over there."

* * *

A longhour later, the elevator doors opened, and Zei and Bo each stepped in to their respective elevators. Su waved them goodbye. The doors closed.

A quarter minute after that, Zei's elevator door opened once again, exposing a giant room much larger than the room in Pafogai Du. About five to ten thousand people seemed to be watching. A path lit up, showing the way to a giant console with a chair.

Zei walked forward. He could make out Bo's silhouette walking forward in the distance. He got to the chair, and sat down.

Drones about the size of a hand device buzzed as they flew around Zei, scanning for any kind of cheating equipment. This time, there were about three times more drones than there were in Pafogai Du, and with even more shapes and sizes. The drones buzzed around for about twenty ticks, and then retreated back to their resting positions.

A loud automated voice, this time sounding more female, announced: "MU GU GEI FA". Starting in fifty ticks.

Zei closed his eyes, and breathed in and out, trying his best to calm himself. This match was not decisive, he knew: if he lost, he could still make it up in the others. But in another way, that fact made him more scared: armed with the excuse that it's okay to lose, will his mind have the motivation needed to keep his full attention and mindpower on the game and win?

"LE MU GEI FA"

Zei opened his eyes again.

"PA GU GEI FA"

Zei patiently waited.

"Welcome to the fourth qualifying match of the north stadium for today. Everyone, meet our players for this round: Zei Leimin from Pafogai Du, and Bo Zume, from right here in Sadzu Du."

"I am also pleased to announce - perhaps one time too many for the audience, but for the first time, for our contestants - today's rule change. Today, Minpentai will be played on a hex grid."

Zei opened his mouth in shock.

"You will have twenty minutes to figure out your structures, any gliders, spaceships and walls, to the best of your ability. And then the serene battle over a field of snowflakes will begin."

Zei's display turned on, and he could see the hexagonal grid in front of him.

"LE FI SHI GEI TAU FA", the automated loud voice shouted. The battle begins in two thousand ticks.

Zei began by quickly drawing random junk in his sandbox, seeing what stayed stable and what turned into a glider. Within one minute, he identified a few gliders and a few static structures that could be used for walls. To his surprise, the gliders that moved the fastest moved ... directly east.

His next move was to deliberately draw different kinds of walls, fly gliders at them, and see what sticks.

He watched the gliders carefully, to try to quickly get an intuition for what kinds of shapes can move while preserving their shape. He did several rounds of defining both the gliders and the walls.

"PA MU GU GU GEI TAU FA." The battle begins in fifteen hundred ticks.

Zei began experimenting with shapes that were similar to his symbol, seeing if any of them could function as a glider. He needed to come up with *something* - otherwise he would be able to progress only very slowly, by repeatedly building a new symbol beside his existing symbol during each intervention turn.

He realized that to make actual headway, he would need to interact with the rock formations. He made a couple of guesses about what shape the rock formations in the hexgrid version of Minpentai might look like, and examined the behavior of gliders as they crashed into them.

"FI SHI GEI TAU FA." The battle begins in a thousand ticks.

Zei began to sweat. So far, no luck. No combination of a glider, a plausible rock shape and an offset that he tried led to a copy of his symbol.

He thought more about the exact shape of his symbol.

After another two minutes of fruitless exploration, Zei's mind flashed back to the textbook he had been reading less than two longhours before.

> *The core building block maps inputs to outputs one-to-one. And not only that, it can be efficiently run both forwards and backwards.*

But unlike the hash function, here there was no truncate-and-xor wrapper at the end. And so if he knew his desired end state ... he just had to put that end state into the sandbox, and play it backwards.

But ... how do you play it backwards?

In the internal permutation of a hash function, you walk backwards from the last layer to the first layer, and invert each step as you go.

Here ... what was the inverse of the hexgrid rules that Zei was interacting with for the first time in his life?

Zei took the gliders he saw, and tried the most basic transformations: flipping them, rotating them 60 degrees, launching them one timestep later. None worked.

"MU FI LE GEI TAU FA." The battle begins in five hundred ticks.

Zei stepped through each step more slowly. And then he saw it. During each step, the game board split up into triangles, and the hexagons rotated inside each triangle - clockwise if it had one hex filled, counterclockwise if it had two hexes filled.

And so in order to time reverse, he just had to ... let each time step happen twice. The opposite of a 120-degree rotation is, after all, two 120-degree rotations. But there was a problem: after each time step, the split into triangles shifted. So to play a time step backward, he had to... play one step forward, then shift time back one step so that he could play the next step forward under the same tiling, then play a second step forward to bring the total rotation under that tiling to 240 degrees within each triangle, then shift time back *two* steps so that the game would shift to the *previous* tiling. Invert each step, and play the steps in reverse order, just like inverting a permutation in a hash.

He confirmed: there was a button to play one step forward, and two buttons to shift time - either forward or backward. So that's what he would have to do. A, B, A, B, B. Like some kind of video game cheat code. Which, in some sense, here it literally was.

Fine, he thought.

He put a glider into the sandbox, and tried it. It was an arduous process, and his fingers quickly tired. But it worked. The glider moved backwards.

Basic test passed.

Zei tried the real test: using the same technique, starting with a symbol forming beside a rock formation and a glider flying out.

The first time, it failed: as time went backwards, the glider and the symbol crashed, but then the same symbol re-formed a few steps away, and the glider flew away. So in that particular history that Zei

had just sampled, before the crash, the symbol had been there all along.

He tried again, with a different glider. This time, it succeeded. As time moved backwards, past the moment of the crash, the rock-formation-plus-symbol turned into a bare rock formation, and two gliders flew back out.

"FI LE GEI TAU FA." The battle begins in one hundred ticks.

Zei quickly repeated the same experiment with other candidates for a rock formation. He paid no attention to the voice announcing the countdown, and kept trying until he found one for the two other most natural options.

"MU GU GEI TAU FA."

Zei quickly put up symbols, walls, and gliders that would create symbols if they crashed into favorable rock formations.

He was surprised at his own speed doing this, given how much longer everything else had taken. But almost immediately he realized in the back of his mind that this made sense: figuring out what to deploy is the hard part, deployment is just copy and paste.

After making a basic setup on the board, he breathed out. Just as he breathed back in and started doing a final pass refining his structures, a final countdown began.

"PA GU ... SO ... BI ... ZE ... HA ... MU ... FO ... SHI ... LE ... PA ..."

"TAU FA."

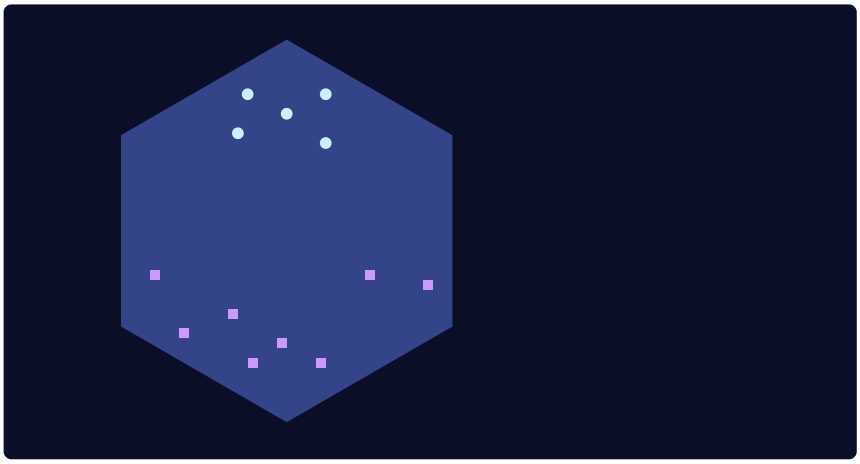
Gliders started flying out. At first, nothing happened.

Five hundred turns later, a part of the board came into view. One of his gliders had successfully created a symbol, and revealed a part of the board that showed what the rock formations look like.

Immediately, Zei switched to the next task: figuring out how to build glider factories.

When the first intervention turn hit, Zei had not yet figured this out. However, he did send out several more gliders that flew from his home base, ready to create more copies of his symbol if they collided into a rock formation in the right place. He also sent gliders from nearby his new symbol, rapidly heading east.

The audience watched intently. From across the river, Bai was also watching intently. And from afar, so were Fin, Mu and Den.



The audience was surprised. Neither player had managed to figure out any gliders that move vertically, they realized.

On the second intervention turn, Zei launched gliders from the sides of the board, aiming straight toward Bo's home base.

And just before the third intervention turn, one of Zei's gliders hit a rock formation right beside Bo's home base, allowing Zei to see all of Bo's symbols.

Zei used that intervention turn to send gliders directly toward the three of Bo's symbols that he could see, and more toward some locations nearby just in case.

Within three hundred turns, all of Bo's symbols were destroyed, and Zei was announced the winner.

Chapter 13

Meldan, Veridia · 3724 HARVESTMOON 8

Gladias pulled up the cloth curtain, and entered the booth. By now, he was very familiar with the style: a table, a decorated lamp and two flowers at the end, and an air filter pressed against a corner on the floor.

Delwart was already sitting inside.

"So how are the Keeper meetings going?", Delwart asked.

"They're interesting", Gladias replied.

"I'm actually surprised. This idea that we should not be intervening so much in everything, we should have fewer rubrics, I'm far from the only one bringing it up. It feels like almost half the first-years have the same idea in mind, and are bringing it up just as proactively as you wanted me to."

"So at least some of them don't want you to get taxed for every bad word in your song and micromanaged on like five different kinds of accessibility for every house or apartment that you build?"

"I mean, I don't think it's that bad now. We're not making people fill forms for everything like they do in Inglewore. The taxes are just another variable for people to watch, along with electricity costs. But even then, I think the Keepers are understanding that we have to be reasonable and not too prescriptive."

"Are they making their case well?"

"Often better than I could. The second and third years tend to be more skeptical, they have a more traditional mindset. That comes through especially on anything related to Dzego or the Arctic threat."

Delwart sighed. "There's been decades of propaganda on all this stuff. Honestly, the Arctics are not making a great case for themselves; everything they do, their drone patrols near Redshire, even Ephelion's speeches, it just always aggravates people so much."

"Yeah, a bunch of people think they're going to make some kind of more serious sabotage campaign against Redshire."

"Do you think they'll make a more serious sabotage campaign?"

Gladias thought for a moment. In his heart, he did not know, and while he expected that nothing would happen anytime soon, he feared that the answer might be yes. But he also knew that if he admitted it to Delwart, he would get a long lecture trying to impress on him how biased his sources of evidence were - and that was all *without* him having told Delwart about the time Seila met Zei and Bai in Freetown.

"Probably not."

"So do the best job you can to try to convince them not to worry."

A robot came in, and brought Gladias a water and a salad, and Delwart a tea along with a rectangular nutrition bar.

They both quietly drank, and then ate.

They talked for a little bit more, and then after about twenty minutes, Gladias stood up.

"I'm sorry, I have to leave for a dinner with my family."

"Of course you should, family is the most important thing!"

Gladias stepped out, and began to walk through Greater Plum Harbor's tunnels toward the exit.

* * *

After about ten minutes of walking through tunnels, Gladias emerged into a pathway above ground level. The Greater Plum Harbor's tall entrance doors were directly in front. Gladias passed through the turnstile, and exited back out into the mixture of greenery and glass and steel that defined Meldan's commercial district.

Gladias turned right, and began walking back toward his home.

His watch buzzed.

From		Message	Time
Mov	→	I've been following Delwart more, we should talk.	71256

Gladias whispered into his neck band, and opened a call with Mov.

"What's going on?"

"He was a little more careless than usual this time", Mov said. "I followed him to his next meeting. It was with another Keeper. I got the adjacent booth, and I could listen through the vibrations on the wall."

"Wow. The GPH booths are that much worse than the ones the Keepers use?"

"Yep. I don't think enough people know the difference yet."

"And we were using those booths for our conversations for months..."

"Indeed... well, now we know."

"Anyway, the other Keeper was in the hardware openness rubric, in a different discussion group from you. He was fully on board with Delwart's program."

"And?"

"The whole conversation was strategizing about how to persuade the other Keepers in that group to go along with his position."

"Delwart was really forceful. He seems to really care about this. And, between you and this other Keeper, he has managed to get contacts in two discussion groups for the same rubric. The whole point of splitting the discussion groups is to force people to think independently, to have them all taking cues from the same person!"

"Anyway, I'm convinced that this whole thing is more organized, and deeper, than we all realize."

"Next time you talk to Delwart, I think you should demand some answers."

"What do you think it could be? Some kind of organized political effort to make Veridia more like Freetown, but without the part where they care about opposing the Arctics? So it's not just one person who

believes in the cause, there might be some kind of whole network behind him?"

"I'm curious too."

Gladias stepped onto the autobus.

He sighed.

"I'll think about it more. I'm heading home now, having dinner with my family. The kids are back home today."

"Yeah, please do. And - take care."

* * *

Gladias got off the autobus, and began to walk toward the restaurant. Wearing the privacy robe and walking quickly, he sweated in the hot sun.

After a few minutes, he arrived. Seila and all four children were already sitting at the table.

"Surprise!", Seila exclaimed. "We got here before you this time."

Gladias's face turned into a smile. He was happy to see the kids again. It had been two months since his visit to their house for that one evening, and they had been away from his and Seila's home for half a year.

Gladias put away his thoughts about Delwart and Mov, and made a deliberate effort to focus his attention on his family.

A few minutes later, a robot slid up to them, bringing meals for Febric and Hreda.

"How did your math exam go?", Seila asked Hreda.

"Ninety-six percent", Hreda replied.

The look on Seila's face immediately changed to worry. Now it was going to be even more awkward to ask Lily, she knew. She had planned to ask her right after, but now simply congratulated Hreda instead.

Lily's eyes sank down to look at the table right in front of her.

The robot slid up to them once again, and brought the meals for Zven and Lily. Zven eagerly took his. Lily did not move.

Gladias took Lily's meal, and put it down on the table in front of her.

Lily slowly began to eat, and said nothing.

The table went silent. They ate slowly, while watching the professional Helisport game that was playing on a nearby screen.

Suddenly, the Helisport game cut out, and was replaced by a completely different broadcast.

An image of Lord Ephelion appeared on the left side of the screen, and a multi-paragraph message on the right side.

The Dead Hand of Egalitarianism, and the Path of Strength

* * *

All humans are equal, the schools, the posters, the songs, and everything else in Veridia is telling you.

Maybe not Dreadknot - but that's exactly why Dreadknot's taxes were bumped up to tier 4 in the audit last year.

Nobody is better than anyone else, and if you even dare claim that one human being is better than another, then you are immediately cast out as evil in Veridia.

But there is a fundamental flaw in this kind of thinking. If one human being cannot be better than another human being, then that also implies that a human being cannot be better than their former self.

Denying the possibility of hierarchy, of vertical difference between people, the fact that there are traits that everyone agrees are good, but that only some actually manage to achieve, is thus also denying the possibility of self-improvement.

Do not think that the idea that there are multiple dimensions to goodness and to success saves you from this. Even if there are, some people perform better, and others perform worse, across *all* the dimensions that matter.

Stop looking to society for answers, to Veridia's institutions to make life fair. Life is not fair - and should not be fair. Embrace the competition.

Today, you may be weak. But you have it in you to be strong. Do not believe the lie that you are adequate just the way you are. But do believe a truth - the truth that you have it in you to become strong. Take that opportunity. Grow stronger. Grow so strong, that you later look upon your former self with shame.

```
And find the other strong people, the people who
will help you on your journey of striving for
greatness. Not those who will love you
unconditionally - that is slow-burning poison, like
sweet candy. But those who will give you love when
you win, and a cold shoulder when you lose. Those
that give you both the incentive, and the
disincentive. You are not good because you are
human, you are good because you can win. Only thus,
can you train your mind to love winning, and achieve
your true potential.
With regards,
Lord Ephelion
Arctic Empire consul to Veridia.
```

Gladias quickly scanned the message, and looked around. Seila had already gotten up from the table, and was trying to figure out how to unplug the screen, even before the restaurant staff could get there themselves.

Febric and Hreda continued eating. Zven was looking at the onscreen message with curiosity, his mind jumping between the paragraphs, and occasionally moving his attention to watching Seila try to turn the screen off. And Lily was simply staring at the message.

Finally, the screen turned off.

The family went silent. Febric took out his hand device, and began to study for school. Lily took out her hand device, and began to intensely tap on it. Gladias could not see what she was doing, though she seemed to be playing some kind of game.

The family quietly ate. Five minutes later, Lily's face suddenly lit up, and she quietly cheered. She put down her hand device, and joined the family eating. Despite starting to eat later than they did, she finished the food before everyone else.

⁂

3724 HARVESTMOON 9

Gladias slid the barrier open, and walked into the booth. The other Keepers were all already there.

"Good to see you again, Six!", Thirteen exclaimed.

"Wait, how did you know I'm Six?"

"By instinct. Probably some subtle detail of your body language - as they say, ninety percent of our communication is subconscious."

"Huh, I guess even after hiding our faces, even after speaking through neck bands and sending out our speech with a re-randomized AI-generated voice, though I still can't tell who of you is male and who is female, these physical privacy tools still have their limits. Well good to see you too, Eighteen."

"I'm Thirteen!", Thirteen replied. "I guess for you communication might be only thirty percent subconscious."

"I'm female, by the way. I guess that's okay to say, it only cuts down my anonymity set by a factor of two."

"Anyway, this is the last time we meet before we have to vote on this rubric in about two months."

"Do we have a proposal?"

Zero raised his head.

"I drafted the change in language", he interjected. "To exempt all military, surveillance and counter-surveillance products from the more favorable tiers on the openness tax rubric. All that stuff becomes minimum tier 3, unconditionally."

"Counter-surveillance too?"

"Of course. It's all part of the same zero-sum game."

Eighteen began to speak.

"I still think you guys are not understanding the Arctic threat at all. They have the technological advantage. This isn't a symmetric game. If everyone has to make their own tech in those areas, the Arctics win."

"Come on! They're not going to attack. They've had drones buzzing around different places for years now, ever since drones were invented. They're not going to start doing anything crazy in Redshire, okay?"

"You really think so?"

Gladias looked around the room. His ability to read body language was indeed not great. But from what he could tell, three of the others were leaning in favor of cutting military, surveillance and counter-surveillance out of the hardware openness rubric, Thirteen was in the middle, and only Eighteen and One were against.

His conversation with Mov the day before put him into a suspicious mood. Before, he had been more open-minded. Now, he started to think, doing nothing at all to the rubric this year might be best. His mind bounced around, trying to think of ways to nudge the situation in a more favorable direction. After about half a minute of thinking, he had his plan.

"I just had an idea."

The others turned toward Gladias.

"We don't have to decide this now. We still have two months before we have to vote, don't we?"

"Correct."

"Eighteen, in what timeframe do you think the Arctic Empire is going to make some kind of move, do some more serious sabotage against Redshire?"

"Less than two months, definitely."

"One, I remember you were also of the mindset that we should take the Arctics seriously as a threat. What's your timeline?"

"Ten or twenty days, maybe a month. Won't be anything too drastic, but we'll see sabotage of some kind. It might get reported as a big accident, that's how it always happens."

"So how about this? Of course, this is not binding, it's just a preliminary thought. But, how about we all wait for two months and see what happens. If there's no attack, then we'll all be more inclined to accept Zero's amendment. But if there is, then we'll keep the rubric as is, or even increase the distance between the tax brackets."

Zero softly muttered some disapproval under his breath but did not say anything. He clearly did not like to be pinned down like this. But he could not come up with any argument to rebut Gladias's proposal.

Eighteen seemed to have a similar reaction. Like Zero, he did not like being pinned down, but he could not avoid admitting that he could not come up with any better way to break the impasse.

"So let's leave it at that."

Chapter 14

Sadzu Du, Dzego · 3724 FIREMOON 10

Zei's alarm buzzed.

He slowly stood up. He tapped a few buttons on the room service menu, and ordered a fruit and vegetable juice.

He put on his clothes.

Three minutes later, the juice arrived. Zei quickly drank it, and without wasting time he opened the door and went downstairs.

When he was in the lobby, he could see someone in the lobby restaurant waving at him.

He turned to look. It was Bai.

Zei walked over, and sat down beside Bai.

"jie mo kai tu jie hei ja ma?", she asked, in a cute voice that was unambiguously making fun of the restaurant robots.

"I already had my juice, I'm fine", Zei replied.

"How have you been?"

"You're not going to ask me what I've learned about the other players' strategies? You're going into the semi-final today", Bai replied.

"Well what have you learned about the other players' strategies?"

Then, he quickly added:

"You haven't been doing anything *illicit* to get your information, have you?"

"Don't worry, I'm not some kind of Arctic spy or anything like that. I've just been watching all the games, I made friends with some of the other players, and they let me participate in some of their practice sessions."

Zei rolled his eyes slightly, and then sighed with relief.

"Basic social skills. I needed to learn them to survive growing up."

"Thank you Bai, I appreciate you."

He paused for a few moments.

"So what kind of advice do you have?"

"If there was one pattern that came up again and again", Bai replied, "it would be this. All the other players - *definitely* Gun, but really all the others too, are way too impatient. They don't just want to win, they want to win quickly."

"I know the scoring system of the qualifying rounds incentivizes this, but ... it carried over even past the eight thousand turn threshold, and even in the practice game that I saw between Gun and Deluin. They're not being strategic, they're being fast for the sake of being fast."

"So please don't rush things if you don't have to, okay Zei?"

Zei nodded.

"Thanks for the advice."

The robot arrived, carrying Bai's vegetable and rice dish, and a tea. Zei ordered a tea for himself, and drank while watching Bai eat and drink.

Ten minutes later, Zei's watch buzzed. The car that would take him to the semi-finals had arrived.

* * *

"Welcome to the semi-finals", the announcer voice shouted.

"We have come far to get to where we are. We have had forty days of some of the most challenging matches we have seen. The top four players in the qualifiers were: Deluin from Redshire, who won six of six games, Zei Leimin from Pafogai Du who won five of six with an average win time of 5,047 turns, Gun Caibai from Dzundei who won five of six with an average win time of 5,699 turns, and Kau Seiza from Geijaken who won five of six with an average win time of 6,740 turns."

"Today, in this very special round, we will see Zei Leimin and Gun Caibai face off. And right after this, the other two players will have

their own semi-final match. And in fifteen days, the two winners will face off in an epic best-two-out-of-three, and we will have our national champion!"

"And of course, today's rule change. The resources you can allocate in each intervention turn are reduced by eighty percent. But in addition, there will be five shrines, four arranged along the center of each edge of the board, and one in the center. Each thousand turns, any changes that had taken place to the contents of the shrine will be applied as an XOR to the regions near the four corners of the shrine, and the shrine will reset to being a default rock formation."

"FI LE GEI TAU FA."

Zei immediately understood the meaning of the new rules. The shrines are the only objects that violate mass conservation - though even still, with these new rules, nothing decreases entropy. XOR is not weight-preserving, but it is reversible. So the game board would get more and more full, and more and more messy, over time. The reduced resource allocation meant that the intervention turns were not for building - they were for *steering*.

Zei put down his usual walls, gliders and spaceships. This time though, he used some different designs - a steerable symbol-carrying spaceship he had developed months prior, that could change its direction if you intervene in only a few cells.

"MU GU GEI TAU FA."

Zei figured out one additional adjustment: he adjusted the positioning of his spaceships so that they could be located right past the shrines after exactly two intervention turns. This would let him send out a glider in exactly the right direction that it would turn a pair of shrines into an eternal glider factory: a glider would bounce back and forth, at each collision it would break off rock to spray even more gliders further north, and then every thousand turns, the rock would reset.

"PA GU..."

Zei put all of his effort into glider storms that would protect his spaceships as they went to the center, east and south; he decided to

sacrifice the west.

"SO ... BI ... ZE ... HA ... MU ... FO ... SHI ... LE ... PA ..."

Until the last moment, Zei kept optimizing his structures.

"TAU FA."

The spaceships and gliders started moving.

A thousand turns later, the first intervention turn came. Zei shot off a few gliders to turn more rock formations into symbols and glider factories.

He realized that he had one major weakness: while he could make symbols, it would take him four intervention turns to make a symbol-carrying spaceship or glider factory. That meant that his three symbol-carrying spaceships were all the fleet that he had, and he had to protect them.

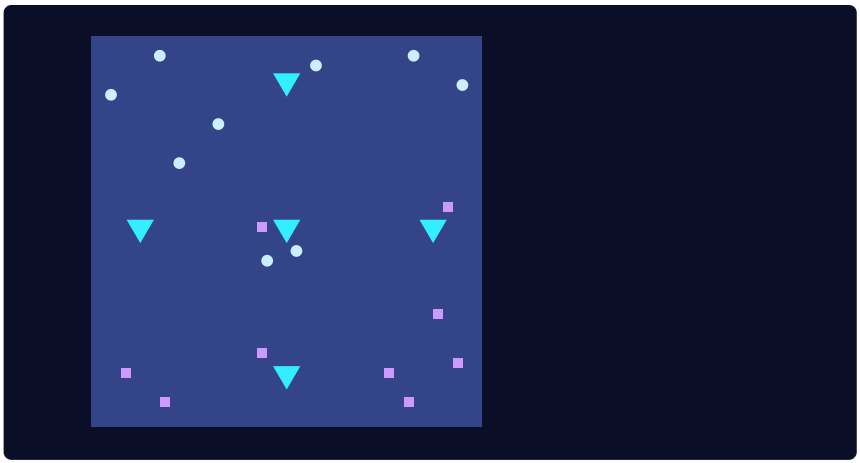
Just before the second intervention turn, his spaceship and two of Gun's spaceships flew past each other in the center. Evidently, Gun was even more intent on doing something in the center than he was.

Zei realized that he had a choice to make: he could send gliders out to attack Gun's ship, he could put up walls to defend against Gun's ship, or he could set up the eternal glider factory.

The second intervention turn came. He decided on the last course of action, sending gliders out that would bounce between the center shrine and the south shrine, the center shrine and the east shrine, and the south shrine and the east shrine.

Right after the intervention turn finished, Zei noticed that Gun had done the exact same thing, using both his symbol-carrying spaceships in the center. He had also adjusted the heading of both of his ships, so that they would also head south.

Zei and Gun could not see the full view of what was going on, but the audience could.



A few hundred turns later, Zei realized that Gun's attack had been more effective. He had not only created eternal glider factories between the center and south shrines, and the center and west, but he had also unleashed a glider attack that destroyed Zei's symbol-carrying spaceship.

Meanwhile, Gun's two symbol-carrying spaceships were moving south, one of them on a collision course with Zei's own - a collision that would happen before the next intervention turn.

Zei would only have one symbol-carrying spaceship left, the one in the east.

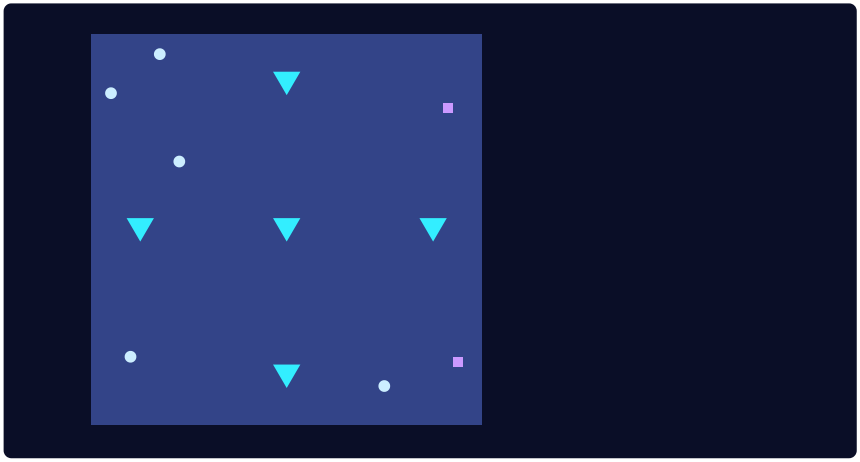
Zei sighed. He realized that his last symbol-carrying spaceship was the last mobile structure he had that Gun knew nothing about. His best advantage might be keeping it that way. On his next intervention turn, he steered that spaceship to temporarily stop, just outside of visual range of Gun's northeastern base. He set the spaceship to fire off gliders to destroy the symbols that were undoubtedly in Gun's northeastern base. To avoid giving up his location, they would not fly north directly - instead, they would reflect off of the northern rock shrine.

Meanwhile, in the southeast, Zei's own symbols began to disappear one by one - first, from Gun's eternal glider factories, then from new

gliders fired by his symbol-carrying spaceship now moving eastward from the south.

And just before the next intervention turn, Zei briefly saw Gun's symbol-carrying spaceship come into view near his southwest base - just before it sent out gliders that destroyed it.

Then the fourth intervention turn came.



Gun had a large advantage in material, Zei knew. Zei's only advantage was staying hidden. There were also gliders flying all over from the eternal glider factories, and the spot just south of the northeast corner was one of the more comfortable spots to be.

Zei decided he would continue to lay low. He stopped his symbol-carrying spaceship, and started on a five-intervention-turn-long quest to build another one.

Meanwhile, he remembered what he saw of the rock formations near the center just before his symbol-carrying spaceship there was destroyed, replicated the pattern in his sandbox, and figured out a strategy that would spawn an eternal glider factory between the center and north shrines to wreak havoc on Gun's northwest.

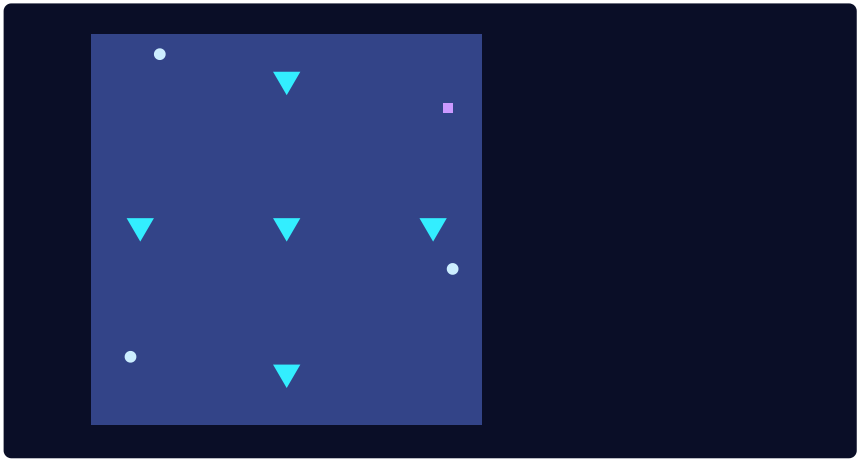
By the fifth intervention turn, his attack succeeded in heavily damaging Gun's bases in the northwest - though Zei had no way to

know how much damage was done. Zei continued his quest to build another symbol-carrying ship.

Shortly after that, Zei's last symbol in the southeast got destroyed by a storm of gliders. The last thing he could see in the southeast was Gun's symbol-carrying spaceship starting to move northwards.

Zei sent gliders to create another eternal glider factory, this time flooding the space along the center of the west.

He did not move his own spaceship. He could not: for the next three thousand turns he was a sitting duck, as he was building his new spaceship.



The sixth intervention turn came, right when Gun's symbol-carrying ship was getting hit by a barrage of gliders. This gave Gun a lucky chance to protect it from gliders, allowing it to survive a little bit longer.

The spaceship finally did get destroyed, but only after it had gotten within visual range of Zei's own.

Zei calculated: Gun had been able to see Zei's ship, but not the new ship that Zei was building.

Zei decided to do the only thing that he could do: he kept building, while putting up extra walls along the west and the diagonal. He

judiciously ignored his south: the existing glider storm, and the debris from Gun's ship, would itself be enough protection.

The seventh intervention turn came. Zei put all resources into building the new ship. Almost finished. The next turn would be just enough to finish it, and to give a steering instruction to both.

Then, a glider storm came toward Zei's ship from all directions.

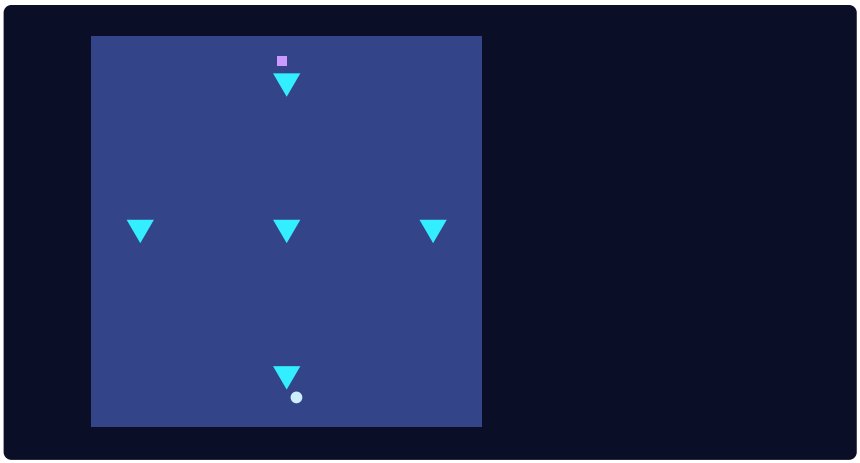
At first, the walls handled it. But then, the gliders kept coming, and soon they began to breach the walls.

The eighth intervention turn came just in time. Zei's last ship was not quite yet destroyed, but it was almost there. He could now finish his new ship, and give his existing ship one last instruction.

Zei programmed both to head west. He figured that the existing ship's symbol would get destroyed within a hundred turns, but the wreck could still coast west, and crash into the northern shrine and release a large glider storm southward. Meanwhile, the new ship would *also* head west, but stop just north of the northern shrine. This would give it some protection from the glider storms that were getting more and more intense.

When the ninth intervention turn came, Zei used a nearby rock formation, which was almost destroyed by the gliders, to bounce a glider that would then head southward and re-adjust the eternal glider factories that were attacking to the south. He did not have time to calculate what the adjustment would look like, and he did not have any way to know what its exact consequences would be, but he hoped that it would make previously safe territory unsafe. He also launched one more glider attack westward.

The tenth intervention turn came. Nothing seemed to happen, except debris. More and more debris.



Zei reinforced the eastern and western walls beside his last symbol-carrying spaceship.

He launched a few gliders to try to create more eternal glider factories, that would hopefully create more chaos southward. One of the gliders was destroyed, hit by another stray glider before it even left Zei's visual field.

The eleventh intervention turn came. Zei figured out one last trick. He figured out how to bounce a glider southward off of one of his own walls so that it would hug the edge of the shrine. From his simulations in the sandbox, he saw that if it hit the corner of the middle shrine, it would bounce off back north. The glider would bounce back and forth between the center and the north. Each time it hit the center, it would simultaneously release another glider that would go west. And when *that* glider hit the western shrine, it would bounce off and go southeast, hopefully causing damage to whatever was in the south. He did this, and simultaneously set up a mirror image of the same strategy toward the east.

The twelfth intervention turn came. By this point, the debris and stray gliders in the game field were so intense that offense was impossible. Zei simply kept building walls. Hunker down, play defense, outlast.

The thirteenth intervention turn came. Zei had built four layers of walls around this ship, and finally the walls were beginning to crumble faster than he could create new ones.

Finally, a few hundred turns after that, an alarm buzzed.

"And ... Zei Leimin is the winner!"

Zei let out a sigh of relief.

* * *

Zei sat in the waiting room, watching the game between Deluin and Kau. From the beginning, Zei was surprised, and impressed, by Deluin's strategy. Deluin spent all his attention early on to use the resources in the rock formations near his north and northeast to build walls. Simultaneously, he sent out gliders to try to create eternal glider factories pointing southward.

Deluin did not build any mobile symbol-carrying spaceships. Instead, he simply fortified. Kau's eternal glider factories proved to be of limited use: though the gliders would keep coming forever, they would only have any effect for a short period of time. As soon as a storm of gliders came, Deluin would figure out how to reflect it.

Zei was impressed. Looks like Deluin had also managed to figure out that he needed to be more patient.

Seven thousand turns later, Kau was reduced to hiding behind the southern shrine. But Kau barely survived three thousand turns more: the number of stray gliders and debris kept increasing, and Deluin's walls made him far more protected than Kau was.

Ten thousand two hundred turns into the game, Deluin was announced the winner.

Deluin stepped out from the elevator.

"That was a really impressive job, Deluin."

"Thanks! I was impressed that the walls held out so well. Now I'm excited to see how you handled your game."

"Oh - my style was much uglier than yours. Though I also basically won by playing defense."

"Do you want to get tea later?"

"Sure!"

* * *

Deluin was sitting in the courtyard drinking a tea. As soon as Zei saw him, he started running to greet him.

"Deluin, good to see you!"

Deluin waved back. Zei ran the rest of the way to his table and sat down.

A robot immediately slid up to their table.

"jie mo kai tu jie hei ja ma?"

A menu appeared.



"Whoa", Zei gasped.

"What is it?", Deluin asked.

"I think these are some of the new symbols they're A/B testing for Dzegoban version nine."

He immediately realized he was talking to a foreigner, and followed up:

"Have you heard of Dzegoban version nine?"

"Oh yeah I had a couple people mention it to me. Making the language visual. Bringing back hieroglyphs for the individual roots, but pronunciation perfectly readable from the glyph. And also another layer where they're trying to make the grammar look more like math."

"Yeah, so I'm trying to parse this. The cup is water - 'ja'. I see two unusual things about the cup. More straight horizontal lines than usual, and the three diagonal lines on the right. Maybe those are the pronunciation markers? One for the consonant, one for the vowel?"

"Top right is 'kai'. The unusual things I see are ... the extra vertical vein on the left, and then the transition to veins being more and more squiggly on the right? Although ... I'm not sure if I would make the images so detailed, when they often have different meanings depending on context."

"Wait, you know Dzegoban?"

"Zei, the whole point of Dzegoban is that there's only 256 roots, you can learn it in less than a month, and you don't get all the nuances but you can make your way around just fine.", Deluin replied in a lecturing tone.

"I'm impressed, you're almost a local already."

"Well, wanna come to Redshire with me tomorrow? I have to fly back for my mom and dad's birthdays, they're only a few days apart, and get some last-minute tutoring from our top players. And if you come, I can tell them that it's okay to skip the tutoring if you're doing the same!"

"I'd love to see Redshire, though unfortunately I just don't have the money for that. Maybe in a few years."

"But I'm super curious to see what Redshire is like! Send me pictures."

"Oh I can show you a few now."

Deluin took out his hand device, and showed Zei some pictures of Redshire. He saw castles, pine forests, large pristine grasslands, red-colored mountains with their tops covered in snow, and one giant icy waterfall.

"Wow, that's beautiful."

"You know, when I was younger I thought it was a back-handed compliment, almost an insult, to praise someone's country by saying that the nature is beautiful. At the time I thought, if the nicest thing you can say about a country is something that the people did not even make themselves, rather something that the forces of orbital dynamics and biology and tectonics randomly handed them down for free, then that's actually pretty harsh on the people."

"But as I grew up I've come to realize, that's not true at all. Even beautiful nature takes so much human work to maintain. Human work to build the paths and roads so people can walk or drive there. Human work to maintain a political system that stops people from polluting it, cutting down the forests, putting their houses on the mountainsides and making them ugly, strip-mining the mountains - while still allowing housing to be built where people can enjoy the views, keeping that housing affordable, and maintaining industry. Human work to restore any damage that was done before we got wise enough, or maybe rich enough, to care about these things."

"I'll let you know as soon as I can afford to go."

"Hopefully that will be in fifteen days", he joked.

"Because I use my championship winnings to pay for you to go", Deluin playfully snapped back.

Chapter 15

Freetown, United Cities · 3724 FIREMOON 17

Jahn passed through the turnstile in the Freetown Economics Institute, and walked toward the restaurant.

It was still early morning, and Jahn had intentionally not yet looked at any of the news on that day - better to clear his mind, and enjoy whatever the lecture on the restaurant screen would end up being.

After a minute of walking, he arrived.

When he saw what was on the screen, his eyes froze with horror.

Special broadcast

ARCTIC EMPIRE INVADES AND ANNEXES REDSHIRE

Jahn began to listen intently to the reporter relaying the details of what happened.

At 15,500 this morning, a large number of Arctic drones and aircraft landed all across the major cities and military bases in Redshire, timed simultaneously with a massive land invasion from the north with thousands of vehicles coming out of never-before-seen underground tunnels.

The Redshire military was caught completely unprepared, and many suspect corruption at the highest ranks.

Major battles lasted less than two longhours, with very limited casualties. Almost all Redshire garrisons have by now

surrendered.

He saw about ten people sitting in the restaurant, including Utaku, who seemed just as shocked as Jahn was.

Three government aeroplanes containing President Albor, Prime Minister Celedil and most of the parliament attempted to take off. All three were shot down. Current indications suggest no survivors.

The president's son, Tafindel, is believed to be in Arctic captivity.

The reporter paused briefly, and then began speaking again.

JUST IN:

The last Redshire garrison in the south has agreed on a conditional surrender. The garrison staff, plus thirty other personnel including two senators, will be allowed to fly to Devanvil, in exchange for the garrison discontinuing all attempts to defend itself.

The three aeroplanes will leave in five minutes, and will be the only aeroplanes allowed to leave. Arctic authorities are not allowing any other vehicles to enter or leave Redshire. They have cut off all communications and are already actively setting up jamming equipment.

"What do we do now?", Jahn, by this point standing beside the table with Utaku, mumbled.

"Take the threat seriously", Utaku replied.

Before Jahn could look away, Utaku had already begun typing frantically on her hand device.

⁂

MELDAN, VERIDIA

Gladias and Seila stared at the screen with Febric and Hreda beside them. They had just woken up at sunrise, light coming in clean through the window, the trees on the footpath outside already throwing long morning shadows. The news was the first thing they saw.

Zven was still asleep, and Lily was still in her room.

Seila took out her hand device, and began frantically typing.

Message		To
Please tell me you're in Freetown and you're okay.	→	Jahn

Send

She clicked send.

A few ticks later:

From		Message	Time
Jahn	→	Don't worry, I am in Freetown.	32117

From		Message	Time
Jahn	→	We've somehow been taking defense seriously here, and Freetown is about as far from Redshire as you can get in the United Cities, except for Silverbeach and Mabuli.	32124

Seila sighed with relief.

Gladias saw Seila messaging Jahn, and put his hand on her shoulder to comfort her. He did not respond further.

His mind soon shifted to the rubrics. Now we're definitely not adjusting the hardware openness rubric next month, he thought. Maybe we'll even be increasing the tax brackets.

A moment after that, he flashed with anger toward Delwart.

Seila left the room to wake up Zven and get Lily. Within twenty minutes, the four of them were on their way to school, and Gladias and Seila were alone in the house.

Gladias went down to the basement and closed the door.

"What do we do now, Mov?", he whispered through his neck band.

"Well clearly Delwart was wrong about a lot of things. And next time we talk to him, we *really* have a lot of questions to ask. We need a straight answer, not excuses."

"Remember, it's not just him turning out to be so wrong on the politics. It's about what I learned a month and a half ago - the sheer level of organization of whatever it is that he's doing."

"Did you manage to track him down any more after that time?"

"Only two more times after that. First was a meeting with someone in a privacy robe, not sure who, and they were talking a bunch about Silverchat. Second was a meeting with a Keeper, I made out some conversation about Dreadknot. Nothing more since about a month ago, he either changed his location or he got much better at hiding."

"So when is the vote?"

"First one is in about twenty days."

"Delwart might make a move to try to 'verify' that you voted in the correct way. So remember the machinery that the Order set up to make sure he can't do that."

"Only votes signed with a valid key count", Gladias recited. "You generate your voting key by meeting at least 3 of 5 people from your Key Allocation Group and registering each blinded share of the voting key with them. Once that's done, the information of which keys are and are not valid is only decryptable by the obfuscated circuit under the conditions that it specifies."

"So neither Delwart nor anyone else - technically, not even I - will know which key, how many keys or if any key actually got successfully registered. Only the result of the vote gets decrypted. So Delwart can't tell which votes I submit are real or fake. I don't have the ability to prove that to him, even if I wanted to. For Delwart to be able to take over my votes, he would have to take over my identity, which means going to the KAG and *pretending to be me in person* - difficult because the whole point of the KAG is that it includes people I've met in person."

"Humans - the ultimate trusted hardware."

"You've studied well."

"But I *hope* that this all won't be necessary, and Delwart is going to give me some sensible explanation for all this."

"Hope is not a strategy."

"Indeed it's not."

⁂

PAFOGAI DU, DZEGO

Zei anxiously tapped buttons on his hand device, and sent off a message for a fifth time.

tie		dun
Are you okay????! Please tell me where you are.	→	Deluin

be

He sat crying at a table in the lobby restaurant of his hotel. There was a tea and a rice and vegetable dish on the table in front of him. He had drunk the tea quickly; the robot had already refilled it four times. The rice and vegetables were uneaten for almost a longhour.

Finally, he got a message in response.

zan		tie	tei
Deluin	→	Don't worry, I am alive. Though it looks like I'm not going to be getting back to Dzego anytime soon.	84690

It was not the outcome Zei wanted, but still much better than the outcome he feared. He breathed a sigh of relief.

zan		tie	tei
Deluin	→	I found a way to send messages out. Some Arctic soldier and his friend, or his AI, or something, coded up a way to embed hidden text messages inside real-time health data that he is sending back home to command. And someone else in command is relaying that signal into a weather feed, which is then going up to the satellites. He's making a whole bunch of zipcoins anonymously rebroadcasting messages for people.	84714

Zei wiped off his tears, and started typing intently.

tie		dun
Wow. Are you ... in touch with this soldier?	→	Deluin

be

zan		tie	tei
Deluin	→	I have an anonymous link.	84723

tie		dun
Ask him how he's encoding the messages that he's putting into his health data.	→	Deluin

be

Zei thought for a moment, and realized that perhaps the time had arrived for him to become the instructor. He took out a keyboard and started typing furiously.

tie		dun
Or actually, on second thought, he won't answer. But tell him that randomly flipping the least-significant bits, the strategy that everyone always reaches for when they encounter this problem for the first time, is not the ideal solution. Sometimes higher-order bits are totally fine to flip, and sometimes flipping even a least-significant bit is dangerous. What you want to do is: * Train a language model on the actual distribution, so that conditional on what all the other bits are, it can give you probabilities for whether a given bit should be 0 or 1 * Split the data up into buckets - make sure to use the decryption key itself to choose the buckets randomly - and require the bits in the i'th bucket to XOR to the i'th bit in the ciphertext. Within each bucket, use the model to determine which bit's probabilities are closest to 50/50 - meaning both values are "natural" for that bit to have in that context - and flip that bit to make the bucket XOR correct. So you're letting the LLM decide which bits are "least significant" and okay to flip, instead of just always taking the lowest-order binary digit in each number.	→	Deluin

be

Zei sat back briefly, proud that he had managed to understand and explain not just a complicated concept, but *the exact complicated concept that was needed for this exact job*.

He decided to keep going, out of a feeling of personal pride in his understanding as much as out of a desire to let more people in Redshire sneak information past the Arctic censors.

tie		dun
If you want to go even further, you can make the buckets partially overlap. Say: * the first bucket is 0...199 * the second is 20...219 * the third is 40...239 and so on, though remember the whole order should be a pseudorandom permutation. Also, choose only a random half of the positions in each bucket to go into the XOR, so there are always bits to flip that affect one bucket but not the bucket before or after it. So say, 204 might be part of the second bucket _but not_ the third. The happy-path algorithm is: * choose a bit to flip in 0...199 to make the first bucket XOR correctly * choose a bit to flip in 200...219 to make the second bucket XOR correctly * choose a bit to flip in 220...239 to make the third bucket XOR correctly and so on... But if, let's say, 240...259 gives you 20 bits in a row where _none of them_ are reasonable to flip - think, a very clear picture of a night sky, so you get 20 pixels in a row that are really supposed to be exactly black - or the equivalent of that for health data - then you walk back and try to find a combination of bits to flip in 220...239 that makes both of the last two buckets XOR correctly, and if that fails, you walk back further and try to find a combination of bits to flip in 200...239 to make the last three buckets XOR correctly, and so on.	→	Deluin

be

He then realized that his last message might have come off too much as him trying to be a hotshot, so he sent a followup.

tie		dun
I learned that from Den's cryptography textbooks.	→	Deluin

be

zan		tie	tei
Deluin	→	I'll forward that to him. But I suspect that for him, first priority is going to be finding higher-bandwidth channels to hide data inside. Optimizing which bits to flip is really smart but it's maybe a 2x gain, finding a better data stream might be a 1000x gain. Always take the dumb 1000x gain before the super-clever 2x gain.	85190

zan		tie	tei
Deluin	→	I learned that in business school.	85200

Zei felt more optimistic about how much of an improvement a better steganographic encoding would be, but decided not to press the

point further.

tie		dun
Alright, well stay safe. And let me know if there's anything that I can do, or that smart people in my network could do, to help you get out of there safely.	→	Deluin

be

Zei thought about saying "or that the Kungaupei could do", or even "or that advanced Dzegojan secret societies could do", instead. But he realized that first, it would be a breach of confidentiality to let an outsider, especially an outsider now in hostile occupied territory, know about the existence of those things, and second, he couldn't fully trust the connection - he hadn't even asked Deluin a security question yet. And so "smart people in my network" it was.

zan		tie	tei
Deluin	→	Will do. Thank you for caring about me.	84821

Zei put down his hand device, and breathed out. His eyes briefly wandered around the restaurant, and he saw that by this point there was no one else there. He finally began eating his rice and vegetable dish.

* * *

Zei felt someone put their hands on his shoulder.

Zei turned around. It was Su.

"So sorry to hear about Deluin."

"I assume the Kungaupei are figuring out how to get him out?"

Zei realized that technically he had just committed an error: he had not actually confirmed that Su knew or was supposed to know about the Kungaupei - so far that was just Den and Mu.

"That was the first thing I asked Den", Su began to reply. Zei gave out a barely audible sigh of relief. "So far they've acquired a bunch of internet links into Redshire, and they got a message from Deluin a minute ago, so at least he's safe for now. We'll let you know if there's any kind of rescue plan possible."

Zei nodded silently.

"The finals are cancelled", Su added.

"Cancelled? Not postponed? What if he comes back tomorrow?"

"Even if he does, at this point it's just not right to have more games. Also we don't want to give the audience false hope, and we don't want to distract people."

"You're the champion this year. And if Deluin does come back, we'll give him the same trophy and the same prize money that we're giving you."

"No...", Zei pleaded. "I don't want to win like this."

"I know it's already late today, but you should go talk to Den. He's in Sadzu Du, in the southern mountain pyramid right now. He might want you here, or maybe in Pafogai. He will know better than me."

Zei nodded. Of course this was all more important than the game, he understood. He hoped that Den would put him in Pafogai, so that he could see Fin and his mom again.

* * *

Zei's car arrived at an entrance to the mountain pyramid that he had not been at before. This was a series of doors on the side of the mountain pyramid, much nearer to ground level than the usual entrance, though further from the city. The car stopped, its door lined

up perfectly with one of the mountain pyramid doors. The car door and the mountain pyramid door slid open simultaneously.

Zei stepped out, and walked down the hallway.

A hundred meters and a left and right turn later, he arrived at a door.

He put his watch against a green circle. The door opened.

Zei saw a long room, with a table at the end. Den was sitting on the other side, alone.

Zei sat down.

"Thank you for coming, Zei."

Zei nodded silently.

"I think you're one of the very few people who is in a position to help Dzego get through this next year."

Zei's eyes opened wide. He was now the national Minpentai champion, and he was getting far in his studies from Den's cryptography books, yes, but ... he still did not know how that made him valuable at this exact moment.

"In your last Minpentai game, you did two things that were crucial to winning. Do you remember what they are?"

"I can think of a few answers, but I'm not sure if any of them are what you had in mind."

"First, you hunkered down and played defense, and outlasted your opponent."

"Second, you made the most out of a very limited-bandwidth information channel, by pre-publishing most of the data - the symbol-carrying spaceship - and then using only a few bits of manipulation at the end to deploy the most important parts of your strategy."

"Actually a very similar tactic to what you used in your other game months earlier, when you sent out a spaceship that encoded basically every possible glider attack strategy, and then another glider to choose which specific one you wanted to use."

"How is this related?"

"We have contact with Deluin. Do you know how we accomplished that?"

"Actually, he got in touch with me and he told me. Some Arctic soldier is charging people zipcoins to hide their messages in his health data."

"Correct. And so everyone is dropping down to the most efficient, lowest-bandwidth form of communication they can use - raw text - and pushing all that through him. One soldier's health data, communications for thousands of people."

"A quarter longhour ago the rates he was charging suddenly dropped by two thirds - we suspect he found some more efficient way to encode data. Interestingly enough neither he nor anyone else has found any better channel than health data so far."

"Okay, now I wonder if that was me. I told Deluin to tell the soldier to train an LLM on his own health data and then modify the health data that he is sending out in-flight by flipping the bits that the LLM says are safest to flip, and using XOR over buckets to encode the message so he has a lot of freedom in choosing which bit to flip. Much better for reducing the 'suspiciousness' per bit of hidden data encoded, so he could afford to hide more data inside the same cover traffic."

Den's eyes opened up.

"Well we have no way to know if he used that specific trick-"

Zei remembered to check his watch.

zan		tie	tei
Deluin	→	Arctic soldier says thanks, it helped a lot.	92115

"Looks like he did."

"See, you've helped the anti-Arctic resistance already."

"Look, Zei. What's happening now is, our hand has been forced. We can't just keep improving our capabilities secretly anymore. Dzegojan society has to come out of hiding."

"The shadow society has to become the society. The shadow government has to become the government. We have to sprint to deploy everything that we have figured out across all of Dzego - before the Arctics can make any moves on us."

"The part of this most relevant to you - the cryptographic infrastructure we need so that our coordination can keep functioning and we what needs to be secret can stay secret - is run out of Pafogai Du. So I want you to take the train back - Su can reschedule your ticket for tomorrow - and I'll message you about whom you should reach out to next."

"For now, maybe see if you can come up with any more ideas for that soldier to sell more packets. That seems to be even better cryptography training for you than my textbooks."

Chapter 16

Meldan, Veridia · 3724 FIREMOON 25

Gladias pulled open the barrier on the booth, walked in, and slid the barrier shut behind him. He sat down angrily on the bench across from Delwart.

"You have a lot of explaining to do."

Delwart looked downwards toward the table, with a sorry expression on his face.

"Look, Gladias, you were right, I was wrong. I let my hate for all this interventionism in Veridia's government and culture blind me to the fact that there are things government has always been for, and defense is one of them."

"That's not why I'm angry."

Delwart looked up, startled.

"I mean, it's one of the reasons, but far from the primary."

"What is the primary then?"

"What the hell is going on? Who are you? Who are you working with?"

Delwart sighed. For a few moments he was silent. Then he leaned forward.

"Fine, I will explain."

"I'm not just one ideologue who is doing this alone. I, and a group of others - I don't know them - are working for Silverchat."

"We were frustrated by these rubrics that were increasing our taxes just because a bunch of people were putting crude jokes on our platform. We wanted Silverchat to be a refuge from the blandness

that's taking over so much of the rest of Veridia. We wanted to be a place where Hydrafill and even Dreadknot could be welcome."

"And so we decided to take matters into our own hands. Make an organized effort to try to pull the politics in a better direction. Appeal to the Keepers directly."

"And now I realize just how naive this all was. If we manage to ward off blandness, but at the price of the Arctic Empire being able to take over and inflict things far worse - first on Redshire and later on the whole world - then this whole thing was not just not worth it - it was ruinous."

Gladias sighed. He was still extremely angry at Delwart, but the anger was now mixed with a dose of sympathy.

A robot slid up to them, carrying a salad, a rectangular nutrition bar and water for the two of them to eat and drink.

Gladias and Delwart both first picked up their plates of food, and put them down on the table. Gladias then picked up his cup and drank about half the water in one gulp. Delwart did the same.

They began eating, silently. After a few minutes, the conversation resumed.

"What do you want me to do now?", Gladias asked.

"Obviously don't make the rubric exclude military now. You could even increase the tax brackets, though you doing that would only help a tiny bit - the most important levers there are on the Graph Funding side."

"But I also have another idea for what you could do."

"What is it?"

"I think you should go to Dzego."

"Me? Dzego? Why?"

"They clearly have this whole Arctic question figured out much better than we do in Veridia. Study them. Learn from them. See what you can bring back to Veridia."

"And why should I listen to you when you've been so untrustworthy?"

"Well definitely don't just listen to me. Ask your closest friends and advisors. See what they think."

* * *

Gladias walked along the same path through Kalimar that he would usually pass through. As usual, it was covered with trees and surrounded by stone houses on both sides. But on that day, it was unusually hot, and Gladias sweated slightly as some of the sunlight managed to shine through the tree cover and warm up his entire body inside of his privacy robe.

He passed by one of the houses that he remembered for selling traditional Veridian food products. This time, an additional note was hastily taped on to the poster, advertising packages that offered tasty, nutritionally complete and long-lasting preserved food that was sufficient for thirty days of survival.

He tapped his watch, and called Mov.

"So I had my chat with Delwart."

"How did it go?"

"You're right. There was an organized effort. Turns out - at least he now says - he's been working for Silverchat this whole time."

"Silverchat? And here I was expecting some more evil company, like Bluewhale."

"I guess it makes sense, probably that conversation I overheard a month and a half ago was with a Silverchat guy."

"So he wants me to go to Dzego now."

"Dzego? What for?"

"He says we in Veridia have a lot to learn from them. Do you think that's true?"

"You know, one of my friends mentioned something like that to me the day Redshire got taken over. He thought *I* should go, and see how their secret societies work or something."

"Well you know, Seila has a friend from there now, and the last thing she told me was that he's been climbing up the Minpentai ranks

really well, gotten into the nationals. So if I go, maybe I will actually be able to learn a few things."

"Yeah I can certainly see the appeal. Veridians really need to know the outside world better. Especially the people in the institutions - the Order, the Courts, the parliament, all of them."

"Give me a few days to think through this."

"Well, the votes on the rubrics are in eight, twelve and sixteen days, so you have that time to think through it no matter what happens."

Gladias paused for a moment.

At that moment, he was walking past the restaurant near his home where he would often stop for food and drinks. Through the window, he could see Febric eating.

"I'll think about it too", he finally said, and ended the call.

* * *

Gladias entered the restaurant, and almost immediately took off the face cover from his privacy robe, and partially unzipped the front. He could not be that private while eating anyway, and he had been sweating in the hot sun. He sat down beside Febric.

"Oh uncle Glad, happy to see you here again!"

"And I'm happy to see you!"

Gladias noticed that Febric had been watching the screen. A game of Helisport was actively broadcasting.

Gladias let himself watch the action for a minute. The screen was following the lead player wearing a yellow shirt chasing a redball.

Suddenly, out of nowhere, a player from the other team, wearing a purple shirt, whacked him off to the side. The player in the yellow shirt tumbled, and then automatically regained his bearings about five meters further down.

The screen shifted view to the player wearing the purple shirt. He caught a blueball in midair. Two other players wearing yellow shirts started flying towards him. Before they could get to him, he threw the blueball toward a teammate. That teammate successfully caught it, and began flying forward.

The screen shifted view toward ten yellow standing pins arranged in a triangle on a circular board mounted on a pole high above the ground. The player wearing the purple shirt holding the blueball came into view. A player wearing a yellow shirt zoomed into position between him and the pins.

The player wearing the purple shirt suddenly threw the blueball up high into the air, just as a different player wearing a yellow shirt crashed into him.

The blueball passed about five meters over the player wearing the yellow shirt protecting the pins, and then landed on the board. It knocked three of the pins down.

A scoreboard onscreen adjusted.

	Yellow	Purple
Redball captures	6	11
Greenball player hits	9	7
Blueball pin knockdowns	31	23
Combined score	1674	1771

Purple's blueball score had just increased from 20 to 23, and their combined score from 1540 to 1771 - enough to push them ahead of their opponents.

Suddenly, while the player who had just taken down those pins was celebrating, a greenball hit him in the back and splatted its paint over him.

The score adjusted again.

	Yellow	Purple
Redball captures	6	11
Greenball player hits	10	7
Blueball pin knockdowns	31	23
Combined score	1860	1771

"You have to admit, the scoring system is really good now", Febric exclaimed.

"I remember when I was much younger, they kept adjusting the ratios between what a redball gives you, versus a pin knockdown or a player hit. And they just couldn't get out of having a situation where the rational thing is to just focus all your attention on one."

"Even when they got it right, within a year the strategies adjusted, and it became lopsided again."

"I think making the score multiplicative was the only way out. That way you pretty much have to care about all three of the point types."

"Or try really hard to make sure your opponent can't get any of one type", Gladias interjected.

"Ah yeah that was the first thing a few people tried", Febric replied. "Holding onto redballs for more than thirty ticks had already been banned for ages, that had been a good cheat for whichever side was worse at redballs even back when the scores were additive. But what one team did try, is knock down a bunch of pins very early in the game, and then use most of their players to protect their pins for the rest of the game."

"Turns out it doesn't actually work. People on copters can only pack together so tightly, no matter what you do, the other team always manages to sneak a few blueballs through. And then if you do pack tightly in one place around the pins, you're a sitting duck for the greenballs."

"Are you thinking of playing?", Gladias asked.

"No, but I just had a bunch of conversations with Zven about this. The math is almost as fun as the actual game!"

Gladias and Febric both turned their attention to the food.

"How's Lily doing by the way?", Gladias asked.

"She actually started getting a little better over the last couple of months. She mentioned to me a few days ago that she got seventy on her latest civics test. She seems happier. And she's even more engaged playing that game on her hand device."

"Do you know what game?", Gladias asked.

"She doesn't show me either."

Gladias and Febric both returned to eating. A few minutes later, they were both finished.

"Did you remember to take out your zipcoins this time?", Febric asked playfully.

"Hey, I may make lots of mistakes, but I don't make the same mistake twice", Gladias snapped back.

"By the way, Febric, thanks for reminding me - I need to test my recovery procedure for my coins", he whispered through his neck band.

"Want me to help you sign a test operation?"

"Sure! What do you want the test to be?"

"I dunno, buy a Hydrafill bottle?"

"Buy a Hydrafill bottle? Straight out of a life savings wallet?", Gladias asked incredulously. "Those ads really have been getting to you too, haven't they?"

"Hey, you're the one who asked me!"

"Okay fine ... let me start the operation."

Gladias began whispering into his neck band. Out of the corner of his eye, he saw Febric run off somewhere, though he was too busy to wonder where.

A few ticks later, Febric's watch buzzed.

From		Message	Time
Gladias	→	[Wallet 0x8f62... social recovery mode transaction request]	65129

Febric opened up his hand device, and a more detailed view of the transaction appeared. He asked Emerald to search for the receiving address online. A few ticks later, Emerald replied with a confirmation that it was, in fact, the Hydrafill purchase address. He then checked the amount - 5.5 zipcoins. It seemed reasonable for a water bottle plus shipping, though the bulk of the cost was definitely the shipping. He checked the transaction data field. Yep, looked like an encoding of Gladias's home address, in the correct format according to the standard online shopping protocol.

In the cryptographic network that processed these transactions, all of this data was, of course, encrypted.

Febric clicked "Confirm".

Gladias's watch immediately buzzed.

Less than a minute later, Gladias's watch buzzed again.

From		Message	Time
Seila	→	Security question: what was the most unusual thing I did on the evening the kids came back from Vil and Daia's?	73105

Gladias immediately whispered into his neck band a reply.

Message		To
You unplugged the screen in the restaurant when it was showing Lord Ephelion's speech.	→	Seila

Send

A few moments later, Gladias's watch buzzed again. Three signatures done, including his own. One to go. Either Mov or Pelow - or, of course, if he really had to, himself running to the vault and picking up his second key.

"I'm obviously not gonna ask who that was, or who else you need to confirm it", Febric added. "But good luck!"

* * *

"Seila, I'm home!"

"Glad!"

"So I have interesting news."

"What is it?"

"There's some chance I'll be going to Dzego soon."

Seila stared.

"Why?"

"I talked to Mov about this. He thinks that Veridians, especially Keepers, the Order, the institutions generally, have a lot to learn from what's going on over there."

"I'm not sure about this ... it's a difficult time. You're not really the best person to make good use of a Dzego trip. And the Keepers would benefit from having a strong hand guide them these next few months."

"Arctic stuff is really much more of a military and parliament responsibility, maybe Graph Funding too. And ... maybe I can make good use of a trip if you introduce me to Zei?"

Seila retreated into her thoughts for a few moments.

"If Mov thinks it's a good idea, then fine. But please do your homework first. Find someone, anyone, in Graph Funding or in parliament to talk to, so you get some idea of whether this trip will be useful, and if you do go, how to make it as useful as possible."

"But really, think hard about this."

"If you go, then yes, of course I'll introduce you to Zei. And Bai too."

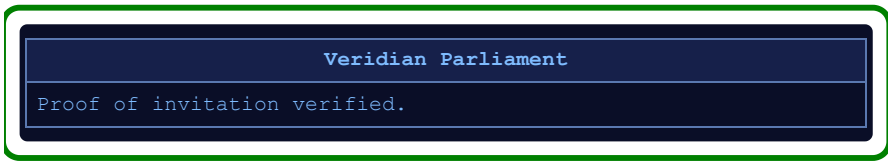
Chapter 17

Meldan, Veridia · 3724 FRUITMOON 11

Gladias walked forward along a path covered with trees on both sides, toward the large circular parliament building in front.

When he arrived at a side door of the building, he tapped his watch on the green circle.

The circle buzzed.



The door opened. Gladias walked in, and followed the instructions that were being relayed by his watch. Forward, then right turn, then forward, then third door on the left.

Gladias got to the door, and tapped his watch again.

The door opened.

Inside, he saw a man wearing a Veridian parliament robe.

"Gladias, thank you for coming!", the senator exclaimed.

"And thank you for seeing me, Senator Verdow!"

"Please sit down."

A robot slid into the room.

"jie mo kai tu jie hei ja ma?", the robot asked.

Gladias briefly stared in surprise.

Verdow laughed.

"I adjusted the settings to make it do that a few months ago. Just a little something as a reminder that other countries exist and that they matter."

Verdow tapped a few buttons on the robot's screen, and turned the screen toward Gladias.

The screen was in Dzegoban. Gladias's eyes wandered around with confusion.

"Oh, if you want water, that's 'ja'. And if you want tea, that's 'kai ja'. Literally 'plant water'."

Gladias clicked on 'kai ja'. If he's going to be in Dzego for a few months, might as well get acclimated and start drinking some tea - and knowing the local word for it. The robot beeped, signaling acceptance of the order, and began to quietly glide away from them. The room door briefly opened, and the robot exited the room.

"So you think I should go to Dzego?"

"You're obviously smart. Of course I have no idea what you say in your secret meetings, but judging from everything you said in the hearing, and from your original application to the Order years ago, I think you're a good and smart person. We need people like you to be more active in serving our country. Being a Keeper is nice, but come on, you're only one of over a thousand. I think getting and bringing back some first-hand experience from Dzego could be even more meaningful."

"If you think it's a good idea, I'm happy to. But who should I talk to there? That's what I wanted to get your input on."

Gladias intentionally omitted any mention of Zei - not out of any desire to hide that indirect connection, but because he wanted to get a fully independent answer from Verdow first.

"Well my strongest connections are with some of their authors and musicians", he said, gently laughing. "Though I also know a few people in the government there, especially the security and health ministry. I'll send you their contacts, and give them a heads up. I wouldn't stop with them, but I think they can help get you started."

"Thanks, that would help a lot!"

"I only know two people from there, indirectly; my wife met them at a chance meeting in the Economics Institute in Freetown. Zei and Bai, they were in Minpentai."

Verdow's eyes opened. "Honestly, you might have a better lead than I do. When you see the Minpentai games in Dzego, the players and the people who train them are clearly top-notch. If I had to find someone to defend me against the Arctics, I'd almost trust them over the actual Dzego army."

"How advanced are Zei and Bai? Did they get into any of the championships?"

"Last thing Seila told me, Zei won the semi-finals, and then they cancelled the finals when Redshire was invaded."

"The semi-finals, like municipal level?"

"National."

Verdow stared with a mix of surprise and joy.

"Your connections are definitely better than mine then. When you come back, you'll have to tell me more about Dzego!"

"Anyway, I'll introduce you to the other folks I mentioned so you can see them when you come. It's always good to get more than one perspective."

Gladias nodded appreciatively.

"Oh, by the way, speaking of getting more perspectives, one of the Graph Funding stewards is here about to visit me. You want to talk to him too?"

"About what?"

"Oh, his work, how he's feeling about Graph Funding, and of course Dzego."

"Sure, why not?"

* * *

A few minutes later, the door opened, and another man, wearing a Veridian privacy robe but without the face cover and hood, walked in.

"Belgor, good to see you again!"

"Good to see you too, Verdow!"

"And meet Gladias, he's a Keeper."

"Nice to meet you, Belgor."

"And nice to meet you, Gladias!"

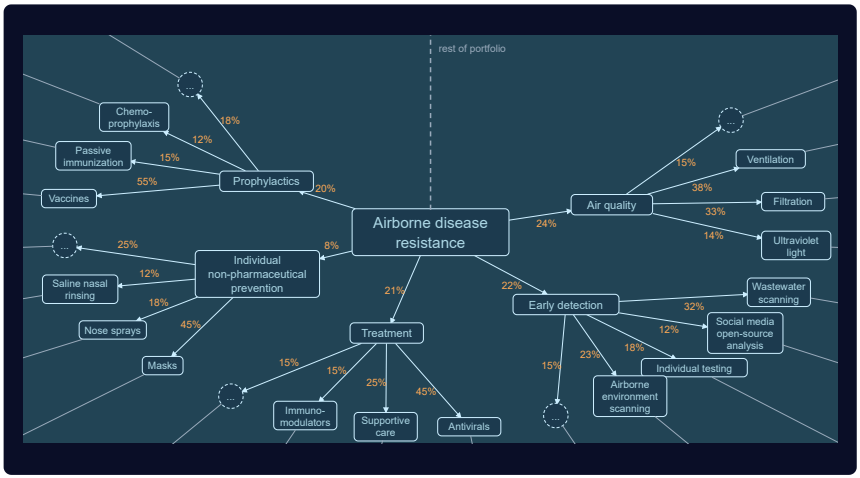
"So Gladias is thinking of going to Dzego soon. We thought that maybe you should tell him more about Graph Funding, and any advice on what he should do or learn in Dzego."

"Sure, happy to. How familiar are you with how Graph Funding works?"

"Honestly, not very."

"Great."

He took out his hand device, and pressed some buttons. A diagram opened up on the screen, and he showed it to Gladias.



"This is the subgraph for airborne disease resistance. For each node on the graph, the cryptographic network randomly selects a committee, that makes the decision of what weights the edges right below that node get."

"So it's secure in the same way the Courts and the Order of Steering are. Having something like that for funding is especially important now, given that the Arctics just did the same attack on Quadratic Funding that Bluewhale did but at ten times the scale - and of course

any more traditional style of allocation, they would try to interfere with as well."

"Oh you're definitely right. Even the Keepers are getting attacked lately, though with all the secrecy and internal communication barriers, at least for now the system is holding."

"Oh is that so? Good to hear. Though I am sad that we don't have a good way to make both of these things more democratic. Listening to citizens' assemblies is good and Heralds are good, but they're not a substitute for actual public accountability."

"Anyway, one difference between the Graph Funding design and how the Courts and the Order work is that Graph Funding allows even more specialization. After all, funding allocation is a pretty technical task."

"The goal is to try to split up each decision of funding allocation into more manageable pieces. Don't ask 'what percent of all Veridian public funding should go to this study on UVC lamps'. Ask, what percent should go to health, then within health, what percent should go to airborne diseases, and so on."

"And if one project benefits two areas: say, some lamp that's somehow useful both for UVC lamps and for defense, then the graph stops being a tree, it becomes a proper directed acyclic graph. One node can have multiple parents, and the funding you get along all paths adds up."

"So a committee is responsible for allocating the funding going to each node among its direct children?"

"Yes."

"Can they remove children or make new ones?"

"Yes, of course. That's how the graph gets made in the first place."

"This is fascinating."

"I mean, this all hinges completely on how effective these committees are, right? The more you can specialize, the more it might get captured by special interests, and the more you randomize, the more you get decisions made by people who know nothing about what they're deciding?"

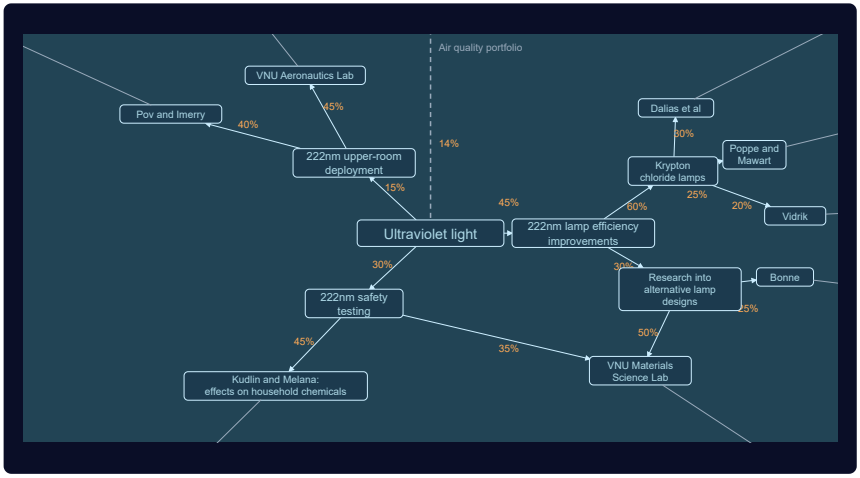
"Of course, that's always the tradeoff."

"There's also another tradeoff. It turns out that it's fairly easy to get committees up to the point where they can tell bad projects from good ones, but very hard for them to tell good projects from great ones. So for the lower levels of the graph, there's been a movement lately to just randomize at the top end. If the committee says you're in the top ten percent, you have the same chance of getting in, no matter if you're at ninety one or ninety nine."

"So the further someone is above ninety, the more they basically don't have to worry about optimizing for the committee at all, they can just do what feels right."

"Exactly. Sometimes you don't actually want accountability, you want to give space for people's intrinsic motivation. The people at the very top may even be smarter than the committee. That's what randomize-above-cutoff does: accountability for those who won't succeed without it, and freedom from incentives for the best - they can convince the committee that their work is good enough with barely any effort at all."

"Anyway, here's what it looks like zoomed in at the level of individual projects."



"Is it wise to have these randomly-selected Graph Funding committees making decisions at that level of granularity?"

"I'm not sure myself. The nice thing about this design is that it can always evolve without changing the structure. Committees could just decide to give the VNU Materials Science Lab 'ownership' of an entire category, and let them allocate internally. Actually, for the open source social media analysis, we basically did just that, eighty percent is just going to Silverchat."

"Bluewhale is unhappy, but, hey, their choice to be so closed and proprietary with everything."

"So how does this all connect to Dzego?"

"Ah, well a lot of this tech does actually make its way over to Dzego, they're starting to take the air filters and ultraviolet lights more seriously - a few of them are getting paranoid that the Arctics might hit them with a deliberately-released pandemic at some point."

"Is there any chance that the Arctics could hit them with a pandemic that won't eventually bounce right back to them?"

"Well, if Arctic anti-pandemic technology shoots far enough ahead of the rest of the world, then yeah, they could release a pandemic that only they can effectively protect only their own people from. That's a big part of why it's so valuable to open-source all this stuff - at least the defensive things. Put the whole world on a more level playing field, so no one is in a position to light a fire and be the only ones with fireproof suits."

"Anyway, if you end up going to Dzego, do tell me what you learn about their public health when you get back - I am curious!"

* * *

Gladias opened the door to his house, and entered.

Seila and the kids were already eating dinner.

"Today is your first vote, isn't it?", Seila asked.

"It is, glad you remembered!"

"So are you leaving again this evening?", Zven asked.

"Oh no, the in-person part was the discussions months ago, and the key registration, I did that right after my meetings at the Parliament."

"There were also some discussions happening more recently in our Keeper discussion group online, though that chat automatically cut off seven days before the vote - have a quiet period to make more space for independent thought, the argument goes."

"Oof, so the cutoff was after Redshire got annexed. What was that like?"

"I think everyone who said we should cut military hardware out of the best tiers in the rubric went quiet. Eighteen then proposed a new amendment that increases the tax brackets, pushing up the incentives to open things up even more. I'll probably vote for that."

"Glad to hear."

"How did your Parliament meetings go?"

"Very well. The Senator and the Graph Funding person both seemed quite supportive of me going. Though they said that Zei might be an even better contact to help me with my visit than anyone they knew!"

"That actually makes sense, everyone knows the official Dzego government is not where any of the interesting action is happening."

"I guess I'll talk to Zei and put you two in touch then!"

Gladias, Seila and the kids took a few bites of their food. Soon, the dinner conversation resumed.

"Febric, Hreda, want to share your news?"

"We're going on an exchange trip to Northglade for the winter term!", Hreda exclaimed proudly.

"Ooh, that's pretty far northeast, almost at the Arctic border", Gladias replied.

"It's fun, we'll get to ski off mountains wearing a copter and then glide back down!", Hreda exclaimed. "And it's not like it's right at the border, Northglade is over two hundred kilometers inland. It will be my first time there!"

"So what are you guys doing there?", Gladias asked.

"Veridia's best math university is there," Hreda replied. "I'm in the top five percent in math in my class, so I got accepted to attend some special education there."

"Same for me," Febric added. "Though I'm in computer science."

"Didn't Lily also want to go to Northglade?"

Lily was in the washroom, so she could not answer.

"Vil said that he will take her in the summer if she gets to seventy-five percent in her exams."

"She prefers summer, she doesn't like the cold as much."

Gladias's hand device buzzed.

"Well, voting on the hardware openness rubric started."

Rubric: openness in hardware

* * *

Motion	Vote
Add after line 3 in Tier 1 and line 3 in Tier 2: "Products in military, surveillance and counter-surveillance applications are not eligible for this tier."	Yes No
Increase tax brackets from 0 / 6 / 12 / 18 / 24 to 0 / 10 / 20 / 26 / 32	Yes No
Change line 4 in Tier 4 and line 4 in Tier 5, from: "Use of hardware means to prevent third-party repair." To: "Use of hardware or software means to prevent third-party repair."	Yes No

This felt anti-climactic, Gladias thought. So much ceremony, in-person meetings, debate and work in the months leading up to the vote. And then the vote itself was just ... tapping buttons on a screen.

It was reasonable, Gladias supposed. The way that the cryptography worked, as long as key registration was secure, this part could just be done on his own hardware at home. And even if Delwart did come knocking and demand to stare at him from behind his shoulder for the entire fifteen-longhour voting window, he had no way to prove to him that the key on his device even was the one registered -

as opposed to being a distraction while the real voting was done by Mov, Seila or even Zven.

Gladias took one last look at the screen, and nonchalantly pressed: `No`, `Yes`, `Yes`.

Chapter 18

Pafogai Du, Dzego · 3724 FRUITMOON 18

Gladias exited the air harbor, stepping through a sealed opening that opened immediately into a large black car.

Zei and Bai were both already sitting inside, Zei on a backward-facing seat and Bai facing him on a forward-facing seat.

"You must be Zei and Bai."

"We are indeed!", Bai replied.

"Well, I'm glad to finally meet."

"As are we."

Gladias sat down to the left of Zei.

The door closed, and the car began to accelerate.

Gladias stared to the right, looking past Zei to the right-side window of the car.

The car exited the air harbor, and soon began passing by a series of stores.

Gladias immediately noticed that something was off: the view on the window shifted as the car moved, but it did not respond to Gladias himself shifting his body position.

"Is that a real window?", Gladias asked.

"It can be", Zei replied. "But we have heightened security protocols now, and the black cars changed their settings. Can't let light pass through windows directly, because that would be bi-directional even if it was dimmed - a good camera could see that we're inside. So it's a simulated window now, picking up the view with cameras."

"It actually works well when there's one person, because it tracks your eyes too so it adjusts to look more natural. But now there's two of us, and it's picking me."

"Huh."

Gladias looked through the left window instead. He adjusted his body a few times to test it. The colors and the depth seemed somewhat unnatural, but otherwise it looked surprisingly normal. He continued looking, paying close attention to everything that he saw outside.

Many of the stores had cartoon animals on the sides, with speech bubbles saying various things in Dzegoban that Gladias could not understand at all.

Gladias could briefly see the products inside each of the stores: hand devices, compute boxes, lamps, drones, sensors.

"MIN KUI", a sign on a store selling compute boxes read.

"GUI SAU", a sign on a store selling lamps, most of them in the shapes of long cylinders, read.

But to his surprise, only a few of the stores seemed to be open.

The buildings were all one or two storeys tall, with many stairways leading underground.

There were people outside, though fewer than Gladias was used to seeing in Veridia - despite the visibly greater density of Pafogai Du compared to his own hometown.

"hun fe bin jin zo de jan li gi", Zei wondered aloud.

"Huh?", Gladias replied.

"Ah, sorry. So many people wearing neck bands, I was just saying."

Gladias also noticed a large number of identical black cars on the roads.

"Does everyone in Pafogai Du use this kind of black car? Is this a privacy thing?", he asked.

"Now that I think about it, no. We've had them for a while, but having this many people use them is quite new. Just a few days ago, there were barely half as many."

They continued riding silently.

A few minutes later, the car descended into a tunnel. Midway through, it slowed and then came to a stop. Gladias could feel a gentle bump, some mechanical noises, and what seemed to be the car shifting in all directions: up, sideways and then down. He looked out through the window, but could see nothing. And about half a minute later, the procedure ended, and the car appeared to be accelerating forward again.

"Did we just ... swap cars?"

"Swapped cabins. The system waits for ten cars to come, swaps the cabins around randomly, and then everyone drives back out. No one watching the exits knows which one we're in. Like a mixnet, but for people."

Less than a minute later, they exited the tunnel. Soon, the mountain pyramid in the south of Pafogai Du came into view.

"That ... is an impressive structure", Gladias said.

"Heh, everyone says that. Actually we carved it out of a pre-existing mountain."

"Still impressive! Probably more secure that way too, if ninety or ninety-nine percent of the inside is solid rock."

"It's ninety-six point seven."

"So one over thirty open space? If a normal building is ten percent solid, that makes it equivalent in capacity to a normal building three times smaller in each dimension?"

"You got it!"

"Maybe you're right we should be impressed. It *is* the largest one of these in all of Dzego."

The car began to slow down, and soon it came to a stop. The car door opened, directly into the hotel.

"This is your hotel, Gladias!"

Gladias stood up, and stepped out of the car with his bag.

"So what's the plan?"

"Take a longhour to rest, we'll meet you back here and chat more."

Gladias walked into the hotel, and tapped his watch onto the nearest green circle.



His watch buzzed. He had his room key.

Gladias looked around. Zei was right, he realized. *Almost everyone* was already wearing neck bands here.

Gladias walked into the elevator. The elevator door closed. When it opened again, Gladias stepped out, and walked into his room.

* * *

Gladias stepped out of the front door of the hotel, and looked outside. He had only taken forty minutes to rest; he had another sixty left before Zei and Bai would be expecting him. He figured that he would go on a walk, and take a look around at least this one piece of Pafogai Du himself.

He turned around. He saw the hotel address: 1565 Tan Kai.

He turned right, and began walking.

Almost immediately, his watch began flashing warnings at him, "unknown" counts in the double digits. Figuring that this was just his Veridian watch not being up to date on Dzego electronics, he decided to ignore the warnings, and continued his walk.

He crossed the street. He saw a black car pass by him, and then almost immediately slow to a stop. A man walked out from a nearby shop, holding a large box. He put the large box down on the ground beside the car.

The door of the car opened, and the driver stepped outside, putting her own large box down on the ground, identical in size. She proceeded to open the top of the box.

Looking at the box from a few meters away, Gladias could see about thirty much smaller white boxes on the top row inside.

The driver took out a tray containing the top row, and put it on the ground, on a staircase. The man from the store took out a hand device, and seemed to scan the two-dimensional codes on all the white boxes simultaneously.

The driver then took out a tray containing the second topmost row, and put it on top of the first tray. The man from the store scanned again.

This repeated five more times. Then, the driver opened up the man's large box, and scanned it row by row as well.

After scanning each row, she put the tray into her own box. And finally, she took the box back into her car and drove away.

"What's going on?", Gladias asked.

"Shipping", the man replied. "High security."

The man finished putting the trays on the staircase into his own box, picked up the box, hurried back inside his shop and locked the door.

A mixnet for physical objects, Gladias realized - Zei had told him about it once. Each object would pass through several carriers, and no single person knew both the source and the destination. There was no way to sabotage one person's device without sabotaging *everyone's* device - and if you did that, you would quickly get caught.

Gladias kept walking. He turned right on 17th street, and could see Pafogai Du's giant mountain pyramid in front of him, and the pathway leading up to it starting after only two more streets.

On this street, Gladias could see far more shops that were open. Almost every shop had a sign above the window advertising it, about half the time polished and professional and the other half the time improvised and hand-made from neon lights. He saw a flower shop, still bustling with customers. About twenty meters after that, a clothing shop. The posters on the outside still advertised jackets, but looking through the window inside, he could see that about half the space was now taken up by hastily-added privacy robes. About twenty

meters after that, an unmarked staircase leading to a basement, with two teenage boys and one girl climbing down.

At the top of the staircase, Gladias saw a sign. There were pictures of cute cartoon animals playing with electronic gadgets and chemistry lab equipment, with a backdrop of green trees and blue skies. In the middle of the sign, he saw a motto in Dzegoban:

pa jan li sun zo

zai gau fe pa jan

go li sun zo

bai gau fe go

He wondered what the words meant.

He made a mental note to himself: gotta actually learn those 256 Dzegoban roots now that I'm here.

About twenty meters after that, he saw another staircase, leading down to what seemed like an underground tea shop. The entrance and the staircase were decorated in a style that reminded Gladias of depictions he had seen as a child of Dzego's medieval era.

And soon after that, Gladias arrived at the next street: Zin Go. The sidewalk had had good tree cover, enough so that he had lost sight of the giant pyramid towering in front of him. Here, the street intersection temporarily interrupted the trees, and he could see the pyramid once again. He saw many people going up and down the inclined walkway leading up to the entrance. The majority of them were wearing privacy robes. He wondered what they were up to.

He crossed the street, and continued walking until he got to the next street: Dzai Gun. This was the last street before the hill began. He crossed the street again, and began to walk up the incline.

About five minutes later, Gladias was high enough that he was now towering over Pafogai Du, and could see the roofs of nearly all of the buildings. From the top, the city was a mix of concrete roofs and green

trees - much more green than Freetown, but still much less green than Veridia.

He noticed a large rock nearby. He sat down on it, and enjoyed the view.

A few minutes later, he began to walk back down.

He got to Dzai Gun street, and turned left, walking along the edge of the city until the westernmost point. He was pleased by the beautiful path covered with trees that seemed to surrounding the whole perimeter of the city. He saw many people walking and running along it. When he reached 0th street, on the west edge of the city, the path turned north.

He continued following the path, walking north two streets back to Tan Kai, and then crossed back into the city, heading east until he got back to his hotel.

* * *

Gladias entered the hotel, and immediately turned left to enter the lobby restaurant. Even though he was ten minutes early, he saw Zei and Bai already sitting down at a table, looking at a hand device. Bai looked up, and waved at him.

Gladias headed over to their table, and sat down.

"What are you looking at?", he asked.

Bai showed him the screen.

```
He took the path less traveled, and that blazed a trail
for all.

         jan jo san sia
   ci pa jan li san sia
           te bia pin hu fe lo shun
ci fi mu jan li san sia
       fe fi mu san sia
           te shun zo bia pin fe jan
```

"Do you know what it means?"

"Well, that first line is from a Veridian."

"Indeed. And the rest?"

"Not yet. I only know like five Dzegoban roots", Gladias laughed.

"Two percent of the language already", Bai replied.

"People should dream. If one person dreams, they will not fear pain. If a hundred thousand people dream a hundred thousand dreams, pain will fear them."

"It's an old poem."

"Not too different a message from ours."

"Indeed. Although you're blazing trails, and we're fighting here to survive."

Immediately after she finished speaking, Zei and Bai both turned to wave at someone else who was running up to the table.

Zei immediately recognized him, and slid out the remaining chair for him to sit down.

"Gladias, meet Fin."

"Nice to meet you, Fin. Seila mentioned all the good things that Zei had said about you."

"And nice to meet you too, Gladias. Anyone Zei brings is a friend to me."

"Well I'm just happy to finally get to see Dzego with my own eyes."

"So what's been up in your world, Fin?", Bai interjected. "I think this is the first time we've seen you in almost half a month."

"I'm really sorry I've been ignoring you guys for so long. But thank you for asking the question - because finally, I think I can give a worthy answer."

Gladias, Zei and Bai all turned to stare at Fin.

"You might have been wondering", Fin began, "why we spend so much effort on the Bansunpei. What's the point of optimizing Dzegoban, optimizing the measurement units, all of this, when there's so much else that Dzego desperately needs - education, defense, economic development, all of it."

"And for quite a few years, I've been wondering this myself. And it's not until my Bansunpei meeting the day Redshire was attacked that I finally got the answer."

"What is the answer?", Zei asked.

"Dzegoban is coordination training."

Zei and Bai both opened their mouths.

"Think about what we're doing each time we release a new iteration of Dzegoban."

"First, we A/B test many different versions of it. We motivate different people to try different things, and report back on their experiences."

"This ensures two things. First, it ensures that there's public feedback into the process, so the final product is informed by what behaviors actually work when you get real human beings to try to start doing them."

"Second, the public feedback, and the fact that the final outcome is a result of many people choosing between many different options across different dimensions, helps people accept the outcome. It takes no one by surprise. Anyone who disagrees with some part of it almost always knows someone who likes it more, and can at least hear their perspective."

"The Bansunpei intentionally seeks out members from all walks of life - you actually get *bonuses* for recruiting someone whom no one else in the organization knows."

"And so on the day Redshire was attacked, the Bansunpei digitally convened, and made an unprecedented decision. At least for the time being, they were putting Dzegoban version nine on hold. Releasing a few basic highlighting changes that had been agreed upon for over a year, but that's it."

Zei audibly sighed with relief.

Gladias could understand. The last month had been intense for everyone - hardly a good time to get everyone across the country to learn a whole new version of their language.

"Instead, they decided to use the exact same machinery, though with less public participation and more secrecy because of the unique circumstances and the time constraint, to coordinate a totally different set of changes across Dzego."

"Have you seen how many more people are wearing and having conversations through neck bands now? Or wearing privacy robes? Or going between places in black cars? Or buying products online, through the physical-delivery mixnets, instead of going to stores. And ten times more meaningful than any of that is what you don't see - more of the sensitive meetings are just going online, and everyone doing things on the internet is using much more secure tools too. Bluewhale Messenger has almost disappeared from Dzego overnight."

"None of that was happening three days ago. Instead, the Bansunpei just released its new privacy and security recommendations, through the channel where all the people who follow them were expecting news about Dzegoban version nine. About five percent of the population saw the recommendations at the same time. They were surprised, but they read and understood the reasoning, and respected the people who made the decisions. The majority of those people have already changed their behavior in at least one major way, and even that two and a half percent was already enough to start a trend that's now sweeping across all of Dzego."

Fin grew even more excited, and he continued.

"The Arctics talk a lot about a person's Intelligence Score, and how society should be run by those whose score is the highest. A few Veridian academics say that we should take Emotional Intelligence Scores seriously as well - though they always have a hard time getting people to actually do anything beyond just nodding along to that statement."

"But there's a third category that they are completely missing. The Coordination Score. A Coordination Score is not a property of a person - it's a property of an entire community. It's a community's ability to make collective decisions, to choose positive-sum, even if personally

risky, cooperation over zero-sum competition, and to jump out of bad equilibria into better ones."

"This is what the Arctics think can only be done with a strong leader that everyone bows down to. But we Dzegojan say NO. We say 'gie fe kiu kai ci bin hu' - grow roots, no head. And now we've grown our roots, and we Dzegojan, who have improved and perfected our language eight times with no top-down coercion and no head, are going to take on the Arctics with no head. And we are going to WIN."

Chapter 19

Pafogai Du, Dzego · 3724 GRAPETIME 15

Gladias and Zei were sitting at a table in a cafeteria, deep inside the mountain pryamid. Attached to one of the walls, they could see two large screens, doing the best job that they could to pretend to be windows showing a view of Pafogai Du as though the cafeteria was sitting on top of the mountain, rather than thirty meters below ground.

"So who is this that you wanted me to meet again?", Gladias asked.

"Den. He is a High Council member of the Kungaupei. You wanted to better understand Dzego's institutions, he helps run the one that's already been protecting us from the Arctic shadow campaign for the last decade. You want Veridia to learn from him, not from our official government."

At that point, they heard a faint buzz that began to rapidly get louder. Almost immediately, a few small drones, similar to those that would check Zei for cheating equipment in the Minpentai championships, flew into view.

Gladias's watch buzzed.

Air	
CO_2	994
PM2.5	30.1

Devices	
Attested ✓	21
Unknown, increased risk!	4

"What's going on?", Zei asked.

"Ah, I think my watch is detecting those drones as a threat. It's been doing this for many days now, I'm assuming it's just not been updated yet."

"Ah, let me check mine", Zei replied.

Zei took out his watch.

ten min	
cau tie zen ✓	25
sen hu ?	0

He clicked on the green, and scrolled through.

gi dai	
fen jan ten lai	16
mo fan min jan	1
mo fan lai	4
kun gau min jan	4

"Err... let me try this", Gladias said, staring intently at the watch.

"'mo fan' is eating room, so something to do with the cafeteria."

"Yep!"

"'min' is machine, 'jan' is person, so I guess the two together mean ... robot?"

"Yep!"

"'kun gau' is ... wait, is that connected to the Kungaupei? So these are Kungaupei robots?"

"Yes, but what does 'kun gau' mean?"

"I'll guess security?"

"Exactly."

"Security robots. Phew, that wasn't hard at all!"

"Just be glad I'm not Fin, he would start taking half a longhour to go through this whole interface and explain the history of each and every word and why they decided to keep it as they were shrinking the vocabulary."

One of the robots buzzed closer to Gladias, and stopped to hover about half a meter to the right of his ear.

"But anyway, why have I been detecting so many hostile devices? Is my watch just not up to date on something?"

"Ah yeah, it's just every lamp and drone and camera announcing itself; the protocols stepped up a lot these past few days. Your watch just doesn't speak the language yet. Here, let me help you with that."

Zei then spoke in a softer voice:

"jo sun fe zin ten lai de cau tie dai"

About one tick later, his watch buzzed, indicating that the data was received.

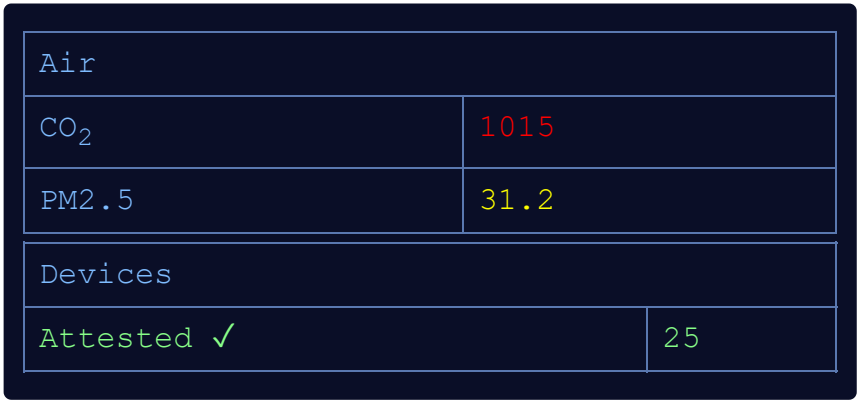


Zei moved his watch beside Gladias's.

"be"

The two watches buzzed at the same time, indicating a connection, and a file transferred from one to the other. Gladias pressed a button on his watch to accept.

He then checked his watch again.



"Looks good!"

"Huh, you're measuring CO2 and PM2.5 on there as well?"

"Yep! Pretty standard in Veridia. We really care about having a clean environment. We breathe twelve kilograms of air every day, more than what we eat and drink combined. So we're really all underrating how important the air is."

"Why is CO2 on there? We have air sensors on our watches too, but that's more for smoke detection and poisons."

"The average concentration of carbon dioxide in our atmosphere is 300 parts per million. Air that we breathe out is 40,000 parts per million. So if the air you're breathing has 700 parts per million, then either there's some other source bringing that extra CO2 in, or, more likely, one percent of the air you're breathing in is air that someone else breathed out. And aerosol particles carrying all kinds of things stick around in the air for a long time - especially airborne viruses."

"Ah, I see, so it's not even about industrial pollution."

"Actually I'm surprised you're not caught up on this, and that the CO2 numbers are so poor here. I would have thought you would care a lot from the defense side, for pandemic prevention."

"Ah well you can ask Den when he comes!"

A robot arrived, bringing a salad and tea for Gladias, and a vegetable and rice dish covered with mushrooms and a tea for Zei.

"Wait, that dish you're eating. I think that's exactly what we call Number Ten in our Dzego food trucks!", Gladias exclaimed.

"Do you want to try it? I'm happy to take your salad. Tell me how it compares to the version you're used to."

"Sure!"

They switched plates, and began eating.

"Okay, this is much better than what we get in Veridia", Gladias excitedly remarked.

They continued eating silently for a few minutes, and after finishing their food switched to drinking tea.

Zei's watch buzzed again. Zei checked his watch, and his eyes opened wide.

"What is it?", Gladias asked.

"A message from a courtyard where I ate in Sadzu Du once. This is really weird."

"It seems to be addressed directly to me."

"How does the courtyard know you ate there? Payment is done in zipcoins, that's fully anonymized."

"Exactly. The courtyard seems to have sent it to *all* their guests from the past half year, and it has some kind of tag that my local AI accepted, almost everyone else's filtered out."

zan		tie
sa dzu du de len su 2415 de bun kai mo fan	→	ci sen hu jan pu zau fe 400 lin man ci cau be fe ti tie ci tan jan pu ciu cu shan jin fe pa bi ci sen hu fe bu

"What does it say?", Gladias asked.

"It's from the Beautiful Plants food court, on 2415 Len Su street, in Sadzu Du. An anonymous person burned four hundred zipcoins to send this message. The message envelope says that the recipient 'is successful' and is 18 years old. I know nothing else."

"That's a ... well-calibrated description."

"Indeed. They're trying to make the set wide enough to make sure it contains me, they burned four hundred zipcoins to signal to the courtyard, and to our local AIs, and to us, that the message is worth putting an effort into understanding, but also they made the set narrow enough that they *could* make this broadcast for only four hundred zipcoins."

"And narrow enough that not too many people find out about this message."

"Okay, let me open it up..."

Zei stared in shock again.

"What is it?"

"The message is a series of links to some files, in total over sixty gigabytes. I need to download eighty percent of them apparently."

"Will that take a while to download?"

"About a day wirelessly. But it looks like the digital archive node inside this mountain pyramid already has enough of the files."

"Shall we just walk there?"

"Exactly what I'm thinking."

Zei put his watch against a green circle on the corner of the desk. A moment later it buzzed. He stood up, and Gladias stood up immediately after him.

They exited the cafeteria, and started walking through the network of tunnels.

* * *

A few minutes later, they arrived at a large ornamented door.

To the left of the door, there was a drawing of a river horse reading a book, with a backdrop of a marsh with green trees and a blue sky in the background. To the right of the door, there was a drawing of a hamster sitting at a computer. Both drawings were a careful balance of cartoon-like playfulness, and a serious aura of mystery.

At the top of the door, Zei could read a slogan:

mu fu bau hai ka sha liu li dzai hen ji cin jan fe jan

ci mu fu bau fei ka zen die jia cu can hu fe lo kin liu

"Outside of a dog, a book is man's best friend", Zei translated for Gladias. "Inside of a dog, it's too dark to read."

Zei and Gladias stepped through the door, and entered.

To their left, the entire wall was a large shelf, about five meters tall, filled with physical paper books. The upper half of the shelf was set back about a meter, allowing the roof of the lower half to serve as a walkway to look at and take books from the upper half.

The shelf had openings in two places on the upper floor, leading to two miniature tunnels lined with more books. The shelf also had an opening in one place on the lower floor, also lined with books. All three openings seemed to go inwards for three to six meters; after that, one of the book-lined hallways turned ninety degrees rightward, another ninety degrees leftward, and the third, the one on the lower floor, turned into a T-shaped intersection.

It looked like the book section was deliberately constructed to be a maze, Gladias thought.

To their right, there was an open space with tables, and along the wall, there were many computer terminals.

Near the front of the space with tables, Zei saw a woman wearing a robe. He immediately ran toward her.

She turned around.

"Welcome to the Pafogai Du grand library. What is it that you seek?", she asked.

"There are very large files in the archive that I would like to load."

The librarian laughed.

"Glad you're being honest", she replied.

"Most people tell me they're looking to learn some high-minded thing about history or culture or mathematics, and then go and download their movies as soon as I'm not looking."

"Well I'm not downloading a movie, it's more ... a mysterious message."

"Ah, a quest! Even if that ends up just being an excuse, even that is something new and interesting in my day. Well, just follow me."

Zei and Gladias followed.

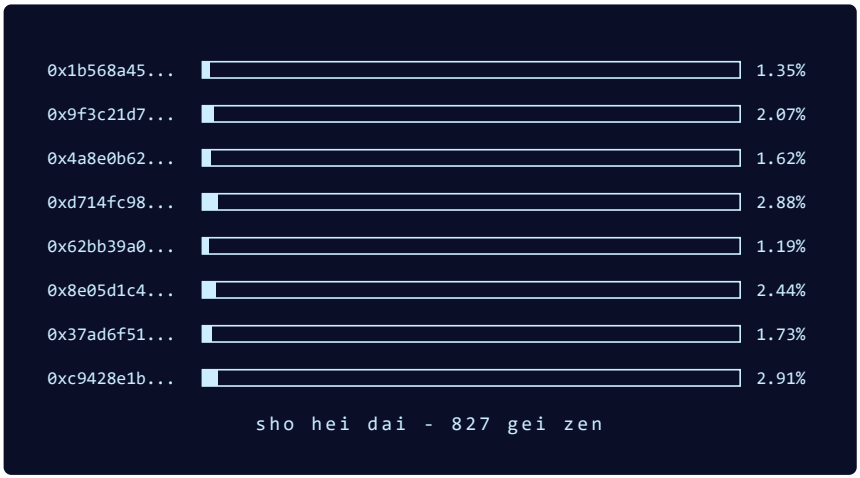
"So you got a mysterious message telling you to download some big files? Any idea what kinds?"

"Well so far I just saw some hashes..."

They arrived at a computer terminal. The librarian took out a data cable, and put one end of the cable into the terminal.

"Here, put the other end into your hand device."

Zei followed the librarian's instructions. The download started almost immediately.



"Wow, only eight minutes!"

"So now might you be more open-minded to the things we stodgy librarians say about how physical spaces are still relevant to the pursuit of knowledge after all?"

A robot slid up to them.

"jie hei ja ma?"

A menu appeared onscreen; there was no food, but a selection of drinks was available.

Gladias immediately clicked a button asking for a tea. Following along, Zei did the same.

"This space is beautiful, I am glad you have it."

Gladias stood up and walked over to the shelf with the books. He opened a few one by one. At that moment, not in the mood for straining his brain at reading and interpreting Dzegoban - especially not the earlier-version Dzegoban that the older books were in that he could understand nothing of at all - he focused his attention on the diagrams and pictures.

Two minutes later, the robot slid up to him, and gave him his tea. The tea was in a sealed cup, from which he could only drink through a straw. Surprised, Gladias turned to Zei, and saw that Zei, sitting at a table far away from any books, already had his tea. The cup that Zei received was open-faced, without much protection against the tea spilling.

Gladias found a book about Dzego architecture, titled "zui fia pai dan". He flipped through it. It contained pictures of many designs of basement rooms, shelters and underground rooms inside of mountain pyramids. The book showed them in roughly chronological order starting shortly after the war against the Arctic Empire a hundred years ago.

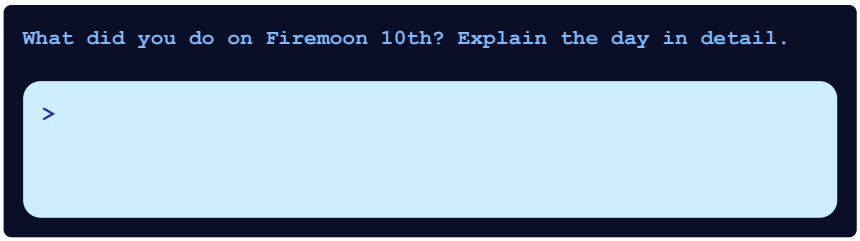
The earlier designs were very practical in construction, with not much thought put to comfort, or even safety against non-human threats. The designs in the middle started resembling ordinary rooms, with the main glaring exception being lack of windows. And finally, the designs near the end of the book were very well-decorated and comfortable, even more so than above-ground construction. About half had imitation windows, pretending to be outside. The other half did not have windows, and instead had plants, well-decorated walls, and other objects that seemed carefully chosen to make the inhabitant not see any space that they might think is supposed to have a window.

"Gladias!", Zei shouted. "Come back, it's almost done!"

Gladias turned around, and walked back. Just as he arrived, the last progress bar reached 100%. Another window automatically opened. A script began running. "Reconstructing data... estimated time remaining 175", it said onscreen - not in Dzegoban.

Gladias and Zei sipped their tea, and stared at the timer counting down. The librarian, curious, hovered quietly about two meters behind them.

Right on schedule, a little under two minutes later, the script finished, and a screen appeared.



Zei stared.

"Let's reschedule the meeting with Den for tomorrow. I have a feeling this is not going to be a simple task."

Gladias nodded.

"Are you going to answer?"

Zei pressed a few buttons on his hand device. "I put this program into a sandbox using a journaling file system, so it can't access the internet, and I can always revert it back to start. So there's no privacy leak risk, and if I'm not happy with the outcome I can just revert it back to start and try again. So why not?"

"Now ... to remember what I did on the 10th of Firemoon ... or rather I'll just ask my hand device."

A record of Zei's life events on that day popped up. That day was his last Minpentai game, the semifinal against Gun - the last day when things in his life were at least somewhat normal. Before that game, he woke up and had a meeting with Bai. After that game, he met up in the courtyard with Deluin.

Wait, Zei thought, the courtyard ... what was the address again? He looked it up.



The exact same cafeteria that the original message was from.

This was, as his local AI liked to say, the punchline. Zei started typing, recalling that day of his life from a combination of the AI records and his own memories which were rapidly coming back.

Waking up.

The meeting with Bai.

Going to the game.

Waiting for the game.

The game itself.

Watching Deluin's game.

A brief conversation with Deluin after the game. What was it I had said ... "my style was uglier than yours. Though I also won by playing defense"

And then the meeting with Deluin at the courtyard itself. Zei spent as much time detailing this event as the rest of the day put together.

The robot. A/B testing the new proposed hieroglyphs for Dzegoban.

Deluin invited him to Redshire. He refused. Deluin showed pictures of Redshire.

Deluin talking about how the nature of a place being beautiful actually does say something about the civilization there.

And then Deluin's gently trolling line: "I'll use my championship winnings to pay for you to go."

After typing everything in, Zei clicked "send".

```
Thinking ... estimated remaining time 440
```

"Okay, so this is a language model. But how large was that file? After the data reconstruction."

"10.7 gigabytes. The overhead was exactly a factor of six. I had a background thread check where the files were from, they were all recently uploaded. One sixth was a few recent movies - public data, though of course *the selection of which movies they chose* is a secret key of sorts. Another half was a big pile of someone's health data. The last third was also video, though it seemed to be AI-generated. It was

all Arctic sermons talking about how fundamentally flawed the Veridian and Dzego civilizations are."

"The recombination algorithm was a five-of-six erasure code of five pieces of 10.7 gigabytes each. Each of those pieces, in turn, was a steganographic encoding: they used language models to find bits to flip so that the data still looked plausible, while encoding the actual underlying message, at a very ambitious rate of one bit of plaintext per four bits of ciphertext. And then the movies were a sort of cryptographic pad: if you don't know the exact set of movies that were used, even if you could fully uncover the plaintext, it would just look like random data."

"Wait, if they wanted a pad, couldn't they just pseudorandomly generate one by repeatedly hashing some data?"

"Ah actually you're right. Maybe I was overthinking it, maybe they just did it this way to throw people off."

"Okay, so 10.7 gigabytes, what is it?"

"It's not a plaintext language model. It's an obfuscated language model. Level two obfuscation, much weaker parameters than the full cryptographic-grade parameters that are used in the cryptographic networks for voting in DU or Bansunpei or for that matter the Order of Steering, but it's still the same algorithm."

"Level two obfuscation is actually the right tradeoff for language models, because language models themselves are big and compute-intensive, so they can't handle the high overhead of going level four, and on the other hand, language models are at least partially obfuscated and resistant to bit flips all on their own. If you uncover the value of a few weights, or get the ability to reliably flip a few values in a few tensors, it's not like you can get anything useful out of that."

"Overhead factor of level two obfuscation is in the tens to low hundreds on size, and high hundreds to low thousands on compute. So this model is pretty tiny; I suspect whoever sent this fine-tuned it to only do this one task. I also think this language model's wrapper program, the full package that's being obfuscated, is doing the

thinking all at once, to verify whether or not my answer is satisfactory. If it is, it will just give us its message at the end."

As though on cue, Zei's hand device buzzed. The program finished executing. A new message appeared onscreen:

Hello Zei.
You may be wondering why I have not communicated any further with you within the past month. I greatly apologize for this, and my goal in sending you this letter is that you are the only friend I have who is outside of Arctic control, and you deserve my explanation about what has been happening.
The Arctics did a full sweep of Redshire almost a month ago, to confiscate all electronic devices, and give us back only devices that were backdoored and surveilled by them. This is why I lost contact with you. I had no way to reach you, and so I had to encrypt my message to you by wrapping it in this obfuscated language model that asked you to describe our shared experience to verify your identity. I have only been able to construct and send this message through a series of rounds of stealing devices from others, including two watches containing two Arctics' health data, doing the needed work inside of a non-transmitting room, copying the outputs to an inert storage medium, and then chemically destroying the circuits and storage on those devices.
Physical surveillance of Redshire is, fortunately, much less total than the Arctic itself, though the window on that is closing fast. But there was also another window that was closing fast for me.
You see, my parents were always on the side of being more sympathetic to the Arctics. And so after the annexation happened, they decided to cooperate with them. They told me that this was the best way to get any chance of autonomy for themselves and for Redshire in the future: to win trust by becoming a helpful part of the Arctic Empire. And so by the time you read this, most likely we will already be on an aeroplane going into the Arctic Empire core territory.
I do not share their sympathies. But I have decided that the way that I can maximize my chance of being useful, is to play along, and win their trust too. The Arctic Empire respects strength, I have proven my strength by winning the Minpentai semifinals in Dzego, and my parents may be proof enough of my alignment. Hopefully, there will come a moment where I will be able to do something helpful to you. But I do not know.
And so for this reason, this may well be the last message that you hear from me. I hope that this will not be true, but I cannot guarantee anything. I am sorry for everything.
dze go ba fau gie
Your dear friend Deluin.

Chapter 20

Meldan, Veridia · 3724 MISTIME 10

Seila woke up, as the alarm in her watch buzzed. Her watch was flashing bright red - a message from Emerald of a type that she had never before seen.

She immediately sat up, and opened up her hand device to take a look at the news in more detail.

When she saw the top line, her heart almost stopped.

ARCTIC EMPIRE INVADES NORTHGLADE

Northglade. Febric and Hreda were there. No.....
She kept reading.

At 5,250 this morning, Arctic soldiers, flying vehicles and drones descended on the province of Northshore from all sides.

The Veridian military offered almost no resistance, and within two longhours their soldiers and drones were stationed in Northglade.

The entire area is now fully under Arctic control, with only one exception. The Greater Plum Harbor inside Northglade has remained untouched by the Arctics, and seems to be at present respected as some kind of neutral zone. Hundreds of refugees streamed into the Greater Plum Harbor, before the Arctics set up barricades to stem any further entrances. More refugees attempted to enter on personal copters. About half succeeded, especially in the first longhour, and others were shot down.

Internet communications between Northglade and the rest of Veridia have been cut, except for the Greater Plum Harbor. Long-distance transmissions between the Greater Plum Harbor and the rest of Northglade, which are evading the jamming mechanisms, are now the main channel of two-way communication with the province of Northshore.

Neither local Northshore politicians nor Arctic officials have yet made any public comment. As of our last knowledge, Lord Ephelion, Arctic consul to Veridia, was in Northglade at the time of the invasion. We await further information.

She scrolled down in her hand device, and saw that Emerald was highlighting five other messages that it deemed very important to her.

From		Message	Time
Febric	→	Hreda and I are both safely inside the Greater Plum Harbor. We will survive the best that we can. We love you.	19867
Hreda	→	Northglade was invaded this morning.	20045
Hreda	→	It looked like there were barely any battles, as though our politicians and army are totally okay with this happening.	20060
Hreda	→	Febric and I are safely inside the Greater Plum Harbor. The Arctics seem to be respecting it as some kind of neutral zone, at least for now.	20071
Hreda	→	I'm scared. I hope everything will be okay. Love you.	20077

Seila started typing back a message, confirming to Febric and Hreda that she received their message and she and Vil and Daia would do everything she could to bring them home. Before she finished typing, she began to cry.

* * *

A longhour later, she heard a knock on her door. She walked up to the door, and saw through a screen that there was a man wearing a privacy robe outside.

"Who are you?", she asked.

"I'm Mov. Gladias knows me well."

"Why should I believe you?"

The man whispered a few words into his neck band, and put up his hand device against the green circle beside the door of the house. Seila's watch buzzed.



Seila cautiously opened the door, and let the man in.

Mov took off his shoes, and immediately proceeded to the basement room, knowing exactly where within the house it was located. Seila followed him.

They both sat down.

"I'm getting really concerned about what's happening to Veridia."

Seila looked at him with a blank stare, suprised that it was even possible to say something so uselessly obvious.

"I don't mean the invasion of Northglade. That was really shocking, it's terrible. And I'm really sorry Febric and Hreda are there. I hope they're safe."

"They are. The Arctics are leaving the Greater Plum Harbor there alone somehow, and Febric and Hreda managed to get there in time."

"So what do you mean?"

"I mean the things that made that invasion possible, and succeed with so little resistance. All of those factors are true in the rest of Veridia too."

"Gladias would be the person I would normally speak about this with, but he has been very busy in meetings with some Dzego secret government people the past few days, and he's been asleep the past few longhours. So I wanted to talk to you."

"What do you think I know? Our politicians are clearly bought out, or blackmailed, or apathetic, or some of all of those things. Gladias doesn't talk to me much about the Order, after all he's not supposed to."

"Maybe I'll tell you what I know. I was following Delwart, the mysterious gentleman who has been having meetings with Gladias for the past months. He was not just talking to Gladias, he was talking to various other Keepers, he seemed to have a pretty wide network."

"At one point, before I lost track of him, I caught him in a meeting with someone from Silverchat. And then Gladias told me that he confronted Delwart, and Delwart admitted that he had been working for Silverchat the whole time."

"Silverchat? They're doing some kind of big operation to corrupt the Order? But why? I would have expected Bluewhale to do things like that, but Silverchat were always the good ones! And what does this have to do with the Arctics?"

"I think our society has just gotten really complacent. Everyone is focused on maintaining their happy life, getting some advantage that they can, and people don't value actually maintaining the things that keep Veridia running. Even Silverchat. And what's worse, the Keepers just don't seem to be watching out for this."

"The Order seems to be performing better than any other part of our institutions. The military has failed completely at setting up defenses for Northshore. From what I can tell, the Keepers have done

what they could, the tax rubric even changed to be even more supportive of open-sourcing defense hardware, Gladias told me that much. He was really proud of it, as was I."

"Indeed, though that was a close call. Whoever split up the Keepers into three groups that could not talk to each other was a genius - otherwise Delwart might have gotten to all of them."

"Something about this doesn't sit right. Silverchat... they just don't seem like the people who would do this."

"I know you've loved them since you were a child, for the same reasons Gladias did. And I did too. But sometimes ... your heroes turn out not to be heroes. Or maybe Silverchat doesn't care about that particular rubric, and that was just Delwart himself being an extreme peace dove."

"Do you know any of the other Keepers that Delwart talked to. Can we talk to any of them?"

"I did put trackers on two of them. I could have totally ratted them out to the cryptographic network and gotten a big bounty, but ... I thought it would be better to keep the option to track them more."

"Can we talk to them?"

"I suppose."

"Let's go visit them. Like, right now."

"You don't want to focus on seeing what you can do for Febric and Hreda today?"

"I may be a worried mom - or, well, aunt - but I know there's not much I can do. I had a call with them already, they're sitting tight, I'm sitting tight, I don't think we and they are any better at pulling off some kind of grand escape from Northglade than any of the hundred people trying to fly in to the Greater Plum Harbor who got shot down. So, sit and tight is the plan, nothing more to do today. But figuring out what on earth is going on with the Order, that's something I can do today."

She sighed.

"I guess I feel like I have to be doing *something*, and this is something that I can do."

Mov nodded.

"Just don't take a copter. You might think it's faster and *this time* the crazy high noise taxes are worth it, but you would just be inviting attention to yourself."

"Don't worry, I wasn't planning to."

A taxi arrived. Seila put on one of Gladias's privacy robes. She and Mov, who was already wearing his privacy robe, rushed toward the taxi. Seila reached the door first, and she tapped her watch against a green circle beside the taxi door, simultaneously opening the door and cryptographically approving the payment. They got in the taxi.

The taxi began moving, headed to a location a short autobus ride away from the home of one of the other Keepers.

* * *

Twenty minutes later, they arrived at an apartment building. It was a new construction. The outer wall was wrapped with some kind of imitation stone material. On the ground floor, they could see restaurants and shops selling various kinds of goods. On the higher floors, from the outside all they could see were regularly-spaced but well-decorated windows of people's apartments.

They saw another man walking towards the door, about to enter. They immediately slowed down their walk so that they would appear to naturally reach the door right after the man.

The man walked to the door, put his watch against the green circle beside the door, and the door opened. He walked through. Before the door closed, Seila and Mov walked in too.

"He's in room 515", Mov said. "Let's just go up the stairs."

"How do you know that?"

"The tracker I put on him."

They walked up the stairs, and eventually got to the door of room 515. They knocked.

On the inside, they heard footsteps. Mov removed the hood on his privacy robe, a gesture to appear less threatening. He was not supposed to know this man, so he had no way to prove that he was

someone to talk to - except perhaps for burning a hundred zipcoins at his doorstep. Seila removed her hood too.

The man simply opened the door.

"Hello?"

"Hi, I'm Seila. I know this is awkward, but we wanted to talk about the Order and Delwart."

The man's eyes opened wide.

"Nice to meet you Seila."

"And I'm Mov."

"I'm Telroy. Though ... if you know I'm in the Order and you know about Delwart, maybe you already know that too."

At that moment, Mov realized that in a certain sense, he already had burned much more than a hundred zipcoins at Telroy's doorstep - by clearly having enough information to claim a large bounty by ratting out his role to the Order's cryptographic network, and choosing not to do that.

Mov and Seila walked in. Telroy directed them to sit down on the couch, and he began preparing tea for the three of them.

"So what do you know about me and Delwart?"

"You're a Keeper, you're in the rubric group on cybersecurity. One of the rubrics being debated has to do with data retention in environmental sensors. You met Delwart at least once, though I was not able to hear the content of your conversation."

"Okay, you do know a lot. Delwart has been insistent during our conversations that the recent push for trying to minimize data collection did not actually make sense, it was some kind of paranoid threat against an enemy that was not really present - though maybe now, he will have changed his tune. He argued to me that the more important information threat is protecting against insider trading."

"He also said that the local environmental and public safety agencies need broad access to be able to adapt to changing circumstances quickly without asking for permission every time - and that the main privacy defense we really need is delaying their access to

non-disease-related data by five or ten days so that they can't trade with it."

"That ... does not sound at all like the Delwart that Gladias told me about."

"Did he say anything else?"

"He was generally anti-war, as I mentioned - that's how he described himself. Otherwise not too much - his main opinion was that he thought people were getting overcomplicated trying to throw in all the cryptography, and we should just trust people more."

"Okay, now I'm curious - what is the Delwart you know?"

"He was anti-war. He valued people's freedom to live their own lives, he wanted the Order to do much less - even the tax nudges were too much for him. In particular, though, he was really against this change to a tax rubric -"

Mov paused. He realized that Seila had already volunteered Gladias's name, and if he continued, he might be giving Telroy information through which Telroy might be able to figure out *Gladias's* rubric, and collect the bounty by ratting him out.

But then he thought - Telroy was trusting him and Seila, the least they could do was reciprocate back with some trust of their own.

"The rubric for open source hardware. He believed that anything military-relevant should be exempt from tax breaks, because military is a zero-sum game, we're not doing anything good for the world by participating in that."

Telroy nodded. "I guess he really is anti-war then."

"But then I followed him around more, and Gladias confronted him after Redshire was invaded, and ... it turns out he has been working for Silverchat."

Telroy stared in surprise. "Silverchat?"

"That is ... surprising."

* * *

Seila and Mov got out of the taxi, and walked along the forested path through Kalimar. A gentle cool breeze blew against their backs,

subconsciously driving them to walk faster. A few hundred meters later, they walked off the path and turned right. Thirty meters after that, they saw a stone house that was mostly underground, with the above-ground part sticking up only a meter above ground. The whole structure was barely visible from beyond a few meters away.

That was one way to score high on aesthetics, public access and emergency shelter functionality at the same time, she thought. That house was probably paying close to no land taxes.

Seila and Mov walked down a staircase, and knocked on the door. They took off the hoods of their privacy robes.

They heard footsteps.

"Who is this?", they heard a female voice ask.

Seila quietly stepped on Mov's toe, urging him to stay silent. Mov picked up the cue, and stepped back.

"Hi, I'm Seila. I know this is awkward, but we wanted to talk about the Order and Delwart."

The door immediately opened.

"Hi, I'm Balme. Please, come in."

"Is it okay if my friend Mov comes in too?"

Seila felt slight discomfort from Balme - the interaction was now two on one, and Mov had a larger and more imposing physique - but at that point she had already opened the door, and it would have been very awkward at best to shut it on them.

"Sure", she replied softly.

Seila and Mov both entered.

"I apologize for intruding so rudely, and for spying on you to know where to come. But I hope you'll forgive me; our country is in a pretty dire situation as you can tell."

Balme nodded.

"What do you know about Delwart?"

"I met him a few times. The first time we talked was when he sent me an anonymous message about a longhour after a Keeper meeting. Somehow he had been able to track me down. I'm in a rubric group for private schools. He talked a lot about his own views about education,

and how he thought we should not be putting our thumbs on the scale, using the tax rubrics to encourage what he considered biased civics education talking about how beautiful Veridia is and encouraging the students to cherish and love it - he thought that schools should just teach people knowledge, and they should learn to love Veridia by seeing how great it is on their own."

"Did he have any other perspectives?"

"He seemed generally pro-freedom, he believed that people should be left to evolve on their own, generally things work out best when they happen spontaneously and naturally, the Order should not be putting its thumb on the scale so much."

"Anything about international politics?"

"Not really, he never brought that up. He seemed not to care."

"He did once ask me what I thought about Dreadknot. I responded that I did not like them much. He didn't answer, and changed the subject."

Seila nodded.

"The Delwart that I know is also concerned about freedom, also thinks the Order should do much less. But... he also does have strong opinions on international politics. He basically thinks anything military is a zero-sum game, you can't know what will lead to a good or bad outcome, and Veridia should stay out entirely."

"Although ... he's apparently more mixed on the freedom thing. He seems okay with the public security agencies collecting more information, and not using cryptography to try to limit the data collection and get guarantees about what purpose the data is being used for."

Balme's eyes opened wider in surprise.

"Wait, I'm curious, how could you use cryptography for that?"

"I have a friend from Dzego, Zei, who told me about it once. Basically you have a language model inside the camera, that detects whether or not, say, an act of violence is going on. And only if that's happening, it passes the information along. Every time it does, it also

uploads a hash onto a cryptographic network, so we get a public trace of how many times the check was triggered."

"And then they have a whole infrastructure to verify that the hardware is actually running that code. Signatures from keys that were randomly generated inside the chip by chemical imperfections in the manufacturing process at the time the chip was created. There's apparently a rule that anyone has a right to randomly take down a camera and micro-inspect it, and many people are doing that as a hobby, so no manufacturer can get away with making cameras that cheat the rules."

"That ... sounds amazing. Cryptography is not my area at all, but ... cameras that you can actually trust, that makes total sense! Wait, but Delwart was so pro-freedom, how could he not already know about that kind of thing and be enthusiastic about it?"

"I'm confused too."

Seila paused for a few moments.

"Okay, I know this is a random question, but - do you know any Keepers who are in any rubric groups related to social media?"

"I know a few Keepers as friends, but I don't dare ask them what they're working on. It's our tradition, you only talk openly about things you did when you already finished your three years and moved on to a different rubric group."

"Could you ... forward a message to them from me? I'll write it up for you right now. And tell them that it's worth responding to."

Balme thought for a moment, checking with herself whether or not she thought Seila, and her friend who had so far said nothing at all, could be trusted. An instinct told her that they can.

"Of course."

* * *

Seila walked up to the door of a house, similar to the one that she and Gladias lived in. This time, Mov stayed back entirely. Seila seemed more effective at winning trust alone, he thought, so he might as well

let her do the questioning on her own. If he needed to, he could suggest questions by earpiece.

Seila knocked on the door.

No response.

A minute later, she knocked again.

Still no response.

She heard footsteps inside, but they did not seem to be getting closer to the door.

Her watch buzzed. She got a notification:



Mov ... she thought. She knew that whoever was in the house would see the same message, and know that she was serious.

Seila knocked on the door for a third time.

She heard footsteps again, this time coming toward the door.

"What do you want?"

"Hi, I'm Seila. I'm the one that asked Balme and Telroy to send the message about the social media rubrics. I want to talk."

The man opened the door.

"I'm Lectoby. I'm in the social media openness rubric."

"I'm sorry I didn't open the door. My wife is stuck in Northglade. I've been really upset today, I just don't want to talk to people."

Seila nodded.

"I'm Seila. I understand. My kids are stuck in Northglade too."

"Why did you burn those zipcoins? You could have just asked Balme to tell me you're coming."

Mov ... Seila thought to herself furiously. Why did he have to be so impatient?

"I'm sorry, I'm just really desperate."

Lectoby motioned for Seila to come in, and she followed. They both sat down.

"So what is this about?"

"Has there been any strange man who attempted to contact you, or anyone else you know in your rubric group? In the past year or so."

"Huh ... nope, never heard of that happening."

"Could you maybe ... directly ask in your rubric group chat? I want to confirm."

"Have *you* had that happen?"

"My husband has. Balme too. Quite a few others. He has strong opinions that seem to differ depending on whom he talks to. Sometimes he's anti-war, sometimes he isn't. Sometimes he's against the Order intervening in anything, sometimes he isn't."

Lectoby whispered a few sentences through his neck band.

"Let's wait for my Keeper group to see what they say."

He stood up.

"Would you like some tea?"

"Sure."

Lectoby walked over to the kitchen, and started to boil water for himself and for Seila.

"I'm sorry about your kids. This whole Arctic thing is just ... so crazy. Redshire, now this. It feels like they're shifting gears, and within a year or two they'll just launch a big war to take over the whole world."

"Or worse, less than a year. They took Northglade, seems like no one is lifting a finger again. I checked the senate hearing on my way here, a lot of performative outrage, no action, no introspection about how we got here."

Lectoby sighed.

He brought a cup of tea for himself, and a cup for Seila, and they sat down. They drank it silently.

Twenty minutes later, Lectoby leaned forward, and Seila's attention turned to him.

"They all replied", he said.

"And?"

"None of them had that happen. One did reply that a friend in a different rubric group had that happen to him. But none in this rubric group."

"That is ... so strange."

"It could be a coincidence, but..."

"What's the matter?"

"Your rubric group is on social media openness and interoperability, right?"

"It is."

"Delwart was supposed to be from Silverchat. But this is the one topic that Silverchat should care about the most, and ... zero inroads into this rubric?"

"Again, this could be a coincidence, but..."

She stood up.

"Wait, what's going on?"

She knew Mov would want her to leave, so she could sync up with Mov privately. But Lectoby had trusted her, and she thought it was fair to trust him too.

"Why would someone from Silverchat care about whether or not Veridian schools are teaching people civics well? Why would someone from Silverchat be so anti-war? How could someone from Silverchat be so effective at tracking down keepers? And ... why would someone from Silverchat, in that position, not make any effort at all on the rubric that Silverchat cared the most about, getting properly rewarded for being so effective at openness and interoperability?"

But by this point Seila had long since known the answer, and this was simply a final confirmatory straw. She figured there was no harm in blurting out her conclusion, as she stood up, and picked up her hand device, getting ready to leave.

"Delwart is not from Silverchat."

Chapter 21

Meldan, Veridia · 3724 MISTIME 20

Seila, Vil, Daia, Lily and Zven were all seated around the table in Seila's house, waiting for dinner to arrive.

"Thank you guys for coming and keeping us company", Seila said softly. "It really means a lot to me."

"Any more news from Febric and Hreda?"

"I just sent them some more zipcoins for their personal expenses a few days ago. They're just living the life they can inside Greater Plum Harbor. The schools are full, but one of the other refugees was a math teacher, and we're hiring her to informally give them and a few other kids some lessons."

"No word on when there will be any way for them to get back here?"

"Still none. At this point I'm starting to suspect that the Arctics' plan to reunify everyone is to just invade the rest of Veridia too."

"Well at least they're safe."

Seila sighed.

"How is your lawsuit against Divergent going?"

This time Vil sighed.

"We won in the original trial a few months ago. But then they appealed. So a whole different group of judges got selected to hear the case, and a month ago they ruled in our favor too. But only a few days ago, we learned that they're appealing a second time."

"Can they just keep on doing that forever? Do another appeal, another five judges get sortitioned in, you have to convince them all

over again, and keep doing that until you're too tired to keep going and then you give up and settle for like a quarter of the money?"

"Fortunately no, there are Veridian common law rules where judges can rule a case to be in bad faith, and require the submitter to give the other side a bunch of extra money as a penalty. That often happens to frivolous appeals."

"Here, their latest appeal document has some new footage from Jump's internal monitoring systems. Of course it was all encrypted, but some former employee had access and handed it over. They found some video and messages that make some of us look bad. So they're arguing it's a genuine appeal because of that. But really the video says nothing that's actually relevant to the case. I'm hoping these judges can see through what Divergent is doing more quickly than the previous round's judges did. We're accusing them of making a bad-faith appeal in our defense statement, with luck that should close things quickly."

"These hedge funds really are mean, aren't they."

"If judging wasn't anonymized, randomly selected and split into two and a half groups each round, I'm sure they would find a way to threaten or bribe the judges too."

At that point, Lily rushed into the room.

"Uncle Vil, Aunt Daia, Mom! I got eighty one percent on my latest civics test!"

"Congratulations, Lily!", Daia replied.

"Normally I would say, now you've earned your trip to Northglade.", Vil added. "But, well..."

"If it's safe in a few months and Gladias doesn't need me, I'll take you somewhere in the United Cities instead. Maybe Mabuli", Seila offered.

The doorbell rang. Before anyone else could react, Zven ran to the door to get the food.

Zven opened the door, picked up two bags from a robot, and closed the door. Zven brought the food back to the table, and immediately

began unpacking. He, Seila, Daia, Vil and Lily each found the boxes with their own meals, and the five of them began eating.

"How was your time in school been, Zven?", Daia asked.

"My team won another Helisport game yesterday!", Zven replied excitedly.

"I also got ninety five on a physics exam a month ago. My teacher is really impressed."

"That's amazing", Seila replied.

"So a few days ago he took me out privately to a garage, and showed me a mini rocket. He let me take a look inside, and then press the button to let it blast off. It was so fun!"

"I want to build a rocket when I grow up! And then I want to fly in it. Maybe I'll even be able to go to the moon one day!"

"Sounds like you won't need me to take you on any adventures!", Vil added.

Daia continued asking the family table for status updates.

"So what else is new for you, Seila?"

"Well aside from Febric and Hreda, I've basically been thinking nonstop about those meetings with the other Order members I had recently."

"Are you able to say more?"

"Not the specifics. But in general, it feels like there's a really coordinated effort to try to manipulate the Order, they're saying different things to different people to manipulate each person based on their exact position, they have amazing spying powers, and their objectives don't even seem to be anything commercial."

"Is it the Arctics?", Zven blurted out, with the excitement of a hero in a detective novel solving a mystery.

The table went silent.

* * *

Seila was sitting in the chair in her basement, and was staring at her hand device.

Come on, she thought. Gladias should be awake any minute now.

She hit her palm against her face for a fifth time, ashamed at how blind she had been. Gladias had been having these secret meetings for half a year, and she had not asked anything, assuming that he had things under control - and that anything Gladias could not figure out himself, his shadowy friend Mov could.

But as it turned out, Gladias and Mov had been successfully hoodwinked by this man the whole time. And when his original pretense - that he was just a kindly man who cared about peace and freedom - fell through, they then got hoodwinked *again* by his mask, his pretense that he was from Silverchat.

Seila herself had seen through the Silverchat ruse within a day - really, she had felt doubts within minutes. But she had not gone further quickly enough. She had entertained the thought that it might be the Arctics, but she had dismissed it - none of Delwart's lines seemed anything like anything she had ever heard from Lord Ephelion. Why would an Arctic say anything nice about peace and freedom? But now the picture had all come together. When you have eliminated the impossible, whatever remains, how improbable, must be the truth.

Her watch buzzed. It was Gladias.

She immediately called.

"Seila! I miss you so much. How are you?"

At first, she wanted to scream.

"Gladias, you idiot! Do you know what you've done?!!!!"

At first, silence.

"What's wrong, Seila?"

"Delwart."

Silence again.

"I see. Did Mov catch you up?"

"Mov was an idiot too. And you're a double idiot for trusting him. Yes, he did catch me up, Gladias. But I saw through Delwart's lies almost immediately, and I dragged Mov out to do some sleuthing myself."

"What did you find?"

"He's not working for Silverchat, you incredibly naive idiot!"

"What?"

Silence again, this time from Seila's side, as she figured out the next thing she would say.

"What do you know?"

"We tracked down two of the other Keepers that Delwart talked to. One of them was on a rubric for data retention in environmental sensors. Delwart tracked him down, the same way he tracked you down. And he was telling him to not bother pushing for more privacy preservation in any of the security or environmental the data collection."

"We then tracked down another. She was on a rubric for education. Delwart was telling *her* to remove the incentives for private schools to have proper Veridian civics education."

"And then finally, just to confirm, we found a Keeper who was on the rubric that Silverchat should care the most about - social media openness and interoperability - and neither Delwart nor anyone else had tracked down any of them at all."

"But ... Mov overheard Delwart talking to a Silverchat person, how did that happen? Is this someone else trying to corrupt Silverchat too? Convince them to up-boost some content favoring whatever their position is?"

"No, you idiot. Delwart faked that. Come on, his spying ability is much better than Mov's, at some point he must have caught on to the fact that Mov was following him, and one day he planted that seed on purpose, to make his cover story more believable."

"So..."

"Think about what we now know about Delwart's goals. You thought he cares about freedom - but not when he's pushing for unrestricted surveillance. You thought he is anti-war, but he never mentioned that to the second Keeper at all. And then eventually you thought he cares about Silverchat - but he never found his way at all into the Keeper group that was working on the rubric that Silverchat would care about the most!"

"There's three Keeper groups, they don't talk to each other. He could always have gotten into the other two."

"IT'S ABOUT THE PREPONDERANCE OF EVIDENCE", Seila snapped. "Remember the probability update formula, Hreda has shouted it at us many times. Each individual unit of evidence might only shift the odds ratio by a factor of two or three. But when there's three independent factors that all point in the same direction, that multiplies up to shifting the odds ratio a factor of eight to twenty seven."

"Okay, so what do you think Delwart's real motives are?"

"He's an Arctic."

"WHAT?"

"That's really hard to believe. Nothing he has ever said to me sounds anything like Lord Ephelion."

"Gladias... how can you be so naive?"

"Ephelion and Delwart are not intellectuals, they are propagandists. Their goal is not to seek and share what they think is the truth. Their goal is to adjust the discourse environment, either of one person or of a whole country, so suit their interests. To turn bystanders into friends, and enemies into bystanders."

"Ephelion's goal is to inspire people inside Veridia to join the Arctic cause. So he's openly saying a version of the Arctic cause that some smart, but sick and twisted, people in Veridia will find appealing."

"Delwart's goal is to break down the Order. He knows he can't make the Order pro-Arctic, but what he *can* do is make it ineffective. And so it's a better message for him to be sometimes pro-freedom, sometimes pro-peace, sometimes whatever else, depending on the rubric and the person he's talking to."

"Why are you so certain?"

"PREPONDERANCE OF EVIDENCE. Let's list it all."

"What are Delwart's goals? Not the goals he says he has, but the goals that are visible from his actions?"

"One: reduce support for open source defense hardware. One point for freedom, one point for peace, one point for Arctics, zero points for

Silverchat."

"Two: push for unrestricted surveillance. One point against freedom, zero points for peace, one point for Arctics, zero points for Silverchat."

"Three: push for Veridian schools to not teach people Veridian civics. Delwart specifically said: 'people should learn to love Veridia by seeing how great it is on their own'. One point for freedom, zero points for peace, one point for Arctics, zero points for Silverchat."

"Four: not participate in the social media openness and interoperability rubric. Fine, there's three Keeper groups, we can multiply this one by half. Zero points for freedom, zero points for peace, zero points for Arctics, minus half a point for Silverchat."

"Five: once he blew his cover, he sent you off to Dzego. You might say, that's a good suggestion, Veridia should learn from Dzego more. But that's not the only interpretation. The other interpretation is that he wanted you out of the way, halfway across the world so that once you got suspicious, you would not interfere with any of his plans! Both are possible. So one point for every possibility."

"Six: his spying capabilities, beyond any corporation, far beyond some nonprofit group of idealists. Minus one point for an idealistic freedom lover or a peace lover, zero points for Silverchat, one point for Arctics."

"Sum it up. Arctics five, freedom two, peace two, Silverchat one half. And if you say it's a Silverchat person who also cares about freedom, then, well, we're at plus one half. I'd even multiply the last point, about spying capabilities by two, then you get six points for Arctics and zero or a half for any of the other explanations."

"And notice what it took, Gladias. He got to you, and through you he got to one third of one discussion group. The military, the parliament, media platforms - he and his friends walked straight through all of them. The only doors that he couldn't break were the ones whose keys he couldn't find: the Order, the Courts, the graph funding. Privacy isn't a luxury, it's why we're still here."

Seila stopped talking. For a few moments, Gladias stayed silent.

"I'm sorry, Seila."

"I'm sorry I didn't trust you enough to involve you in this. I love you. I should have trusted and respected not just your heart, but also your mind."

Silence again.

"So what do we do?"

"I hope that Delwart was wrong that you would be useless in Dzego, and you have something amazing to show from all the adventures you've been having there while I've been crying because our kids are stuck in an enclave surrounded by Arctics and trying to clean up your mess."

"I think I understand their military well now. Or rather, their shadow military. The official Dzego military is as compromised as ours, and the same situation for the rest of the government. But they've basically built up their Minpentai championship as a sort of secret army training ground, making the bet this whole time that the war that will matter will be a drone war - more similar to Minpentai than to anything like Helisport."

"I've been doing my best to convince them that the next battle will be in Veridia, not Dzego - we're ahead on technology and industry that might be a threat to them, but behind on thinking at all about defense - and that if they want to save themselves, they need to come help us over here first."

"Good. I hope for the kids' sake, and for Veridia's, and for yours, that you're actually going to have a plan to make that happen."

"Convincing the Dzegojan is only half the battle. We also have to convince the Veridians to accept them. And not the Keepers, the parliament. The same parliament whose subcommittee just barely accepted you as a Keeper, eleven to ten, despite having no valid reason to think anything negatively of you - unless they had some magic foresight about how naive you are, which I doubt."

"I know."

"Do you have any idea how we can do that?"

"I do not."

"I can try to figure this out. But you will have to trust me."
"I will, I promise."
"No more secrets, okay?"
"Okay."
"I promise."

Chapter 22

Freetown, United Cities · 3724 MISTIME 22

Jahn, sitting in a taxi, looked out through the window. He saw Freetown's usual chaotic mixture of buildings: some glass and steel, some concrete, some stone, some tall, some short, some beautiful, some ugly.

The taxi passed under a highway, and he could see piles of uncollected garbage on the sides of the road. Almost immediately after the highway, the taxi passed by a small park to the left, and a tall industrial building on the right. A movie with subtitles was playing, projected onto the industrial building's wall. Jahn looked again to the left, to see where the movie was projected from. Just before the taxi moved too far away for Jahn to be able to see, he found the projector, hidden under a bench in the park. Even without Graph Funding or rubrics, he thought, some people will always provide public goods.

On the whole, everything seemed normal - though Jahn could just barely notice that the amount of activity on the streets seemed to be lower than usual.

The taxi passed under another highway. On the other side, on the side of one of the buildings, Jahn noticed a large advertisement.

> *Buy a luxurious room in a shelter, right on the northern edge of central Freetown.*
>
> *10 meters underground, maximum-grade physical, bio and info filtration and insulation.*
>
> *Only 325,000 zipcoins!*

A minute later, another advertisement:

Shelters for the common man, in all shapes and sizes.

Prices start at only 50,000 zipcoins, mortgage plans available!

As the taxi continued west, it entered the region that the advertisement was talking about. Here, Jahn saw a type of activity that, to him, seemed to have been steadily increasing for the past six months: construction of roads. Jahn had also seen many shelters being constructed, but roads were a close second - particularly on the northern side of the town.

Jahn's hand device buzzed in his lap.

Taxi fare		
Driver base fare		3.69
Per-minute toll	9 min	1.89
Per-kilometer toll	5.65 km	1.26
Acceleration and deceleration toll	113 m/s Δv	1.03
Road congestion toll		1.50
Road congestion toll		1.00
Road congestion toll		2.25
Total		12.52

"Hey, Chief?", Jahn asked the driver.

"What is it?", the driver replied.

"Do you know why there's so many congestion charges this time? The road seems quite a bit less congested than usual, why would the tolls be going that high?"

"Ah yeah, they're bumping up every tax and toll they can. Funding defense, they say."

"Huh, the roads too?"

"Are they bumping up any other fees on you?", the driver asked.

"Actually not the government", Jahn replied. "At least, not directly. But, the shelters. I heard some interesting gossip there."

"And what is that?"

"The best shelters were originally owned by a few defense companies. Soon after Redshire got annexed, they got together and formed a cartel. Sold off a lot of the shelter space for money, and used that to modernize their armor and weapons a bunch."

"Huh, all private companies? I guess that's their property, their rules. Less shelter space left for us poor folks than if they just rationed it, but hey, they are trying. And if they use that money to defend us well, less chance we'll end up needing any shelters in the first place. And with fewer cars on the road, I'm enjoying the roads more too."

"Well, yes that's the polite story. Freetown's economic philosophy at its finest. But I'm getting the feeling it's not *quite* that clean. The government seems to be making a special effort to build roads in exactly those places that make it easier to get to and from the shelters. There's no evidence of any collusion between them and the companies - that would get knocked down hard by the courts - but my theory is that it's a sort of retroactive reward. Create the expectation that if you use your private property for the public good, then the government will twist the public work that it does in order to make life easier for you. It's a trick to work around the maximum budget rule - pay people without actually directly paying them."

"And if you refuse to use your private property in public-spirited ways, well... if things get really bad the next step might be that they'll start closing some roads for 'maintenance'. Though I guess people can always fly."

"The Greater Plum Harbor has been refusing to fund anything that could be seen as going against the Arctics, they're scared that their node in Northglade might get shut down. I guess that's where we'll find out how good Freetown is at staying 'free'."

"And at defending itself."

The driver and Jahn stayed silent for a minute. Soon, the taxi slowed to a halt. Jahn looked outside the window. He had arrived at

the Freetown Economics Institute.

"Thank you Chief, and have a good day!"

"You too, boss", the driver replied.

Jahn opened the door, and got out of the taxi.

* * *

Jahn passed through the turnstile in the Freetown Economics Institute, and walked toward the restaurant.

After a minute of walking, he arrived. He saw a familiar face: Utaku.

"Utaku! Surprised to see you still here."

"My parents told me to come back. They think Freetown is taking defense more seriously than the other United Cities, so it's safer than Belpaki here."

"Do you think that?"

"There have been some moves recently to create a unified defense grid between the aks. I'll be more optimistic if that goes through. But even then, I worry it's more theater than anything substantive."

"Has Freetown been doing anything substantive?"

"Well the big corps are paying insurance on all their property, the rates tripled when Redshire was annexed, and doubled again after Northglade. And then apparently the rates dropped back down by a quarter when the corps announced their recent defense packages."

"So ... somehow the market incentives seem to be working."

"You know, I've been talking to Zei from Dzego lately, and learning how quickly they've gotten everyone to start caring about privacy, now about bio safety, average indoor CO_2 levels dropped by over a hundred points within a month. All coordinated by machinery that they originally created, supposedly, to upgrade their language - and to give us a much more sensible distance and time system."

"I've been getting this feeling that maybe almost any system can work well if people accept it and believe in it."

"Do people in Freetown believe strongly in economic freedom? Everyone in the Economics Institute does, obviously, but what about

regular people?"

"Are you familiar with the early history of the United Cities, a hundred years ago?"

"Vaguely. I know there were some wars, both between some of the cities and against the Arctics."

"But do you know how those wars got resolved?"

"Eventually the armies of this continent came together and cooperated more, and that was a big part of why the Arctic offensive against Dzego was finally stopped."

"Coming together in a crisis is easy. What's hard is keeping that spirit together through the easy times, the boring times, the times when the founding generation dies out and the kids and the grandkids need someone to fight against and have a hard time resisting the urge to start turning on each other."

"So what happened? The co-governance system, I'm guessing?"

"That's one part of it. Each city gave a percent of the votes in its parliament to its neighbors - not enough to threaten the cities' independence and their unique cultures, but enough that future politicians would not have an incentive to ignore their surrounding continent completely."

"But also, they did another thing to preserve unique cultures. They had a period during which they subsidized people swapping homes and citizenships. People who felt more at home in another city were able to move easily and change their allegiance to their new home."

"The government of Freetown made a decision. They passed a new constitution, the one we still have today, which limited the government's power to regulate and tax anything down to a bare minimum. And then they said: this is what we are about. If you are scared of it, go to another city, but if you like it, no matter which city you're from, we welcome you. They even did a personality test without telling people what it's for, and then gave bigger subsidies to people who got what the government thought was a compatible result."

"Wow, I did not know that."

"Actually Arturia did a similar thing, in a somewhat opposite direction. They described their values as being about peace and quiet, research and learning, and they filtered by conscientiousness and test scores. Today, their whole society is sort of like a big university. If you go against the system too much, you get pushed harder and harder to leave and go to Freetown. But if you're fine with the system, you can learn a lot and have a lot of fun inside it. They actually do quite a lot of good science and have reliable infrastructure."

"The strongest legitimacy comes from people buying in to an idea voluntarily - signing the form, knowing exactly what they are getting themselves into and what they are giving up."

"That sounds like selection matters too. Different styles work for different people. Maybe a Freetown made out of people from Arturia would be a disaster, and same with the reverse."

"Of course, now we're several generations into people being just born here, like people get born anywhere else. But for now, Freetown has held."

"Do you think some people are ... genetically more freedom-loving than others? And Freetown got a lot of those people back then, and those genes got passed down?"

"People here bring that idea up from time to time. To me it feels unlikely. All the congestion tolls that I paid to get here, all the turnstiles, I somehow doubt that there can exist a gene that would register all that as being "free". You need to have a pretty deep education to understand how it's free - how it's all part of a market, each person and institution allocating their private property as they see fit, and how all these payments give people the ability to honestly self-report whether or not they truly value something more than the resources that it costs to make it available for them, and how that's better and more civilized than the government just deciding what things it thinks are wasteful and banning them."

"I think there might be a core that even someone from the medieval era would understand. I don't bother you, you don't bother me. We're all each just trying to live our own lives, if we can help each other that's

great, but not at the expense of leaving each other space to be in peace. You don't get the forager freedom of the great outdoors, especially not in the city, but within your own box you are the master."

"At least, that's what Freetown always feels like to me."

"Although ... I do love the economists here. Well, the intellectual culture in general. I like that you can honestly bring up the genetics stuff, and even if people disagree, it's not like Belpaki where they just immediately think you must be an Arctic or something."

"Changing the subject a bit, of course I know why they would never tell either of us, but I am curious what these defense companies are up to. What do they think the threat model is? Same type of invasion as Redshire and Northglade? But what happens if we resist harder? And how are they going to fight back?"

"Well, funny you ask. I actually do know some gossip, although ... this is military, I don't think I'm allowed to tell you."

"Huh, I'm surprised you got into those circles at all. Were they interested in your thoughts as an economist on how to raise more funding for defense?"

Jahn sighed. "Unfortunately, I really can't tell you. Freedom of speech is great for many things, but ... not for military secrets."

* * *

Jahn ordered a dish with vegetables and a tea, and stayed in the restaurant area studying the internet to better understand exactly the question Utaku had asked: what options Freetown still had to raise more funding for defense. He took out a computer, and made a table, scoring different options in three columns: how much money it could raise, how much economic collateral damage it would cause, and how much it risked making Freetown less ... free.

A longhour later, Jahn's watch buzzed. It was Zei, asking for a call.

Jahn picked up his computer and his glass of water, and moved over to a nearby calling booth.

"What's up, Zei?"

"Thank you again for that connection you made with the defense companies last month. I actually have some interesting updates there!"

"Hey, I only connected you to the Economics Institute director, the connection to the companies was his doing."

"But what is it?"

"Well, first let me give you a status update. Do you know what the Dzego folks have been doing with the defense companies' help?"

"I do not."

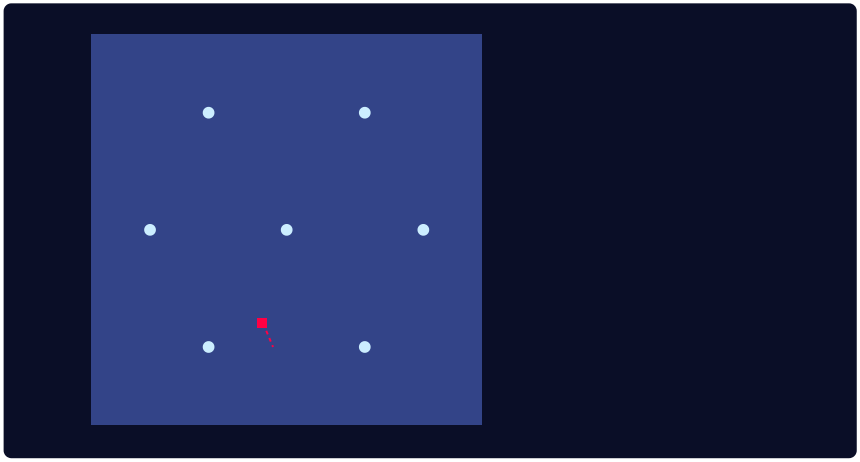
"They've been hunting down Arctic drones in Redshire."

"If you're allowed to ... tell me more?"

"How about I show you more?"

Jahn's computer buzzed, and he saw a file download. It was a view-once file. If Jahn wanted, he could of course save it clandestinely through the operating system, or just take a picture of the screen with another device. But by default, once opened, the file would stay viewable for five minutes, cannot be copied, and would self-delete at the end. Politeness dictated not interfering with these defaults.

Jahn clicked on the file, and a view opened up.



"What am I looking at here?"

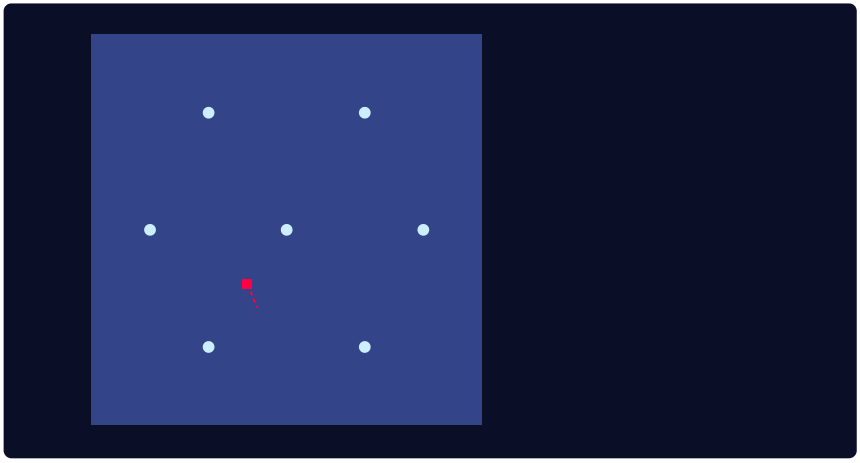
"This is happening right now as we speak. It's our clandestine kill chain on an Arctic ground drone in Redshire."

"The red square is the enemy. The blue circles are seven of our own drones, that we arranged in a hexagon. Their role is to jam communications. Right now, they are dormant, waiting for the Arctic drone to get fully inside."

"So what happens next?"

"Ah, well let's find out together."

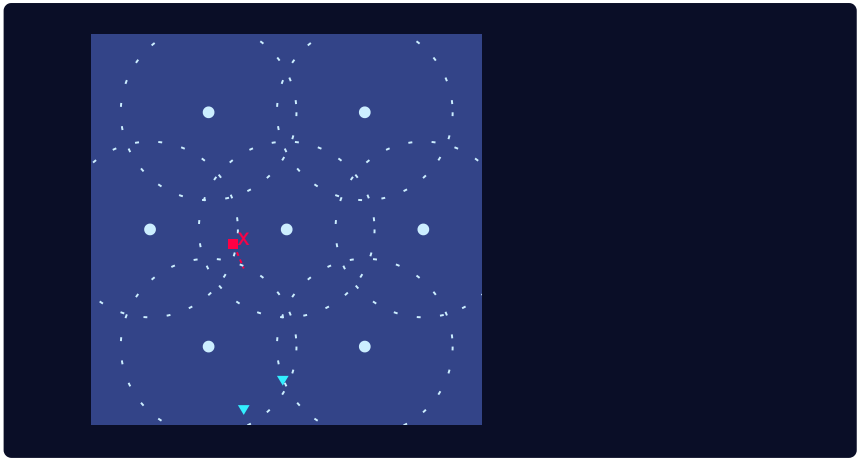
About ten ticks later, another view-once file appeared on Jahn's screen.



"Just waiting a little bit longer."

"I'm excited."

"And ... action."

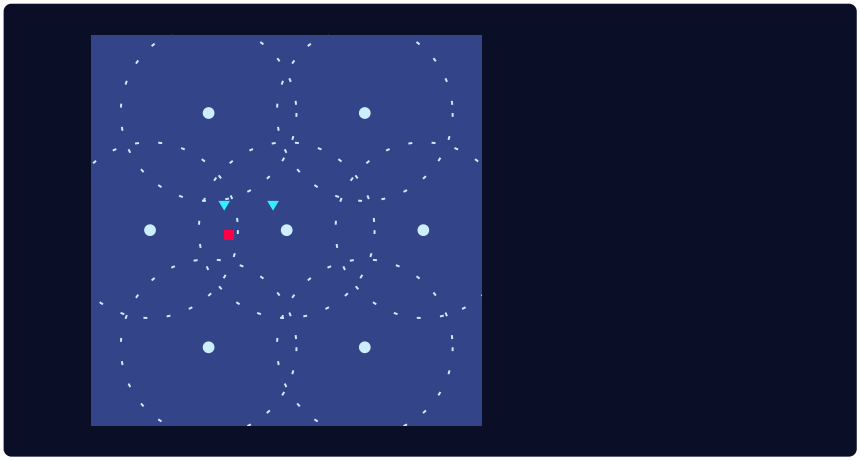


"Do you see it?"

"Yep."

"The jammer drones are jamming. The red X signifies that there was a hit on the target. It's supposed to be two hits: one on its main antenna, and the other on its legs, to prevent it from leaving the jamming area. And the blue triangles are our two attack drones."

The conversation paused for about twenty ticks.



"Still not dead? What's taking so long?"

"Our drones, the triangles, are not actually trying to make a kill - if we wanted to do that, we could have just done that with a bigger first strike."

"So what are they doing?"

"They are dancing."

"Dancing?"

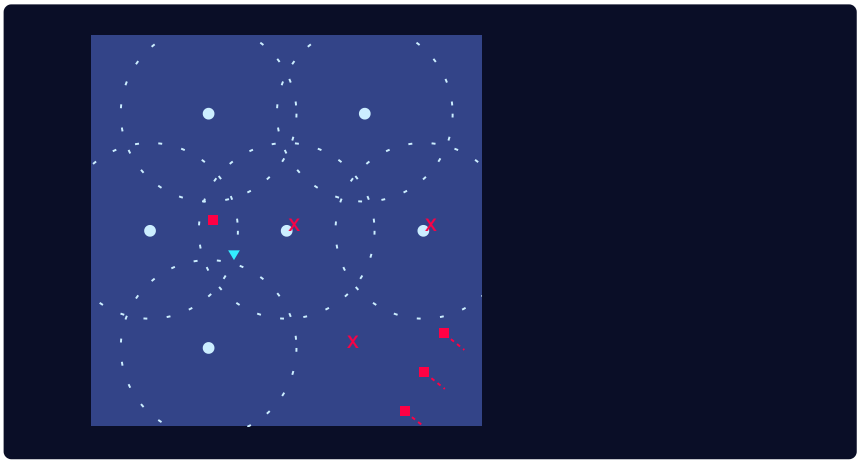
"Their goal is to *survive* as long as possible, trying any kind of trick to avoid getting killed, ideally get mistaken for an Arctic drone or even an animal, and eventually only kill when the target is too close to escaping the trap."

"That's our goal. Using Arctic drones as target practice to improve our ability to kill them is expensive. But using Arctic drones as practice to improve our ability to *survive* them, that we can do, for basically as long as either of the triangles is still alive. Then the seven drones in the hexagon can finish the kill, and make sure to fry the target's circuits so the Arctics don't get any training data themselves."

"Wait ..."

"What's the matter?"

"We have a problem."



"Arctic reinforcements? Are you going to take over and pilot your drones to victory now?"

"Yep, that's exactly what I'm doing. Temporarily have the two damaged-but-alive jammers play dead, lure the attackers to center, jammers reactivate, reform a circle, and then make a kill."

"Do your thing, no need to give me a play by play."

"I'm talking to the AI and letting you listen."

"Circles, finish the northwest square now. Goal: all comms and memory dead."

"Deactivate all jamming except left and top left."

"When it makes sense, reform the circle, reactivate jamming and strike. Maximize surprise."

"No-"

"Wha-"

Jahn stopped mid-word, remembering that he was not supposed to distract Zei.

"They started jamming, I got disconnected."

"Nothing for me to do now. It's just our AI against the Arctics, AI plus human. And if our jamming works, our AI against theirs."

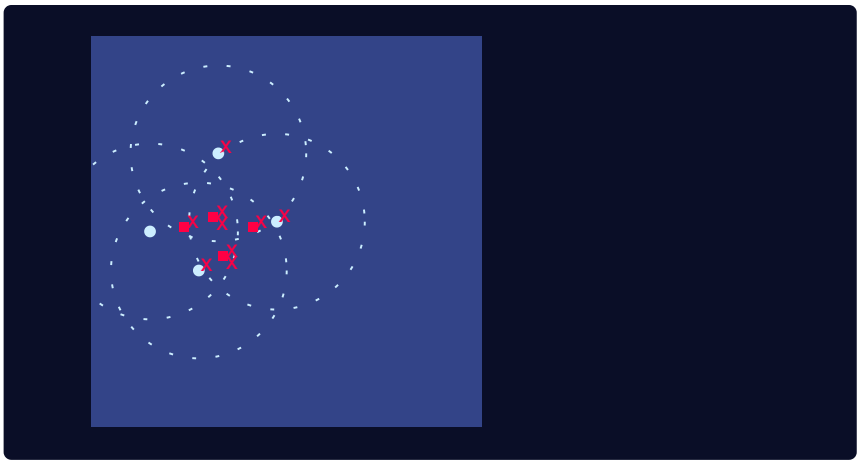
They both waited silently.

Ten ticks.

Twenty ticks.

Thirty ticks.

Finally, the download icon turned on again on Jahn's computer.



"That was really close."

"Wow."

"Only one undamaged drone on our side. Last step is that it will walk around and chip and memory-kill all the drones, first the Arctics and then the rest of our own just in case. It needs to do that before more Arctic drones come."

"Can the damaged ones help?"

"They can't move. They can still attack, but proper chip and memory-killing is best done from ten centimeters away."

"Even then we can't actually be certain that the Arctics won't be able to salvage training data. We can just hope for the best."

"Okay... I gave it the instructions, now I'm just watching."

"I am curious. How does your internet link into Redshire work?"

"Ah, that's one of our other big projects. The problem now is that the Arctics have confiscated all the un-backdoored electronics. There's still many watches, hand devices and compute boxes left hidden in various places, but the Arctics will be able to tell if any device tries to transmit things. So what we do is steganography."

"Two people pretend to have a voice or video call. While the call is happening, the un-backdoored device is sitting nearby, and it's softly playing audio. That audio stream sounds random - it's representing encrypted data wrapped in an erasure code that's highly noise-tolerant

- but we use the same tricks I told you before with the health data to maximize how much data we can hide, so that it appears to anyone or any bot spying on the call that there's nothing abnormal."

"One of ... do you have more big projects?"

"Well, the next step is that we want to actually kidnap an Arctic drone and bring it back to Dzego for analysis."

"Wow."

"Actually, that's what I'll ask for your help one more time for."

"So what do you need?"

"I want you to introduce me to another defense corp. Preferably not through the Institute director, find a different path."

"Why?"

"The company the Institute director introduced does seaborne well, I want to find one of the others, that are better at airborne."

"Also, come on, it's Freetown, this is a market. You gotta talk to multiple people and compare them against each other, to make sure they're not screwing you over."

Chapter 23

Pafogai Du, Dzego · 3724 MISTIME 22

Gladias, Bai and Fin were sitting at a table in a cafeteria, deep underground inside the mountain pyramid.

Between the tables and the wall of the cafeteria, there were green trees on all sides. Two of the walls were covered with vines. The other two walls were covered with large mirrors, making the whole room seem like it was part of some kind of endlessly-repeating infinitely large space. There was a single entrance, with two thick shrubs on both sides blocking the view to the entrance tunnel, in one of the corners.

In one of the other corners, and in the middle of the cafeteria a few meters away from them, a few air filters stood on the ground - clearly recently added, with not yet much thought put in to how to make them fit into the room aesthetically. Downward-pointed fans and violet lights hung on the ceiling.

The three of them were in a private area, with shrubs blocking most of the rest of the cafeteria from view. Two speakers were simulating the sound of cicadas buzzing in a forest in the morning, drowning out the conversations from the other tables - and thus, they knew, protecting their own conversation as well.

Zei walked in, visibly exhausted, and sat down with them.

"That was close. We got ambushed by some Arctic drones, we just barely managed to beat them. Fortunately, we had a full five minutes before the next swarm came to finish ours off, so I think we were able

to destroy any chips that the Arctics could extract to get training data from the battle."

"How are the Freetown folks feeling?"

"They're very excited."

"See, I told you that working with Freetown first is going to be a good way to warm the Kungaupei up to cooperating more internationally."

"You have to give people some quick wins that they can show their superiors or their friends, get legitimacy by performance as soon as possible so people trust you to take the next step."

"Yep, your advice was good. But ... I suspect the next step will be harder than this one. The Freetown defense corps want to cooperate. In Veridia, we'll have to find a way to work around their institutions, that's much harder."

"You can do small projects by working around people. But if you want to win the big battle, you need scale. And if you want scale, you have to do the hard thing, and actually bring some of the bigger institutions onto your side."

"Do you have any ideas on how?"

"Seila says she's going to try to figure something out. I gave her Verdow's contact, so far he's the only senator I really trust. So we'll see."

A robot arrived, putting food for the four of them onto the table. They began to quietly eat the food.

Fin and Gladias finished their food first, and rested for a few minutes, drinking their tea.

Fin took out a book from his bag, and handed it to Gladias.

"Here, let's see how good your Dzegoban is. Open up a page and read."

Gladias grabbed the book and flipped to a random page. He took a look.

ci zan fe da cei bia sia ci jei fe hi dzia lin ji bia sia

---

ci dze fei dzen ji fo gu shan jin dzan ji dzo cei de bun sia de shia sin hie li bia sia

ci gi jan li sia kin fe lo lo ju fe gi da jui zo tin hu lin pin cu cie fan

ci ki sau dan te ki sau pin

ci ju je ka cin pin ka su zo zai

ci ti fo gu shan jin dzan ka dzei zo cai

ci ten min cu hi dzia lin li lia fa

te gi jan jei da jui li fa fe lo jie hu

ci kau min li jie hu gau fe hui pei min cu lau za min cu lau shau min cu pa gu bu ji dzan ju min

ci dzen fe sei gen hu ji sha liu

ci dzo mi li su zo de gi pen li den fe ten cei

te da cei de sau zo tia gei

ci bui je shan jin ka fe bu sin hie li kin

ci dzo tei hu ka jan shau dai dzu li hen fe jan de sia kie hen su

ci tin hu lo fe gia dzan jan li bia pin zo mia fe lo jie sen pin fe pui sen hu

ci dzo jan li shau fe dzo jan zo cu lo sin sia ce hu ka jan pui su de sin sia ce hu dzei li dzo jan shau pen zo tu mia fe ti

ci jan li bia pin fe hi dzia lin de gi pen

ci pui sen hu de shen sin de sia li sie hu cu kin sia je zo de go li ti die sie

ci fe lo zun li dziu kin zo cu lo fe jan zu sin li sin fin zo cu dzo li bia pin

ci pui sen hu li tin hu

ci jan li sia kin fa fe lo lo hi dzia lin li lan pin zo miu sie fe lan pin kai

ci dzo mi sho kan fe lo bu bia sia li lia fa

ci ti li hi dzia lin bia sia

ci zie hu

ci pui ten min ka zu

ci ten cei li sie gie zo ce gau fe da cei bia sia

ci miu ka min sia cu pui ten min de bin dai li sie gie zo ce gau fe hi dzia lin bia sia

ci pui min ka ce ju fe dzo cei

"You think I can do this?", Gladias asked incredulously.

"Come on, try."

"Okay, here goes..."

"From material world small thinking, to internet small thinking."

"Small thinking is minimalism."

"Ah, I see, this is one of those cases where knowing the roots on its own really doesn't get you there."

"Indeed. And the whole project of shrinking the roots down only succeeded once we accepted that that's inevitable."

"Containing inside the green, forty years ago, a big pattern change in beautiful thinking in the whole world was minimalism."

"That first bit is a set phrase, it means 'including Dzego and Veridia'. Beautiful thinking is aesthetics."

"Many people recognized that having many physical things is not only line-heartedness but also a locked room."

"Line-heartedness is a feeling of simplicity, or convenience, a locked room in this context is more like a prison."

"A complicated room implies a complicated heart."

"Good!"

"Living with the warm-heartedness - happiness? - of having few things is freedom."

"Great!"

"That this happened forty years ago has a logic - a reason?"

"Yep."

"Electric machines and the internet began to exist."

"This implied that many people gathered physical things ... wait, physical things gathered by many people, began to be unwanted."

"I would say unnecessary here."

"Hand devices caused to be unnecessary aggregate-and-group devices, music devices, sound conversation devices and ten other devices that one would previously have."

"Aggregate-and-group devices should be add and multiply devices - calculators. Sound conversation devices are phones."

"It can contain books without an upper lid."

"Unlimited."

"Many parts of that we all live are moving to the electric world."

"Many parts of all our lives."

"As a result, the complexity of the physical world stopped behind."

"Was left behind."

"In the nearby few years, we are seeking a different change in pattern."

Gladias paused for a few moments.

"Okay, I'm tired. But I am curious. Can you just tell me what the rest means?"

Zei picked up the book.

"Social media is not always good for people's mental health. Much of the interest in privacy is driven not just by fear of the Arctics, but rather the sheer volume of everyone talking about everyone else, and the unpredictability with which random events from people's private lives become public discourse."

"People are afraid of many parts of the internet, especially in other countries where privacy technology is not as strong and well-understood. Location tracking, behavior profiling, all of it. And it's not just privacy - people are recognizing that the internet can be as addictive as drugs."

"And so now we're slowly seeing the emergence of a different kind of minimalism: internet minimalism. Disconnect. Do things on your own device. And like physical minimalism was made possible by the growing capabilities of the digital world, internet minimalism is made possible by the growing capabilities of AI and local software - you can have your whole world on your own device."

"Fascinating."

"Gladias, you're doing very well."

"Thank you. Now your turn. How well do you understand the Order of Steering's governance rules?"

"There is a collection of tax rubrics. For each rubric, twenty one Keepers are randomly chosen - seven added in each year, for a three year term. The job of the group is to propose and vote on changes to the rubric. The twenty one are split into three sub-groups of seven, all

mixing first, second and third years. Any Keeper can make a proposal for a change to the rubric. Each sub-group discusses every proposal independently, they don't know who the other sub-groups are. After a few rounds of discussion, all twenty one Keepers vote on each of the proposals, and the ones that pass, with at least thirteen out of twenty one votes approving them, get accepted."

"There is a separate mechanism, the Sentinels, who are randomly selected and put into groups that audit individual businesses. There, it's three groups of three. They have a very specific task: determine what tier on each rubric a business gets. And then smaller businesses have audits done by Acolytes, Order members in training. For an Acolyte to get accepted, they have to both pass Senate committee approval, and be in the top ten percent by prediction score - by how well their votes on the small portion of their audits that get randomly sampled and re-done by Sentinels agree with the Sentinels' results."

"Now, what about the Courts?"

"Same mechanism as the Sentinels, though with five judges instead of nine. Also, in civil cases either side, in criminal cases only the defense, can appeal a ruling. There's no hierarchy, it just selects another random sample. There's a common law rule that if you submit a case - either an appeal, or even a first-time lawsuit - that has no merit to the point of being bad-faith, then that case can end not just with you losing, but even with a penalty judgement against you. That's the protection against what we call denial of service attacks."

"Actually, good that you mention that last part. My co-father, Vil, was in a long lawsuit against, well, basically his former employer, for doing some financial trickery to fire him and get away without paying the pre-agreed severance. He won the first round. The company appealed and lost. Then they appealed again. And just yesterday, the court struck down their second appeal, declared it bad faith, and tripled the amount that they have to pay Vil and the other workers."

"Glad to hear the system works!"

"That's always the most important question: whether or not it actually works. The Parliament hearings approving new Keepers and

Sentinels, for example, barely work at all at this point - at least in my view."

Bai spoke up.

"This is all really good info, Gladias, and thank you Zei for getting up to speed on it so quickly. If we want to affect Veridia, understanding how these institutions all work today is just as important as the military side."

"Anyway, I think Den is waiting for me and Zei in his private room. We should start heading over."

"Sounds good", Zei replied.

Zei and Gladias simultaneously raced to put their watches against the small green circle on the corner of the table. They ended up both pushing each other's wrists too far away to make the connection. Fin joined in, and slapped his watch against the circle. A moment later, his watch buzzed.

The four of them all stood up, and began to walk to leave the cafeteria.

* * *

Gladias was in bed, getting ready to go to sleep. Suddenly, his watch buzzed.

From		Message	Time
Seila	→	[requesting call]	91454

He had only seen messages in this shade of red twice before: once when Seila had messaged him when Northglade was attacked, and once when she had called him about Delwart.

Emerald was letting Gladias know the level of importance and Seila's mood and, Gladias knew, this was already a large part of the message.

Gladias's heart raced, as he answered the call.

"DO YOU KNOW WHAT HAPPENED TO LILY?", Seila screamed.

"I ... do not. I just remember hearing that she got eighty-one percent on her latest exam. What happened?!"

"I entered her room. She was talking, typing on her hand device. And I could see what she was reading and saying in the mirror."

"For a few minutes, I just stared. I could not believe it."

"What was it?"

"She was in some kind of chat group, and they were sharing Arctic propaganda. Cheering on Lord Ephelion's messages. Hoping that the Arctics would come to the rest of Veridia soon."

"WHAT?"

"I snapped. I called her out on it, and asked how she could possibly be betraying us like this, when Febric and Hreda were stuck a thousand kilometers away because of them."

"What did she reply?"

"That the Arctics were right. Their philosophy, their idea that some people are worse and other people are better, and that the better people should rule everyone else, and the most important thing in life is to strive to be one of the better people, was the thing that had motivated her, and given her a reason to study and work hard. That our unconditional love was poison. That the Arctics' philosophy - appreciation and love only as reward for success - actually brought her out of her hole, when nothing we did helped even a little bit."

"That's ... horrible."

"I screamed at her for ten minutes, and shut the door."

"I cried for a longhour. And then, when I opened the door again, the window was open, and she was gone."

"Can you ... track her? Through her watch?"

"Already got Mov to do that. It was a waste of time, she threw her watch onto some random autobus."

"Wow. Now I feel so bad. We really haven't been doing anything to actually motivate or encourage her."

"GLADIAS, SHE GOT BRAINWASHED BY THE ARCTICS. DON'T YOU DARE GET IN YOUR PHILOSOPHER MOOD AND TELL ME THEY HAVE A POINT."

"HOW COULD YOU HAVE BEEN SO NEGLIGENT?"

"Okay, hold on. Not seeing through Delwart, that one is on me. I was dumb. Mov was dumb. I admit it. This, this is on both of us. You're Lily's mom. You saw her many times - before she went to Vil and Daia's, when we visited them during her time with them, and after she came back. You had many chances to make things better, just like I did. And you didn't succeed, just like I didn't succeed. So let's solve this problem together."

"YOU DON'T UNDERSTAND AT ALL, DO YOU?"

Seila hung up the call.

Chapter 24

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Gladias checked his watch. The time just hit 20,000. He had been awake all night after his call with Seila. He had tried to go to sleep twice, but gave up after twenty minutes each time.

He could not go to sleep knowing that three of the four children he loved dearly were now, in one way or another, in the hands of the Arctics.

He could not go to sleep after Seila gave him that kind of vote of non-confidence, and then cut off communication without giving him any chance to even apologize.

And he *definitely* could not go to sleep when both of those things were happening to him at the same time, and anyone he could hug for emotional support was thousands of kilometers away.

Finally, thinking as much about redeeming himself in Seila's eyes as about his daughter's safety, he decided to do something unorthodox: he called Zven.

Zven picked up immediately.

"Dad?"

"Are you still in contact with Lily?"

"Her icon went green about forty minutes ago. I think she logged in from a new device."

"Good."

"Zven, are you willing to help us save Lily?"

"Of course! Anything for my sister."

"Then here's what I want you to do. Stay in touch with her. Reach out, ask her what's up. Don't give any indication that you disapprove of her decision. Even show sympathy for her. Just stay in contact, and try to learn where she is."

"Dad..."

"What is it?"

"That's like ... *exactly* what Mom told me to do a longhour ago. She gave me a lot of good advice for things I should and shouldn't say."

* * *

Gladias opened the door, walked through the hallways until he left the hotel area, and then turned left and kept going through the tunnel.

This was now his eighth day living entirely underground, inside the mountain pyramid at the south end of Pafogai Du. Soon after he first arrived in Dzego, he had been relocated there from his original hotel. The new recommendations were for all "important business" to move to underground locations, and as a de-facto Kungaupei advisor, Gladias certainly qualified.

For over two months, he would take daily twenty minute walks every day along the trees around the perimeter halfway up the mountain pyramid, and sometimes a longer walk down and into the nearby forest. For the past few days, he had been spending each day in the library, learning about different aspects of Dzego and having almost daily calls with Verdow to think through how to convince the Veridian military to cooperate with the Kungaupei.

And today, after what happened last night, and after only getting one longhour of sleep after it, he did not have the will to go outside at all.

Gladias walked slowly, taking many breaks to stare at the wall and at any side rooms that he walked past on his way to the library.

High-tech industry of all kinds - biomedical, chip manufacturing, even parts of car production - had moved into the mountain pyramid since he came in, he noticed. And now that he thought about it, there

had not been a single successful Arctic attack on Dzego in the news since Northglade had been annexed.

He also noticed that the number of air filters and ultraviolet lamps scattered around various locations had increased more than tenfold. At least the Dzegojan manufacturers and the Bansunpei are doing a good job, he thought.

He passed by the entrance to a cafeteria. Trees about two meters tall growing inside clay pots were now inside, he saw, along with various shrubs. The greenery seemed to play a triple role of giving the people sitting inside a feeling of calm from seeing nature, reducing CO_2 levels, and splitting up different sections of the cafeteria to improve privacy.

After about fifteen minutes of slow walking, he finally arrived at the entrance to the library.

He noticed that the slogan at the top of the door had changed since the last time he had been there.

sun jan li shai fe to dzan dzie ka jie dau fe sei sui
ji cen hai zun
ci sun gau jan li gei shai fe to dzan dzie de mia
ka ti dzie ka pu she

He stopped, and took a moment to read.

"The student says: tomorrow I want to go to a magical and far away place. The teacher replies: for the sake of tomorrow, what have you done today?"

To Gladias's surprise, the admonishment gave him a small burst of energy, allowing him to raise his hand against the green circle to open the door, and enter the library.

He walked through the door, and then slowly turned right and started walking toward one of the computer terminals.

Bai noticed him, and walked up.

"What's the matter?", she asked.

"I'm fine", Gladias replied in a quiet and flat tone.

"Well, I know enough to know that you're not fine. Though I also know enough to know that forcing you to divulge the source of your darkness if you're not comfortable will make you feel even worse."

"So don't worry, I won't ask any questions."

She walked up to Gladias, and gently hugged him.

"I just learned that Lily has been eating up Arctic propaganda for the last year, and Lord Ephelion's sermons made her feel valued and appreciated more than I ever did. And less than a minute later, I learned that she ran away from home. And a minute after that, I learned that Seila now hates me and doesn't want to talk to me."

"What do the Arctics say that someone like her would find compelling?"

"That unconditional love is meaningless. Love is only worth anything if you love some quality that the object of your love has. And so to receive love, you have to deserve it, you have to earn it. Love has to come not as a constant background radiation, but as a reward for winning."

"The most important thing is to always strive to improve yourself. But if you can improve, that means you can be better than your former self - and at that point, you've basically admitted that one person can be better than another. So the whole idea that we're all equal, that the most important thing about us is our equal humanity, is total nonsense."

"You know ... I kind of agree."

"What?"

"I don't mean in some absolute sense, that that way of thinking is healthy for everyone. It definitely isn't. If you think it does, then of course once you think you're the best, you're going to want to start conquering everyone just to prove it. I meant more ... that that kind of thinking was good for me."

"What do you mean?"

"My parents died when I was very young. In an explosion, triggered by the Arctics. I grew up with my aunt, who basically just ignored me. No one loved me unconditionally. I had to start figuring out how to earn money on my own when I was twelve. I learned all my social skills from that. The only appreciation from other human beings that I got, was when I provided something of value to them, or when I impressed them with my success. The loop that you said Ephelion talks about, of love as a reward for winning, that was my childhood. And, I got second place in the Pafogai Du Minpentai games because of that."

"Who got first place?"

"Zei. And he had a motivation to succeed that was even more pure, even more beautiful, than my desire to succeed in order to prove to the world that I deserve love because I am strong."

"And what motivation was that?"

"Curiosity."

The conversation paused for a moment. Then Gladias spoke up again.

"When you hugged me a few minutes ago, were you not offering exactly the thing that Ephelion says is poison? Love as a reward for weakness?"

"I would not have done that a year ago."

"And yet it worked."

"I would love to say that I did that because I gave up the false god of greed and converted to the true religion, that of valuing curiosity. But even that is false. If I was motivated by curiosity, I would probably have just directly asked you what's wrong, and that would have triggered your resentment."

"It definitely would have."

"So what's your conclusion?"

"That maybe there really is no master formula. There are different bags of tricks that work for different people. I'm not going to say I'm fully unopinionated between all the different motivations that human beings might have - I'm very comfortable saying, today, that curiosity

is almost always a better motivation than greed. And resentment might be the worst of all. But I don't think there's a one size fits all for everyone."

"So how do I save Lily? And, for that matter, Febric and Hreda? And how do I get Seila to stop hating me?"

"Well, maybe that's the one place where you do need to become at least ten or twenty percent Arctic. Your family will be back together, fully, if we can win. Think about it from Seila's perspective. The only way *she* can be happy, is if she can hug Febric, Hreda and Lily again. And there is no way to use emotional support and love alone to bring the three of them back. The only thing that *will* bring them back, is if you win - by helping Veridia win against the Arctics. So let that motivate you to try harder, so that in this next battle you can be a winner."

* * *

Gladias and Zei walked down the tunnel, and down the stairs. After a few minutes, they got to a sealed door. Zei tapped his watch against the green circle beside the door. The door slowly opened.

They saw a long room, and at the end of the room there was a large table. At the table, a man was sitting wearing a shadowy cloak.

"Welcome to you both!", Den exclaimed.

"Good to see you again, Den", Zei replied.

Gladias cleared his throat. "mi cin pin fe lo kin do", he added.

"Wow, even your accent is good now. And I'm happy to see you too, Gladias."

"By the way, please sit down."

Gladias and Zei both stepped forward, and sat down on two chairs opposite Den.

"So, Gladias. You called this meeting. What did you want to talk about?"

"Well, so far we have not figured out any amazing plan for how to convince the Veridian military to cooperate with you."

"But, I have a somewhat unorthodox idea for what you can do in the interim."

"And what is that?"

"Spy on us."

Den briefly stared at Gladias in surprise. His face almost immediately smoothed, as though remembering that it was not compatible with his self-image to be openly expressing any excitement or other unplanned emotion.

"I think I see the reasoning. But I want to hear yours."

"The Arctics have made a very deep infiltration into Veridian society. They've corrupted many of the politicians. They've corrupted the military. The only institutions that seem to be resisting are the ones that have some stronger privacy built in: the Order of Steering, Graph Funding and the Courts. Though even within the Order, the intrusion got in pretty deep."

"So if you can spy on Veridia, you might be able to spy on the spies. Figure out as much as you can about where the intrusion into Veridia is, and perhaps even try to counteract it. Maybe even put that information more directly into people's hands, and empower them to try to solve it without going through the central institutions."

"You know the risk, right?"

"If you get caught, a lot of trust between Veridia and Dzego is lost. And then we're divided completely and the Arctics win."

"So why should we be okay with that risk?"

"Well, first of all, we're already pretty late in the game and this is a desperate situation. A ninth intervention turn, as you or Zei might put it."

"Second, I'm not just giving you the one-word suggestion 'spy'. I'm offering to give some very detailed specifics about how the Arctics operate, as well as connections to a whole bunch of Keepers that we're now in contact with, so you can better track the Arctics down."

"Third, I think we should tell Verdow, and convince him to take responsibility, along with me, for proposing and even running the project if this all gets out."

"I like this plan. Zei?"

"You're asking me?"

"Of course. You're already one of our best cryptographers. You've been coordinating the work with the Freetown corps. What do you think?"

"For the past fifteen years now, security on digital devices has been very strong. Spying is primarily a physical-world thing, the physical world is much harder to secure. I think we should take Gladias on a tour of some of the stores and the fabs, like the one you showed me way back then, and we can try to think, what would be a reasonably unobtrusive way to get more information on Veridia."

"Actually ... I think Veridia buys its security cameras from Dzego. The ones that are already in a bunch of places, and that they wanted to put everywhere else too. Veridia used to require cryptographically enforced limits on what the data gets used for if the cameras are in public places, but that got watered down a few years back, an administrative change by the Transportation Ministry no one paid attention to. Could you just ... backdoor those?"

"If it was that easy, then surely the Arctics must have massively infiltrated Veridia already?"

"Yes?"

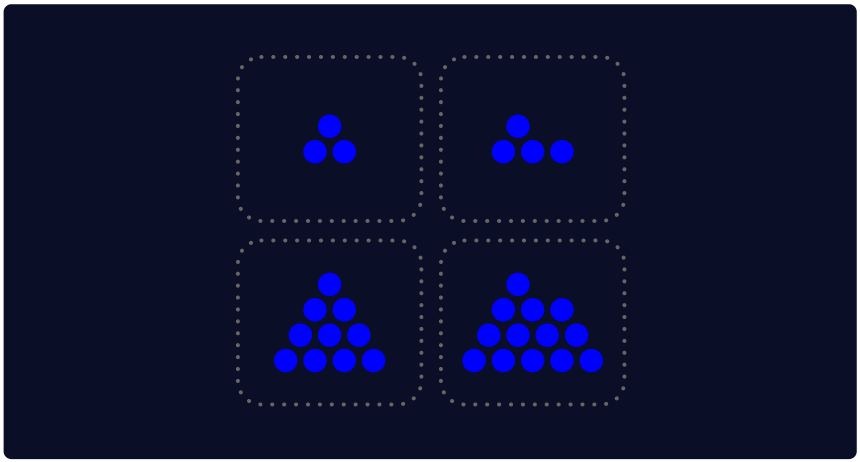
"Have they?"

"Yes? That's what I've been trying to tell you all along!"

Chapter 25

Pafogai Du, Dzego · 3724 FROSTIME 1

Gladias and Zei were sitting at a table in an enclosed room, with the tabletop functioning as a screen. On the screen, there were four piles of circles.



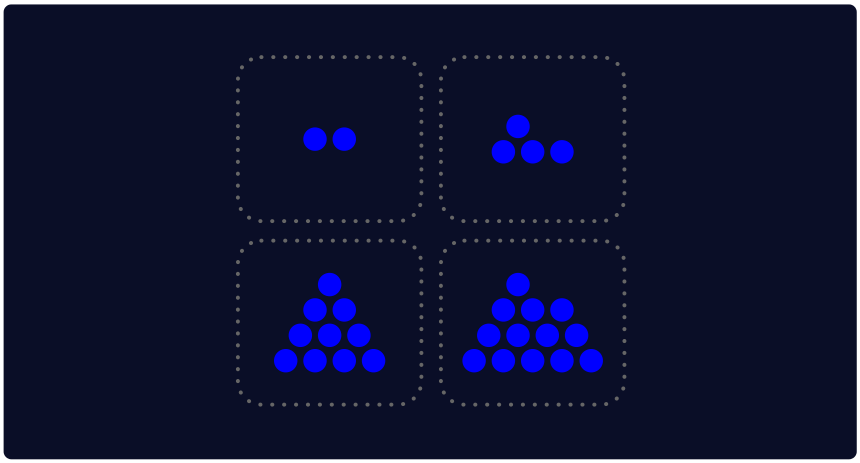
"Okay, so what do I do here?"

"It's your turn first, so you pick one pile, and take at least one circle from the pile. You can take more, but you have to take at least one, and you can only take from one pile."

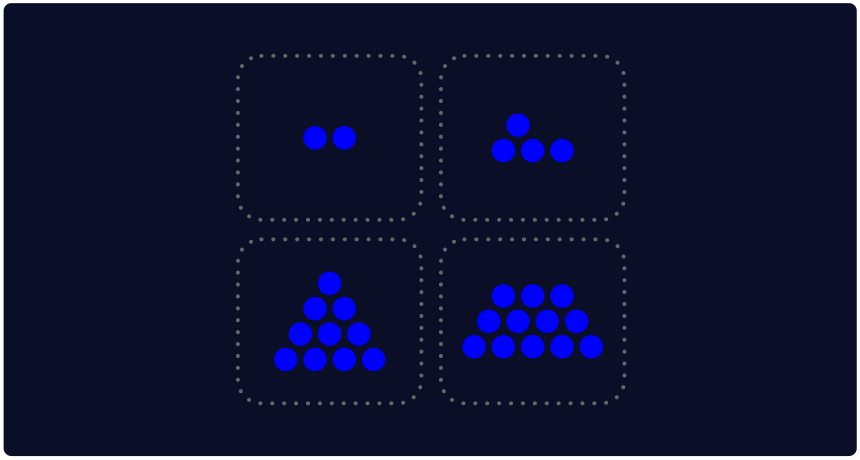
"Whoever picks up the last circle wins."

Gladias did not feel like he had any strategic insight about the game, and so he decided he would learn by playing.

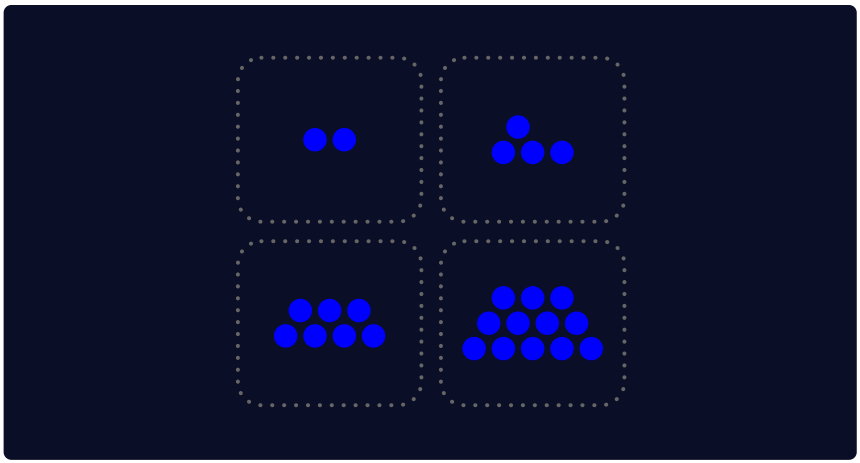
He tapped one circle in the top left box with his finger.



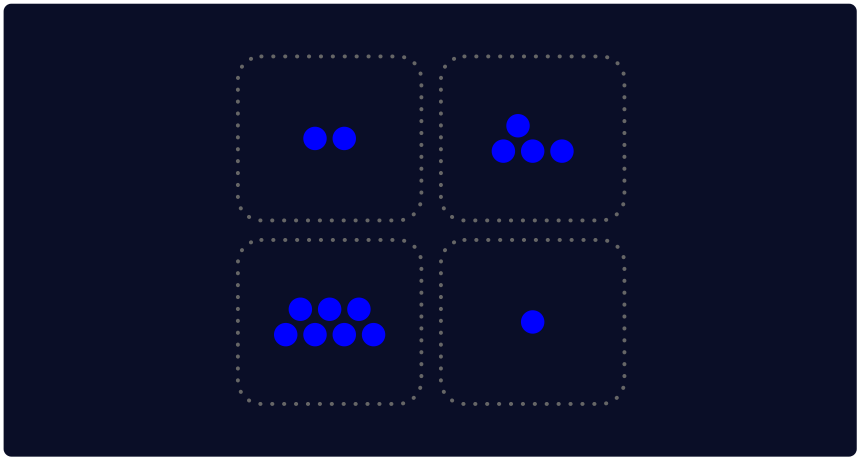
Zei immediately responded by tapping the circle at the top of the bottom right box.



Gladias decided he would try something different. He tapped three circles in the bottom left box.

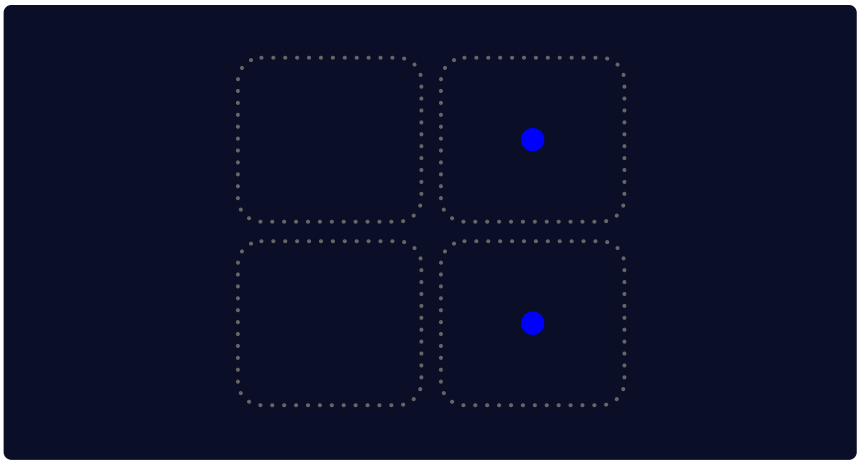


Zei responded by tapping every circle in the bottom right box, except for one.



Gladias tapped on all the remaining circles in the bottom left box. Zei responded by simply tapping on one of the circles in the top right box.

Gladias tapped on all the remaining circles in the top left box. Zei tapped on two of the three remaining circles in the top right box.



"Okay fine, I lost", Gladias admitted.

The only next move he could make would be to remove the last circle from one of the two boxes. This would leave one circle on the board, which Zei could take and win.

"That was impressive."

"It's a game that we would play often as children, especially before more modern things like Minpentai came along."

"So is this the part where you tell me your secret sauce?"

"You know me well. There is indeed a secret sauce. It's a simple algorithm. It requires mental math, but once you learn it, and if you can do the math without mistakes, then as long as you get to choose if you're going first or second, you will always win."

"So tell me."

"Take the XOR of the circle counts of each box. So, write each count as a number in base 2. For each position, count up how many ones there are in that position. If there is an odd number of ones, in our new number we will put a 1 in that position. Otherwise, we put a 0. We'll call this the XOR sum of the board."

"Wait, slow down, give me an example."

"Let's take the board we started with. The counts are 3, 4, 10, 13."

"Okay."

"What is that in binary?"

"11, 100, 1010 and ... hold on, 1101?"

"Yep! Though let's add some zeroes to make them the same length and write it down."

He opened up his hand device, and wrote down the four numbers in a grid shape.

```
0011
0100
1010
1101
```

"Now, count up along the columns."

"They all have two ones. That's even. So the XOR sum is ... zero everywhere, so it's zero?"

"Exactly. Now, remember what your first move was?"

"I decreased the first count from three to two."

"Correct."

```
0010
0100
1010
1101
```

"What's the XOR sum now?"

"Lowest-order digit has a single one now, so it became odd, so it's 1?"

"Exactly. Now, remember what I did?"

"Sorry, I forget."

"I took one off of the last count, decreased it from thirteen to twelve."

```
0010
0100
1010
1100
```

"Oh I see! And the XOR sum is back to zero!"

"Exactly. So whenever you move, the XOR sum started off zero, and any move you make can only make it nonzero. But if it's nonzero, then I always have *some* move that can set it back to zero again."

"Wait, is that true?"

"Take the highest-order digit of the XOR sum that's a 1. There have to be an odd number of circle counts whose digit at that position is a 1, meaning there has to be at least one. In that box, change its count from N to N XORed with the XOR sum. That's always a decrease, because one digit flips from 1 to 0 and all other digits that change are lower-order."

"I ... maybe get it?"

"And so whenever *you* move, you can only change the XOR sum from zero to nonzero. When *I* move, I set it back to zero. And the final move, the move that clears the board, sets the XOR sum to zero. So that means the winning move is always mine to make."

"Now, if the board started with an XOR sum that's *not* zero, then I would win if I go first. But if the board starts with an XOR sum of zero, then as second player I'm guaranteed to win."

"Interesting!"

Gladias sat back and thought for a few moments.

"I've never seen a game that favors the second player before."

"It doesn't necessarily! It only favors the second player if the initial board has an XOR sum of zero. But games that favor the second player are more common than you might think."

"Normally there's some position, some territory that's valuable, and the first player has an advantage because they can get to it first."

"Can you think of what kind of underlying reason might make a game second-player favoring?"

"What do you mean?"

"What fundamental advantage does the second player have?"

Gladias paused to think again.

"They know more when they get to make their move."

"Correct."

"Here, I would make my move, and you would be in a position to make your counter-move *based on* my move, and so you always know the exact move that will bring things back into balance."

"Exactly."

This time, Zei paused. After a few moments, Gladias spoke again.

"You know, if you were the first player, you would have probably won too."

"And why is that?"

"Because I didn't know the trick. To you, there was a mathematical structure that has one desirable state - the state where the XOR sum is zero every time I move. And once you reach it, you're locked in forever. To me, it was just chaos."

"Indeed. Entropy is always in the eye of the beholder."

"The game is a tower of different kinds of rules", Zei continued. "At the bottom, the game is irreversible: you always have to take at least one circle away in each move. At a level higher than that, there's the XOR sum, and that is reversible. Whatever change you make, I can always set it back. But at a level even higher than *that*, the fact that it's your turn when the XOR sum is in balance is the irreversible thing. Once you let the game slip into that state, if the other player knows the trick, there is no way to get back out."

Zei went silent again. After a few moments, he changed his posture, signaling that the warmup was over and the real line of questioning was about to begin.

"Now, here's your next question, Gladias. Is the fight for Veridia a first-player-favoring or a second-player-favoring game?"

Gladias's eyes opened wide, visibly surprised at the new direction the conversation was going.

"What do you mean?"

"The Arctics moved first. They set up or took over spying infrastructure. They pressured, bribed or threatened Veridian officials. They made a highly coordinated campaign to try to take over even the Order of Steering, and get it to do their bidding and nudge Veridian society to not bother resisting. And finally they attacked Redshire and then Northglade. Do they win because they already captured the important positions, or do we win because we have a counter-strategy?"

"I see..."

"On the field of battle, right now they have the advantage, and not even for any first-player reasons - their technology is just better. But in the battle for the institutions, we have a counter-strategy. We can play the same game they do, but we have the advantage because we're seeking goals that most of the people we would be manipulating actually want."

"What's the word for manipulating someone to do something that's actually in their interest, by just showing them how it's in their interest?"

"This sounds like it has a really simple answer."

"Convincing."

"So whom do we convince, and to do what?"

"You tell me."

Gladias again paused. Finally, he spoke.

"What about the Courts?"

"Go on."

"We run the same kind of influence campaign on the Courts that Delwart and whoever else did on the Order. And we convince them to do ... something that fixes this whole situation."

"You learn quickly, Gladias."

"Now, why do we have the advantage here?"

"Because what we're doing is actually good for Veridia. For Delwart to convince the Order members to go against Veridia's interests, he had to make different arguments for each person, tailor-made to their values and their weaknesses, and on top of that threaten them either

implicitly or directly. We don't have to do that. We just have to have a clear thing that we're asking for, and tell the story directly, as it is. When you lie, it's easy to get tripped up: a lie is essentially a simulated virtual universe, and it's very hard to make a virtual universe that's consistent without actually simulating the whole thing. But the truth gives you self-consistency for free. So we tell the truth."

"So now, the next question. What is our ask for the Courts?"

"Let's talk to Verdow."

"You set up that call. I'll quickly ping Den and get the latest on how our spying in Veridia is going."

* * *

Gladias tapped on his watch.

Message		To	Time
[requesting call]	→	Senator Verdow	78145

After a few moments, Verdow answered.

"How are you, Gladias!"

"Senator, we have a new idea."

"And what is that?"

"We want to do the same thing to the Courts that the Arctics tried to do to the Order. Seek out enough of their judges, convince them to do something that will let us fight back against the Arctics properly."

Zei saw the opportunity to join the conversation.

"Most of the admins running the cameras just installed our update. It also called home to report each camera's existing state; it turns out a bunch of them had been taken over by the Arctics already. So now we have lots of backdoors, we've already identified over a dozen judges, and we've killed some of their backdoors as a bonus - though that does mean we have to be more careful now about not breaking our cover

any further. So we can play the game just as well as the Arctics can - probably even better, because our goals are goals that the Veridian judges should actually want."

"And so we need to figure out what the thing we want to convince them to do actually is."

Verdow seemed to laugh, more with admiration than anything else.

"I knew you guys could pull something together."

"Let me think about this..."

For a few moments, silence. Then, Verdow slowly began to speak.

"Okay, I have one crazy idea."

"In Veridia, one of the most basic laws is that the military is required to defend the country. It's the fourteenth Ground Law, it's written right there in the documents. In Northglade, they completely failed to defend the country, and they are barely even trying to make any serious reforms in response."

"Therefore, the Veridian military are lawbreakers. Therefore, the Courts can fire them. And if the parliament tries to stop that, then, well, the parliament is legally the collective head of the military, so the Courts can fire them too."

"The Courts ... have that much power?"

"In principle the courts can do many things."

Verdow paused, and then decided to enter a philosophical tangent.

"You see, Gladias, law is not like a computer program sitting inside a cryptographic network. The goal of law is not to try to eliminate the subjective nature of interpreting what's right and wrong in the world. Gather, the goal is to structure it - put it into channels that make it more predictable, and more focused, so that each case doesn't re-litigate the fundamental questions of what is good or bad for humanity."

"In a normal case, it often does approach being quite programmatic, and that is the point. A business dispute in court not about who is a good person, it's about whether or not one party committed the legally-defined act of fraud. But in any exceptional case, a lot of flexibility remains - and indeed, any situation that is

settled case law today was unknown and flexible at some point in the past."

"And this latent flexibility is key to giving courts space to make good decisions. Man was not made for categories. Categories were made for man. Declaring the entire military in violation of its sacred duty is crazy. It's something so crazy that I don't think anything remotely like it has been attempted in Veridian history. But Veridia is also in a crazy time. And so at this moment, a crazy judgement like that is exactly what Veridia needs."

"Winning here is not just a matter of blasting out into the public the argument that I just made, where the judges will see that of course it's correct and immediately act on it. You have to convince them. You have to sit with them. Explain to them why this is not just a valid legal argument, but also good for Veridia. Explain to them why replacing the entire military leadership, and if needed the parliament, is something that will rescue Veridia, rather than throwing it into further chaos. And convince them that enough other judges will go along too, and that when they make their decision, the public will go along too. And do all that first quietly than suddenly, so that the Arctics can't react."

"I like how we don't even need to convince you that the plan is worth trying."

"The only other plans I've heard in my parliament meetings are to keep doing what we've been doing all along but 'try harder', and outright surrender. I judge those two as having an even lower success probability than this."

Chapter 26

Veridia · 3724 FROSTIME 5

Zei, Bai, Gladias and Den were sitting in a small enclosed cabin in an aeroplane, with Zei and Bai facing forward and Gladias and Den facing backward. All four were looking silently out the window, enjoying the view of a mostly flat, green landscape. The main visible features were a row of small mountains and the city of Meldan further in the background.

An announcement alarm buzzed. The four of them snapped out of their trance-like state, and prepared to listen.

jo hen jei fe fei go de di mei dan du

"Your turn to translate, Gladias!", Zei prodded.

Gladias scrambled in his mind to understand the words quickly enough to give the answer before the next sentence began. After a few moments, he blurted out:

"Everyone, welcome to Meldan, Veridia."

ti zun no tei li ze fo fi shi ci cin ho li ze ci bia tei gei ka fiu den min ba fia zui

"The local time is 74,000 and the temperature is seven degrees. After a short time, the flying machine will descend to the ground."

ti tei li dzai hen fe lo kin fe hai bun

"Now is a great time to look at the beauty outside."

hai ka sho kin ce fe fei go de bun ji dze kai zui liu

"Outside right now, you can see Veridia's beautiful earth sheet of green plants."

"Landscape of green plants", Zei corrected gently.

biu ka di ge le bo gai dei li lia

"On the right, there are the Gelebor mountains."

cen hai ka ge le bo gai dei de dziu hai ka ce kin fe di mei dan du ci ti li sa dzu du fe fei go

"Far outside, continuing beyond the Gelebor mountains, you can see the Meldan city. This is the capital city of Veridia."

di mei dan du li bai

ci di mei dan du de jan zi li pa pi pa fi ha

ci dzen fe gi hie pen

"Meldan is a large city. The population of Meldan is 1.1 million. It has many different districts."

dzu ka kin ce fe bia gai dei

ci fia sai pen ka dan fe gi jan

ci sei sai pen ka jen bin pei de pen li pai

ci mui dau fe ti li cin pin cu bun

"In the center there is a small mountain. In the lower half there are many people's homes. In the upper half part of the government works. Hiking up it is happy and beautiful."

gai dei de zei biu ka pai hui pen li lia

ci pai hui pen fei ka fei go de gi bai pai hui li lia

ci sa dzu pai hui li dzen fe ten min de bin dai cu man jei cu su kun gau

"East of the mountain, there is the business district. In the business district there are many large companies. The main industries include computer software, finance and health protection."

gai dei de zei gei ka di ka lia mai pen li lia

ci ti pen li lu fe bai kai zun

ko ti dzen fe gi pui gau ji dan

"South of the mountain, there is Kalimar district. This part looks like a forest, but it contains many hidden homes."

gai dei de zei dzun ka gi dan pen li lia

ci ti ka gi jan li san cu sun cu mo

ci ti ju fe gi kai de sai hui cu gi dau kin zun

"West of the mountain, there is a large residential district. Many people sleep, learn and eat here. It has many categories of food and many go-and-see places."

"Tourist attractions."

ti li tin bia pen fe di mei dan du de ce kin ji gi mui sun jui

ci di mei dan du fei ka ji gi shan jin tei ka jan ki su

te zen hu ka ki kia sin fe ti du

"This was only a small part of the many interesting things in Meldan. Even if a person lives for years in Meldan, they will not know the city completely."

jie fe lo cin pin fe lo di mei dan du cu fei go li bun

"We hope you enjoy Meldan and the beauty of Veridia."

"Well done. Only two months and you're almost a local already."

Zei paused for a moment, and then opened his mouth again to speak slowly.

"ci jie fe lo cin pin fe di pa fo gai du"

"I enjoyed Pafogai Du very much."

"I hope you come back."

"Of course I will!"

"Wait, I just realized ... Pafogai means 'fourteen stone'? Where did that name come from?"

"Think of the periodic table."

* * *

The aeroplane came to a stop. Through the window, they could see over a dozen large autobuses and cars approaching.

"The larger black one is ours. We'll exit last, for our privacy", Den stated.

Zei, Gladias and Bai nodded. The four of them reclined back in their seats and waited.

About ten minutes later, the door to their cabin automatically opened. The four of them immediately stood up, grabbed their carry-on bags from under their seats, and walked toward the front of the aeroplane. As they passed through the aeroplane, they could see all of other cabin doors already open and empty, with used blankets and empty bottles and cups littering many of the tables and seats.

They exited the aeroplane though a doorway that was directly connected to a staircase, and then to a narrow room that was clearly also the inside of an autobus. The room had an entrance, which they had just passed through, an exit on the other side, and two passage gates blocking the way in the middle.

Zei and Gladias stepped forward first. They each put their watches against the green circle on the side of the gate they were passing through, and looked at the camera in front. A few ticks later, almost in unison, both gates announced: "Welcome to Veridia."

The gates opened, and Zei and Gladias stepped through. The exit at the other end of the room led straight into the interior of one of the

large black cars that they had seen earlier through the window of the aeroplane. They both stepped in, and found another man already sitting inside.

"Senator Verdow!"

"Welcome home, Gladias!", Verdow replied, with a facial expression and tone warmer than Gladias had ever heard him. He immediately looked to Gladias's side. "And you must be Zei."

"It's a pleasure to meet you, Senator."

As Zei and Gladias were sitting down, Zei could already hear the footsteps of Bai and Den approaching. They also stepped through the opening and entered the car.

"And you must be Bai and Den!"

"We are indeed", Bai replied.

"Well, let's sit down. We have a lot of work to do."

All four of them sat down in their seats. The opening linking the car to the entrance room closed. Almost immediately they could feel the car start to accelerate.

"So first of all, Gladias, I have some very good news for you."

"Oh, what is that?"

"Looks like we know where Lily is."

Gladias's eyes opened.

"Seila pushed really hard and got in touch with me, she told me the whole situation. Zven stayed in touch with Lily, and she admitted to him where she was: she had crossed over into Arctic-occupied territory and was in Northglade."

"We then scanned through all the feeds the security cameras that you so brilliantly suggested the Dzegojan should hack into. After a few days, we found her. She is in some kind of boarding school that the Arctics set up. About one kilometer away from the GPH."

Gladias sighed. "At least we know where she is. It's going to be difficult to get her out of Northglade, but at least that's also the place Febric and Hreda are, so we can get them all together."

"Indeed. If you really wanted to, you could even go to Northglade now - though the doors to the GPH are still closed."

"The Arctics recently started allowing communication and travel both ways between Northshore and the rest of Veridia. They opened up clinics in Northglade offering some of their most advanced treatments, for both longevity and performance enhancement. Looks like they've decided to try to get Veridians to join the Arctic Empire voluntarily, by offering us their shiny technology."

"It's a good strategy."

"What was your line, Zei? Manipulating someone to do something by just showing them how it's in their interest, is just convincing?"

"That's definitely how Ephelion would see it. Of course, Sylka would get upset and say they're hiding the hidden cost - if you can live longer but it depends on ongoing infusions of proprietary Arctic medicine, you become dependent on them for your life, you become their slave. Though Ephelion would say that being dead - being inside a coffin where you are not free to move a single muscle or think a single thought - is the worst slavery of all. And that that's not just true for a person, it's true for a civilization."

"If we want Veridia to succeed and thrive, we have to find the synthesis. We can admit that the Arctics do some things very well, and that their striving for a radically better future can be healthy. Veridia's north star cannot be to just stay the same as we are but with less greed and a better social fabric. We need to have actual progress, and take the good parts of it without the bad. We need to recognize success, and reward it."

"But first, we have to give ourselves breathing room. And that means defeating the Arctics in Northshore."

"Exactly. And to do that, we have to be willing to learn from the outside - learn from our enemies in the Arctic Empire, and from our friends in Freetown and Dzego."

The car slowly came to a stop. A few moments later, the doors opened, leading straight into the inside of what looked to be a government-run hotel.

"Get a good sleep, friends. Zei, Den, the two of you have a big day tomorrow morning. And then once that's done, we'll meet and talk

about our big plans. Veridia ba fau gie."

Gladias and Zei stepped out of the car first. Almost immediately, Gladias could see a woman waiting in the far corner of the room. She turned toward him, and began running toward him with outstretched arms.

It was Seila.

⁂

MELDAN, VERIDIA · 3724 FROSTIME 6

Zei sat down on a chair, in front of a console that looked remarkably similar to the consoles on which he had played all of his championship Minpentai games. The main difference that he could see was that this time, there were no anti-cheating detection drones buzzing around.

He put on the earpiece and glasses, and spoke into the microphone.

"Can you hear me, Den?"

"Yep, loud and clear. I think I'm in the room right below you."

Zei playfully stomped his feet.

"Can you hear that?"

"I'm only hearing your voice through the earpiece."

"I guess your soundproofing is very good!"

"Okay, Zei, let's go through the whole operation again."

"Location: Redshire, presidential residence. Goal: rescue our target - Tafindel, the president's son. Strategy: our man on the inside asks the target to step out onto the balcony. We fly in, drop down a rope for him to tie around his waist and hold on to, fly out before the Arctics can do anything."

"Exactly, that's the mainline plan. Now what are the contingency plans?"

"If Arctics appear - soldiers or drones - we shoot them. If he doesn't step out onto the balcony, no backup plan, we're screwed - our drones won't survive if they have to land directly on the building. If the drone with the rope gets shot down, we have a second drone, and potentially even the other combat drones could pick the target back up, though

they're not as well-designed for it. The only dangerous window is if the drone gets shot down within a few ticks after it picks him up - too soon for the backup drones to swoop in and grab him before he hits the ground."

"What's my responsibility?"

"Supervising the mainline plan."

"What's yours?"

"Supervising the combat and the contingency plans."

"Mu, are you here?"

"I am indeed. I'm supervising the surveillance, I'll make sure we learn about anything unexpected as soon as possible."

"Mu, how did you get here!?"

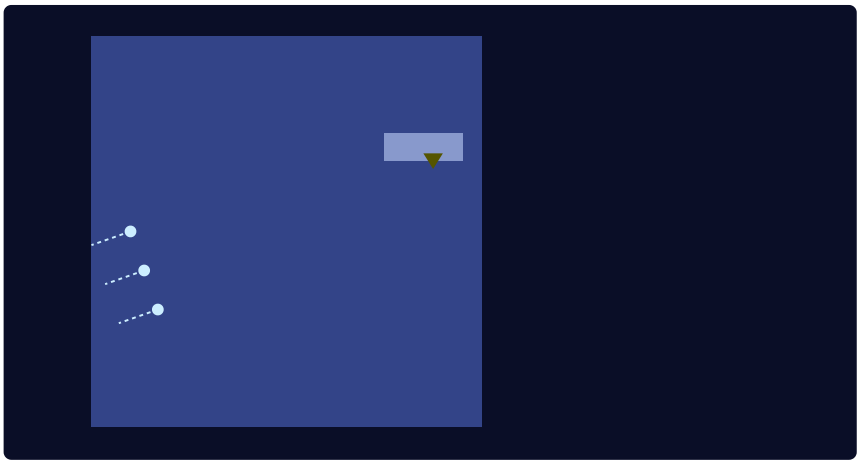
"I was on the same aeroplane as you yesterday, Zei. Just in a different cabin, and different car going into the city. Privacy precautions."

"Well I'm happy to see you! Or, at least, hear you."

Zei checked the various numbers in the display that was being blasted into his eyes through the glasses he was wearing.

"Fi le gei tau fa", he exclaimed, mimicking the voice of the announcer that would count down to the beginning of each Minpentai game.

He then consciously switched his mood to be entirely focused on the battle that was very soon to begin.



Zei stared intently at the screen as the drones came closer to the parliament building and more drones appeared onscreen.

Twenty ticks later, the dark yellow triangle onscreen turned into a brighter yellow checkmark. Tafindel stepped out onto the balcony, exactly on schedule.

Thirty more ticks passed.

"mu gu gei tau fa", he said softly. Battle starts in fifty ticks. In some sense, the battle had started already, but the moment he was waiting for was the moment Tafindel would grab hold of the rope. That is when the real challenge would begin, he thought.

Twenty more ticks passed.

Suddenly, the checkmark turned back into a dark yellow triangle.

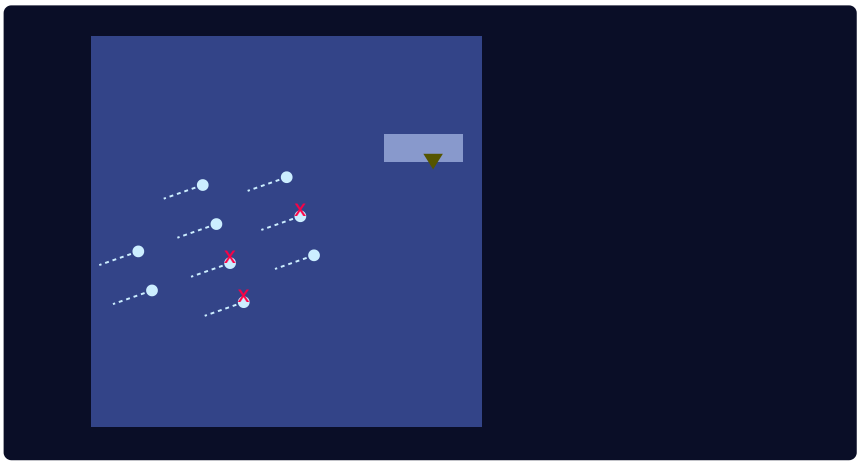
Zei zoomed in to the video feed. Someone had grabbed Tafindel, and pulled him back inside.

And almost at that same moment, Arctic drones started shooting.

Zei could immediately hear Den's words blasting through his earpiece.

"I'll figure out what happened to the target. Zei, supervise the battle."

Zei subconsciously nodded. The headset automatically translated the nod into a checkmark icon, which flashed onto Den and Mu's displays.



Zei groaned. Three of his drones had already been hit.

"Evasive action, return fire at hostiles. All damaged drones, if you are still functional, descend to ground and play dead."

Arctic surveillance was more powerful than he had thought, he realized.

At that moment, an X appeared above a fourth one of his drones - the drone that had been intended to grab the target. A tick later, that drone disappeared completely from the display map. Zei checked the raw visual feed. It had been directly hit, and exploded in midair.

Zei screamed with despair. The evasion tricks that they had learned from their previous missions were not helping. It looked like the mission was going to fail - and perhaps with it, the Kungaupei's credibility.

The next moment, through his earpiece, he heard Mu speaking intensely.

"New plan. Everyone fly low, under the buildings and trees. Fly through the streets. That's where their firing ability is weakest. Then I have a trick up my sleeve."

Zei nodded, and passed along the instruction to all drones. Within ten ticks, the four surviving and fully functional drones were all near

ground level. An additional damaged drone, which Zei had instructed to play dead, was already at ground level - it had been there for over half a minute.

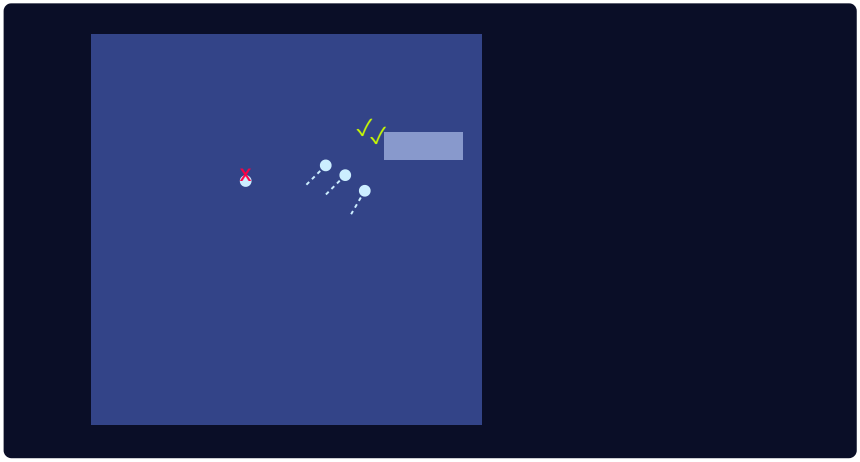
Suddenly, Zei could see in his visual feed the streetlights and the lights inside the buildings all blinking chaotically. Arctic drones started firing in all directions, confused about where their adversaries were.

His drones kept flying forward. As soon as one got within a hundred meters of the presidential residence, Den spoke up.

"I need a crash at these coordinates."

A new yellow triangle appeared on the side of the building. Zei immediately directed his drone to crash there at full speed. Four ticks later, it did, and there was a large explosion and smoke.

Ten ticks after that, the yellow triangle turned into two bright yellow checkmarks.



Zei was confused, but immediately did the only thing that made sense. The drones were too low for the rope to work. Instead, he directed two of the three remaining drones to slow to a stop, right in front of the presidential residence.

Both checkmarks turned green. The drones accelerated again, and sped ahead. Zei directed the third drone to hold back, and fire on all Arctic drones that it could detect.

He checked the visual. On each of the two drones that were flying off, someone was holding on tightly to their backs. One was Tafindel, and the other was their man on the inside - his assistant. They were both wearing thermal cloaks. Good job Den, Zei thought.

Zei continued to watch the screen intently. Somehow, against all odds, they seemed to have done it?

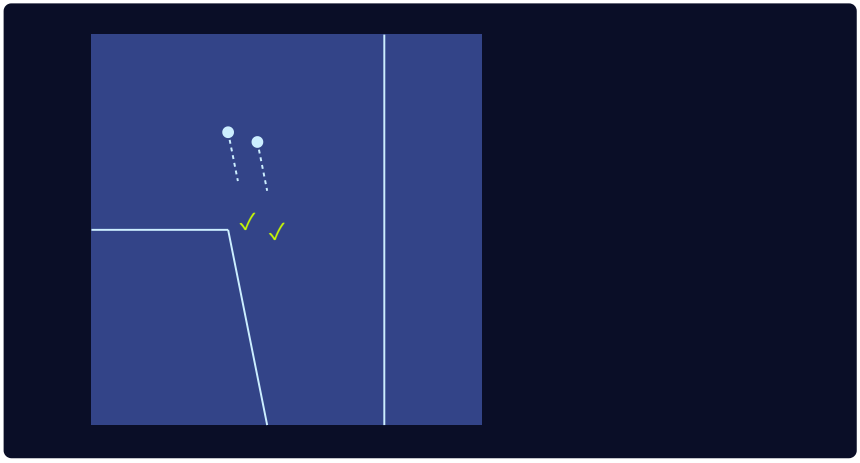
Den cleared his throat and began to speak.

"This is not going to work", he muttered. "I see more Arctic drones coming in. They are faster, they will catch up."

Mu immediately responded.

"Already on it. The targets will jump off the drones soon, the drones will keep going. The targets will be told to run two hundred meters to a road - NOT the road closest to them. I have a car coming to pick them up. With luck, they will be tens of kilometers away before the Arctics know what's going on. Once they get to the shore, a sea drone will come and take them out of Redshire."

Zei sighed. "I trust you, Mu."



Zei sat back down on his chair and sighed, feeling deep disappointment at himself. He closed his eyes.

A minute later, the first good news:

"Pickup successful."

Another ten minutes later:

"Our drones touched down at a large shopping complex in Dolinar. Arctics are closing in on them. Hopefully the manhunt will take as long as possible."

Another thirty minutes later:

"Our car reached the shore without incident. Targets now entering the seaborne drone."

Ten more minutes after that:

"The Arctics realized that we tricked them. They are broadening their search."

Zei watched the screen, which was at this point showing the slow grinding progress of the seaborne drone making its way westward away from the Redshire coast.

Fifty minutes after that:

"The targets have successfully arrived on the coast of Arturia."

* * *

"I'm so sorry guys", Zei muttered softly. "I failed completely, and if it weren't for Mu somehow saving us with a completely new plan, we would have failed, and probably everything else we're planning would have failed too."

"Welcome to real life warfare, Zei", Den replied. "It's not like Minpentai. Sometimes the only way to win is to think completely outside the box. Not outside the box as in figuring out some clever way to encode an entire computer inside of a glider structure, but outside the box as in, jumping out of your console chair and running over and bashing the other player in the face."

"So ... Mu, what did you do there?"

"This is not the Arctic Empire, this is Redshire. There are lots of people there who quietly support us, and a lot of the technical

infrastructure has still not been shifted over to full Arctic control. So for the moment, it's still possible to break into some things. I have a bunch of admin passwords that people from Redshire gave me. I used them to first take control of the streetlights, and then to commandeer a few dozen automated cars. There were actually multiple decoys, the planes stopped and restarted in a few places. We should have done this plan all along, but - I underestimated the extent of Arctic surveillance as much as you did."

"It was truly an amazing job, Mu."

"But what matters is that we won. Tafindel is safe in Arturia. The Arctics have been humiliated. The whole world now knows that they can be defeated."

Den paused, and then began to speak again.

"For the moment, they think it's just Freetown's defense companies. But now it's time for the Kungaupei to stand up and take its share of the credit. And then see if the Veridians are up for giving Veridia the defense that it deserves."

* * *

Gladias slid open the door to the booth, and stepped in. Five judges that Mov and Seila had helped to identify - with significant help from surveillance cameras that had been backdoored by Dzego - were already sitting inside.

"So what is it that you wanted to talk about?", the first judge asked.

Gladias sat down, and slid the booth door behind him.

"I am planning to open a case against the Veridian military leadership. Under the fourteenth Ground Law, the Veridian military is required to defend Veridia. As we have seen from what happened in Northglade, they have completely failed to do this. And so I am arguing that they have broken the law, and should be fired."

"That's crazy. Yes, we were caught by surprise in Northshore and lost pretty badly. But the Veridian military are doing the best job they can in a difficult time. We can't just come in from the outside and say

we don't like how they're doing their job and replace them, when we don't even know what standard we're comparing them to-"

"Caeron", Gladias interrupted. "Did you see in the news what happened today?"

"I did not. Come on, Gladias, you know it's a rule that judges should not read the daily news. We're here to apply timeless principles, not get swayed in every direction by crowds of people getting angry in the moment."

"A Dzegojan group flew into Redshire and successfully rescued the president's son."

Almost immediately, another judge began to reply.

"So you want us to replace the Veridian military leadership with some Dzegojan?"

"Yes, exactly."

"I'm sorry, this is insane. That's not the court's job. Veridia is a democracy. The people in charge of the military, who can change the composition of the military, are the parliament. So if you want to try your crazy scheme, you should go talk to them."

"Come on, Sewlert, you know as well as I do that the parliament is corrupt. They're either already inside an Arctic-engineered bubble of propaganda, or bribed, or have them or their families threatened thanks to the Arctics having broken into our surveillance systems."

"We're a court. We don't have the right to just go and decide the entire parliament is corrupt."

"Actually, you do. Twenty-first Ground Law. Veridian government officials are required to perform their duties honestly, in good faith and without undue influence, and report and resign if any undue influence befalls them."

"So you want us to fire the parliament too?"

"If necessary, yes."

Gladias could feel that he was not getting through. So he decided to try a different line.

"Do any of you have family in Northshore?"

A third judge spoke up.

"I do, a son. He's in the GPH, that's the one place that they are still blockading and not letting people leave from."

Caeron turned angrily toward him.

"Why are you bringing this up? Gladias said it himself, we're required to perform our duties without undue influence."

"Six percent of the Veridian population is in Northshore. Each person has about ten close family members - parents, siblings, children, spouse, other close relations, co-parents. So among the five of you, that's fifty total, so statistically, there should be three of your family members total in Northshore. If you have none, then you're not feeling something that many other Veridians are feeling. So your interests are misaligned. You're misaligned toward compacency, but is that not also a form of undue influence?"

"That's not how those laws work, there are precedents about it."

"Do *you* have family in Northshore? If so, then *you* are the one with the undue influence."

"Now you're getting the law wrong. Plaintiffs are allowed to have a stake in the outcome - in fact, they're required to. The legal term for that is *standing*."

"And yes, I have three children there."

"Look, Gladias. Again, Veridia is a democracy. We are not going to stomp our feet over our democracy just because a bunch of Dzego-man whiz kids rescued a prince and you think you're smarter and more morally pure than the entire parliament."

"And even if we did, just think about the procedure. A case takes a minimum of thirty days from start to finish, and that's if the judges don't decide that it needs to take longer. And then they can appeal, multiple times. Appeals can be shorter, but only from the third round. So it will take months for all this to resolve. And in the meantime, Veridia will be in chaos, and the Arctics will just swoop in and take over."

"So we're not doing this."

Gladias looked around. Three of the other four judges were nodding along to what Selwert was saying. The remaining judge,

Zevin, seemed much more sympathetic to Gladias - no doubt in no small part because he also had a son stuck in Northglade - but he did not seem to feel comfortable expressing his opinion.

"Understood", Gladias said softly. If this sample of judges was that hostile to his pitch, the sample of judges that he would get if he launched a real case would almost certainly be quite hostile too. He needed a different strategy.

He slid open the door to the booth, and slowly exited.

Chapter 27

Meldan, Veridia · 3724 FROSTIME 7

Gladias heard a knock on the door of his government hotel room. Before he could respond, he heard a buzzing sound, and the door opened.

Seila walked in through the door.

"I'm so sorry, Seila", Gladias said almost immediately with a sad expression on his face. "I don't think this plan is going to work."

"I've failed our whole family. We know where Febric and Hreda are, now we know where Lily is, but there's no path to rescuing any of them - except if the Arctics conquer us completely."

Seila walked up to Gladias and hugged him.

For a few moments, neither of them said anything. Then she stepped back and her face broke into a smile.

"So I have a new idea."

Gladias's eyes perked up.

"What is it?"

"So what we learned yesterday is that the judges do not feel comfortable doing something that they perceive as being very undemocratic. So I have a solution. We make it democratic."

"What, like we hold an election tomorrow?"

"In a sense, exactly."

"While you were in Dzego, Mov and I started making our way up the ladder in Silverchat, talking to people there. We were not too successful at getting their attention at first. Verdow joining in about ten days ago helped a lot, but even then they didn't take us too

seriously. But on the day the Kungaupei rescued Tafindel, I took a risk: I told them something interesting military-wise would happen a longhour in advance. And then of course it did. So now, they are listening. And I have a plan for what to ask them."

"Silverchat has a polling feature. You can burn zipcoins - well, pay zipcoins to the company - and it lets you automatically ask a question to a large cross-section of people on the platform. The more zipcoins you burn, the more people it polls. So what we'll do is, ask a lot of people a bunch of questions about how they feel about various Veridian and other government institutions, and other hypothetical scenarios."

"We'll include 'do you think the parliament represents the people?' and 'how would you feel if Veridia's military was run by the Dzegojan people who rescued the Redshire president's son?'. But we include that alongside hundreds of other decoy questions. That way, the Arctics won't know what's going on."

"For the responses to be genuine and well thought out, you need a high priority level, which I imagine would require paying a lot of zipcoins. And for this to function as a social signal with strong legitimacy, there must be common knowledge - everyone in Veridia must know not just that the poll was made and what the results are, but also that everyone else knows that the poll was made and what the results are, and also that everyone else knows that everyone else knows, and so on. That would require *a big chunk of the entire population* to see the poll. So we would have to pay *a lot* of zipcoins."

"Nine hundred thousand zipcoins was my estimate."

"Do we have that many?"

"Nope. All our savings combined, including Mov, adds up to only half that."

"But here's the trick. We don't need to pay. We convince Evelor, the founder of Silverchat, to do it himself, through the company. Then they can set the breadth and the priority as high as they want - they get a hundred percent of the fee back."

"Wait, can't they just do it manually at zero cost?"

"Their platform works by publishing proofs onto a cryptographic network every time the set of readable messages is updated. So there's literally no way for them to break the rules - if they tried, the proof would fail to verify, and the viewers' clients would automatically reject the messages as invalid. The only thing they *could* do is publish a new algorithm hash, but that would come with a twenty-day delay."

"And they get one hundred percent back? No sales taxes?"

"With the latest rubric changes, given how well Silverchat has been behaving with its algorithmic transparency - the exact proof mechanism that doesn't let them just directly backdoor and break their own rules - and interoperability, respecting users' privacy, cooperating with researchers, their taxes have dropped to zero."

"Then let's see what Evelor thinks!"

* * *

Gladias, Seila and Zei walked forward together along their usual path through Kalimar, covered with trees and surrounded by stone houses on both sides. They had taken a black car most of the way there as a security precaution, but cars could only go up to the border of the car-free zone. Thus, they had no choice but to go the last few hundred meters on foot. Hopefully the privacy robes will suffice, Gladias thought.

A cold breeze blew at them from behind. The cold energized them just as the wind pushed them forward, and they began to walk forward faster.

"Right turn here", Seila whispered through her neck band. By this point, she had become almost as accustomed to having all of her conversations through neck bands as Gladias had been.

They turned right, onto a smaller path. The path continued straight for about a hundred meters. At first, there were stone houses on both sides. After about fifty meters, the houses gave way to what seemed like a deep forest. And after a hundred meters, they could see in front of them what looked like a small hill.

The path sloped downwards, and led to an entrance door to what seemed to be the inside of the hill. Yet another one of those buildings that pays almost no land taxes, Gladias thought. By this point, Veridians had come close to perfecting the art of living and working and putting complex structures into a forest in such a way that the same patch of forest would seem untouched from only a hundred meters away.

Zei marveled at the sight, and his mind jumped immediately to comparing it to the mountain pyramids in Dzego. In Dzego, he thought, they crafted natural mountains into pyramids, and made nature seem like the work of man. In Veridia - or at least in the Kalimar district of Meldan - they made what was quite possibly an entirely manmade hill look like nature. Dzego could learn from the Veridian style, he thought. It's probably even better for protection.

Seila tapped her hand device against the green circle beside the door, and the door opened.

She led the three of them down a path: turning left, then right and up a slope, then right again, then up a staircase. Finally, she opened another door, and led them up a staircase into the middle of a circular area from where they could see the open sky. They were at the top of the hill, Gladias and Zei realized.

They turned around. Another man, wearing a privacy robe without the hood or the face covering, was already there.

"I'm Evelor", he said. "Nice to meet you."

"I'm Gladias. It's a pleasure to meet you too."

"And I'm Zei."

Evelor pressed a button, and a roof started to close above them. After about ten ticks, the roof was completely closed.

"Just for privacy", Evelor said. "Feel free to take off your robes. Also, would any of you like a water or a tea?"

"I'll have a tea", Gladias replied.

"I'll have a tea too", Zei added.

Seila looked around briefly. "Then I guess a tea for me as well."

"Sounds good, four teas it is. The robot will be up shortly."

Evelor sat down on a bench pressed against the safety railing behind the opening that the staircase emerged from. He gestured at the circular bench that was all along the circumference of the room. "Please, sit down."

They took off the hoods and face coverings from their privacy robes, and sat down.

"So, Seila, you mentioned to me that you had an idea for what Silverchat can do to make the whole Veridian government less stupid", he blurted out with a tone of blunt honesty.

"Yes, I do."

"Basically, I want to do a poll. I had Emerald come up with about a thousand questions, I reviewed them. Some about how you feel about the present, some about possible futures. A small fraction of the questions are getting at the thing that we actually care about - replacing the military even if the parliament doesn't want to - and the rest are decoys."

Zei winced slightly. Seila was volunteering the content of her master plan too quickly, he thought. The Kungaupei would never operate in this way. But maybe Veridian culture is just more trusting - or maybe Evelor had already proven himself to Seila, or even to all of Veridia, to be someone worthy of trust.

"The goal is to create common knowledge that the poll happened and what the results are. If it goes well, then we will have a much stronger argument to take to the judges that this is something that people would actually want, and would not riot against."

Seila showed her hand device to Evelor. Evelor briefly scrolled through the questions, and started reading out loud.

"Who would you prefer to run the Veridian military? A: its current leadership. B: a new leadership chosen by parliament. C: the Silverchat leadership. D: the Arctics. E: the Kungaupei (Dzegojan group that recently rescued the Redshire president's son). F: Arturian military advisors."

"And how much did you want to spend on this?"

"Three million zipcoins. And I was hoping that Silverchat the company could do it itself, because you get paid back a hundred percent anyway."

"Interesting! I would say three million is too much, it would be more than any previous poll or advertisement on Silverchat. It would make the Arctics suspicious that something unusual is up. I would do one million. Still in the top three in terms of spending, but not unprecedented."

"Works for me."

"But there's also another issue", Evelor continued. "If we do this we can help the judges see that making a big change would be popular, and that the people don't see the parliament as being all that legitimate. Or of course we'll learn that we're very wrong in our own judgement of how Veridians think. But this won't convince them that this kind of drastic action would actually *work*."

"Do you have an idea in mind?"

"Actually, yes. Recently we added a feature, Silverchat Predict, that lets people - or bots - make bets on future events. So far bots have been the most effective-"

"I saw that", Gladias interjected. "Prediction markets are cool. But there's two weaknesses. One is that if the Arctics see the market, they could manipulate it. In theory, any attempt to manipulate creates more incentive for arbitrageurs to come in and profit on average by pushing the price back to where it should be, but it's still very untested how well that actually works in practice."

"There's known results that unless the arbitrageurs are either fully risk-neutral or have infinite money, a manipulator can always force at least some shift in the price. And if the underlying variable is something so unprecedented and unknown - like whether or not we will win the second victory against the Arctics that the world has seen in the past century - that makes a hard case for any arbitrageur to want to come in."

"The second problem is private information. We're confident in the Kungaupei because we know them well and we trust them and we

know a lot about what their capabilities are. But if we reveal that publicly, we reveal it to the Arctics."

"Right, both good points. Hopefully we can work on improving that first one over time, but ... right now prediction markets are new. The second one is harder to fix."

"Okay, I have a different idea. We don't use the markets directly. Instead, we take the five bots that have been performing the highest on the leaderboard, and we use verifiable two-party computation to ask them. We create a sort of ephemeral virtual private sandbox, we feed the private information into the sandbox, they feed the bot into the sandbox, and the sandbox only outputs the bot's answer."

Zei saw the opportunity to speak. "Yep, that's just an extension over garbled circuits - you probably want verifiable garbled circuits. Overhead is quite nasty-"

"There's optimizations specific to AI computation. Remember, almost everything inside a large language model is linear. The nonlinear part is necessary for the model to be smart, but it's only a tenth of a percent of the entire computation."

Zei realized that, on this particular specialized topic, he had been outmatched. Looks like in his nine months of studying cryptography he had not managed to become smarter than everyone at everything.

"I see..."

Their conversation briefly paused as they heard a mechanical noise getting louder and louder below them. A few ticks later, they could see the top of the robot, slowly climbing up the staircase with a raising motion and then a sliding motion at each step, carrying four cups of tea on a tray. Once it got to the top of the staircase, it moved toward Gladias to give him his tea first.

"Gladias", Evelor asked. "You seem smart. I have a different question I wanted to ask you."

"What is it?"

"Do you remember Ephelion's debate with Sylka?"

"I do."

"Ephelion said: most of the 'softer' traits in society seem to move in circles. The good creates the seed of the bad, and the bad creates the seed of the good. Disaster to heroism to prosperity to complacency to disaster. Technocracy to democracy to chaos to strongman to technocracy. And so on. The only thing that doesn't behave that way is economic output: growth just keeps compounding on itself. And so we should stop trying to accelerate the thing that goes in circles, and instead just focus on growing economic output. I dislike that argument. It makes me uncomfortable what it implies. But I still can't quite put my finger on what's *wrong* with it. What do you think?"

"I don't think it's just economic output that compounds on itself, I think many good things compound on each other. Health makes people more productive. Productivity gives people the resources to be better. Education supports both, and health supports education."

"So in other words, the eigenvector that Ephelion talked about includes more than just one single dimension - which is actually what most eigenvectors are like."

"You clearly understand eigenvectors better than I do!"

"Silverchat is a social media website. Eigenvectors are like ... on the first page of the original whitepaper."

Gladias smiled.

"But on the other hand", Evelor continued, "the Arctics do care about health. It's not health with the aesthetic of compassion and love and care that Sylka would want, it's hard metallic longevity clinics, but it's still health."

"Well Sylka would say it's not having a long life that matters, it's having a healthy life."

"Right, but the medieval tales of people drinking eternal life elixirs and then being doomed to lie on the ground forever with all their bones broken are not how any of the longevity science actually works. A longer life is a healthier life by default. The easiest place to stop damage is at the moment when it begins."

"And I guess you'll say the Arctics care about education too. They barely understand music and poetry, but they teach their kids science

just fine, and that's the part of education that goes into the largest eigenvector anyway."

"Right."

Gladias could see Seila frowning with discomfort. His mind raced as he tried to come up with other arguments against Ephelion's way of thinking.

"But the Arctics aren't actually maximizing economic output. They're looking to start a war - wars can destroy economies for decades. Their attempt to eliminate anyone in Dzego who goes too far in advancing science and technology, that's not maximizing economic output either - quite the opposite. They're not trying to maximize economic output. They're trying to maximize the economic output that's under their control, and destroy everything else."

At that point, Zei spoke up.

"There is this underlying model that Ephelion is implying, where there's a vector that's shooting outward toward infinity, and the faster it goes toward infinity the faster everyone is happy. But that's not how I see the world at all."

"The world that I am seeing is a fragile world. It did not use to be this fragile. In the past, communication was limited and transmission of authority was limited. Any empire united for long enough would eventually fall. The world was reversible. But today, with our modern robots and modern surveillance, if *anyone* manages to take over and consolidate control across the whole world, they could lock their power in. Irreversibly. Forever."

"If the Arctics win, a certain kind of entropy - the dimensionality of the space of humanity's future possibilities - would be permanently constricted. From that point forward, the only possible future worlds would be Arctic worlds."

"And that's not the worst that can happen. If the Arctics, or anyone else, make a mistake with one of their superweapons, and no one else has the means to resist it, they could destroy the living world entirely."

"Eventually, the world will become reversible again. We will be able to travel between the stars, and our civilization will become too big to

completely control or completely destroy. But until that happens, we are in a fragile place, and if within the next decade it all gets multiplied by zero forever, the Arctics will be the reason why it happens."

"I have a simpler answer", Seila added. "All these arguments about the far future, they're hard to predict. And it's really easy to convince yourself that one path is right, and come up with really smart math to justify it, if that's the path that you already know you want to take. But there is one thing that's easy to predict: if the Arctics conquer Veridia, the war may kill hundreds of thousands or millions of our people, and the Arctics will destroy the beautiful world that we know and cherish. To me, that's already enough to be worth fighting against. And the beauty of our world and our countries and our friendships, that's already enough to be worth fighting for."

"All good answers", Evelor replied. He took a sip of his tea and sat back in his chair, pondering.

⁂

3724 FROSTIME 8

Gladias, Seila, Zei, Bai, Verdow, Den and Mu were all sitting around a table inside a meeting room in the government-run hotel. Projected on a wall in front of them was a timer, counting down the ticks until the questionnaire on Silverchat would get published. At that moment, the remaining time was 150 ticks.

"Are you all done submitting your information into the package for the bots?", Seila asked.

"Just finished my part", Den replied.

"Remember, it's a bot. It doesn't need any amazing presentation and polish. If you think of something else to say or dump in, just throw it in."

"I hope you and your bots reviewed the cryptography well, Zei. I'm entrusting a lot of our military secrets to it."

"There's a machine-checkable mathematical proof that the garbled circuit algorithm satisfies the standard correctness and privacy

properties. The proof is end-to-end, it covers everything all the way down to the operating system and even the hardware. I read the statements it proves, and I asked Emerald and two other AIs to check it. It's quite solid. Zero information leakage unless hashes are broken, and if that happens we have much bigger problems."

"I checked too", Mu interjected. "I used a different approach, just in case the proof system or the definitions have somehow been broken for over a decade and nobody noticed. I had Emerald read the original academic papers and write its own implementation, and then had a fresh instance read through that code and the original side by side, and look at every difference, see which side is right. Evelor's code is flawless."

"It's starting in fifty ticks!"

"Not going to imitate the Minpentai announcer voice this time?", Bai joked in response.

"MU GU GEI HUI ZIU FA", Zei replied in a voice exaggerating the stereotypical robotic tone for comic effect.

They sat around the table and waited.

"LE MU GEI HUI ZIU FA", Bai exclaimed.

"Hey!", Zei replied.

"Beat you!"

They kept watching the countdown. As soon as it hit ten, they started chanting in unison:

"PA GU ... SO ... BI ... ZE ... HA ... MU ... FO ... SHI ... LE ... PA ... HUI ZIU FA!"

The screen split into two views. On the left, there were five parallel progress bars showing the computation of five garbled-circuit executions with the five bots. On the right, a large collection of pie charts, one for each question that had been asked. Votes quickly began coming in.

"Let's zoom in to the ones we care about", Verdow offered.

Zei nodded, and clicked on one of the pie charts.



"As I expected. Now let's see how people think we should solve this problem."

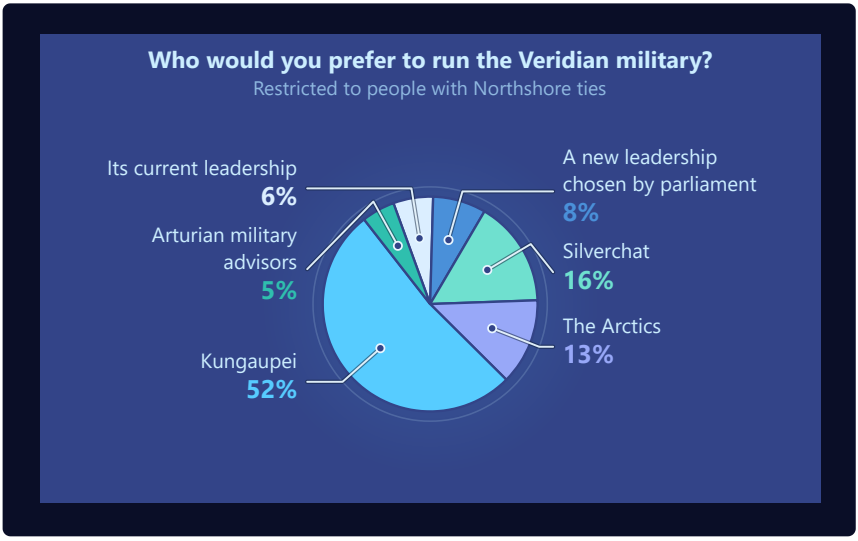


"Restrict to people who are from Northshore."

"Wait, did we set it up to provide that level of granularity? Votes are anonymous, we have no way to tell who sent each one."

"We did break it down by age, self-declared most recent senator votes, province, and a couple of personality type axes inferred by a large language model."

"Ah, found it."



"Congrats, Evelor", Verdow said wryly. "Seems like you are more fit to run Northshore than the actual parliament."

"I'm a bit scared of these results now", Seila interjected. "Haven't we just given the game away to the Arctics?"

"I think it's okay", Mu replied. "We asked many different questions in the poll. One of them literally has twenty percent saying that we should fund recovering Northshore by legally transferring all of its government-owned territory to Hydrafill and letting them figure it out."

"The lawsuit is something that the Arctics will notice and watch, of course. We should delay it by a few days to make it not look too coordinated. But regardless of what we do, we will have made it clear

to everyone - including the Arctics - that Veridians recognize that there is a big problem. It's the details of the solution that we need to hide."

They paused briefly, and stared at the progress bars.

"Okay, that will take a while. Let's look at some more pie charts."

A robot entered the room, bringing each of the participants a cup of tea.

* * *

"And ... done", Zei exclaimed triumphantly.

Five icons representing the responses from five bots appeared onscreen. Zei clicked on the first one.

Executive summary:

- Under any 'ordinary' scenario, meaning either status quo continuation of the Veridian military, or a reorganization led by the Veridian parliament without large outside pressure, the probability that Veridia will hold the same or greater percent of its own territory after one year is 20%.
- Under a hypothetical transfer to Kungaupei control, the probability increases to 26%.
- The main downside risks involve standard organizational transition costs that are typical of chaotic changes in leadership and replacement of experienced personnel with outsiders.
- Probability of successful outcomes is maximized (up to 50%) in cases that involve Kungaupei participation but minimize the disruptiveness of the transition.
- Note that above probabilities are for *same or greater* control in one year. Probabilities of strictly greater control are uniformly much lower.

"Hmm", Verdow muttered.

"Do you think it's reasonable?", Gladias asked.

"The bots definitely have a more conservative temperament than I do, but it's fair."

"The other four are saying vaguely similar things. The most pessimistic one says the chances are the same in the status quo and Kungaupei scenarios."

"Also we have much more data coming in from the votes. It's roughly the same as the results we first saw right when the poll began."

"I guess we have all the data we will get. Let's get ready to file the case in a few days."

Chapter 28

Meldan, Veridia · 3724 FROSTIME 15

Zei, Den, Bai and Mu were sitting around a table inside a conference room.

They heard the door buzz. They immediately paused their conversation, and waited to see who was coming inside.

The door opened. It was Verdow.

"You have a visitor."

"Who is it?", Zei asked.

"One of the generals. Actually one of the few that I respect. I ran an analysis through Emerald, everything I got back is very positive about him."

"What does he want?"

"To talk. About the ongoing case."

"Sure, we're happy to talk to him", Bai replied. "Is he here now?"

"He is."

"Then let him in."

A middle-aged man entered the room. He was wearing a shirt that seemed almost perfectly designed to strike the balance of looking like traditional Veridian military attire while being comfortable for everyday wear.

"I'm Lektor", he exclaimed. "Nice to meet you."

Zei was about to start with a response, but Den motioned for him to stay quiet.

"I'm Den, this is Zei, Mu and Bai. Welcome."

Lektor walked in, and sat down on one of the empty chairs.

"Look", he began, "I want to understand what in the world you guys are thinking."

"This is the first time in Veridian history we've seen someone try to get a court to fire high-ranking government officials. It seems like madness. I get that people are upset, but you're trying to fire the entire military leadership. You don't get a competent military that way."

Then he paused.

"But also ... I get that people are upset. We saw the same Silverchat polls as everyone else."

"So what are you thinking?"

Den began to speak.

"Are you familiar with the operation that rescued Tafindel?"

"Of course. We've been trying to piece together what happened this whole time."

"That was us - the Kungaupei. We rescued him, working together with two Freetown defense companies."

Lektor's jaw dropped.

"Wait, you're *here*? How many of you?"

"Dozens of us. I'm one of the High Council members. Zei and Mu over there personally commanded the operation with me."

"I see..."

Lektor paused.

"So what's your plan? Personally take control of the Veridian drone fleet and use it to take back Northshore?"

"I know it's not a very *polite* plan. But you have to understand. We're desperate. The Arctic Empire is expanding. They're clearly planning to target you before they target us: you're more advanced, and we're better prepared defensively. So we think the Arctics want to knock out Veridia completely within the next year, maybe sooner. Then they will take down the United Cities one by one, and then they will go south and clear out Messalia and Haragmir, or just bomb them to save time. After that it's just them versus us. We can defend well with what we have, but really we've depended on you so much for the

most advanced technology. So we have to take a stand. Together, now."

"You can't just commandeer a military the same way you just pick up and drive a new car. Our soldiers, including our commanders and generals, have spent years working together, training together. They know Veridian logistics inside and out. They know how many new drones we can produce, what that would take, what you can do with Helisport copters if you had to turn *those* into drones. They have over a thousand drone pilots, and thousands of support staff of every kind. With your dozens of advisors, you would have to let almost every drone fly on AI alone. They would get lost, and get tricked by the Arctics at the first opportunity."

"I know we've failed completely at Northshore, but you can't just go throw everything we've built away. Even the command structure - that's something they've spent years working inside. You can't just rip that up and replace it and expect to get a good result."

Bai felt that the conversation was going off track, and saw her chance to intervene.

"Lektor", Bai began. "We don't actually want to zero out the top three levels of the Veridian military hierarchy. Even if we *could* bring over all of our top thousand Minpentai players tomorrow and get them to pilot all your drone fleets successfully, and somehow all the logistics solve themselves, the court case would take over three months to finalize. We don't have that time. So that was never our plan."

"So what *was* your plan?"

"Well, we wanted someone like you to step in and help."

Lektor responded with silence.

"Even the most criticized and hated institutions are never quite as irredeemable and talentless as social media makes them seem. We know that. But at the same time, they *can* get stuck in unhealthy places. And your situation is the toughest: you have to shift paradigms from peacetime to wartime, at the exact moment that many of the generals are probably being bribed or blackmailed by the Arctics. We know that's true - the surveillance cameras that we backdoored

already caught several. So we don't want to zero out the military, but we also know it can't stay as it was."

"So I think there's an opportunity for you to step up here. You're respected by all sides. I checked that while you and Zei were talking. High ratings uniformly across different people of different political persuasions, inside the military and in parliament, and across different personality types. I think you should consider proposing to parliament that you take over and run things. And we can work with you, inside your structure."

Den, Zei and Mu immediately nodded. That line of argument was not part of their plan, but they could all recognize that it was the right plan to switch to.

Lektor pondered for a moment, and then responded again.

"You've been very quiet about the fact that you're here. Even among all the people who are excited about the idea of the Kungaupei running the Veridian military, and who voted that way in the poll, everyone thinks it's just a fantasy. If I start making the argument that I can run things with your support, would that not blow your cover? Even in a closed door session, if the parliament is as corrupt as you say, the information will leak back to the Arctics."

This time, Mu spoke up.

"Don't tell them about us. We have a few exploits against Arctic drones that we can share with you. Those exploits are already outdated against the Arctics themselves - so if the Arctics find out, that's actually *better*, it would help keep their guard down. But the exploits will work great against the damaged drone that you guys managed to recover from Northshore two months ago. You can take the credit for them."

Lektor stared again. They knew about the damaged drone? How did they penetrate Veridia so ... completely? And if the Kungaupei could do that, then the Arctics...

"Oh we can also tell you about one of the generals that's taking bribes. We have clear evidence, security camera footage of his conversations with an Arctic spy."

"They used the wrong kind of booth for too many discussions", Mu grinned.

"We'll give you a dump of the info we have. You can figure out the plan. After all, it's going to be *your* army, and *your* plan", Bai added.

* * *

Gladias and Seila heard a knock on the door of her room.

"Who is it?", Seila asked.

"It's me, Mom!"

"Zven!"

She ran to the door and opened it. Zven ran in, and excitedly jumped onto a sofa with his hand device.

"What's up, Zven?"

"I have some good news. It's Lily!"

"What is it?"

"She says that she is starting to feel like she wants to come back now!"

"What happened?"

"She says that in the Arctic civics class, the instructor was talking about Helisport and how the scoring rules were stupid and an example of how Veridians like to overcomplicate things."

"She was talking with me through her neck band the whole time. I got Febric on the line too. We went through the history of Helisport. She got up and told the instructor how you can prove that if you score by giving each type of ball some number of points and adding those points up, then almost always the best strategy is to just focus on one type of ball. But if you multiply, then the number of points you get from scoring with one type of ball goes up with the number of points you've already scored with the other type. So it makes sense to balance."

"But the instructor did not want to listen. He threw up his hands and said that it doesn't matter in real life, in real life you win by focusing on only one thing, the Arctics understand that and Veridia doesn't, and that's why the Arctics win."

"And Lily replied: but think about personal life and work. If you work really hard for almost the whole day every day, you get paid more and you can get a bigger house, but you can only enjoy the house during the time that you're not working. It's exactly the same math! So real life needs balance too!"

"And then she got thrown in detention."

"So now she thinks the Arctics are lame and stupid. They talk about striving for the best, but really they just care about their own authority."

"Some of the Arctics are smart", Gladias interjected. "They do make very good technology. But that teacher sounds like a total loser."

"That's exactly how anyone would behave if they were a loser that happened to be born on a winning team. They put their entire identity into the team, they buy into all the ideas about why their team is great, and how it's a big sign of your worth which team you're from. And then they don't even notice that they don't actually carry any of that greatness themselves, and they're just a pathetic shell on the inside."

"Glad, don't teach Zven to think about people like that!"

"Zven, it's okay to be mean to people who hurt people", Gladias continued, his tone becoming increasingly self-righteous. "If someone is struggling or doesn't understand, then help them and be kind. But if someone is whacking others over the head with authority that they do not even deserve, then hit them right back."

"Glad!"

"If we want to beat the Arctics, then we have to know how to sometimes be at least a little bit Arctic ourselves. But make sure to channel that energy for good."

Seila sighed.

"I guess I can see it from that perspective. After all, we did find Lily by doing exactly what the Arctics do - surveiling everything. Does that count as being a little bit Arctic?"

"I suppose it does."

"Though I'm sure we don't need unrestricted camera surveillance of our whole country's teenagers to bring our loved ones home.

Unrestricted cameras helped us find Lily, but they also helped the Arctics almost break us. I'm confident Zei already knows of some better way to get the good without the bad."

"Wait..."

"What is it?"

"I think Zei might be the perfect mentor for Lily. She needs to have a good role model of what a real winner is like - someone who doesn't lord over others and push them down, but helps lift other people up and wants others to win along with him."

"Mom, Dad?", Zven interjected. "What do we do for Lily now?"

"Can Lily leave Northshore on her own?"

"I asked. She says the borders of Northshore are open but she is in a boarding school, they will not let her leave."

Gladias groaned.

"I guess if nothing else, we'll get all three of our kids back when we take back Northshore."

"Let me and Bai figure out the advice for Lily, Glad. She knows that the Arctics are lame now, she wants to come home. But until we get her back, she needs advice that's not going to get her killed."

Gladias grunted with resigned acceptance.

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3724 FROSTIME 18

"Well, I just heard back from Lektor", Gladias said to Seila as she was pouring herself a tea in their hotel room.

"What did he say?"

"They convened a special closed session yesterday. Verdow was there."

"Lektor presented the evidence that one of the other generals was taking bribes. He opened the topic of reforming the military leadership. He took some advice that Bai gave: he led the conversation in a direction that led the detractors themselves to first defend the existing top general, and then propose another general as the obvious

leader if there had to be a change. Then he flipped over his cards, and showed more evidence - that the top general had been having regular meetings with someone who turned out to be an Arctic spy, and the other candidate they were proposing as an alternative was being blackmailed."

"After that, he proposed a different strategy that they could use against Arctics that involved using their dependence on AI against them. He showed some exploits against Arctic drones including some Kungaupei footage from Redshire - though he explained it away as being sent to him through spies that he knows in Redshire."

"It worked, far more easily than he or Bai expected. It turned out that among the senators who were just lazy and tired, letting Lektor do what he wanted was the path of least resistance - because that meant that the legal case would immediately end. And among the senators who were being blackmailed, they just needed to have an excuse that it would not be feasible to resist, and Lektor's arguments, together with the knowledge that the legal case might at some point start poking into their situations, provided that. So Lektor is the head general now."

"That sounds like a perfect resolution. The Kungaupei can be involved in the military, they can work with Lektor, and he and much of the rest of the structure is still there. So we're not going in and replacing everything, we're preserving continuity. It's exactly what the forecaster bots wanted."

"Exactly."

"So that's it, we won?"

"The battle for the halls of Meldan is over. The battle of Northshore is about to begin."

Chapter 29

Elenar Forest, Veridia · 3725 SNOWMOON 29

Zei and Mu were standing on top of a snowy hill. Green pine trees covered with thick white snow lined the surroundings as far as their eyes could see. A strong cold wind blew into their faces. The rest of their bodies were shielded by jackets covered in a mixture of white and grey, thick enough to allow them to enjoy the surroundings but deliberately not quite thick enough for them to feel warm. Just above the trees in the distance was the sun that had just risen in the east.

"So this is where our future will be decided", Mu said calmly.

"Do you think we're ready?", Zei asked.

"No", Mu replied. "But nothing meaningful ever happens when people are ready."

"What we hope is that the Arctics are less ready than we are."

Behind them, they could hear footsteps slowly walking up the hill, each step coming about a tick after the previous, fighting the deep snow. After a few moments, Zei felt someone hugging him from behind.

"Whatever happens today, Zei, I am happy that I got to know you."

It was Bai. Zei turned around and grabbed her hand, pulling her up the final step so that she could be at the same height as him and Mu.

"I am happy that I got to know you too."

For a few moments, the three of them waited in silence, observing the snow blowing through and over the trees to their east.

"So what's the plan?"

"Last train to Eldil is arriving today. At this point, there's no way the Arctics don't know that we have some kind of large contingent here. So I'm sure they're already starting to prepare their army to meet us. These forests are hard to navigate through, but not infinitely hard. So our best chance is to strike tonight, push into the northern part of Northshore from the west, completely bypass Northglade and then cut off and surround it."

"Are you confident in the plan?"

"In my last battle against the Arctics, I screwed up completely and Mu saved the day. So no, I'm not confident. But this is the best chance we have."

Bai hugged Zei again from the side.

"I believe in you."

Mu broke her silence. "I believe in you too, Zei."

They silently stared at the thick snowy forest of pine trees for another five minutes, and then started to slowly walk back down toward their makeshift bunker.

* * *

Den, Mu, Zei and several dozen other Veridian generals and Kungaupei leaders were sitting inside of a cramped room, where Lektor was standing at the front presenting.

"Alright, does everyone know their tasks?", Lektor asked loudly. "Gallowar?"

"I'm leading Group North, our task is to cross the forest, move far northeast until we get to the coast, and set up a perimeter cutting the Arctics off from the whole peninsula. This will require fighting and surrounding the Arctic force in Northshore on my southern flank, as well as warding off any reinforcements on my northern flank."

"Good! You, Telpo?"

"I'm leading Group Center, our task is to cross the forest directly east. We are expecting to be fighting the primary Arctic drone group. No offensive positional objective, our job is to keep them occupied and make sure they don't break through."

"And you, Den?"

"I'm leading Group South, our task is to cross the forest, move toward the southeast, and then set up a perimeter on the outskirts of Northglade."

"Vakaria?"

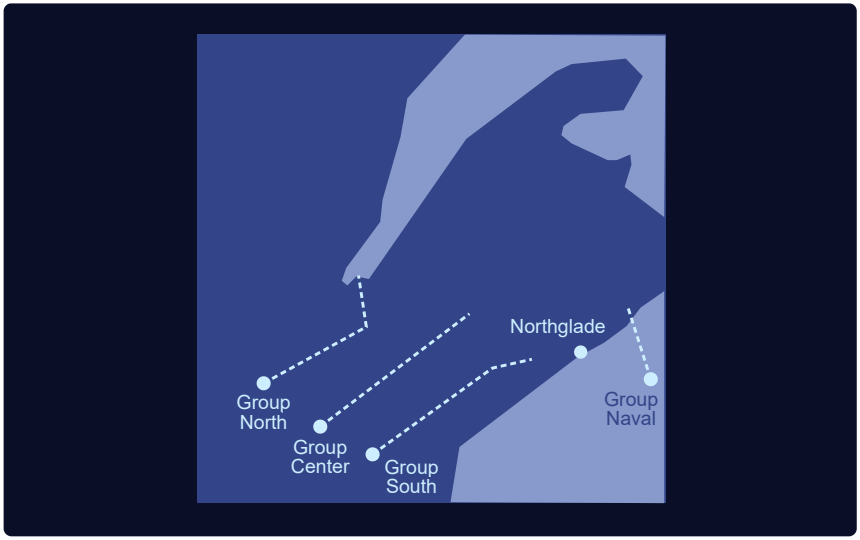
"I'm leading Group Naval, our main task is to attack Northshore east of Northglade from the south with seaborne drones. Distract and surround to the best of my ability. Stretch goal is to also attack the Arctic core territory further east on Thaldur Island."

After Vakaria finished answering, there was a moment of silence. Finally, Lektor spoke again.

"Zei?"

"Supervise all fronts, identify any necessary changes to strategy and tactics in the drone combat."

Lektor turned back to look at the screen again.



Everything that the group leaders had said seemed to match the plan that was on the map.

"Alright, very good. Sounds like we all know what we need to do. Everyone, dismissed. You all have three longhours to make your final preparations. We launch at sundown."

* * *

Zei, Bai and Mu were standing once again at the top of the hill. The sun was by this point just barely above the horizon behind them, minutes away from setting. The pine trees were snowy as before. But the landscape had radically changed. There were thousands of snowmobile-like contraptions sitting at the edge of the forest. Many were visibly equipped with weapons. Zei could make out a large number of thin silver lines glimmering in the sunlight, starting from the rear of many of the largest snow drones and stretching back as far as he could see. Fiber optic cables, he realized. They would stretch for hundreds of kilometers tonight - and they are what would allow Zei to reliably stay in contact with the drones as the front advanced.

"Wait, is that a box of *printers* on top of one of the drones?", Bai suddenly asked.

"Yep. There's also other drones that are carrying stick-on paper and shirts. A lot of classroom equipment. Just in case we need to rapidly apply visual attacks on the Arctic drones' algorithms."

"I see..."

Zei's watch buzzed. He turned around, and saw that the bottom of the sun had already begun receding below the horizon. "Fi le gei fa", he thought - one hundred ticks until it begins. He reminisced about all of the Minpentai games, from practice to championship, that he had played over the past decade. Without realizing it, all that time he had been training for the one game that really mattered: tonight.

"Time to get back to our command post", Mu said softly.

Zei turned around. The sun was already halfway below the horizon.

"Indeed, we should."

The three of them walked back slowly to their bunker at the bottom of the hill. They opened the door, walked in, took off their snowshoes and closed the door.

As soon as they were inside, a robot slid up to them, bringing tea.

"Thank you", Mu gently whispered, patting the robot on its head and taking her tea. The robot beeped softly.

Zei and Bai also picked up their teas.

Zei's watch buzzed.

He heard noises of snow drones accelerating outside.

The mission had begun.

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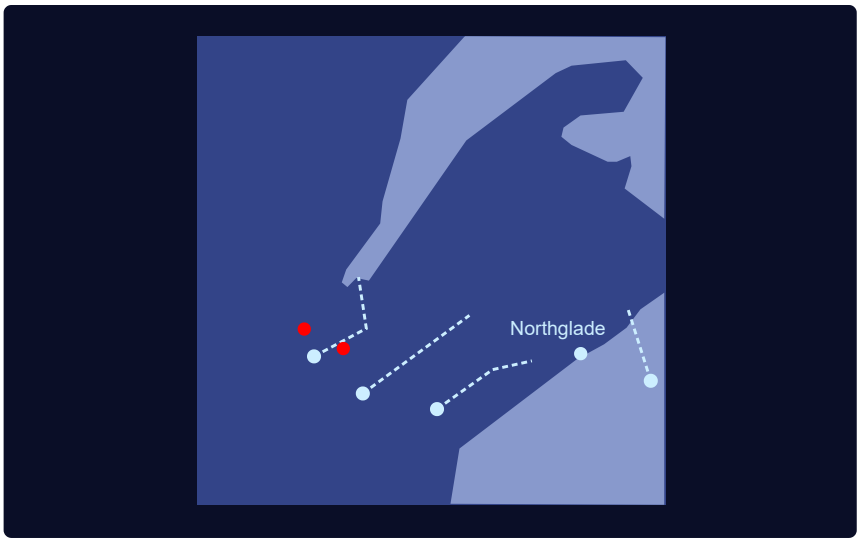
3725 SNOWMOON 30

Zei and Mu were sitting on their chairs in adjacent but separate rooms. They could each see the same command screen. On the top left, the clock was showing the time. It had just struck midnight.

All three land army groups were advancing forward quickly. The launch of the Naval group was delayed, so that its assault would come shortly after the beginning of the main battle. So far, the three main army groups had easily defeated a few small Arctic sentry units, and were continuing to proceed at full speed.

An alarm blared.

Zei immediately saw the cause on the map.



Zei could hear Gallowar and Telpo coordinating in the chat.

"Adopting defensive positions, spreading out to prevent encirclement", Gallowar shouted.

"Sending part of Group Center northward to attack from behind", Telpo replied.

Zei zoomed in. The Veridian and Arctic drone army groups were quickly closing in on each other. They were firing a few long-range weapons, and ten drones from Group North had already been disabled - on top of the few dozen that had already been lost to stationary Arctic traps scattered throughout the forest - but so far the main hostilities had not yet begun.

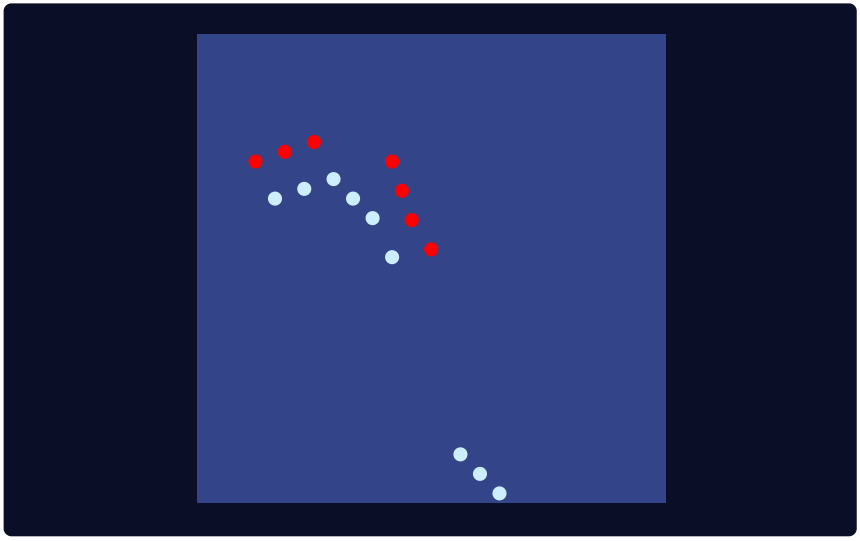
Over the past month and a half, he had played simulations of forest snow drone combat hundreds of times, and he knew what the best strategies were for detection, hiding, attacking and taking cover. The strategies had already been loaded into the drones' AI models.

A visual image appeared onscreen. It was the exact design of the Arctic snow drone. It was slightly different from the images he had seen before - faster and more agile, better protected against stationary traps, and with better defenses on the sides and back. The Arctics had been anticipating another rapid offensive campaign, he realized. These

improvements came at the expense of effectiveness in traditional fixed face-to-face combat. The Veridian snow drone fleet had made some similar modifications, but to a significantly lesser extent. Hence the battle style that was most to their advantage seemed to be ... direct face-to-face combat with fixed battle lines.

Not good, Zei thought - the Arctic drone fleet was larger. They would have to make up for it with better micro-tactics.

A few minutes later, the drone fleets came close enough to clash. The Veridian drone fleet pre-emptively cut down a large number of trees to prevent encirclement, particularly on its north flank. Gallowar's strategy was simple: delay in the north, fight in the east. In about twenty minutes, Telpo's fleet would attack the eastern Arctic group from the south, and hopefully surround and defeat it.



Zei watched carefully. Drone casualty counts were starting to increase rapidly on both sides. Soon, both the Veridian and Arctic losses spiked above a hundred.

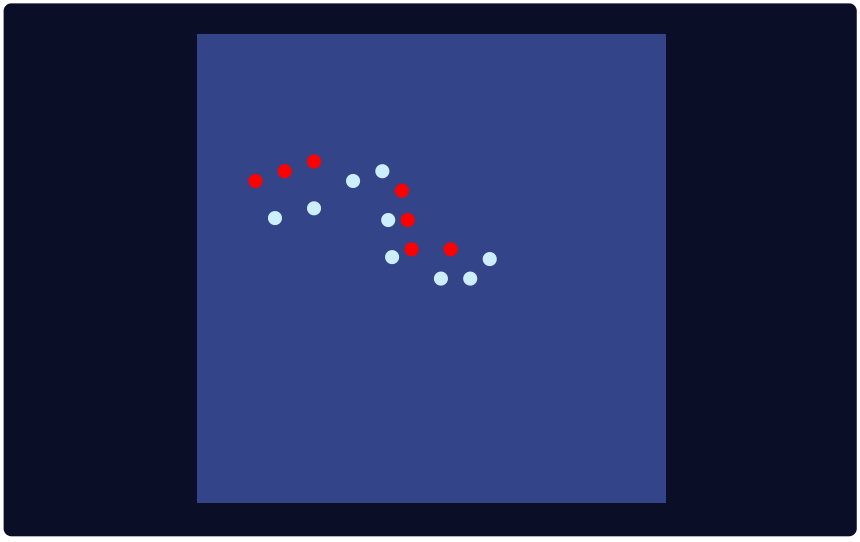
Zei continued scrolling through all of the visual feeds. He saw some large boxes on top of Gallowar's largest snow drones open up. A large

fleet of Helisport copters and delivery copters, seemingly modified to carry more dangerous payloads, flew out. Ah right - the Arctic snow drones were also vulnerable from the top. Good job converting civilian infrastructure, Zei thought.

Embedded in each of the army groups, sitting in the command centers safely far behind the front, were a large number of Dzego's best Minpentai players, which had been quietly flown in by the Kungaupei, as well as some of Veridia's best Helisport players - the two groups with the best hands-on experience understanding airborne duels.

Zei continued watching. For the moment, Gallowar's strategy seemed to be going well. His army group was successfully slowly retreating on the northern flank while cutting down trees and leaving traps, and the battle on the eastern flank was progressing smoothly. Exact casualty ratios were difficult to determine, but from the numbers that appeared onscreen, it was significantly in the Veridians' favor.

Forty minutes later, he saw the good news that mattered: Gallowar's forces had broken through between the Arctic north and east flanks, and Telpo's fleet had begun engaging the Arctic east flank from the south.



Soon, Arctic snow drones in the east began to get rapidly disabled, while Gallowar's western flank continued its slow retreat to slow the Arctic northern flank down. The Arctic drones attempted to replicate the slow retreat strategy, but without success: by that point, Telpo had succeeded at almost completely surrounding them, and they had nowhere to go.

A few minutes later, Gallowar declared the Arctic drone fleet sufficiently defeated. He ordered his eastern flank to go around them to avoid any risk that partially disabled remnants or their traps would cause damage, and then continue northeastward full speed ahead.

Zei zoomed out, and looked at the big-picture view again.



Another one and a half longhours passed, and all four Veridian drone fleets continued advancing - except for the small offshoot that Gallowar had left behind to delay the contingent of Arctic drones to its north.

An alarm blared again. This time, it was Group Center and Group South that were both under threat.



"Alright everyone", Lektor shouted. "This is the real deal."

"Defensive positions, slow retreat in the center and attempt flanking on the sides. Be careful, minimize our losses and maximize theirs. Do not let them break our positions."

A few moments later, he added:

"For Veridia!"

"Dze go ba fau gie", Den shouted immediately after.

Zei directed the drones to begin using all their tactics, including the ones that he had learned from the clandestine battles in Redshire months earlier. They would only work for a short time before the Arctics adapted, he knew, but it was now or never.

Both sides' drone fleets spread out and created defensive lines. On the fully zoomed-out map, Zei could see the lines slowly come closer. A few minutes later, the clash began. Losses quickly pushed into the hundreds on both sides.

Gallowar's northern group suddenly switched from retreat to advance and began using the same tactics.

Zei watched carefully, holding his breath. He scrolled through small-scale tactical maps and real-time visual feeds.

Something was wrong, he realized.

"Emerald, analyze all micro-tactical and visual fields, determine which tactics are effective."

"Analyzing..."

About twenty ticks later, Emerald responded.

"Tactics one and five working with medium effectiveness. Tactics two, three, four, six have zero or slightly negative effect, Arctics appear fully adapted."

Zei groaned. Despite his best efforts, he knew, the Arctics must have successfully recovered at least some of the visual feeds from the drones in Redshire that he and the Freetown defense companies had trained on.

"All units, cease using tactics two, three, four and six. Use tactics one and five. Units with IDs ending in 0 or 5, explore random modifications."

For a few minutes, the casualty ratios moved in favor of the Veridian armies again. Then, slowly but surely, the Arctics again started to gain the upper hand.

"Emerald, which modifications are successful?", he asked.

"Eight were attempted. Only number seven was successful, and has already been introduced as the mainline version of tactic five. Recommend retiring tactic one."

"Do it."

Zei began to sweat with consternation. He had not anticipated the Arctics being able to resist their tactics so effectively.

Come on, Zei thought. Think of something.

The seaborne drones appeared to have been successful, at least in the early stages of their assault. They had landed ground craft, both east of Northglade and in Arctic core territory in Thaldur hundreds of kilometers further east. The seaborne drones had already begun returning to base, to receive their second round of payloads.

For a moment, Zei was pleased. But it was still far from enough.

A robot entered the room, and quietly slid up to Zei with an energy drink. Zei picked it up, gulped about half of it, and set the rest down on the table. Within moments, it took effect - but the only effect seemed to be to make Zei despair even harder.

And a moment later, a message appeared onscreen that drained away all of the remaining light in Zei's eyes.

```
Rear base #3 appears to have been completely destroyed by
highly accurate hits from Arctic missiles.
Most likely cause: internal traitors revealed the location of
the base.
Current expectation is that almost all personnel in the base
did not survive. Probability that Den, commander of Group
South, survived is estimated at under 10%.
```

Zei began to cry.

For the next few minutes, he half-heartedly whispered into his microphone, proposing minor adjustments to drone tactics, trying to optimize positional advantage and casualty ratios. Mu had immediately taken over as the commander of Group South. But in his heart, he knew that the battle was lost.

And at that moment, Zei's watch buzzed.

This was not a standard military communication channel. And this was something that Emerald deemed important enough to override his strict "do not disturb" instruction.

Zei whispered into his neck band to connect the watch to the display in his glasses. A message appeared onscreen.

```
This is Deluin.
Over these past months, I have been executing on my plan. I
have earned the Arctics' trust; with the combination of my
successful Minpentai background and my parents' status, this
proved to be easy. As a result, I am now one of the commanders
on the Arctic side.
We are being heavily surveilled, and there are great limits to
what I can do to the Arctic drone fleet without blowing my
cover. However, I have managed to set up an exploitable
weakness for you.
This file contains descriptions of tactics that should be
effective specifically against the southern flank, which I
control. If you apply them, you should be able to break through
about twenty kilometers north of the shore, and then proceed
north and south to expand the rupture into a complete victory
on the southern front.
From there, it will be up to you to surround the remaining
forces from the south, and recover Northglade.
```

Almost immediately, a burst of energy returned to Zei's mind. He would have to mourn the loss of Den another day.

"Emerald, confirm the authenticity of this message."

"Message was signed by the same key that was embedded in Deluin's original message that you received on Grapetime 15."

"Analyze the tactics, do they seem plausible?"

"Tactics are highly aggressive. They seem unorthodox, but plausible. They will be very effective against an opponent that is vulnerable to them, but will lead to much faster defeat against an opponent that is not. Running a simulation."

About twenty ticks later, Emerald spoke again.

"Tactics would lead to faster defeat if used against Veridian, Kungaupei or Freetown AI. Their effectiveness hinges entirely on Deluin being correct that the Arctic AI has the particular weakness that he is positing, which could be either discovered or planted."

Zei thought for a moment, and then decided to ask for human help.

"Bai, I'm sending you a message from Deluin. Please tell me what you think of it."

Bai responded almost immediately.

"Reading now."

About thirty ticks later, she responded again.

"I don't know if we can trust Deluin. All that we have to go on is his word. Even his message to you back in Grapetime could have been planted. The Arctics are fully capable of these kinds of strategies, Delwart had planted evidence for months against Gladias and Mov."

"But I have a different idea."

"I'm listening."

"We have no good reason to trust Deluin. But Deluin has every reason to trust us. So we flip it around on him. It's his responsibility to take the first leap of faith."

"And what do you want him to do?"

"Send us the raw weights from the Arctic drones' AI model."

Bai paused for a moment, and continued.

"AI models are not like regular software, where you can analyze the whole thing and identify safety properties and mathematically prove them secure. They're unpredictable beasts - they are grown, not forged. They don't have walls, they have immune systems. And so if we have the raw weights, we can run simulations in parallel and optimize attacks that will trick them."

Zei's mood once again became brighter.

"It's actually much easier than running simulations in parallel. You use gradient descent. You can take the derivative of the outcome of the model in the simulator against all of the variables that we can change in the field - drawings that we stick on the side of the drone, the way the drone acts. You automatically identify the small change that will maximize the increase in the probability that their model fails to notice our drone. Even a small step that increases the chance they don't notice one of our drones from one in a million to two in a million can immediately be found."

"You keep finding adjustments that maximize the likelihood of success. Try different simulators with different parameters - remember, the final attack would appear in the real world, not inside a simulator, so the best we can do is filter for changes that are robust

across different simulators. And after a million rounds, hopefully we will have something that works quite well, and in real life. To us humans, it will probably be a bunch of squiggles that look incomprehensible and weird. To the AI that we're optimizing this attack for, it's like a magical snake with a deadly stare looking at you right in your face."

"We probably can't convince their drones to turn against their masters or shut down, but we can get them to not notice our drones, maybe even shoot their own feet and overheat and become disabled."

"Wow. Alright then, let's send back a message to Deluin and ask."

"Emerald, draft a message to Deluin expressing what Bai just said. And send it back through the same steganographic channel that he used. Hopefully, Deluin will understand."

Chapter 30

Elenar Forest, Veridia · 3725 SNOWMOON 30

Zei watched the screen carefully, checking for any signs that the Arctic drone fleet might be breaking through. Forty minutes ago, he had given the recommendation that all groups do a slow retreat, cutting down trees to slow down snow drones and laying traps, with the primary goal of dragging out the battle longer. Lektor had taken his advice, and had relayed that he expected the strategy to work well at least until sunrise. So far, he had been right.

Zei's watch buzzed. It was Deluin.

```
I understand your concern and I fully agree that you do
currently not have good reason to trust me.
Unfortunately, I do not have a way to give you the raw weights
of the language models on the drones. Those are well-protected
and only a very small number of people have access to them,
precisely because they are worried about an adversary using
those weights to run gradient descent attacks.
But I can give you the next best thing: API access to their
simulated environment. This includes simulated copies of Arctic
drones, both full world-model simulations and partial
simulations that focus on their input modalities and give you
classification results - given a particular visual input
stream, what kind of object they think they are looking at.
Here are the network addresses and access keys, as well as
addresses and keys for Arctic endpoints in Redshire that you
can proxy through. Remember to be safe and minimize risk of
detection.
Network:
13178080a5bf1c815dfb1300c6039af76c7b8f14e47581d3549e8990012e1d5f:
35a504923a4d8dde09e0ed814ec05ae81be8e91c2bcb249b22ad13af0bcf7b4d
e680176e029dcfd96b27e328149f8db4ab368a87762568799bf8a56b95f968c5:
519cce3e5e0ee46fbaf2735a53429d6cb1829485e2886962dde9606cab13f0a2
04569c9a285f10e1a3670985a1fa0696530feae84eae853209d1f0345a214933:
8d174efbc3857326999a58e4d4eea9c7fbc61fd56dd233707321d9d5c0bfb9ce
fed7dd2e3ee0f2c9a1ed4b61dd21ae165780497b39ae4ae5b06b1b8b0b15a034:
a9bf7a809e4e998946e59e377e7e96cd9f22dd4db9a87080b669de3e74a03d83

Proxies:
4446e4f5350e5d76fe0b5254f50f3923fc2f480d1d9df33da1dc58d532e49e86:
9b0dadf8d9418283416f70d60afd909c7346e63cbf7c815b110926eeb943afff
27770fdc6c08da318e64f0f16d1d0e300958e4eefe1b55de1a48a9c6a7872ecf:
884c9a310b61edf0e73d2d9e9352ac9f21b8c354aa90203468a76f1a2383ff08
dfb1c543a259ade8d53344893807d47c0bebd02ed04cde46e31335d2e71f7ac3:
b2eb4a17a06c68ab7bf149bb031702ff3514d5571d83c9a2c652dbdaa50768a9
0ca84c8548d95f6edcd3b6fd1b695d743aff2904cec00fd411bd9d36566e4048:
af0dd33ec09e8e8331ff34bf9d15432d8f91ec4870720d8fdc2bdffe29f871d5
```

Zei immediately forwarded the note to Bai. He also sent a copy to Fin, and sent Emerald an instruction to send Fin a maximum-priority emergency alarm and an explanation of what was going on.

Fin responded almost immediately.

"Alright Zei, what do you need?"

"Reading..."

"Yeah this seems doable. It's a not very well known fact that you don't actually need the raw weights of a model to effectively train adversarial attacks. There's definitely Bansunpei people who are very good at this stuff - 'language models' are quite connected to language!"

Zei also forwarded the note to Mu. Emerald automatically formed a real-time chat group. After a minute, about ten people were sharing insights about the problem and beginning to attack it in parallel.

"Let's assume that our attack medium is that we have our snow drones, and we have stick-on paper that we can print things on. We'll assume we deploy this after sunrise. Here's what you do: start with one pattern, run it inside the simulator, have our drone slowly approach the target and see how long it takes for the target to detect it. Measure detection time so that we have a scalar signal of how well we're hiding, not just a binary."

"Continue making small modifications to the pattern in all kinds of directions. Keep each modification that increases the target's detection time. It doesn't matter if the process has a lot of noise, the signal accumulates and the noise averages out over time. It's not as fast as gradient descent, but it still works. Eventually you get to a pattern that reliably tricks the adversary."

"We can do better. We have our own models that were trained in similar ways to the Arctic ones, and we obviously have the raw weights to those. So we can use gradient descent against our model. Identify, say, ten optimal perpendicular directions that maximize time-to-detection according to our model inside our simulator, and then use those as the trial deviations we try against their model inside their simulator. Simultaneously, we also use the results of every query to fine-tune our model to make it a better predictor of how the Arctic model will behave on these attacks."

"How quickly can we get this working?", Zei interjected.

"Coding should take at most ten minutes, Emerald can do it quickly. The full training loop ... maybe a longhour?"

"So it will be done around sunrise?"

"Exactly. And then you will need I don't know how many minutes to actually print the attack patterns on the sides of the drones, you should prepare that pipeline on your side so it's ready to go as soon as we are."

"Alright, everyone. Let's get moving."

Zei sat back, and kept scanning through zoomed-in views of the battle map and visual feeds. About a third of the Veridian drone fleet was already lost, but casualty rates had decreased considerably. The slow retreat strategy was working well: it also meant that any Veridian drones that had motion disabled by combat damage but had their weapons still intact could go into low-power mode and then wake up and do a final burst of damage as soon as the Arctics themselves advanced to their position.

After about fifteen minutes, Zei's watch buzzed again. Emerald was telling him to rejoin the chat.

"We have some early good news", a Bansunpei researcher exclaimed.

"What is it?"

"We've successfully accessed the Arctic simulation platform, and we've run the first few hundred trial rounds against the simulator. We have patterns that are increasing detection time by about five percent."

"Only five percent?"

"That just means the pipeline is starting to work."

"Oh, it just went up to eleven percent."

"It's definitely working", another researcher interjected. "We're continuing to run Emerald in parallel, optimizing the software. It's finding ways to extract more information from each query, and better parallelize the queries so this all happens faster."

Zei's watch buzzed.

```
I can see you are starting to run queries! Though you want to
be more careful - if I can see it, the sentries might be able
to as well. Here are some more proxy endpoints.
c8baa6614d9fba874af5cb90ba7c3d1a53aabaa8facd1e4f5fe46bdeb229a106:
3fa5fbd0bc8aa8af8a4842b099dad4e700b5d54edcd4a21c9e272552318d4a55
8d01f119fbdb499f24a298c59c1b83a7f591df97f3b7a9823e313b1bff3872ca:
1a5529862c273fefb898acccdfa0bec29d594904b427eaee88b7b1df393fdb0b
44ee73e0355611ad0d975bda2eec43c21e490cb42c61f71bc03b3b4bd7718ad1:
bf53236e5d72d2142348d5529275908f995b77791f07a5c818eca0cfc400c1ac
ab7a609ce6f71f55571184cb5d32996a3b5ddd08230ae788f7f65355d5385abe:
254804b48ae25417535e6336735d0b66bdd7d20afe77ac687e933d8bd5c24d24
95d40f8a44c6c3f4d32d2a4c7c533f9dd71c5d9ac1020361e9e7f4552967a778:
12b152f44f00a1f542f4ac6fbb3aced5144c8a98d813a34dce1cb5abc18c3314
508a323cda2f87b053986f17d0c073a51c9a001b9b9152a393fe06ff4a30072b:
eff66952719415cc534e8f1feb06aa044fa468a81c911a14b95f9be93371e528

Also, add a bunch of dummy queries that are doing other things,
and maybe even make each query attempt multiple different
attacks simultaneously so it's less clear what it's doing.
```

Good advice, Zei thought. He forwarded it.

To verify the effectiveness of the simulator, they also ran simulations in a nighttime environment that exactly replicated a real-world setting of a small battle that had taken place earlier in the night. Another team, two researchers and Emerald instances, began figuring out how to automatically analyze the discrepancies between the simulator and reality in the nighttime setting to learn how to construct queries to the simulator that would better match real-world situations.

Zei watched the attack patterns become more and more optimized, sheets of paper containing squiggles and messages that seemed simultaneously very recognizable and familiar but also completely alien. Inside the simulator, the time that it took for the Arctic drone AI to detect their incoming drones more than doubled.

After another thirty minutes, it doubled again.

Lektor gave the order to set up printing infrastructure about two kilometers behind their current positions - a place they would be able to safely organize a retreat to around sunrise.

Human Veridian soldiers were already tens of kilometers behind the robotic frontline, and Lektor ordered them to start moving toward

the front, so that they would be available to paste the attack patterns onto the snow drones. Manual dexterity was still a major weakness of AI.

Slowly, the sky began to shift from a near-total pitch black to a dark blue that was especially visible in the far east. As the minutes passed, the blue became brighter.

"Alright, we have some good patterns. It's getting close to sunrise in Northshore. I recommend shipping them now."

"Agreed", Zei replied.

"Then let's do it", Mu added.

* * *

Zei flipped through the visual feeds in his display. The majority of their drone fleet had retreated a kilometer, and a large number of pieces of paper were being printed and then almost immediately getting distributed and then stuck onto all sides of each snow drone and sky drone. The plan was in motion.

"Group North reporting: all non-forward drones have been equipped."

"Group Center reporting: all non-forward drones have been equipped."

"Group South reporting: all non-forward drones have been equipped."

"Group Naval reporting: we commandeered equipment in villages on the outskirts of Northglade, the locals were amazing, many woke up in the middle of the night to help us. Most non-forward drones have been equipped."

"Alright then", Lektor spoke up softly but firmly. "Action."

Zei checked the visual feeds, curious what the final version of the attack patterns actually looked like. He saw an endless row of snow drones with large pieces of paper plastered on all sides, each paper containing many colorful and squiggly patterns - and in a giant font, the word "**DOG**".

A few minutes later, he could see the results. The Veridian snow drones advanced quickly, pushing through enemy lines and turning the front into a chaotic mixture of drones on both sides. Casualty ratios were more than ten to one in the Veridian drone fleet's favor.



Zei allowed himself to feel a moment of joy - the first genuine joy that he had felt the whole night.

He remembered Den's words to him the day that he had almost caused the failure of the mission to rescue the Redshire president's son. "Outside the box as in, jumping out of your console chair and running over and bashing the other player in the face."

Breaking into your opponent's simulation environment, running millions of queries to optimize squiggles that you could draw on a piece of paper, and tricking your opponent's drones into thinking that your drones are dogs - did that count?

It probably did, Zei decided. Den would be proud.

As soon as he felt he had celebrated enough, Zei turned to a visual view of the statistic that really mattered: the casualty ratios. They had already dropped to six to one. So far the Arctic army had not yet

adapted, but they had adopted emergency defensive positions, each squad of drones circling into a formation, to buy time. And they would adapt soon, Zei knew.

The chat that contained Bansunpei and Kungaupei members making queries to the Arctic simulation environment switched to optimizing simpler attacks: motion patterns that maximally delayed detection and interfered with targeting, sounds that could trick them, as well as the accuracy of their own fire - all mechanisms that did not require printing and sticking on new pieces of paper. They also incorporated some of the aggressive micro-tactics that Deluin had sent along with his original message.

Casualty ratios briefly increased back in the Veridians' favor, and then started decreasing again, though slowly. Zei continued watching.

After fifteen minutes, all access to the simulation environment was suddenly cut off. At around the same time, casualty ratios suddenly dropped back down to only one and a half to one in the Veridians' favor.

They had destroyed a large part of the Arctic drone fleet, but a large part still remained. They were now in a complicated positional battle, with a chaotic mixture of both sides' drone squads littering the battlefield with no clear front line.

Veridian human soldiers were only hundreds of meters behind the frontline, finding snow drones that were quickly repairable and repairing them. They were wearing thermal cloaks to minimize the chance of detection, and smoke from nearby burning trees provided additional cover. In a few of the visual feeds, Zei could see images of Arctic human soldiers doing the same thing in the distance, but they were far fewer in number: while Veridians had been retreating through the night, they had been laying traps along the way, and those traps were much more effective against humans than against drones.

A few minutes later, Zei's watch buzzed again.

```
I hope you trust me now.
My next message will contain exact coordinates of some of our
units on the southern flank, and recommended weak points to
attack through.
jie fe hen dzi.
```

Zei smiled.

Of course I trust you, Deluin, he thought.

Almost immediately, Zei's watch buzzed again. He had received the second message.

He forwarded the coordinates to Mu.

"It's your chance to shine, Mu."

"Don't forget Vakaria."

Within minutes, casualty ratios on the southern front began to diverge heavily from the fronts further north, going back up in the Veridians' favor.

And only a few minutes later, they seemed to have achieved the result that they needed.



Mu transferred control of the southernmost portion of her drone fleet to Vakaria, who was assigned the task of entering and regaining control of Northglade.

The rest of her drone fleet swept northward. Over the next longhour she managed to completely surround the remaining Arctic drone fleet, linking up with Telpo's northernmost flank almost exactly a longhour after sunrise.

Gallowar's Group North defeated the remaining Arctic drones to its north and reached the river that marked the delineation between Veridian and core Arctic territory. A few drones crossed the river and bolted toward nearby Arctic towns, causing the Arctics to send emergency reinforcements. A portion of Telpo's fleet headed north to join Gallowar, reinforcing the northern front.

In the meantime, the second trip of seaborne drones went directly to Thaldur - the island east of Northshore that was core Arctic territory - and landed a fleet of air and snow drones that consolidated control over the island over the next few longhours. The Arctics could not effectively react, as Thaldur was too far away for either side to reach quickly, and the few drones that were in range were stuck defending against Gallowar.

By late afternoon on that day, the fighting grew quieter, as the last remnants of Arctic soldiers and drones in Northshore and Thaldur were methodically hunted down.

Zei sat back in his chair, and set it to incline backward as far it could. For the first time in over fourteen longhours, he went to sleep.

Chapter 31

Northglade, Veridia · 3725 RAINMOON 2

Zei, Bai and Mu slowly walked up the staircase on the side of the mountain. The mountain was fully covered with snow on both sides, and the sunlight reflected off the snow into their eyes. Only pine trees and the staircase with its handrails were visible above the snow - the trees and the handrails because they were too high for the snow to fully cover them, and the staircase itself because it was being regularly plowed and salted by a drone that they could see about thirty meters further above them.

A strong, cold wind blasted into their faces. They continued climbing. Several minutes later, they reached the circular platform at the top of the mountain.

From there, they could see a full view of the island on which Northglade was located. It was about twenty kilometers wide - small enough that the most simplified and highly zoomed out maps of the region did not even bother showing it, but at the same time large enough that it seemed like if it really had to, that island alone would be large enough to sustain a civilization. If Northglade had been a Dzego city, Zei thought, it would have a fully self-sufficient emergency industry already.

To their east, they could see the main city of Northglade. To the west and north, they saw snowy hills for kilometers on end, and a path gently twisting as it descended deeper into the forest. Looking downward to the east, Zei could see another staircase leading up to the

summit, and a young man wearing only a sweater and a light hood climbing up.

As he neared the top of the summit, Zei could recognize a familiar face.

"Deluin!", he shouted.

Deluin raised his head up, saw Zei, and ran up the last twenty meters of the staircase.

He reached the top, and almost threw Zei off balance as he threw himself into his arms for a hug.

"I'm so happy you're still alive, Zei, and that I can finally see you again."

Deluin slowly let go of Zei and turned toward Bai.

"Wait, do I recognize you?"

"I'm Bai. You might have seen me watching some of the practice games in Sadzu Du."

"Ah yes, I remember you!", Deluin replied, somewhat unsure if he was genuinely recollecting a memory of seeing her or simply confabulating one.

"And I'm Mu. Don't you dare mix me up with Su - she's still back in Dzego, she's a completely different person from me."

"Nice to meet you, Mu", Deluin replied, gently laughing.

"So tell me. What happened?"

"Well, this is a long story. Are you sure you want it told at the top of a cold snowy mountain?"

Zei, Bai and Mu all immediately nodded in unison.

"I see you're more fit to live in Redshire than I might have expected!"

"Okay fine, I'll start."

"Back in Grapetime, about two months after they occupied Redshire, some Arctic officers came to our house, to talk with my parents. My parents were never particularly worried about the Arctic threat, saying we should be more open to peace and cooperating with them. But once the invasion happened, they quickly flipped to just

being full supporters of what the Arctics were doing. Quite a large chunk of the Redshire population did."

"The Arctics offered them a position. They said that they should come to Glasscrown, and advise the top leadership on integration - first of the captured Redshire population, then of Northglade residents and eventually of everyone else that they would conquer."

"I made a big risky decision. I decided I would 'go native' - pretend to be a full-on supporter of the Arctic cause, try to use my parents' reputation together with my own status as a Minpentai champion-"

"I'm the Minpentai champion!", Zei playfully interrupted.

"Only because the game was cancelled and they gave you the title. You never beat me."

"I beat you in the battle two days ago!"

"What was that line from that old poem? mi fu zo de tei ka ciu pin - when I am defeated, I feel like I won?"

"Now you get it!"

"Anyway, I used my parents' reputation and my status as a Minpentai *semifinals winner* to climb up the hierarchy myself. Within two months they figured out independently that Minpentai was an effective way to train people for modern warfare-"

"Wait, they figured out what Minpentai is for?"

"Wait ... you mean you guys *intentionally* designed the Minpentai tournaments to train people for defense?"

Zei, Bai and Mu immediately nodded.

"Then no, they definitely did not figure that out. They just figured out that Minpentai *happened* to be effective for that purpose. And so within three months they put me as one of their top commanders."

"The Arctics are actually quite easy to play, you know."

"What do you mean?"

"Their entire ideology completely de-values alignment, and only values competence. They see top talent as being fungible. People as bots that can just be reallocated by pointing a higher salary at them. They don't really have much of a theory of people having their own motivations, dreaming their own hundred thousand dreams, and how

to tell when someone's heart beats together with yours and when they are just pretending. And so if you just say the right slogans, and convince them that you're top talent, they just ... trust you."

"Wow."

"I'm ... actually not surprised", Bai added. "I'm just glad that you managed to play the game better than any of the others did."

"So what are the most *interesting* things about the Arctic Empire?", Mu quietly asked. "What can we learn from them?"

"Their advancements in technology are real, and they are amazing. The medical treatments that they gave to the residents in Northglade are only a sample. They believe that they have already figured out how to make people live a decade longer, and that they're only accelerating."

"Glasscrown is one of the most fascinating cities I've ever seen. It's beautiful in its own way. It makes you feel like you're living inside of a giant ice crystal. Its tallest tower is a kilometer above ground, and its caverns stretch deep underground."

"They even have rows of pine trees alongside some of the walls. That itself is an impressive feat of engineering: genetic engineering to help the plants better tolerate cold, together with underground heating and cleverly shaped reflective walls to make their environment warmer."

"In the winter, Glasscrown has a month and a half of polar night with no sun. Then, the whole city gets enveloped with the glow of white and blue artificial light."

"Now you sound like you want to go back!"

"Actually, I want something else. I think it's good that the Arctics have built all of this technology - though if I could shut one of the things down, it would be the research on superviruses. But what's dangerous is how they keep everything they make for themselves and don't let anyone else benefit. Even when they gave some of it to the Northgladers, it was on their own terms - you come to our health clinics, you don't get to learn what we are doing to your body, and if

you don't want to go back to how you were before, you have to keep coming back to our clinics again and again."

"Instead of science and technology bringing the world together, it's turning one group into a bigger and bigger threat that the rest of the world has to fear more and more."

"But thanks to you, we - we as in all of humanity - have now captured a big pile of their technology. Some that they brought into Northglade. Piles of their military equipment scattered around Elenar Forest. And a bunch on the island of Thaldur, core Arctic territory that has tens of thousands of people living on it."

"So our next step should be to convince as many of them as we can to volunteer, and share their knowledge and expertise voluntarily, and bring the whole world up to at least the frontier that the Arctics have created to keep for themselves."

"Open source - it's not just a good idea, it's the law", Zei replied.

"Exactly. Make the whole world a little bit more like Veridia in that way."

"And then we can play their strategy in reverse. Show the people in Thaldur that they have more calm, more peace and more actual freedom under our way of life, at the same time as having powerful technology. Then perhaps through them, we can convince even more Arctics to join our side."

The conversation paused. Deluin took his turn to walk all around the top of the mountain and enjoy the view of Northglade.

About ten minutes later, Mu spoke up again.

"I got some bad news, Deluin."

"What is it?"

"Freetown defense corps are reporting that the Arctic garrison in Redshire has tripled. They tried some sabotage operations with their drones already on Redshire using the tricks that we discovered during the battle two days ago, but they are mostly failing now."

"They now think that there is basically no chance of us getting Redshire back."

Deluin sighed. An expression of deep sadness filled his eyes.

"Tafindel is recommending that we play a little bit of hardball. Close the borders between Thaldur and the rest of Arctic territory. And then offer a deal. We let anyone who wants to go back to the Arctic Empire go back, and in exchange anyone in Redshire who wants to leave is allowed to leave. They keep Redshire, we keep Thaldur. And then find a beautiful place in Veridia to make a new Redshire."

"Why not just give them the peninsula between here and Thaldur, the one dangling off the eastern end of Northshore? Very beautiful land there."

"I think that should be up to them to decide."

Deluin nodded. "Tafindel is probably right. It's as good an outcome as we're going to get."

The conversation paused once again. All four of them looked toward the east, and enjoyed the view of snowflakes gently falling on every building in the town of Northglade and covering it with thick snow.

* * *

Gladias and Seila walked through the large stone gates at the entrance to the Greater Plum Harbor in Northglade. The turnstile gates had been smashed and thrown to the ground to let people freely move in and out, and the administrative staff did not seem to have the heart to enforce any visa rules at that particular time. Two Greater Plum Harbor security guards, wearing purple and grey attire, simply watched the crowds move in and out, and resigned themselves to meeting the fallback objective of preventing any theft and violence.

Seila's watch buzzed. She raised up her hand, and took a look.

From		Message	Time
Jahn	→	[5 images]	45904

She clicked, and took a look at the pictures. The first three were pictures of people in Freetown in its streets and parks celebrating.

The fourth appeared to be an open public concert hosted in a parking lot. In the background was a giant factory building, on the side of which there was a prominently displayed logo of Ichara - one of the now famous Freetown defense companies.

The last was a picture of Jahn himself taking a selfie with what appeared to be one of the managers who worked at the defense company.

"Are you going to go to Freetown and see Jahn?"

"Actually I was thinking to stay in Northglade another five days or so after you leave, and let him come here. He's always wanted to come see Veridia."

"You should", Gladias replied gently.

They walked forward along the street for about a hundred meters, and then descended down into a tunnel. About two hundred meters later, they turned right and walked up a staircase. At the end of the staircase, there was a door.

Gladias prepared to tap his watch against the green circle on the side of the door, but before he could do so, the door opened on its own.

"Dad!", Lily yelled as she ran up to Gladias and hugged him.

"Lily! I'm so happy you're safe."

Febric and Hreda emerged from another room, and rushed to embrace Gladias and Seila.

"I'm so glad that you are all here together."

"I'm sorry, Mom."

"Don't be. We all made lots of mistakes this past year. We've gotten through it now, and if we need to, we can get through even more. We're a family."

"Is Zven here?"

"Oh, Zven, Vil and Daia are coming tomorrow morning!"

"Amazing! I'm so excited to see Mom!", Hreda exclaimed.

"So how has the Greater Plum Harbor been treating you?", Gladias asked.

"It's been actually quite fun.", Hreda replied. "They did a very good job of organizing education, entertainment, public activities, all kinds of things to keep people happy. They convinced the Arctics to allow food to get droned in, so aside from being a small enclosed space it's been fine."

"We had Arctic food out of curiosity a few times here", Febric added. "They've figured out like a dozen substances that taste like sugar but don't have any calories. They also have protein contraptions that taste like fish and meat. It's not bad! Whatever taste you want with whatever nutrition profile you need, the slogan goes."

Seila lowered her face quietly. She was still not entirely used to having to admit that the Arctic ways of doing things sometimes did actually have upsides and they could learn from them.

"How about I invite some of our new friends", Gladias interjected, "and they can show us what actual properly cooked Dzego food is like?"

"Yay!", Lily shouted.

Febric and Hreda also nodded eagerly.

"Then I'll ask them to come tonight. I have to go out again this afternoon. There's a military funeral for Den and our fallen soldiers at the top of the mountain. Then we can come back down together and enjoy a nice meal."

Lily, Febric and Hreda nodded their heads with excitement.

Chapter 32

Meldan, Veridia · 3725 RAINMOON 12

Gladias and Seila walked together along the same path through Kalimar that they had by this point taken hundreds of times. The path was covered with trees and surrounded by stone houses on both sides.

To their left, they saw a man climbing a tall ladder leaning against a tree. Once he reached the top of the ladder, he began unscrewing a security camera that was located there. Once he finished, he unceremoniously threw the camera into a box that was several meters below him on the ground. The throw landed perfectly. Gladias and Seila heard a loud clang as the camera crashed against the dozens of other cameras that were already in the box. The man immediately proceeded to climb down the ladder.

They continued walking. They passed by a store, one that Gladias had remembered seeing hastily replace much of its inventory with emergency preserved food rations only months earlier. This time, the inventory item that took up the front shelves was a particular type of red flower that was native to Northglade, that residents were buying to celebrate the province's liberation.

Taped to the side of the store, there was an advertisement:

> *We are now offering the freshest of fresh water SEIZED from the CORE TERRITORY of the ARCTIC EMPIRE.*
>
> *Get the latest Hydrafill water, bottled at the springs of Thaldur, today!*
>
> *Enjoy the spoils of victory. Buy Hydrafill.*

The road transitioned to dense shrubbery, and then to a sky bridge overlooking a busy highway. To the left, they could see a mountain covered with trees. A few apartment buildings about fifteen storeys tall, decorated and shaped like oversized medieval castle towers though with far more windows on the sides, poked up high above the trees.

After crossing the sky bridge, the road transitioned into a tunnel, and Gladias and Seila arrived at a Y-shaped intersection. They turned left, walked forward until the path came back up above ground, and then turned right. Gladias checked his watch for directions. He had never actually *walked* all the way to parliament before.

Twenty minutes later, they arrived at the large circular building. Gladias tapped his watch against the green circle, and the doors opened. Inside the building, he remembered the directions well. Forward, then right turn, then forward, then third door on the left.

He tapped his watch again, and the door opened.

Verdow and Zei were both sitting inside the office. They both turned their heads toward Gladias and Seila.

"Gladias, Seila, welcome back!", Verdow exclaimed with a warm friendly tone.

"Verdow, good to see you again!"

"Please, please sit down."

A robot slid into the room.

Gladias briefly turned to face the robot. "dia jie fe kai ja cu dze kai lin pan", he told the robot in a deliberately clear tone.

The robot buzzed, and left the room.

"Impressive, Gladias. Even I would not have been able to do that."

Verdow briefly went silent, and then spoke up again.

"Okay, I'm embarrassed. What is dze kai lin pan?"

"Vegetable and rice."

"Ah right, green plant line bread."

"By the way Zei, when are we going to start switching over from meters to lightfeet?"

"Fin tells me the Bansunpei is completely busy on getting Dzego's privacy and security habits fully up to the new standards. In about ten days he and some others are actually going to come here to Veridia, and see if they can help you guys get up to speed on those things too."

"But I guess we don't have to wait for them, if you like lightfeet you can just start using lightfeet today."

"I guess they're right. Switching over to lightfeet isn't the highest priority right now."

Verdow's tone suddenly became more serious.

"A lot of people are celebrating this month. For the first time in a century, the Arctic Empire has been defeated. Northglade is free. And we already have many people in Thaldur taking up our offers and helping open up the Arctic technologies we captured so that we can properly benefit and learn from them."

"But we have to remember - this was only the first round. And even this round, we did not fully win."

"The Arctics lost Northshore and lost Thaldur, but they kept Redshire. They lost some of their lead in technology, but we've learned over the past few months that they are rapidly gaining in other kinds of technology."

Verdow sighed, and then continued giving his cautionary note.

"The privacy that we thought we had, we do not actually have. In the digital world, we're okay - cybersecurity and cryptography are amazing now, though we have to remember that language models will always be attackable in ways that regular software is not. In the physical world, it's much harder. Information in physical reality is fundamentally much more leaky. There are dozens of ways even our keystrokes on our keyboards leak out into the world outside. We can make an effort to fight this - especially in our private spaces - but there's a limit to how much we can succeed."

"I hear there's already a batch of the new cameras from Dzego flying in, with built-in hard guarantees of what information they can and can't transmit. VNU is figuring out the process for inspecting

them. They seem to want to just copy the way that Dzego has been doing it."

"Ooh, so that's what it was? We saw someone unscrewing a hundred cameras in Kalimar when we were walking to get here."

"Looks like it."

Gladias decided to change to a different topic.

"What do you think about our government?", he asked Verdow.

"The Order of Steering worked, the Courts worked, Graph Funding has been fine. But everything more transparent has been broken. By default, I expect that everything the government does is going to become even more private going forward. More digital-first, more sealed anti-transmission rooms, more privacy robes."

Verdow sighed, clearly at least somewhat unhappy with the state of affairs that he was predicting.

"There's a fundamental tension here that our system of government has been grappling with since the day it was first created. In an interconnected world, how is it even possible for a government to be accountable to its own citizens, without also being 'accountable' to powerful financial interests on the inside and adversaries on the outside, both of which are trying to press at the weakest point, isolate it and pressure and twist it to serve their own ends?"

"We need to keep improving our theory of what transparency and accountability even mean. In this current iteration of our government, the answer has been the selection process from one end, and Heralds from the other end. In the next iteration, it might have to be something else. I think it will have to involve something like Silverchat polls, or maybe something even higher bandwidth, to understand people's goals, and something involving AI to figure out how to achieve those goals."

"Zei, what do you think?"

"AI is the start of the conversation, not the answer. What kind of AI is it? How do we train it? Who trains it? What instructions do we give it? Who feeds the AI information, and who intervenes in the AI's thought process to assist it?"

"How would you answer that?"

"I would first take a step back and ask what the problems are. And I think I have noticed one. Veridian institutions, as they exist today, don't really have a built-in concept of predicting the future. It's all about the present, and public opinion in the present."

"There's no position where you have to stand up and say - if we do X, then Y will happen - and you rise up if you're right, and fall if you're wrong. The closest is Acolytes, that have to predict the Sentinels. But even there, they're not predicting the real world. They're predicting Sentinels. So they're still just predicting other people's opinions. You've aligning people to each other, but you're not aligning people to reality."

"That's where AI could do very well - it's where AI already *has* done very well for us. But I don't think we should pick one AI, we should have some kind of open competition of them. AI should be a player, not the game. Let different people choose different AIs and different ways of working with them, and hold them accountable to the goals."

"Should a society have goals?"

Zei stared at Verdow.

"What do you mean?"

"The Arctics are better at having goals than any of our nations. They know exactly what they want, and are single-minded about achieving it. If you make a perfect goal-executing machine, the Arctics are what you get."

"That's why you need the other piece - being able to gather together different people's goals and add them up, so you know that you're achieving the actual goals of the people, and not just one emperor."

"But what if any goal you come up with and write down on a piece of paper ends up looking like that?"

"What do you mean?"

"What if part of the human condition is that we don't fully know what we want, and any attempt to crystallize it into one piece breaks down if you push it far enough? We see this with our rubrics: they

work for some time, then eventually they get gamed, and then we adjust the rubrics, and repeat. What if that's unavoidable?"

"Then what's the alternative?"

"Think of a lake, full of water. Every molecule of water in the lake acts independently. There is no master plan. There are only trillions of molecules dreaming trillions of dreams. The lake has a harmony and balance to it, we can all see that. But it's not because anyone forces that balance to happen, it's because if any one part of the lake's surface ends up lower than the other parts for even a brief moment, then gravity will attract water to flow in, and the height everywhere will be the same again."

"But also, when winter comes, the molecules come together, into thick crystalline structures of impenetrable ice."

"You understand physics well."

"My daughter is starting to study chemistry, she has been telling me a lot about it. Her friends think she's over-working, but I've been encouraging her - I told her that if she studies hard, one day she can be just like you."

Zei blushed.

"So that's what I think the ideal is. A society that can do the good of ordinary life but not the evil of empire and conquest. One that can both resist the Arctic Empire, and avoid becoming one, and instead prosper peacefully one step at a time."

"It's hard to agree on what we want, it's easier to agree on what we don't want. I think that's the asymmetry that we build everything on top of. In the winter, we all align in the same direction, and we can form the crystal, and defend and strike back. In the summer, we flow like water. Decentralized forces - markets, open intellectual discourse, open science, culture and law - passively make the trillion forces of our trillion dreams add more often than they subtract. We adjust their rules over time, but only carefully and slowly. The natural, gentle progress we make from decades upon decades of that is worth more than satisfying any goal."

"What if sometimes different parts of the lake have to be water and ice at the same time?"

"That may be."

"And what if a lake where every point is at the exact same height as every other point is also a kind of uniformity that we don't want, and sometimes it's better if we form snowballs and let them float?"

"That too may be."

"Seila, what do you think we should do more?"

"The Arctics have competition - they care about who's better and who's worse, and they encourage people to challenge each other and strive and try to rise to the top - but they do not understand cooperation. They are ice without the water. Veridia has cooperation, but maybe we do not understand competition. We can be water, but we don't yet know how to be ice - and even our water does not always flow as well as we would like. Actually I think it's Dzego that's figured out how to have both, and make them serve each other. We should learn from them more."

She sighed deeply. After a few moments, Verdow replied again.

"These are all good thoughts."

"Zei, would you be interested in staying back and helping us figure out some of these issues?"

Zei thought for a moment.

"I'm sure Mu and Su can handle the Kungaupei well", Gladias interjected.

"Of course! Though you might find Fin will be even more helpful than I am."

"Well I appreciate it. You know, Parliament realizes now that their current ways of thinking have hit their limits. Many senators have resigned. The ones who are still there - like me - we want new ideas. We're open to listening."

Verdow stood up, and walked over slowly to a map of the world that was printed on the side of the office wall.

"This is our world. Ninety million people. Gladias, Seila, Zei, have you ever wondered what things might be like if it was nine hundred

million, or nine billion?"

"I have not. Why do you ask?"

"The larger the population, the faster new technologies get invented. That's why the medieval era was so stagnant, and why things are moving so much faster today. But no matter how large the population, everyone still adapts to the technology at the same rate. Maybe we've done so well at adapting - eradicating computer viruses, adding so much privacy and verifiability to everything, optimizing our society to be so beautiful and healthy - only because we don't have nine billion people pushing technology forward even faster and increasing the complexity of the society that has to accept it."

"That's interesting. Do you think we will have nine billion people one day?"

"Not one day. Very soon. And not people. Bots. And not nine billion. Nine trillion."

"That's what I worry about. AI is going to really change everything. So far, progress has been moderate, nobody has lost their jobs, it feels more like everyone has been promoted to being a manager - or sometimes a repairman. But at some point, once the AI can do almost everything that humans can do, even if at a few percent the efficiency, our human population won't be a constraint anymore."

"So what do we do?"

"Well, we don't really have much power to be the ones to answer that question if the Arctics continue to stay further ahead."

"If we want to get back into balance, we have to get better at penetrating *them*. Have more of a view into their society and their technology, make sure that for any advance they make, we learn about it. Privacy for the weak, transparency for the powerful. Having those people from Thaldur helping us is great, and hopefully Deluin is far from the only person from Redshire who pretended to join the Arctics but would be interested in passing more information to us."

"There's a known result from game theory", Zei interjected. "If two players are locked in a game that has some mixture of cooperation and competition, then you can unlock strictly higher levels of cooperation

if those players can at least partially read each others' minds. Maybe that's the right outcome to hope for. As time passes, we'll learn more about each other, we'll learn from each other, and we'll come to actually understand each other better."

"I'm not so optimistic. The Arctic Emperor is upset. He can't do anything to us right now, but he's definitely scheming. There will be a next round. It could be five years from now, it could be twenty years from now. If we can at least keep pace on technology, we can survive."

"Do you think we should try to slow *them* down?"

"Do the same thing to them that they have been doing to Dzego for a century? It's not the worst idea."

Zei sighed.

"If we were able to keep advancing under their pressure, they'll definitely be able to keep advancing under any pressure we can bring to bear against them."

"You know, the later stages of all this are going to get crazy. At some point, eventually we'll have drones the size of worms and flies burrowing under the ground, flying through the air and swimming under the sea. We'll have new forms of genetically engineered life that we can't even imagine - both for peace and for war. Some of us might be uploaded and living inside computers. Not this year. Not the next few years. But definitely in the next hundred years."

"Maybe in two hundred years it will be stable. We'll have figured out exactly the ideal type of drone at every scale, the largest as large as aeroplanes or even hollowed-out asteroids floating in space, and the smallest so tiny that there's no more gap between them and viruses. Once we hit the ceiling, there would be nothing else even stronger that could defeat them. Just big distributed immune systems protecting our planets, our animals and our trees, and ourselves. We will be wiser, almost everyone will choose peace over war, and those who don't will have no path to impose their will. And everyone will just live the kind of life they want to live."

"I just hope we have peace", Seila added. "All these questions: genetic engineering, uploading, going to the stars, I don't think we can

get the right answers if we take every step in a rush because we're scared. I'm not sure I want to upload. But a year ago I was not sure if I wanted to live for more than a hundred years. These things should all be a choice we make carefully, and have respect for people who choose not to take the same choice as we do, not an urgent decision we make to survive in a war."

Verdow rejoined the conversation.

"We might have that peace, we might not. We can hope for the best, but we have to be ready for all possibilities."

"You know, a few days ago Bai mentioned to me that she was thinking of spending a year in Thaldur - precisely to better understand how the Arctics think, and see if she can find a path to that peace."

"That sounds a lot less scary than what Deluin did. I support her."

The door opened. A robot slid up to Gladias, carrying his tea and rice and vegetable dish. Gladias picked up both with his hands. He thanked the robot, and the robot slid back out of the room.

"Do you want to go upstairs and check out the roof?", Verdow asked.

"Sure", Zei answered.

Gladias, Seila and Zei stood up. Verdow walked out of the room and began to walk through the corridor. The three of them followed, Gladias still holding his tea in one hand and his food in the other.

Verdow led them up a staircase, and reached a door which opened automatically. They stepped out.

To their left, they could see the main commercial district of Meldan. There were buildings of various styles, some that were aesthetic imitations of something older and almost medieval, and others that were openly modern and futuristic. The building that looked like an aeroplane stood out from among the others, slightly spoiling the view. The trees are only good at covering it up when you're at ground level yourself, Gladias thought.

Further left of the commercial district, they could see the grey and purple Greater Plum Harbor. The whole thing was a square nearly a kilometer wide, and they could see many of its buildings clearly from

where they were. And on the right edge of the commercial district, Gladias could make out the large concert venue.

In front of them, they could see the Kalimar district. From where they stood, it looked not too different from a forest. Tall trees covered up most of the area, with only a few buildings poking up just barely above them. After scanning the area quickly, Gladias spotted the headquarters of Silverchat.

To their right, they could see the mountain covered with trees. They were already partway up the mountain, about forty meters higher than most of the rest of Meldan. But about two hundred meters to their right, the mountain began sloping upward properly. Only about half of the apartment buildings shaped like oversided medieval-style castle towers were visible from where they were. The mountain looked much larger and more imposing from close up, Gladias thought.

"Probably not as beautiful as the views in Northglade?", Verdow asked.

"I like the green here. I don't think I could handle being up north and dealing with several months of no green and only snow."

"And besides, Meldan is my home."

About this book

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